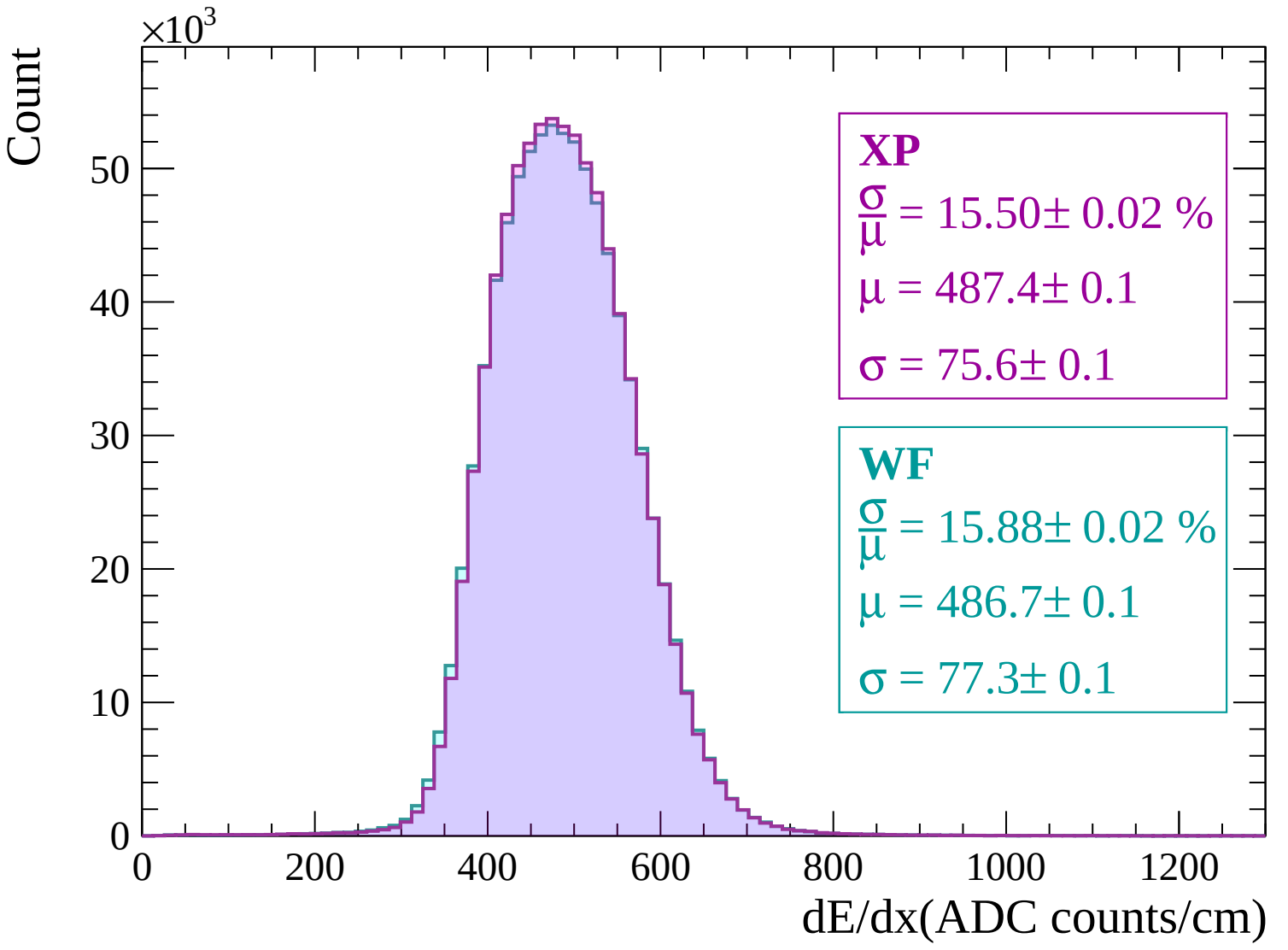
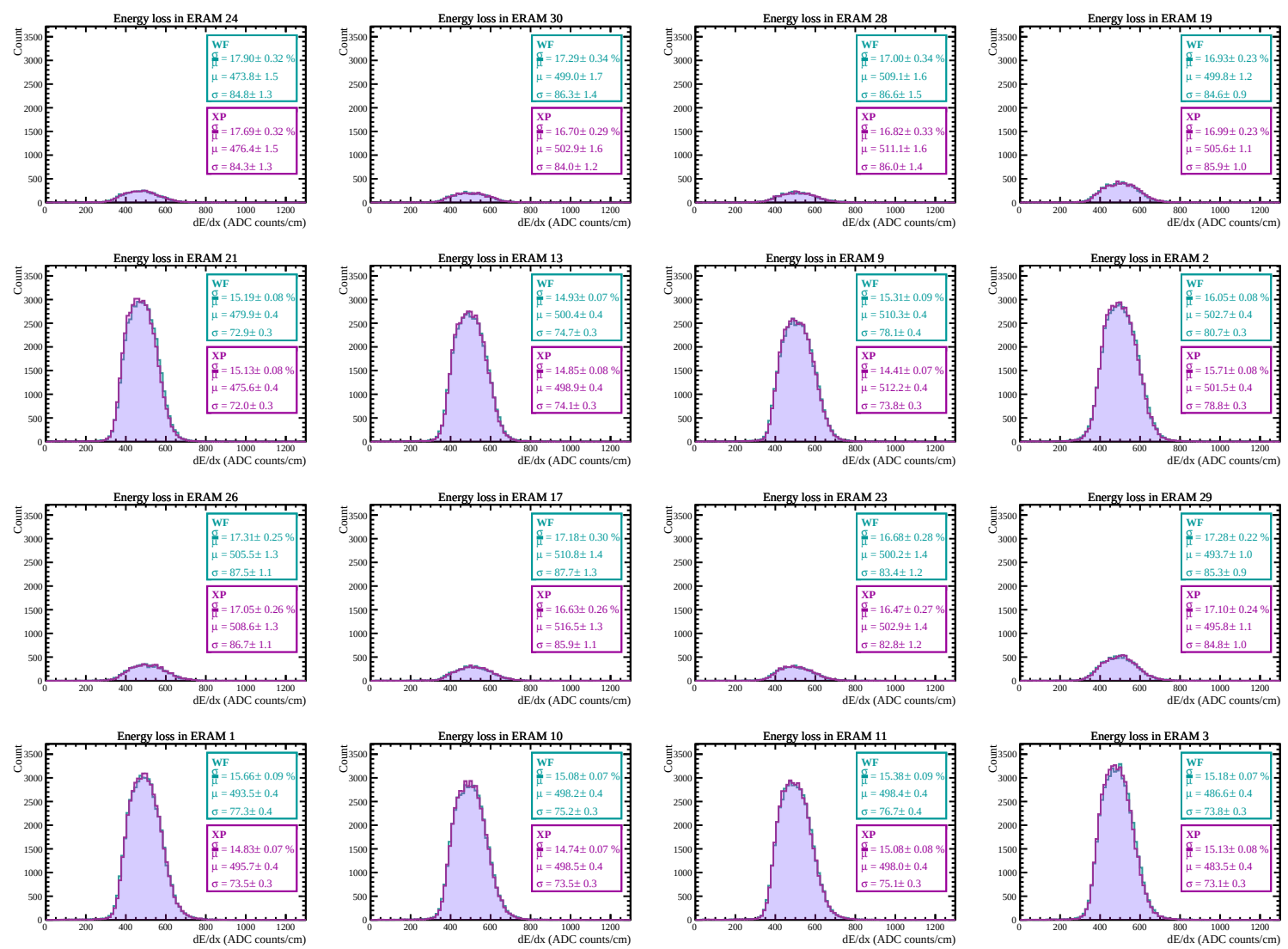
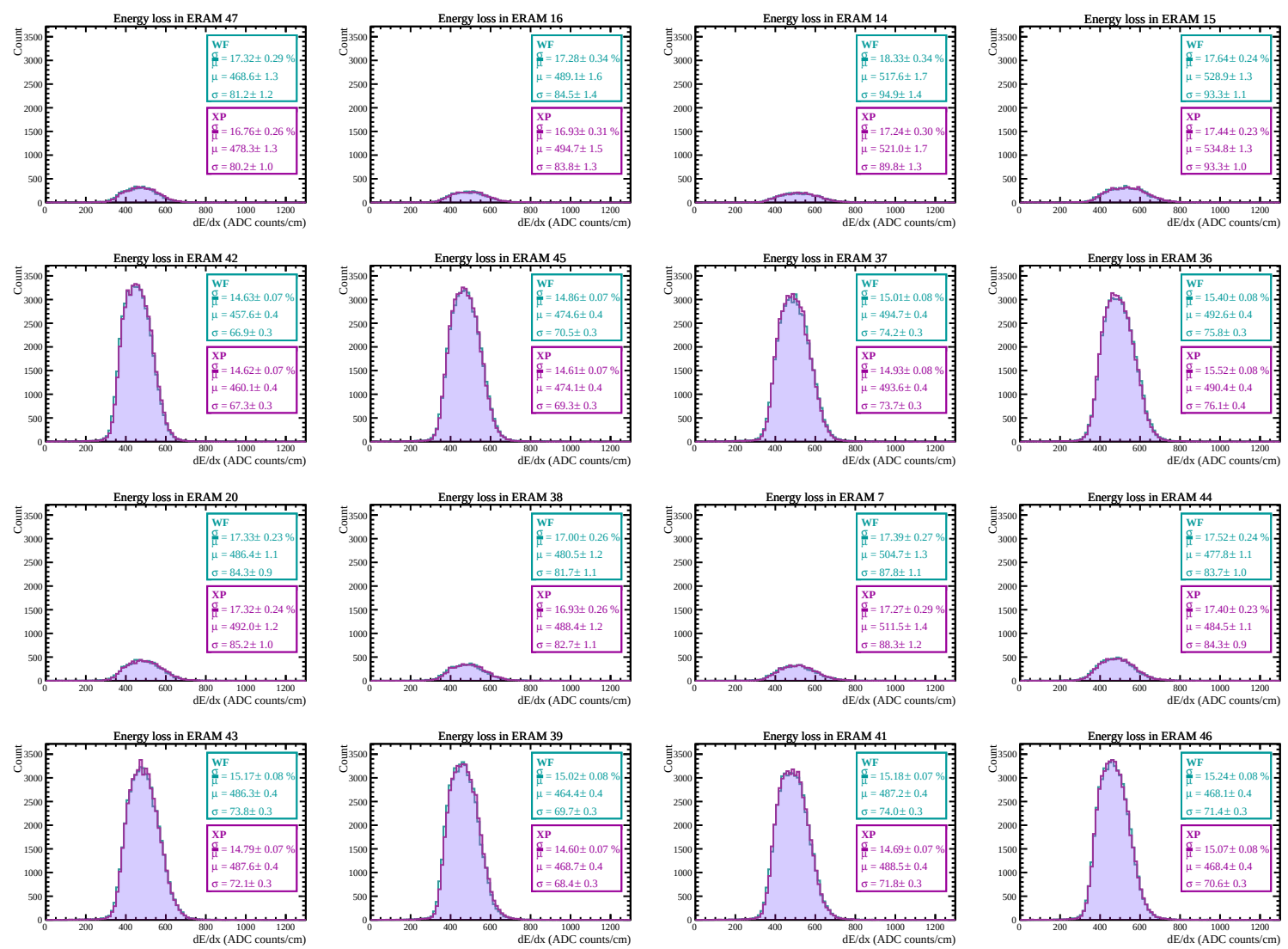
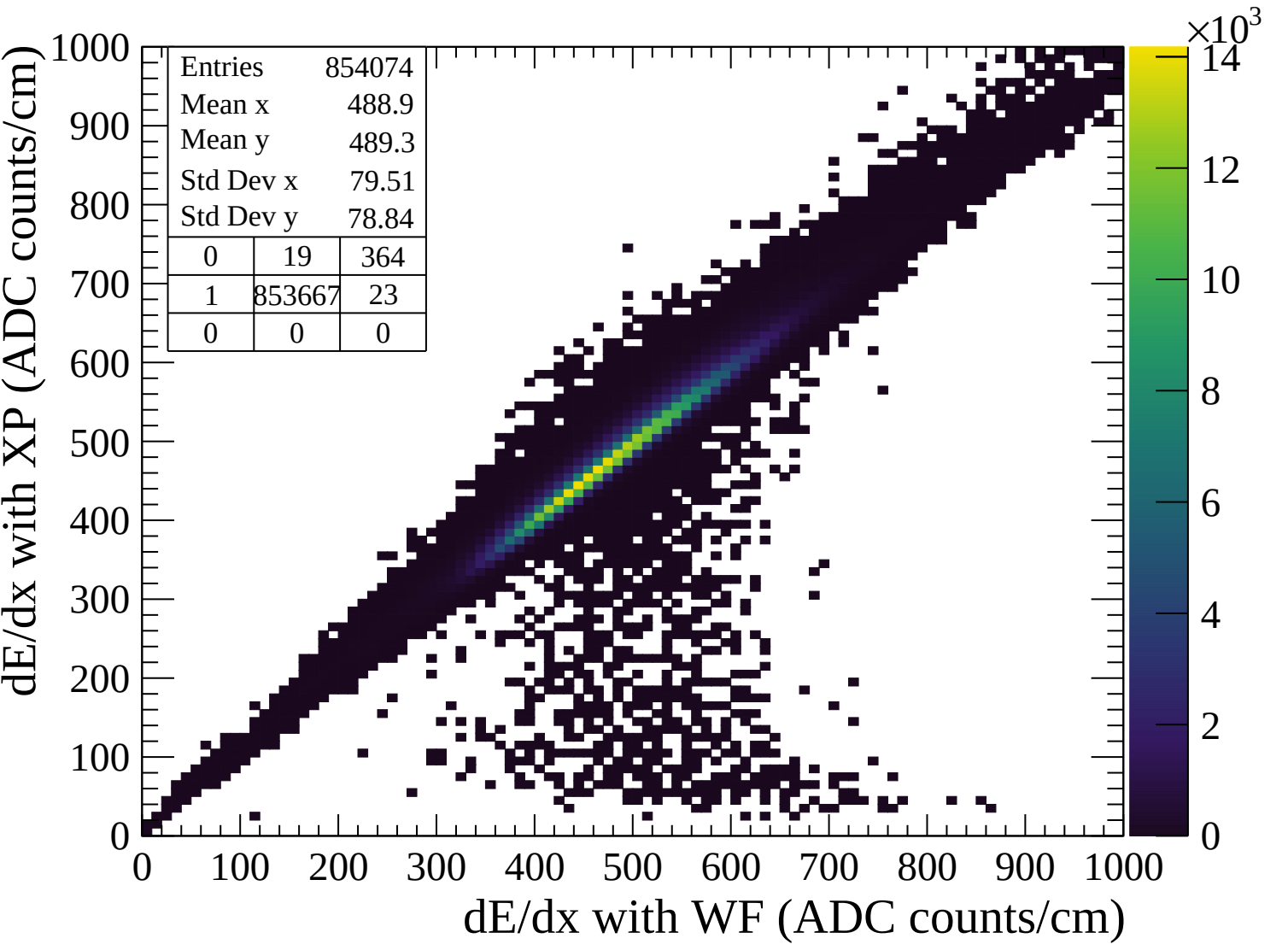
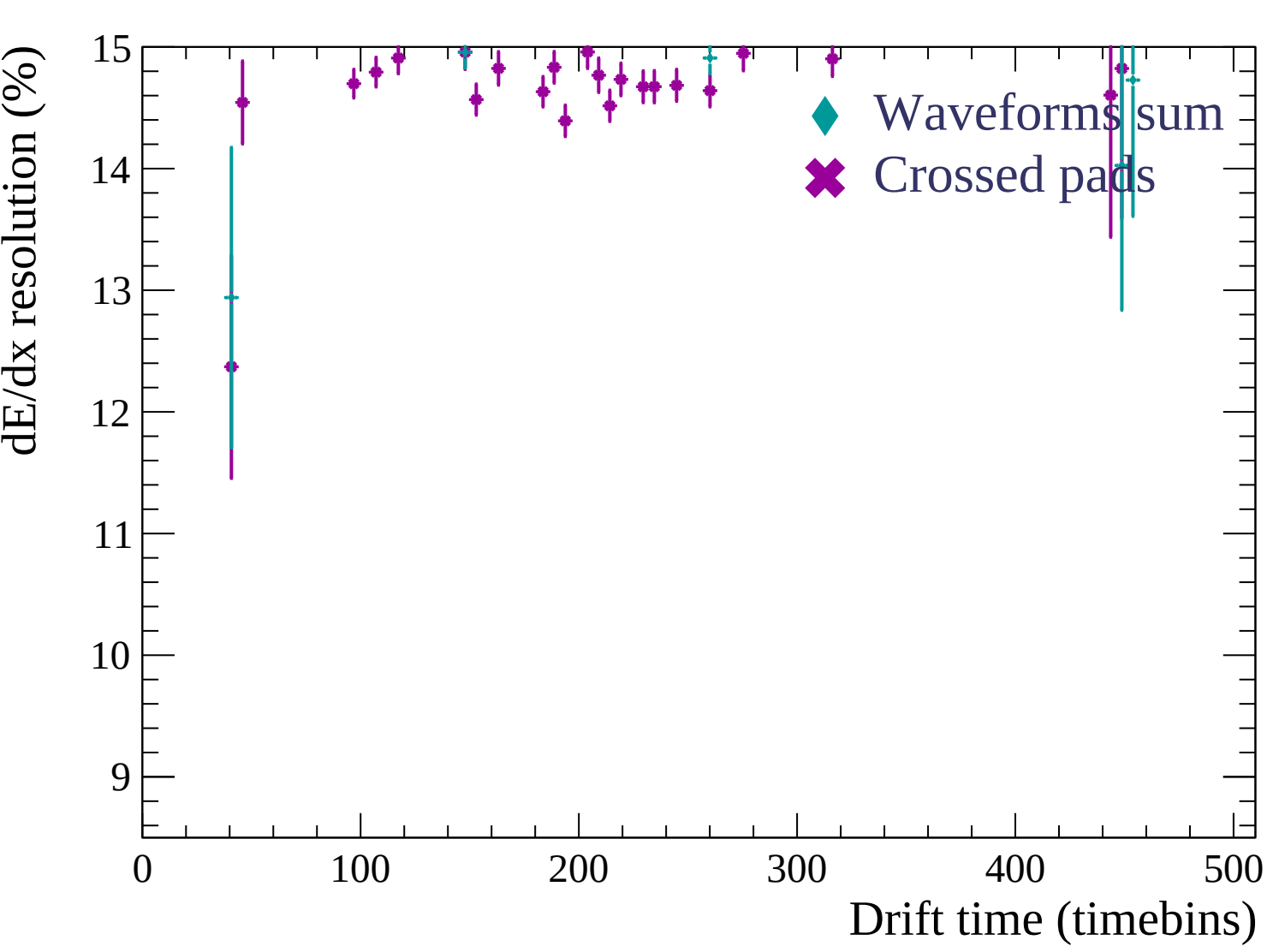


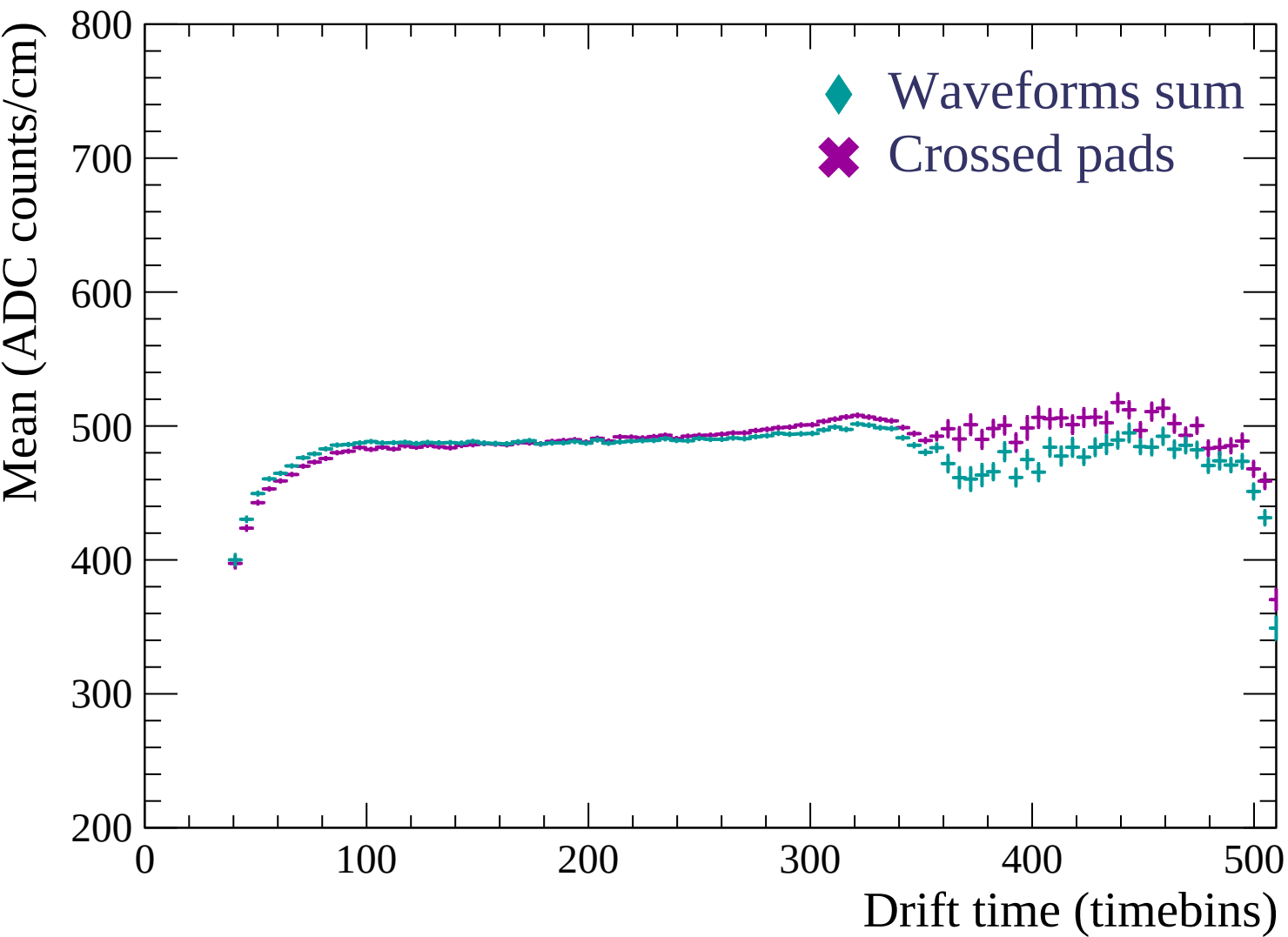


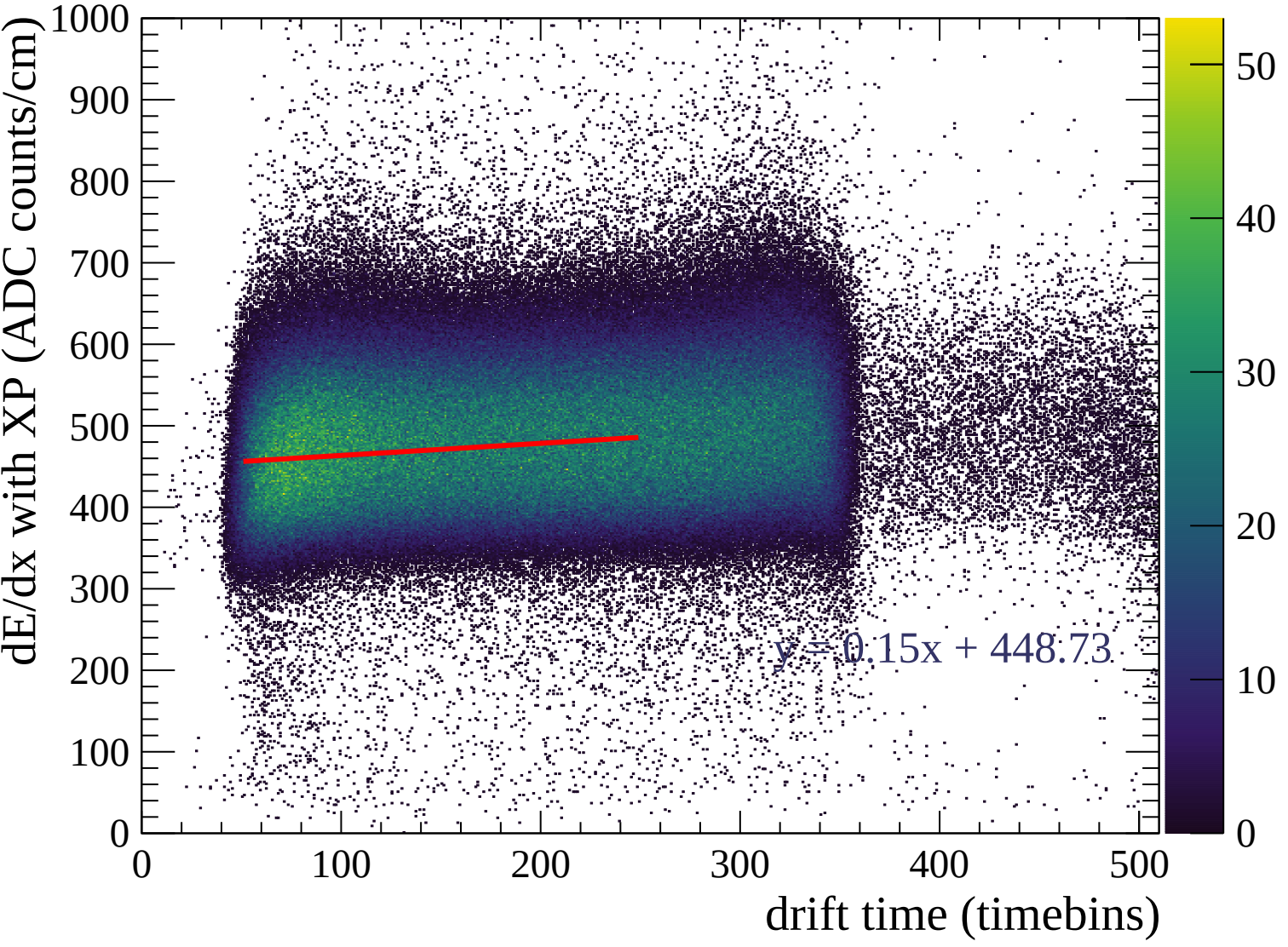


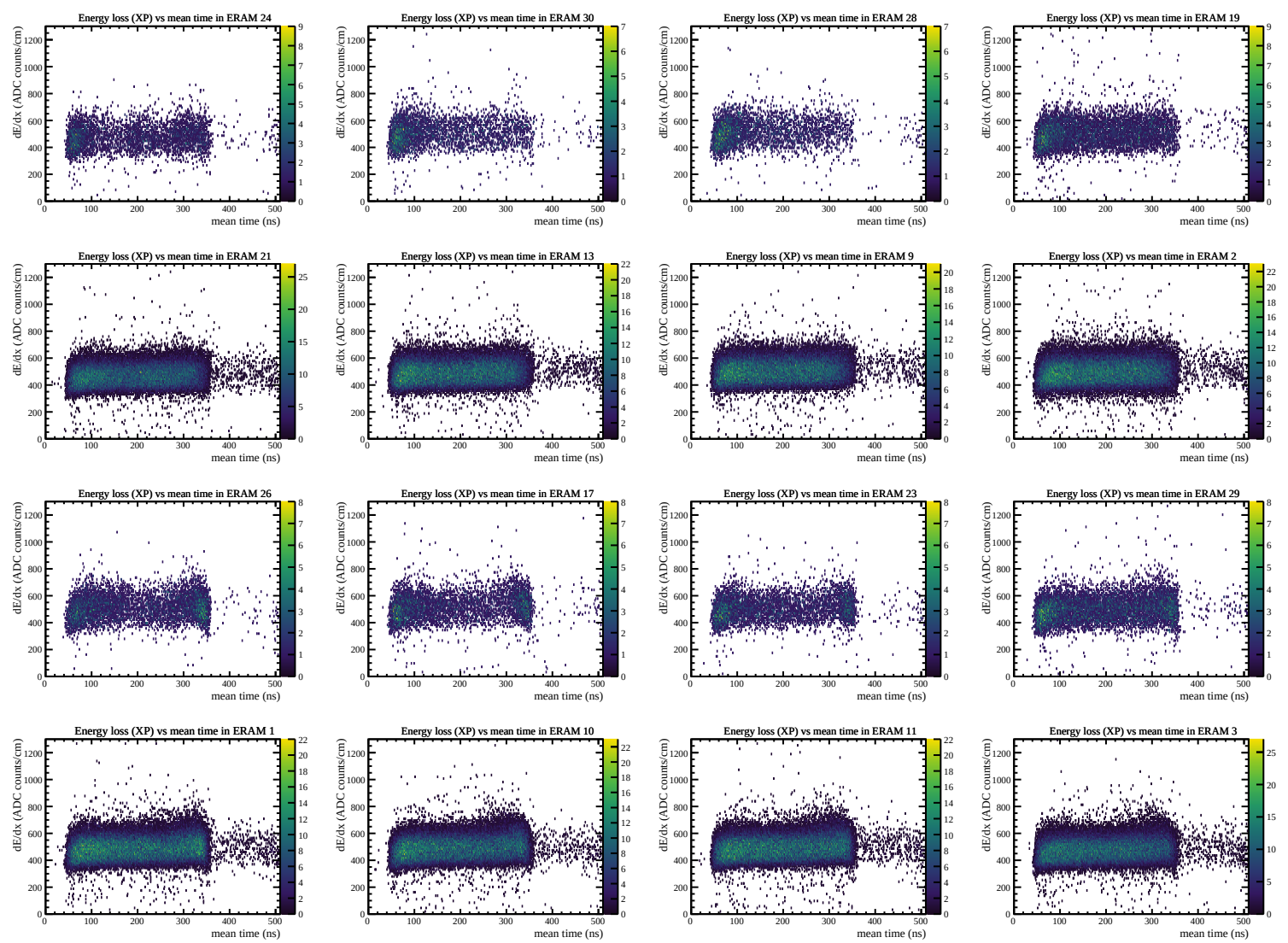


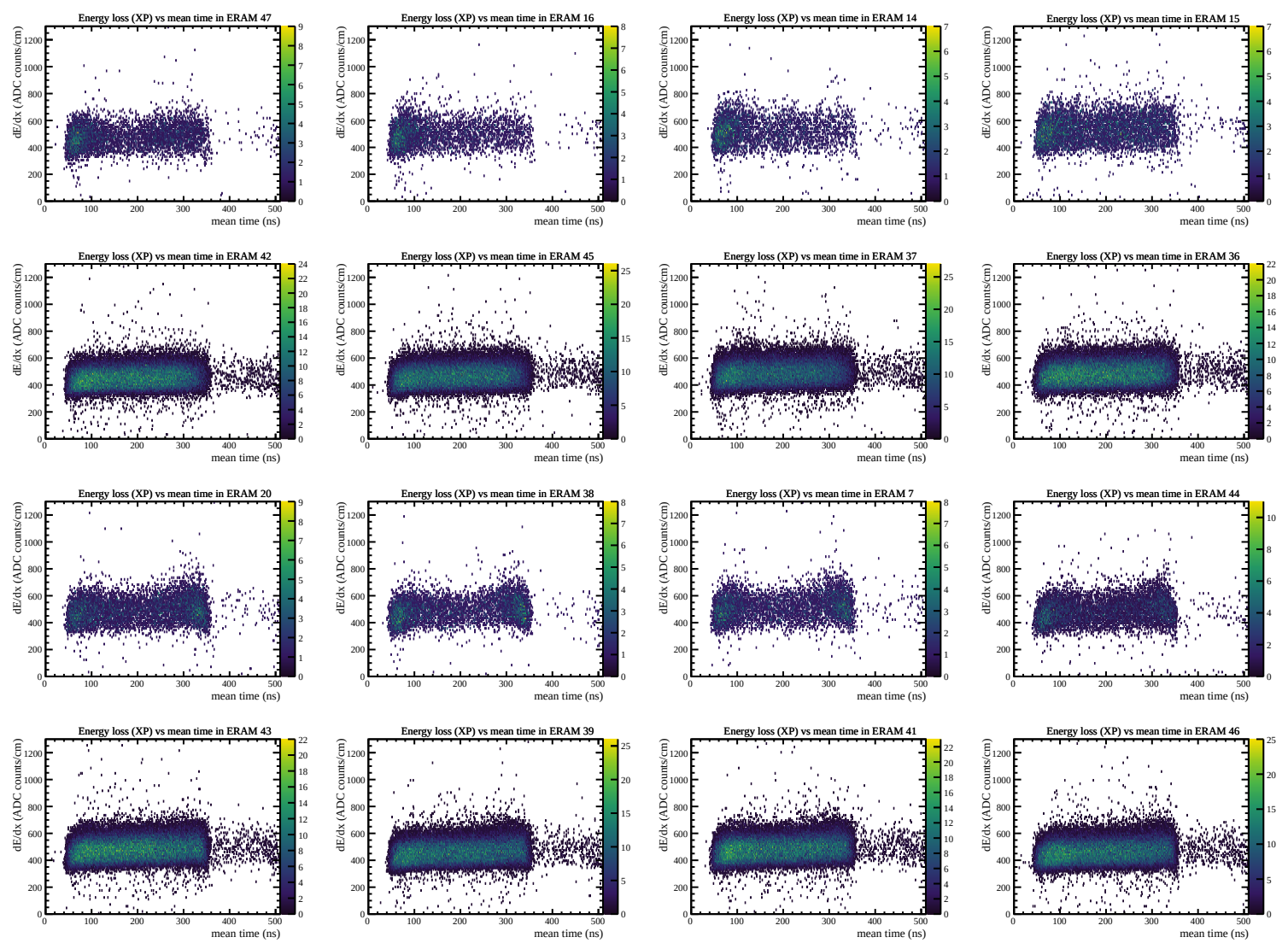


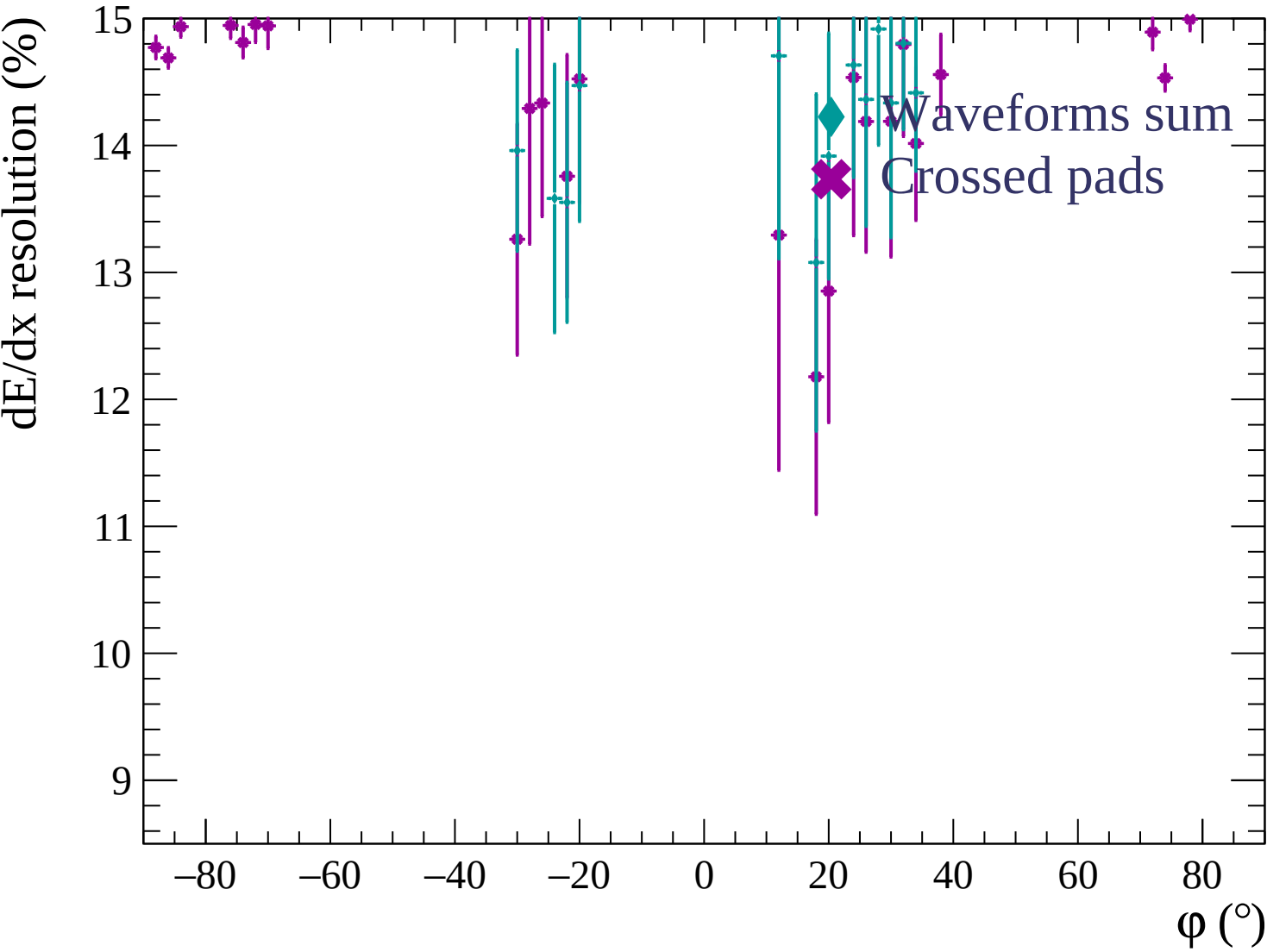


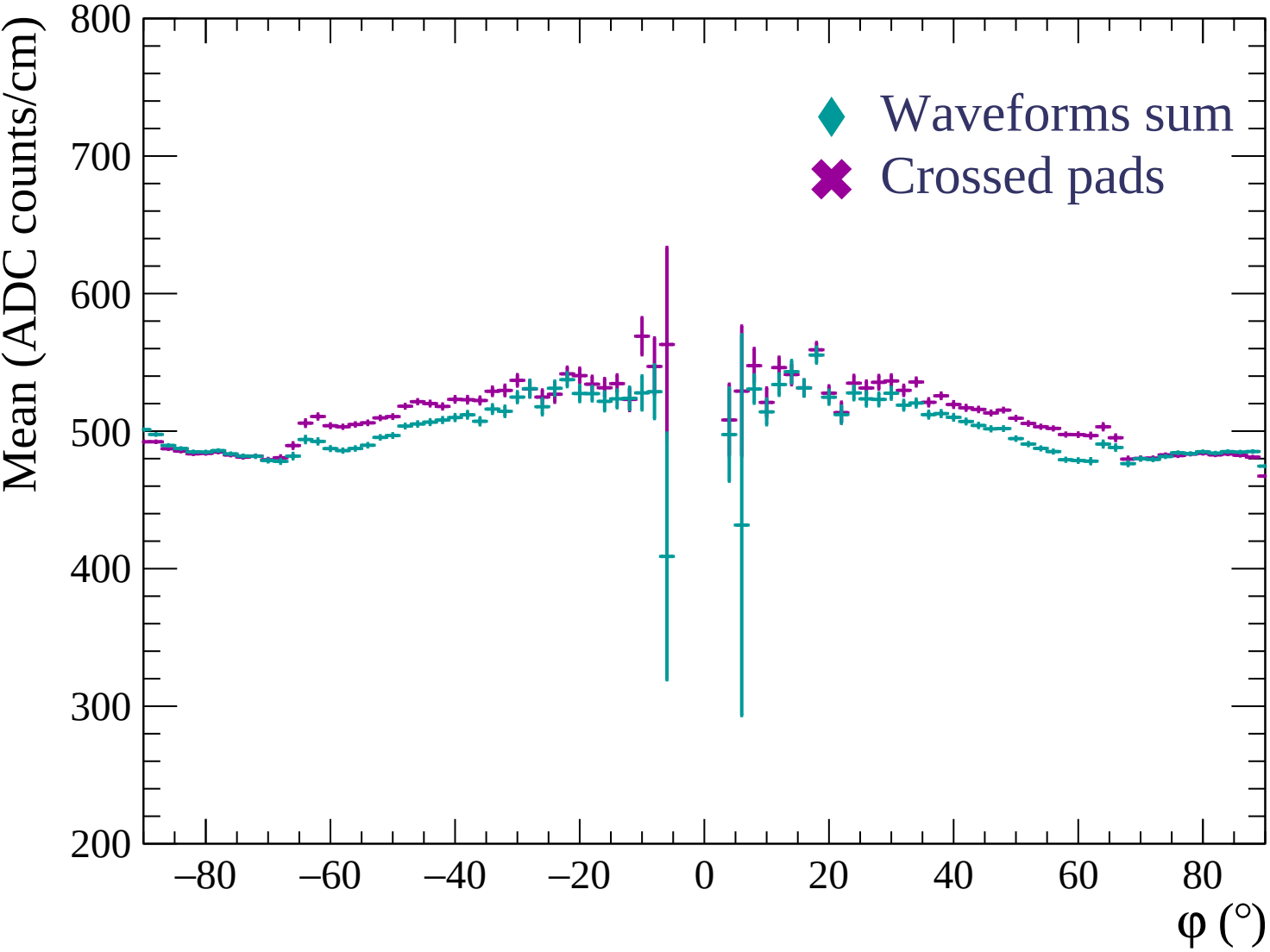


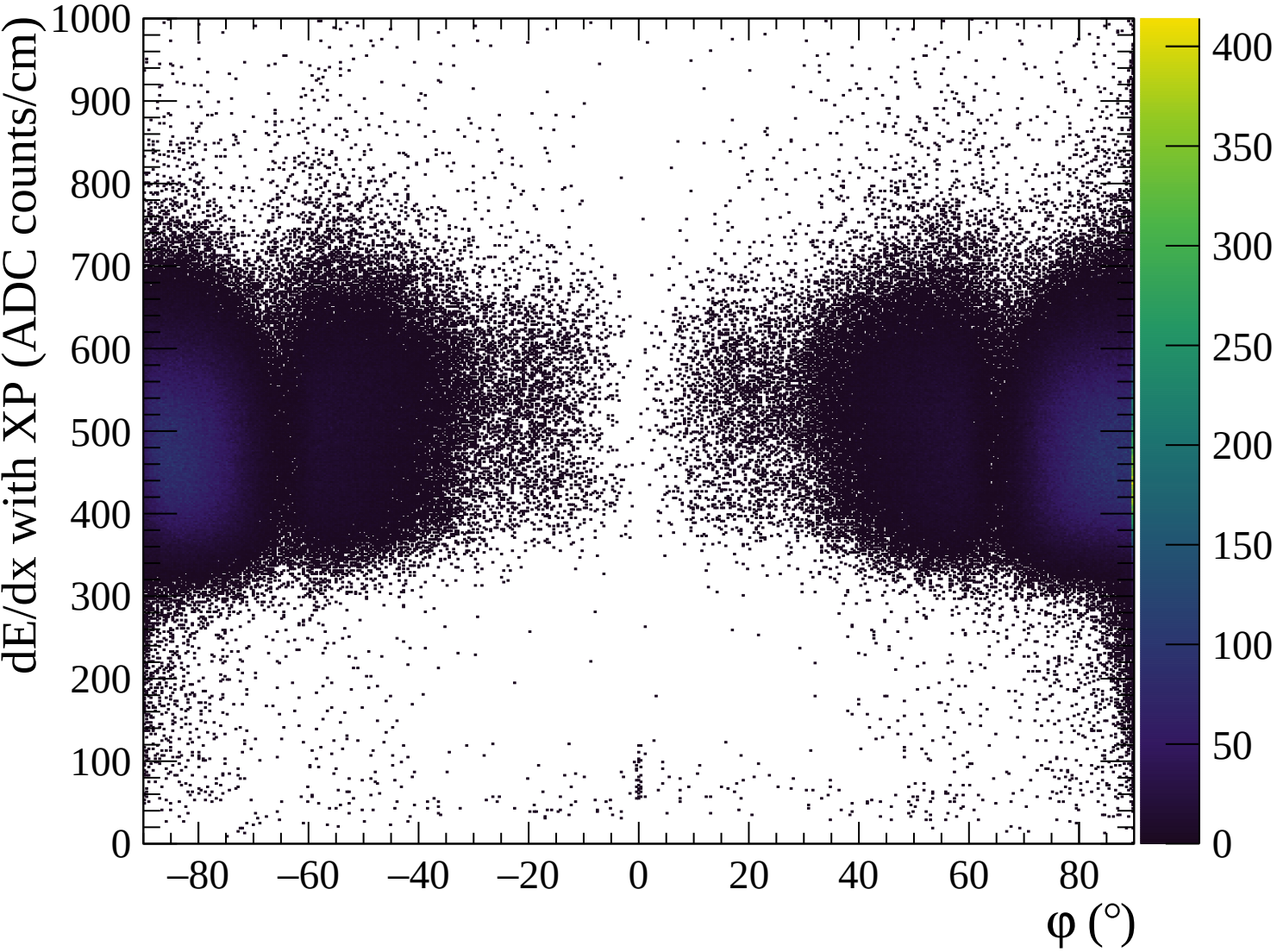


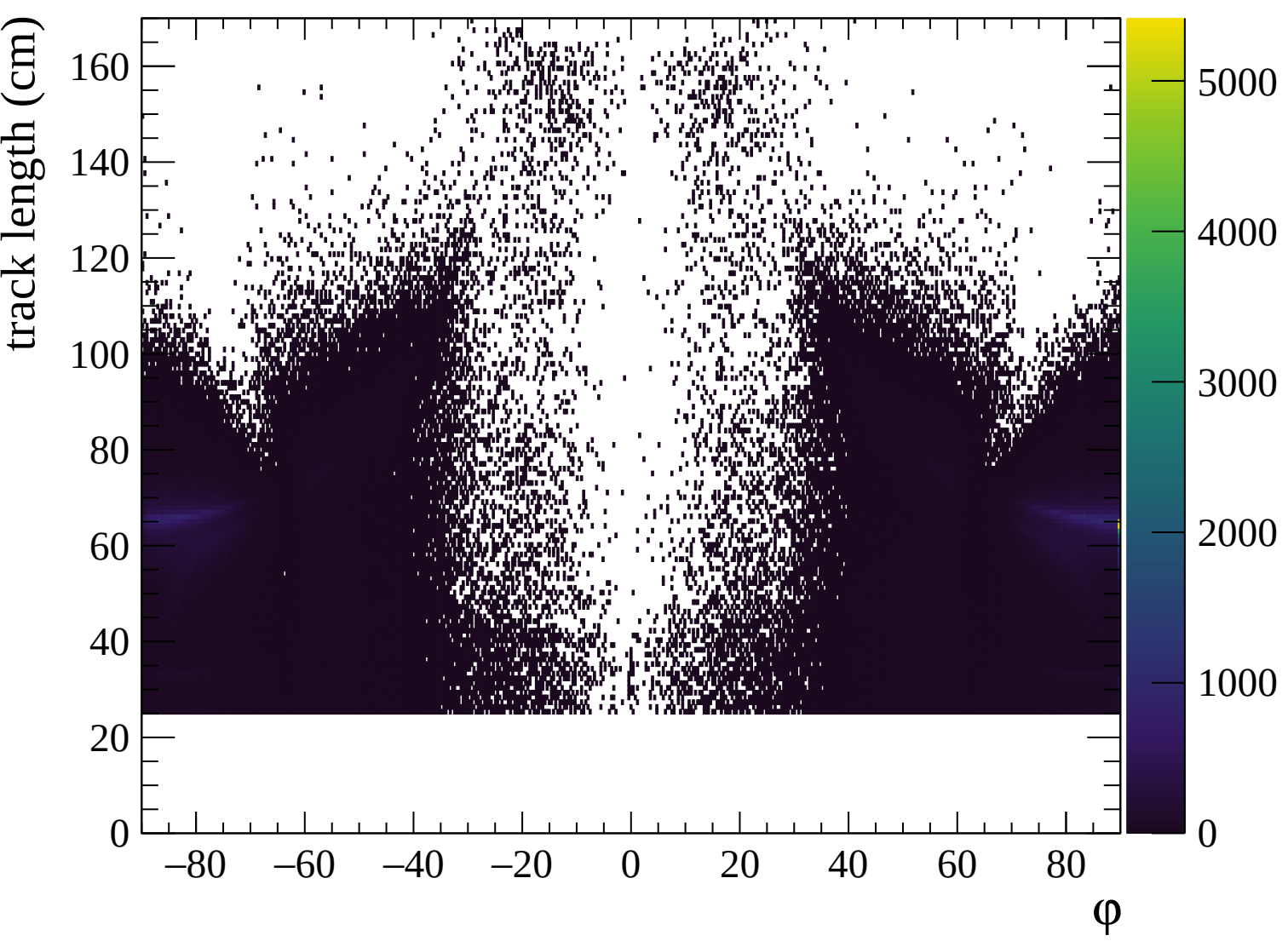


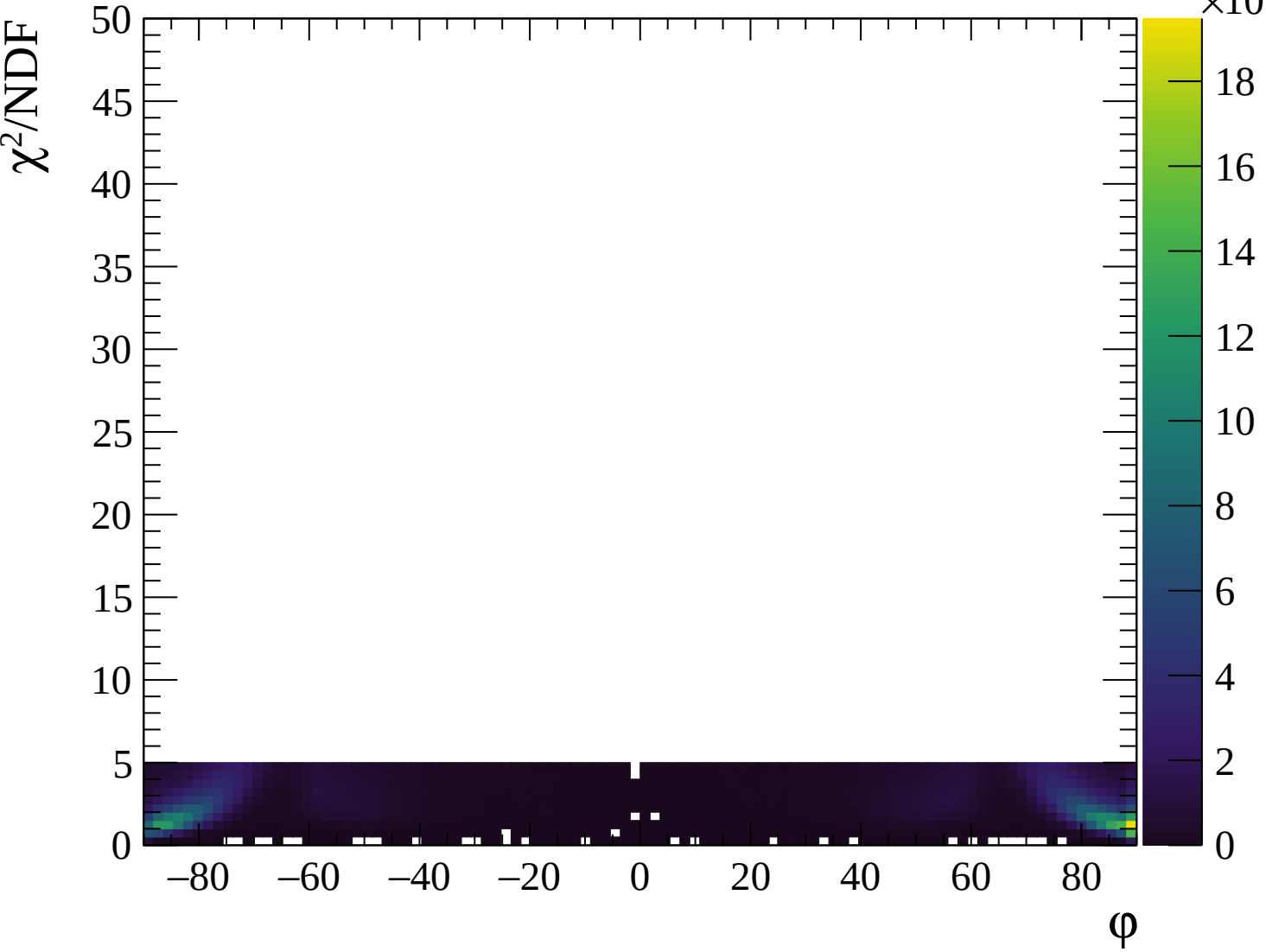


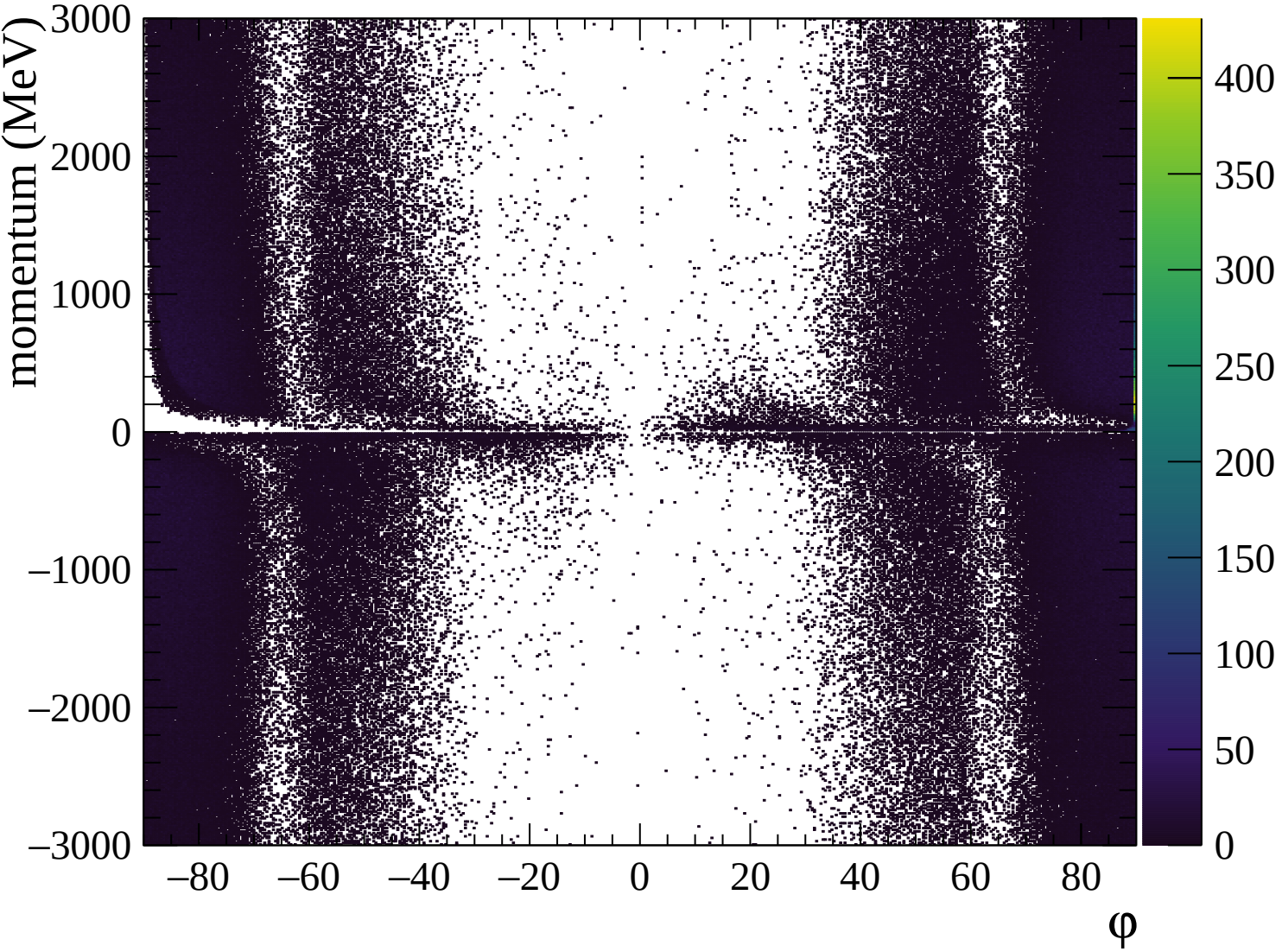


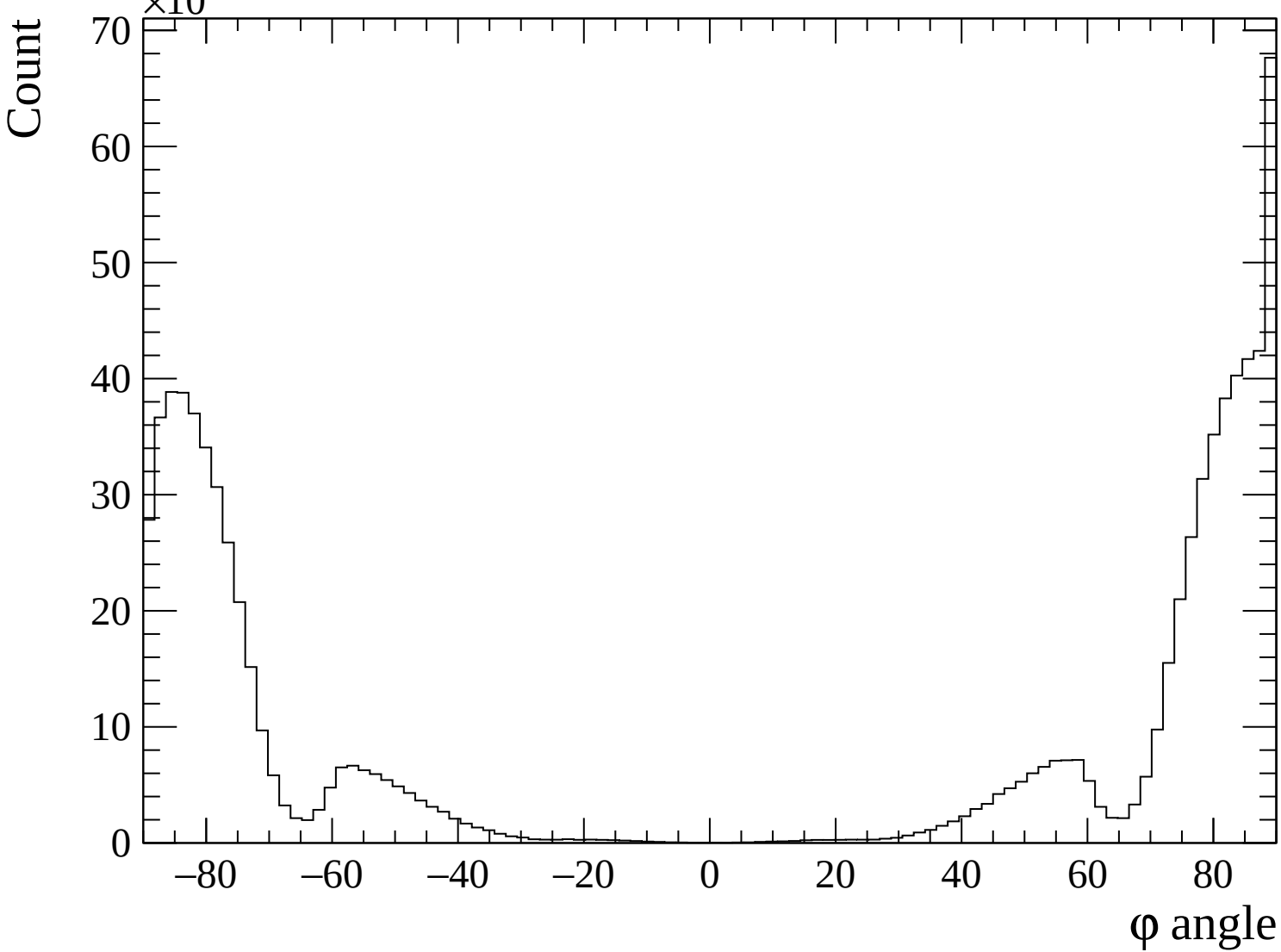


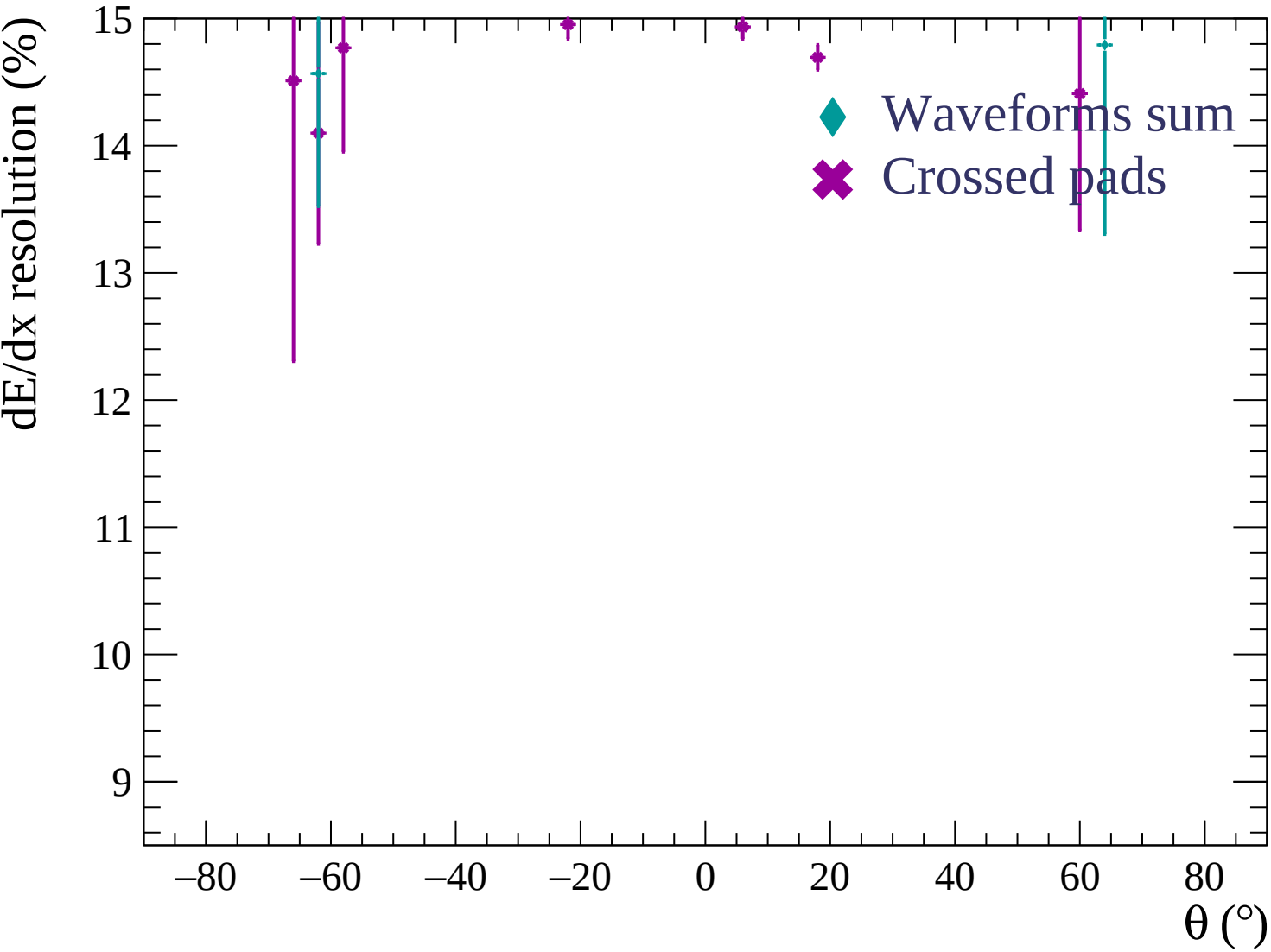


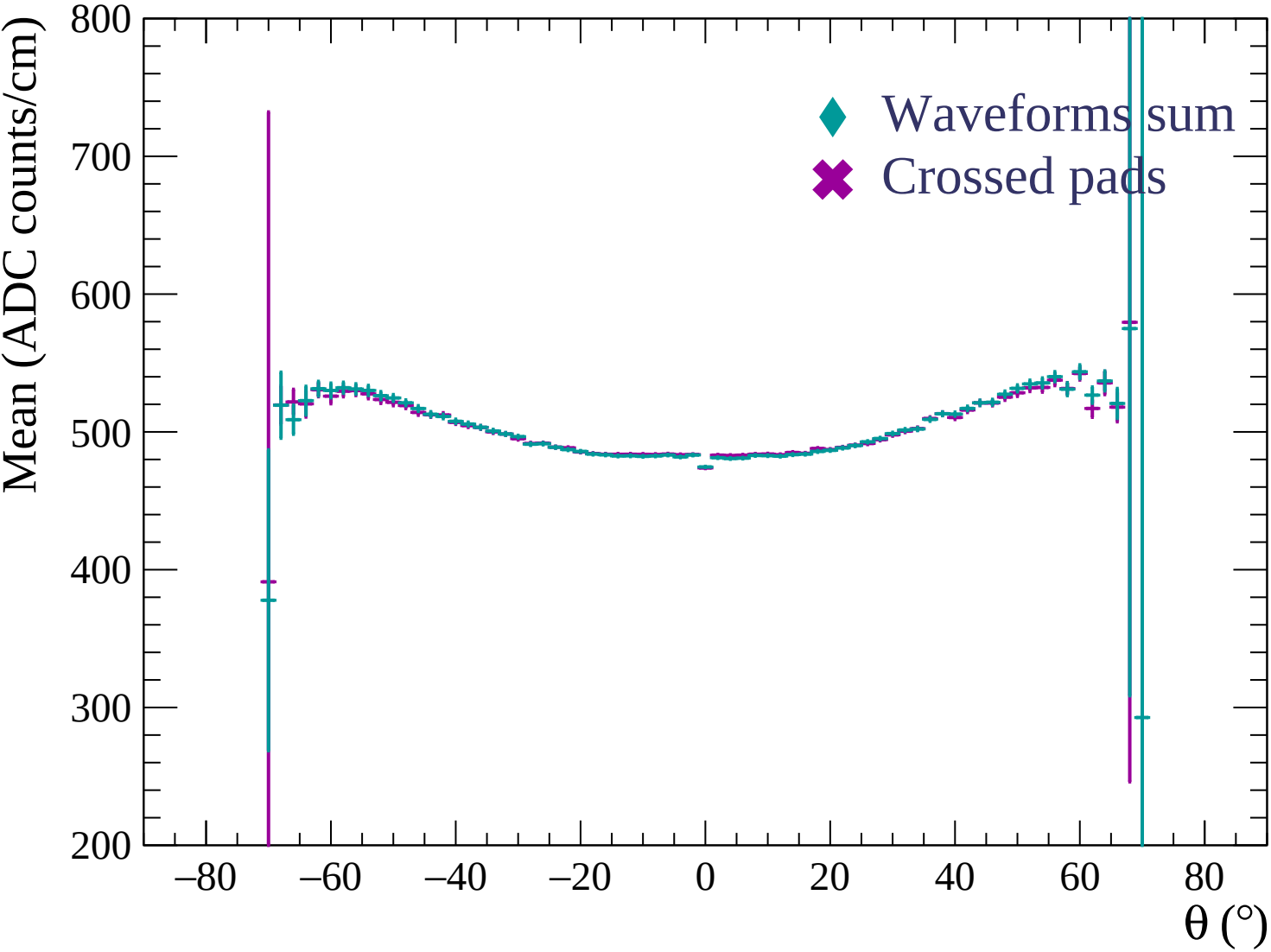


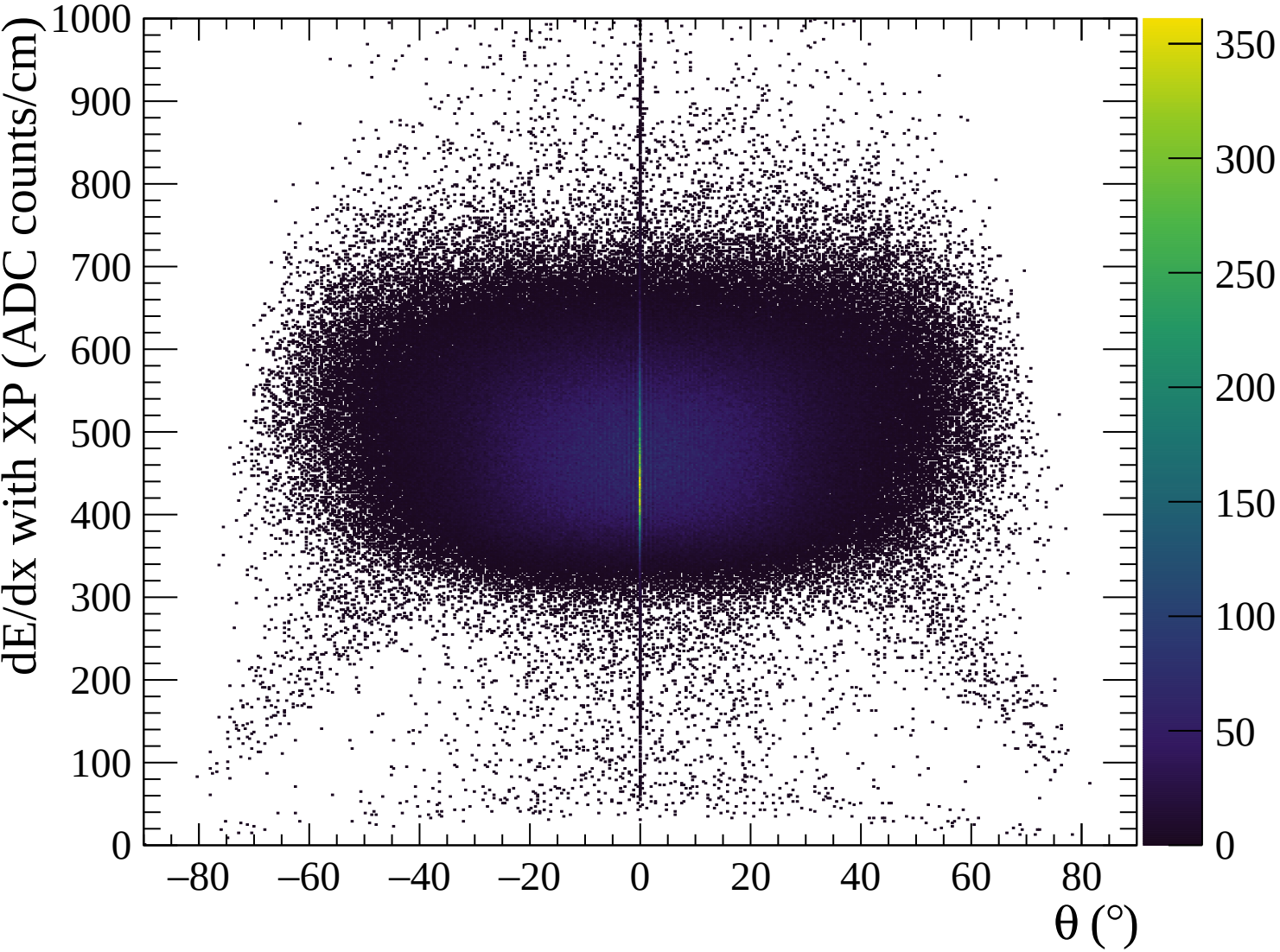


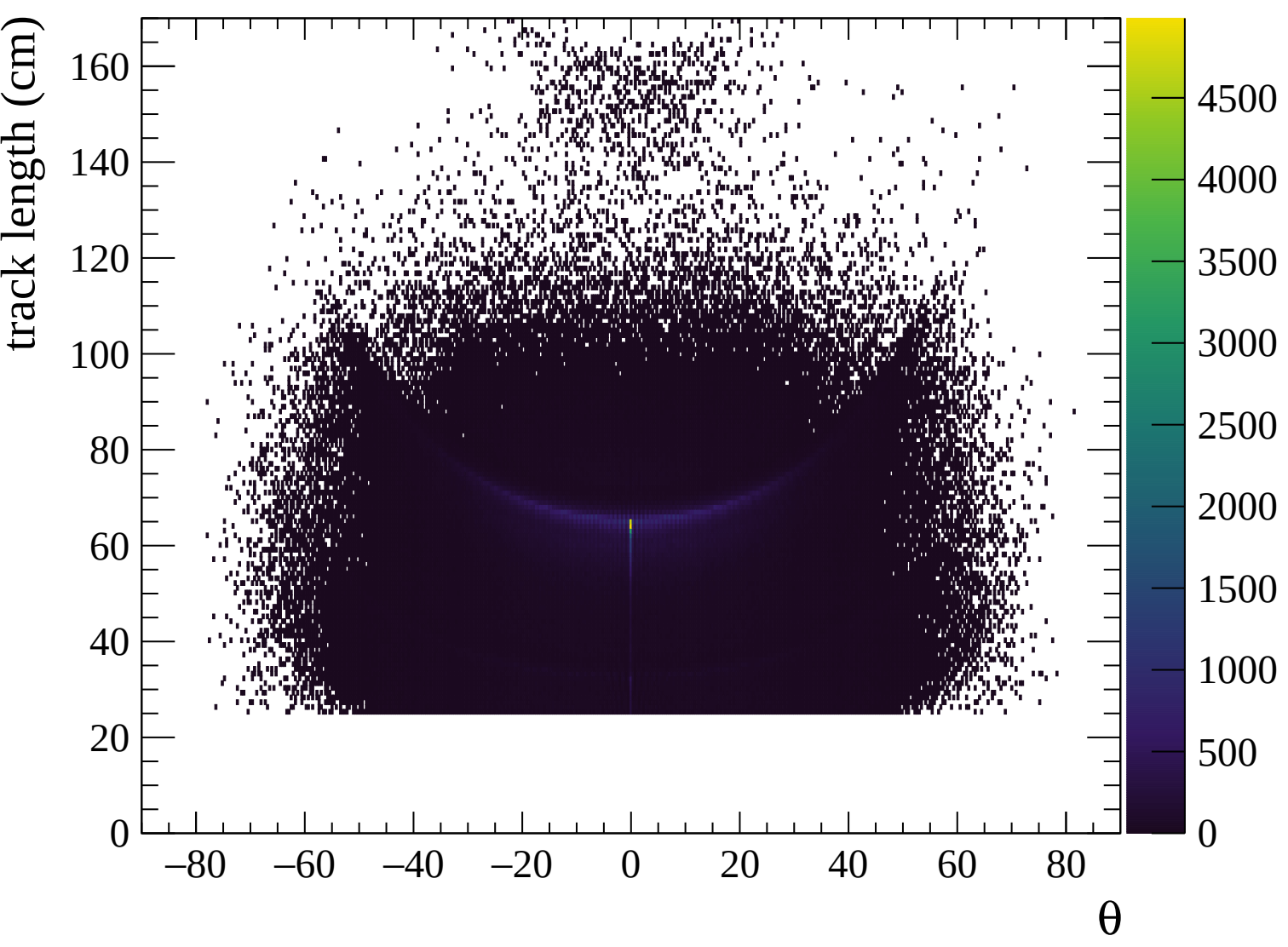


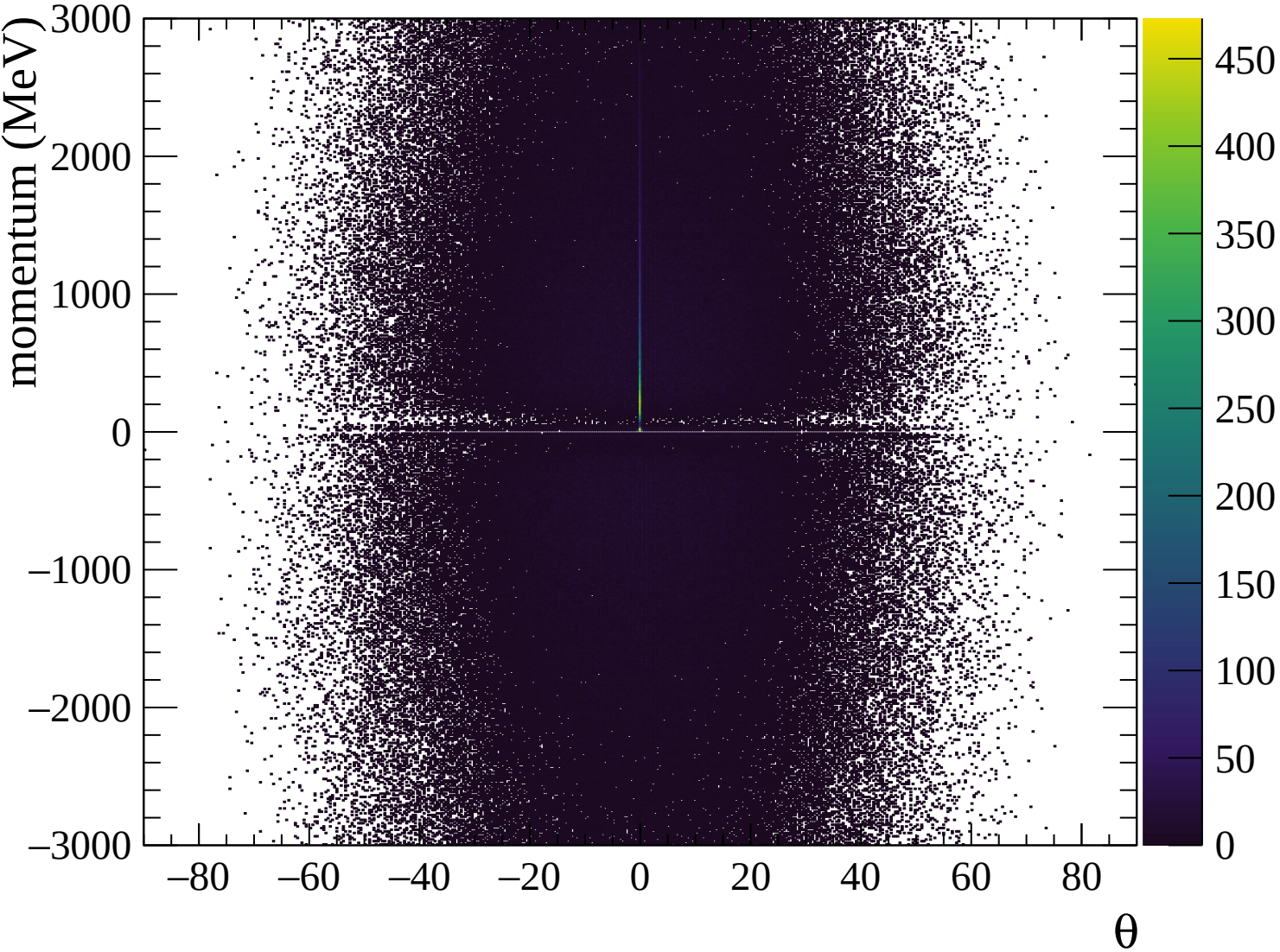


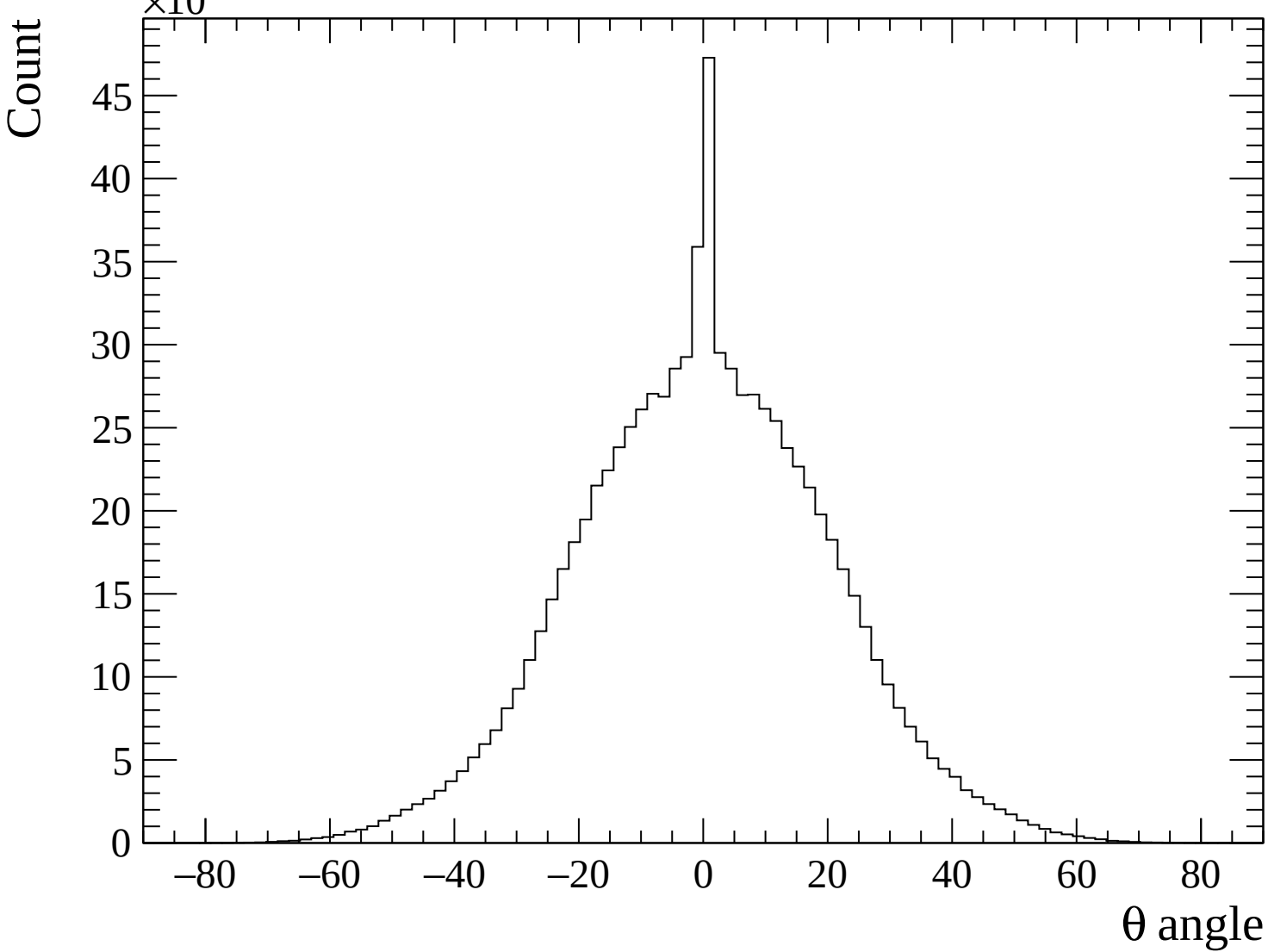


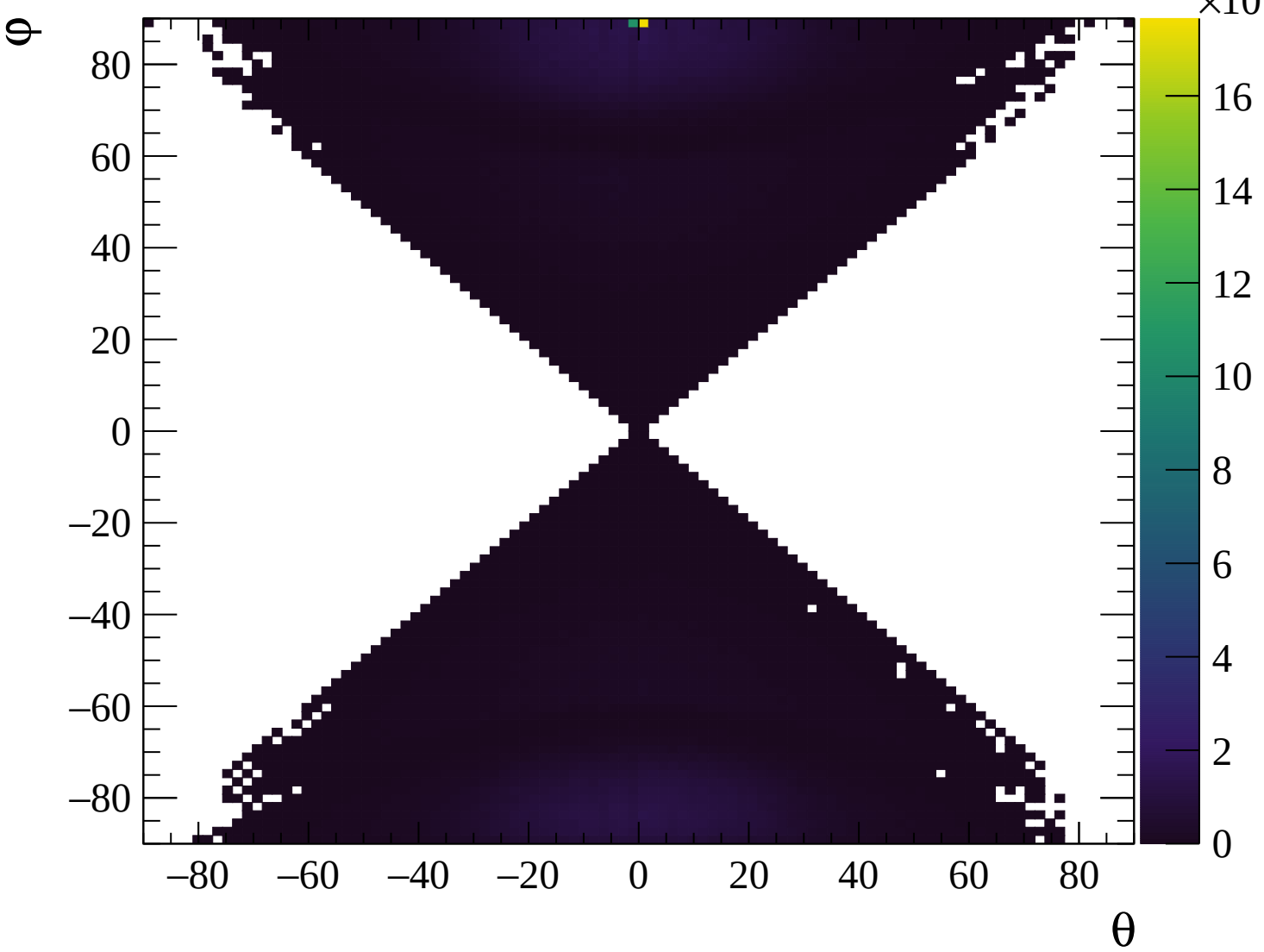


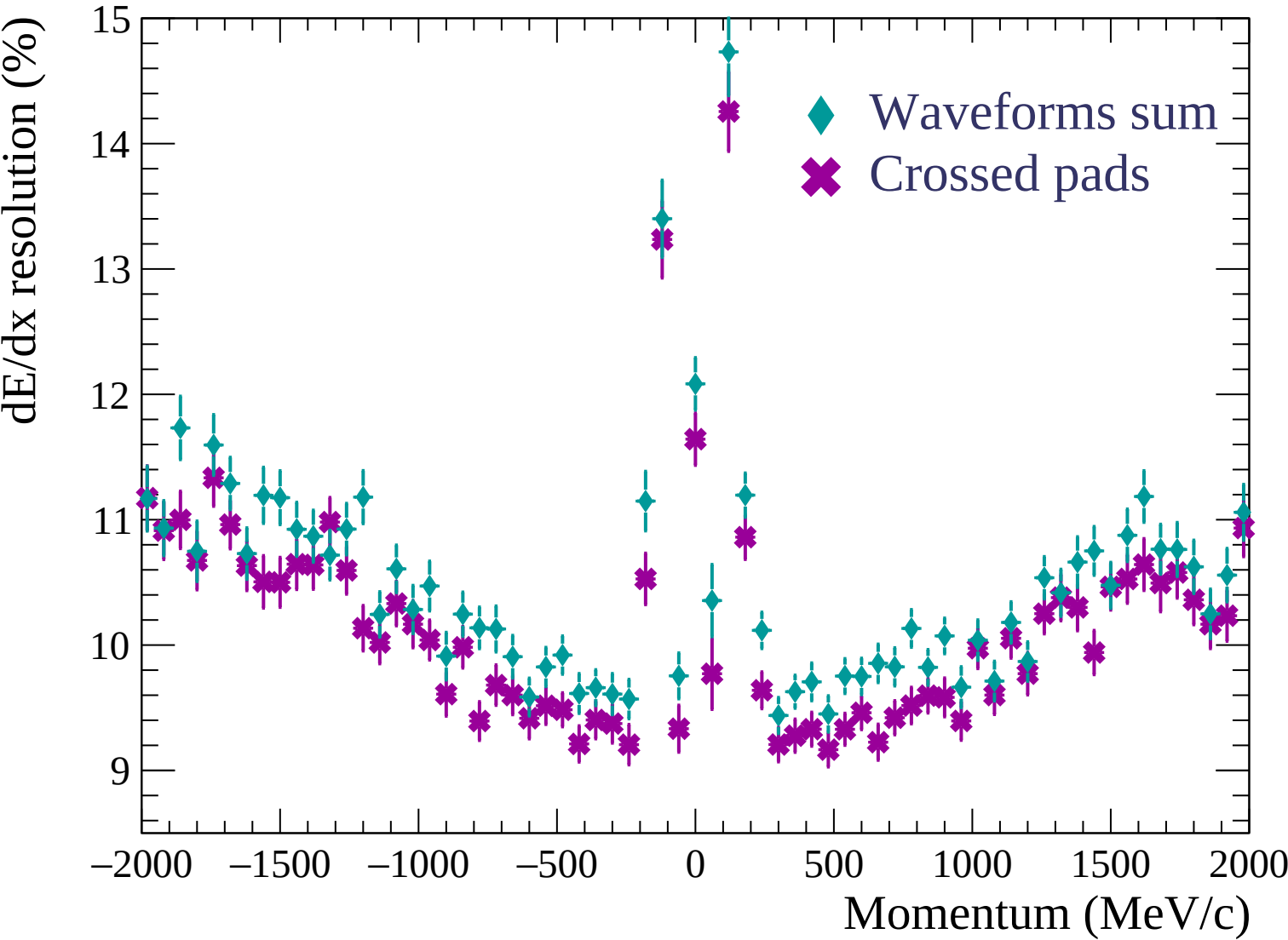


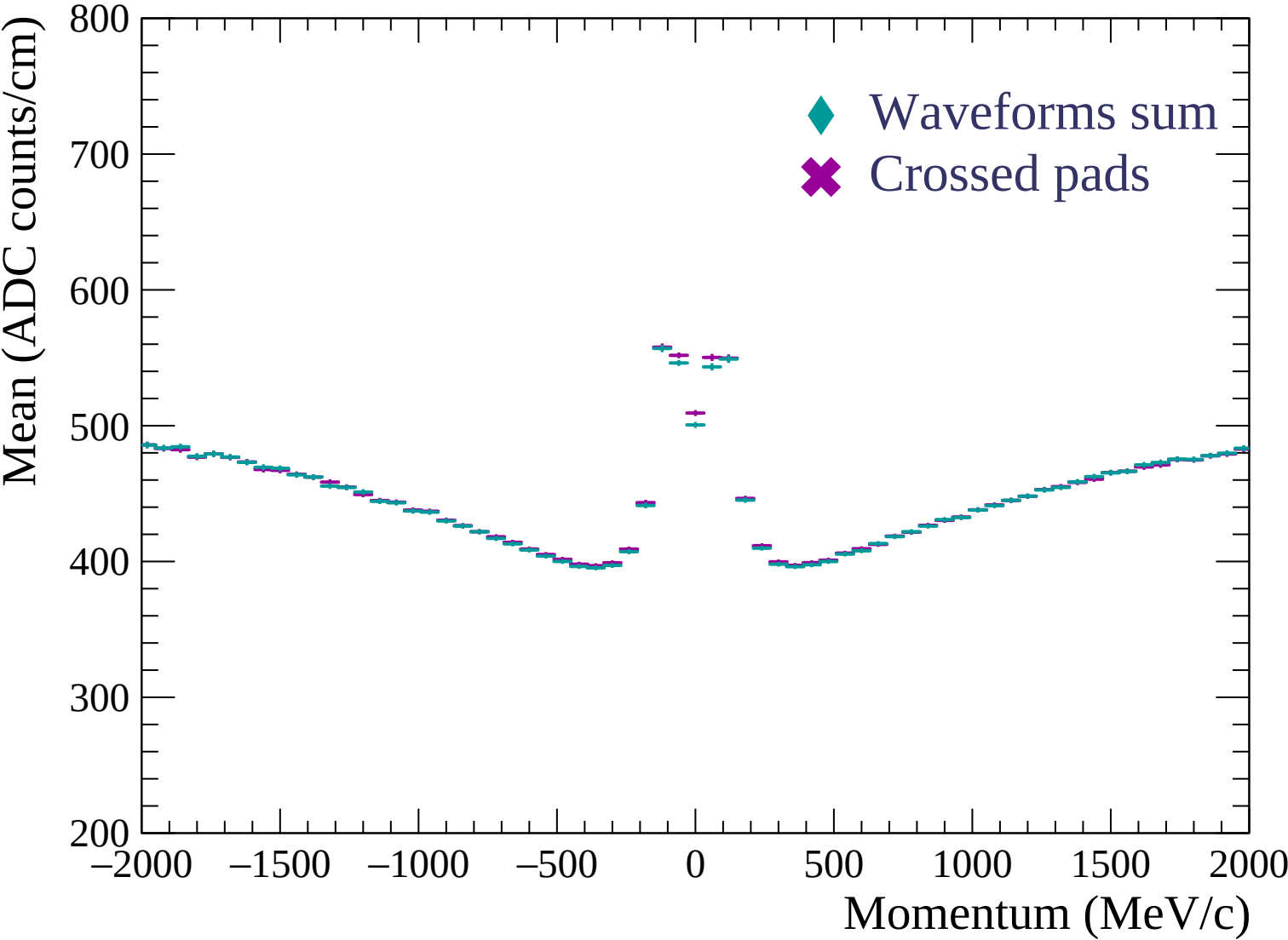


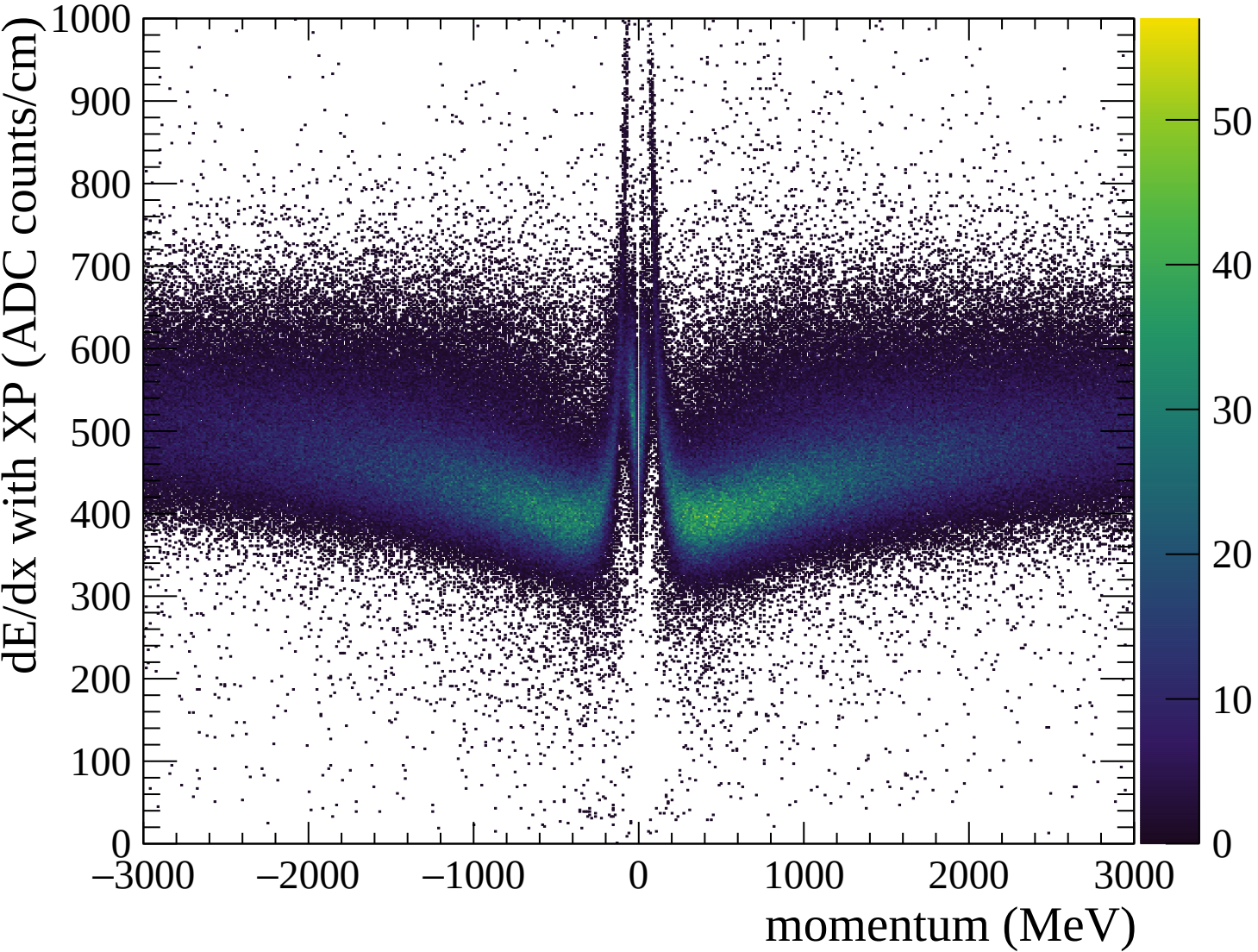


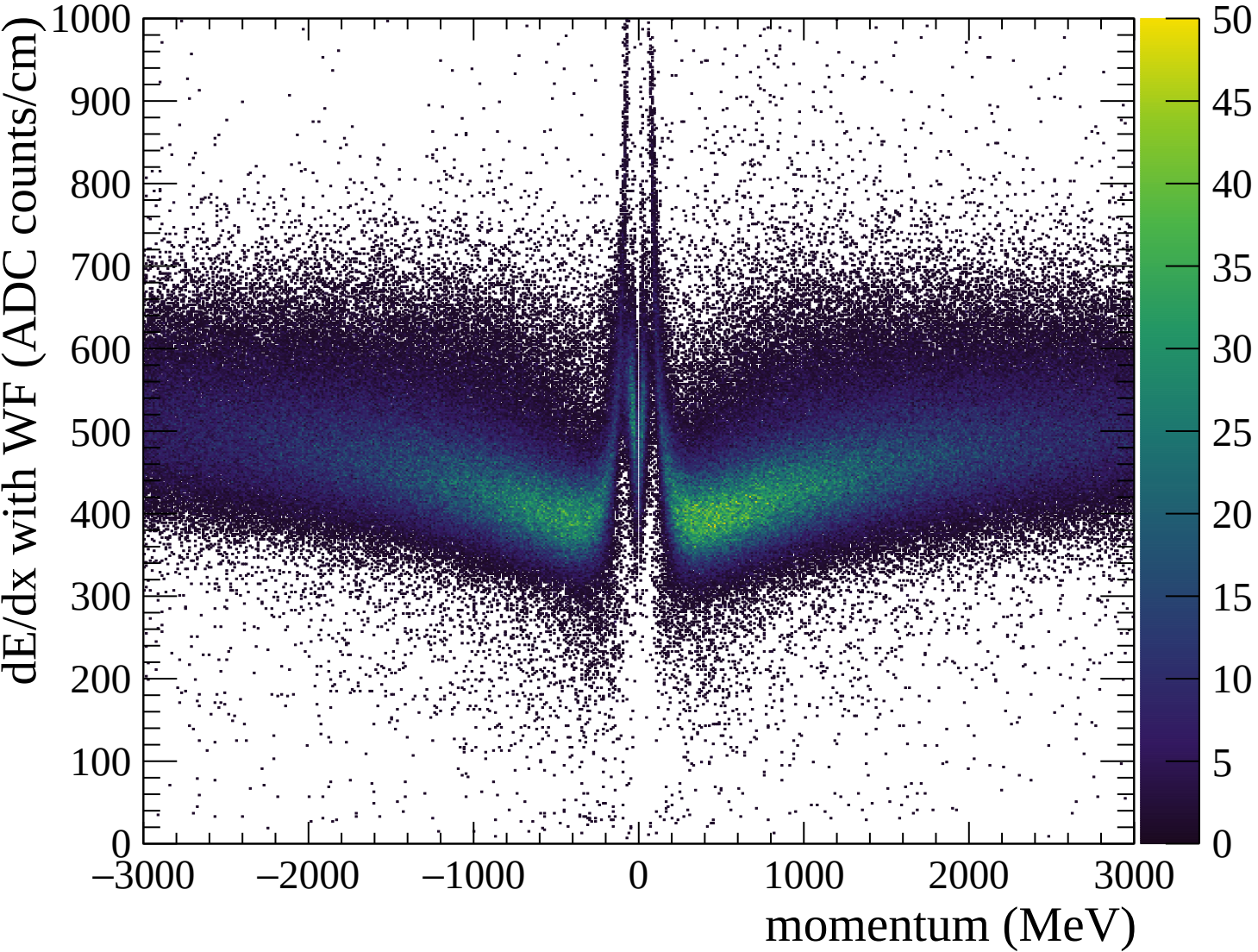


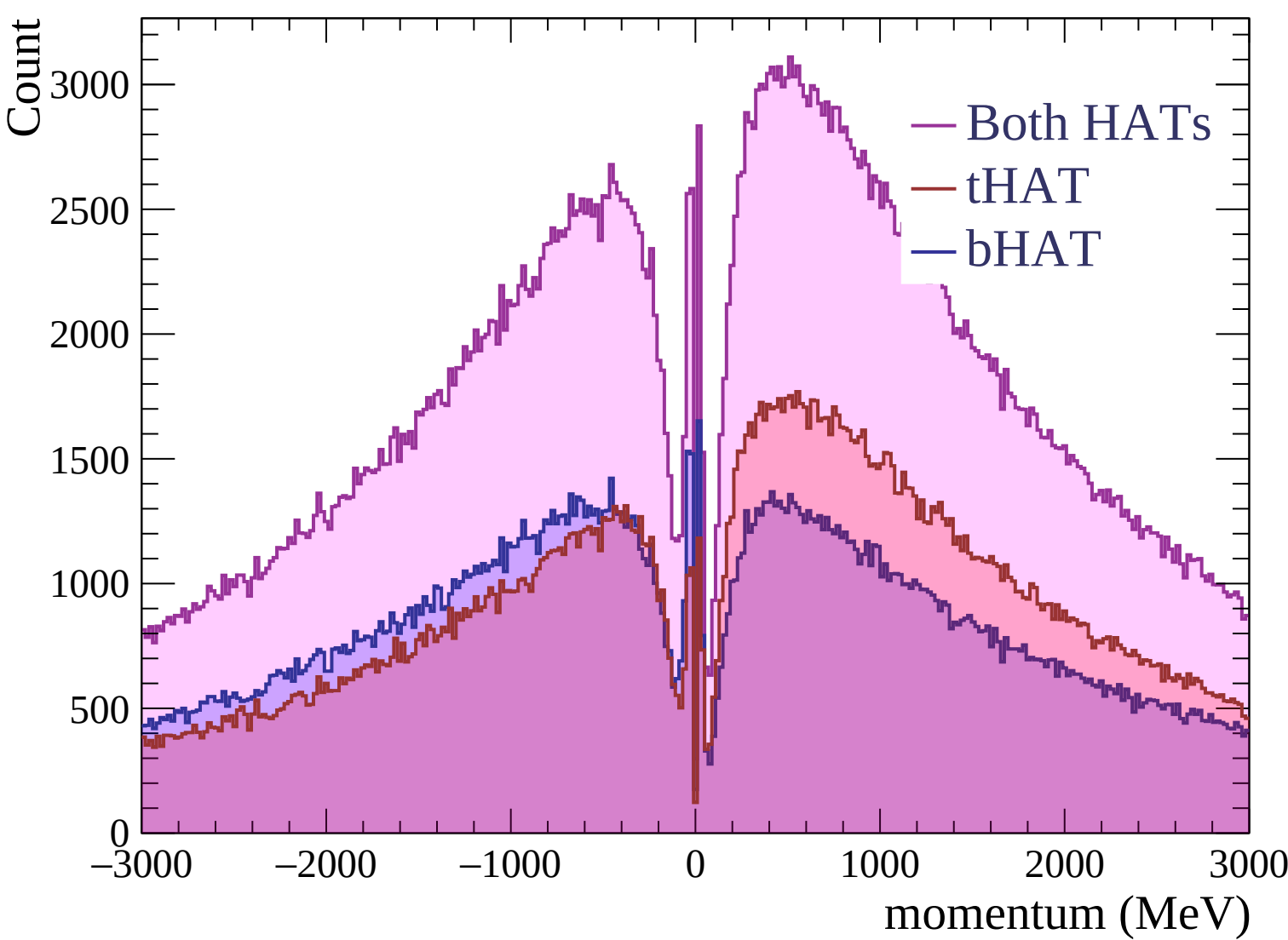


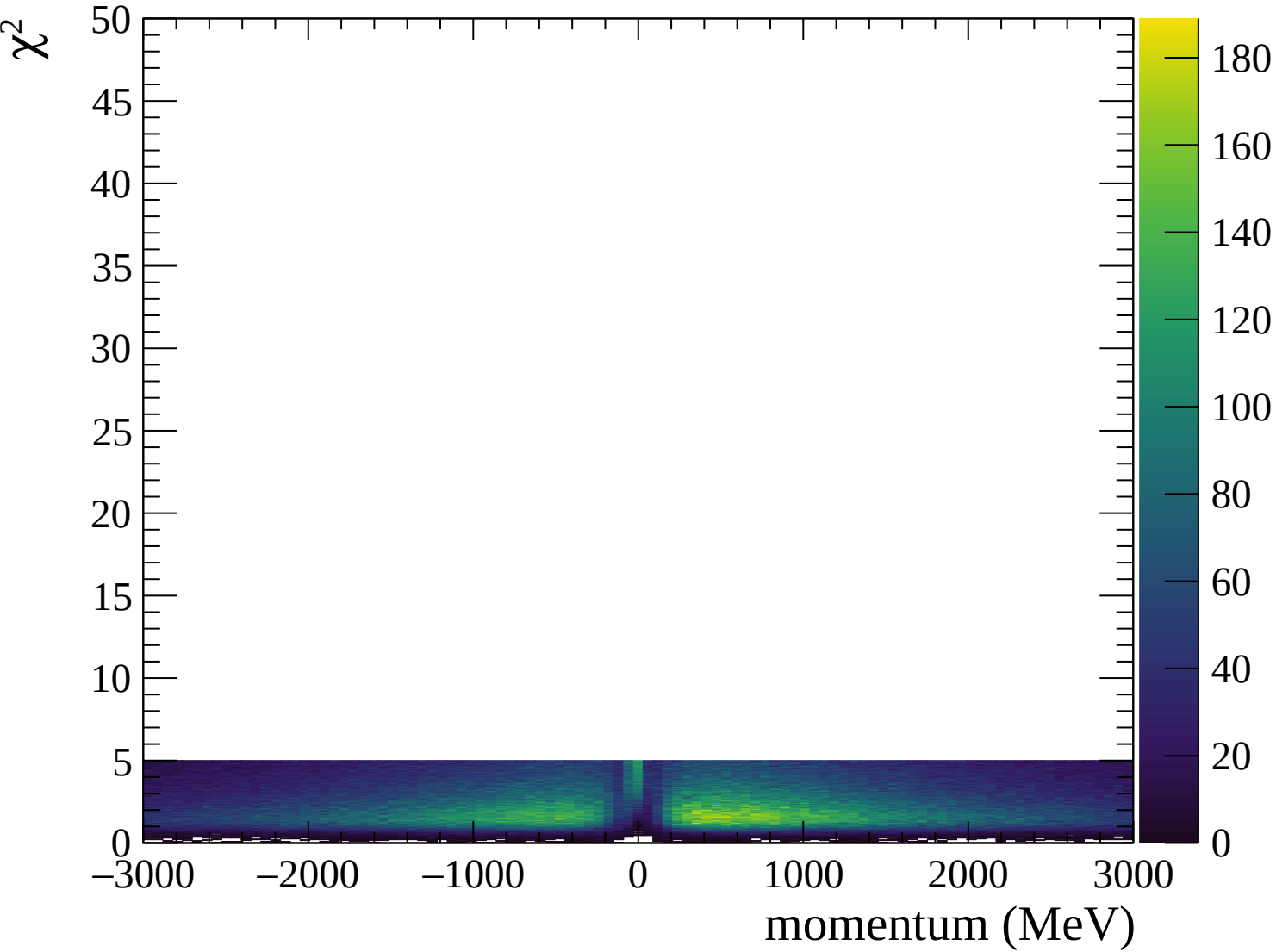


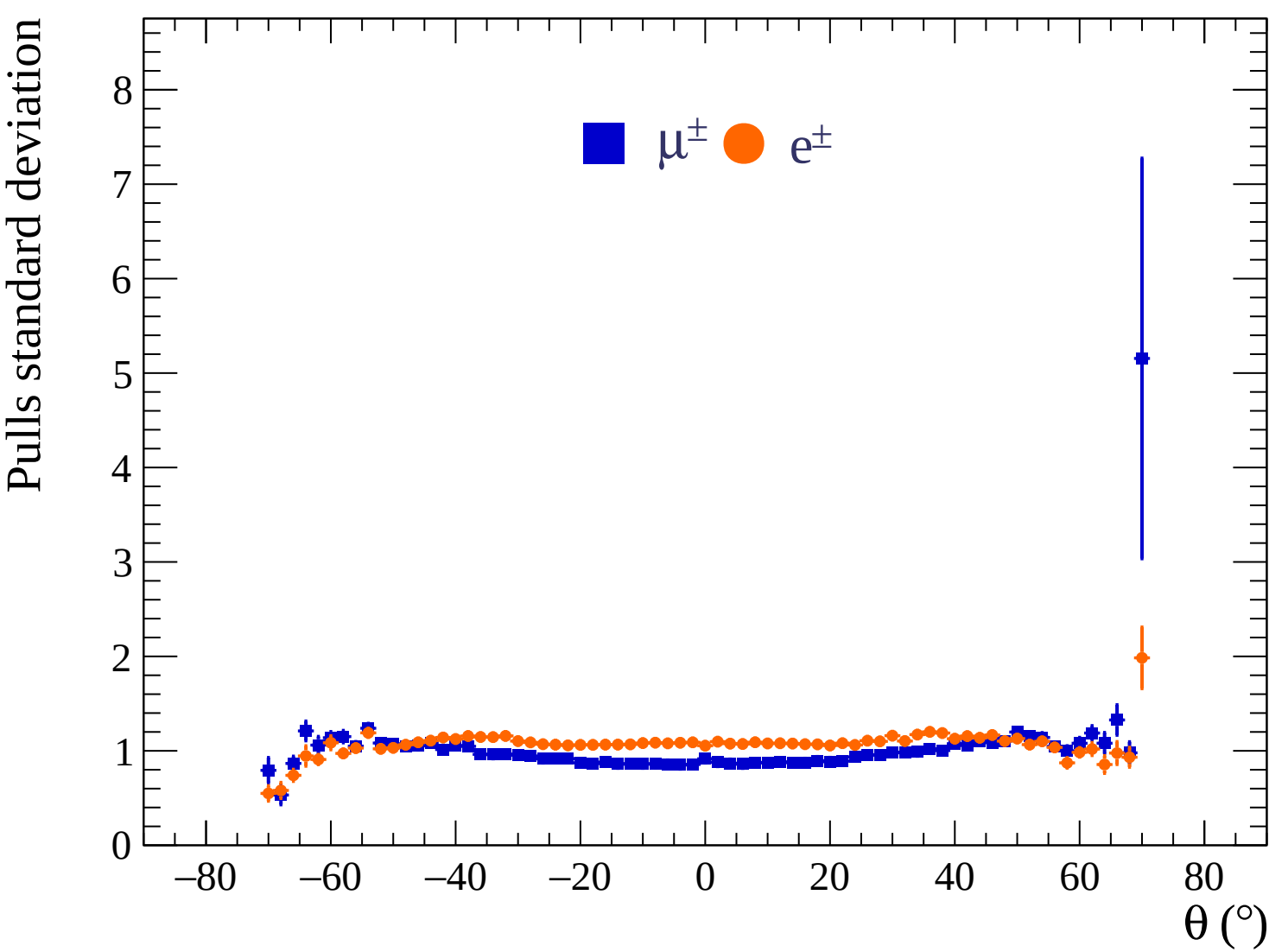


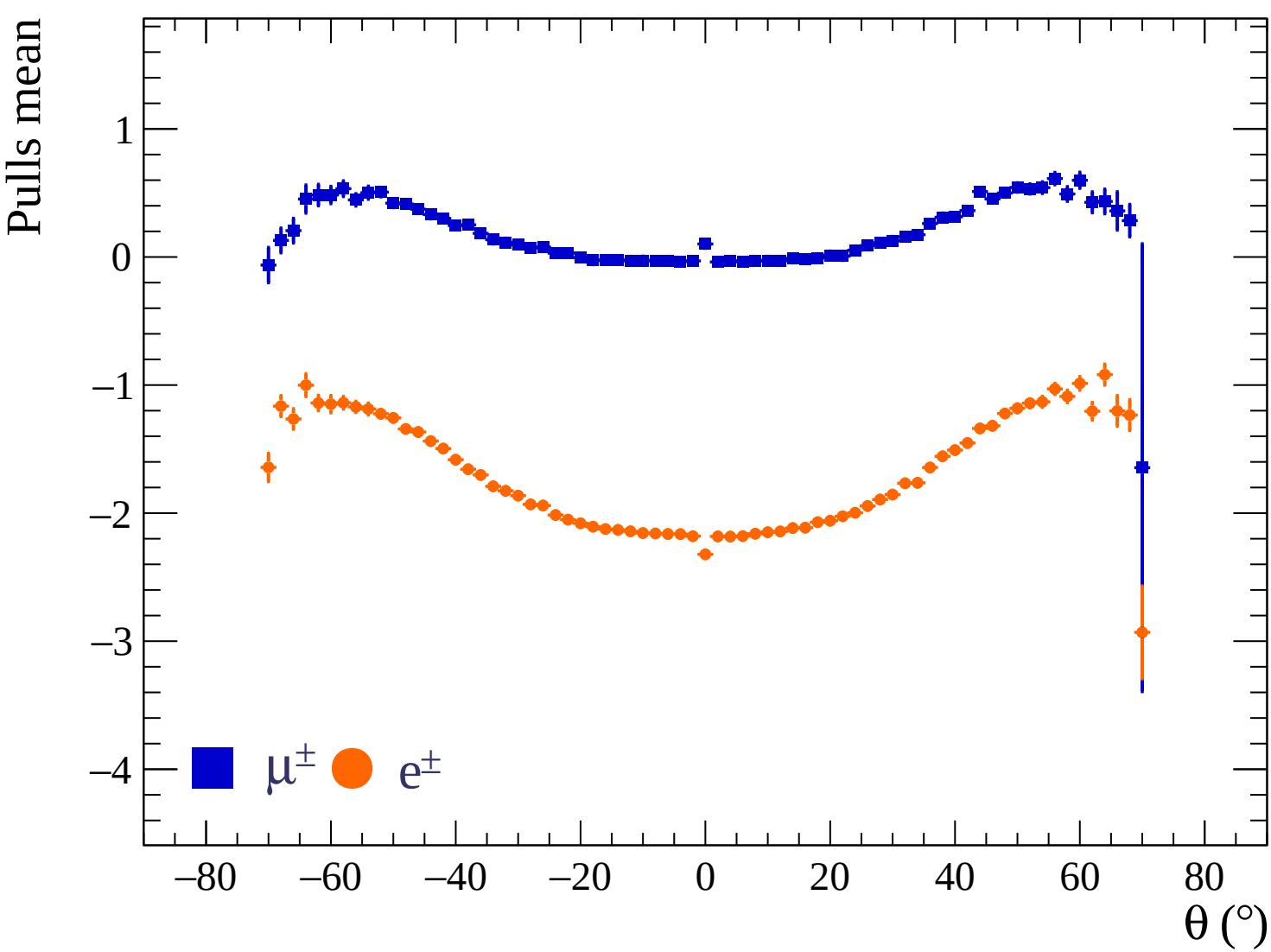


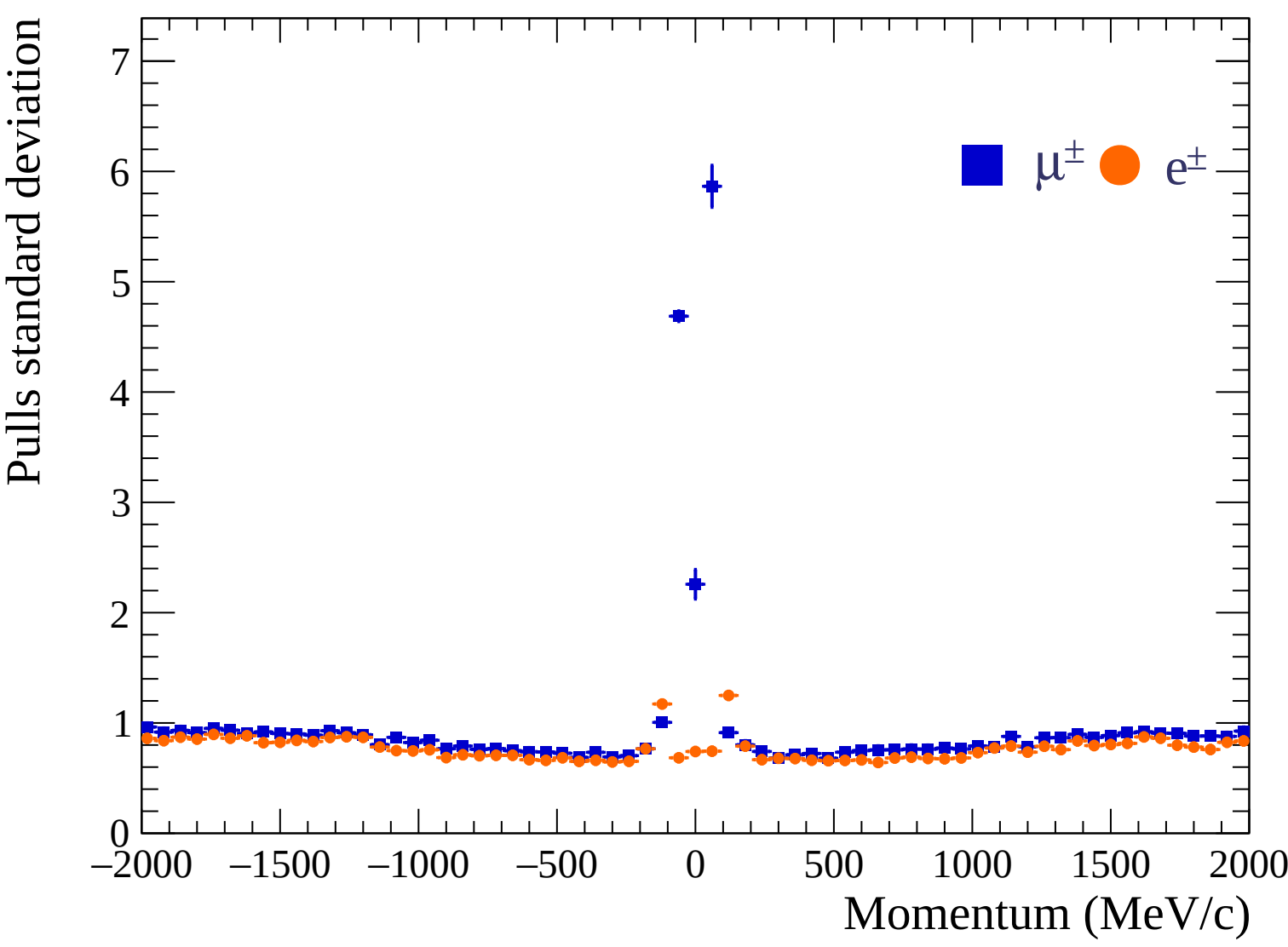


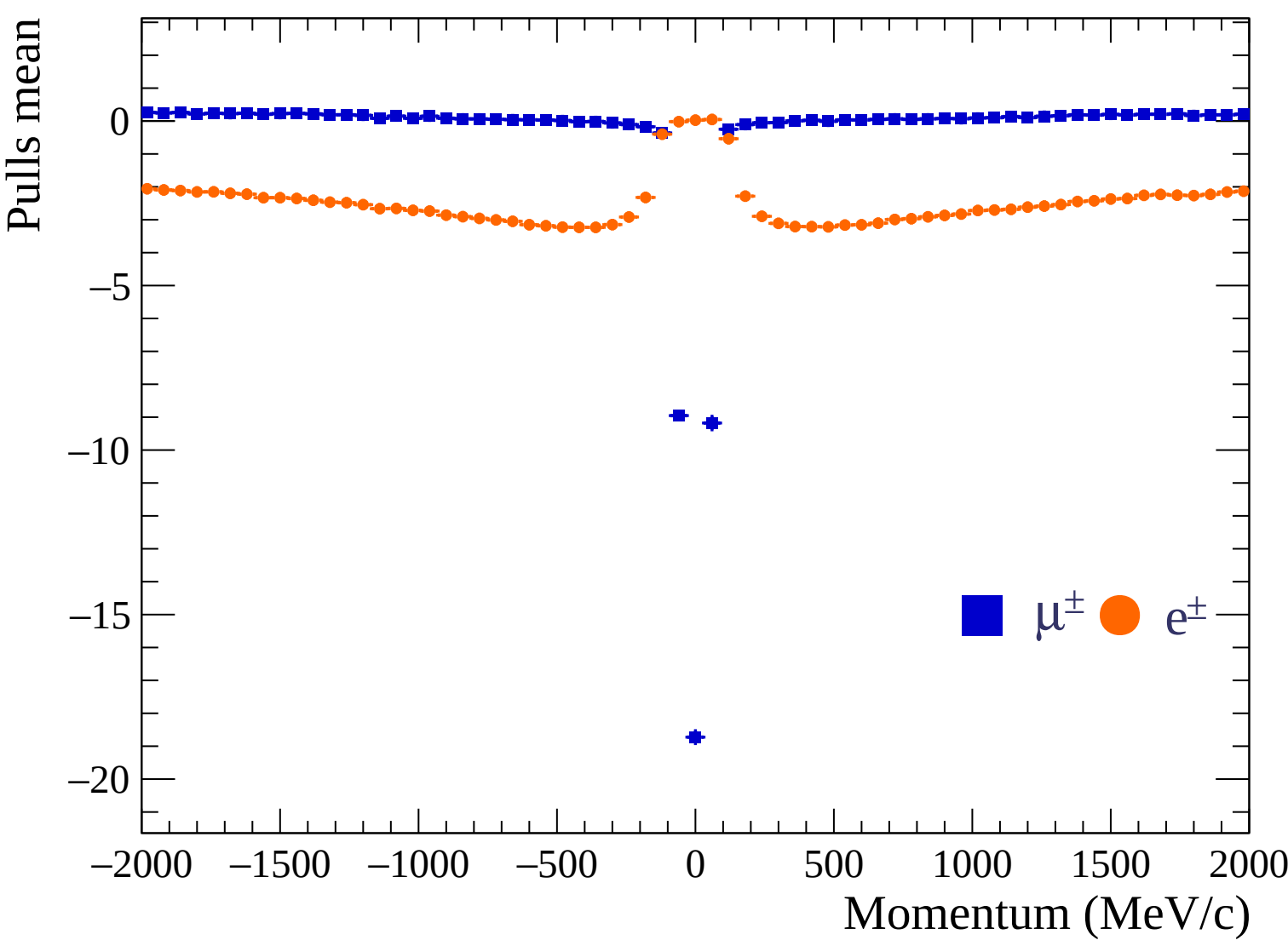


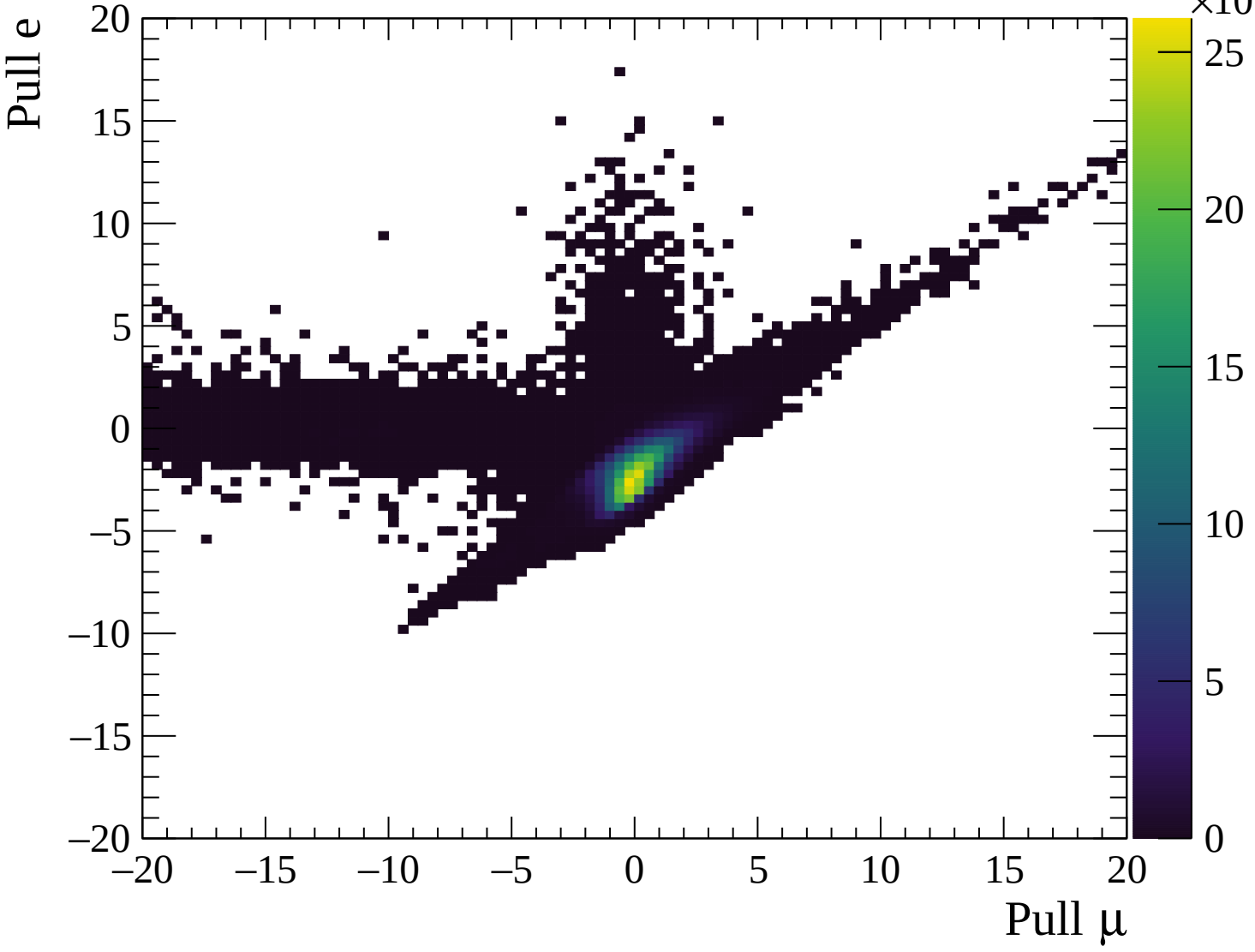


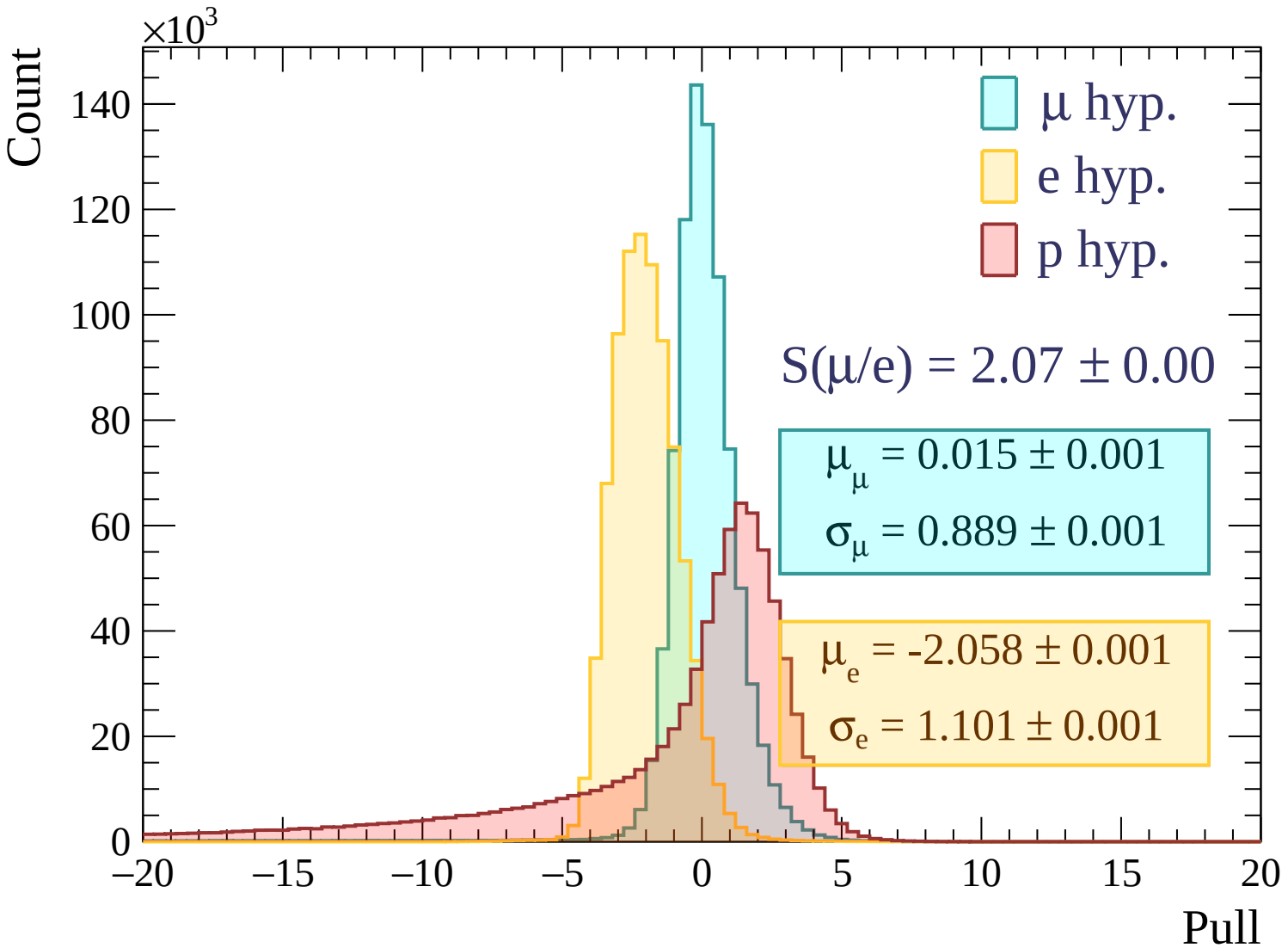


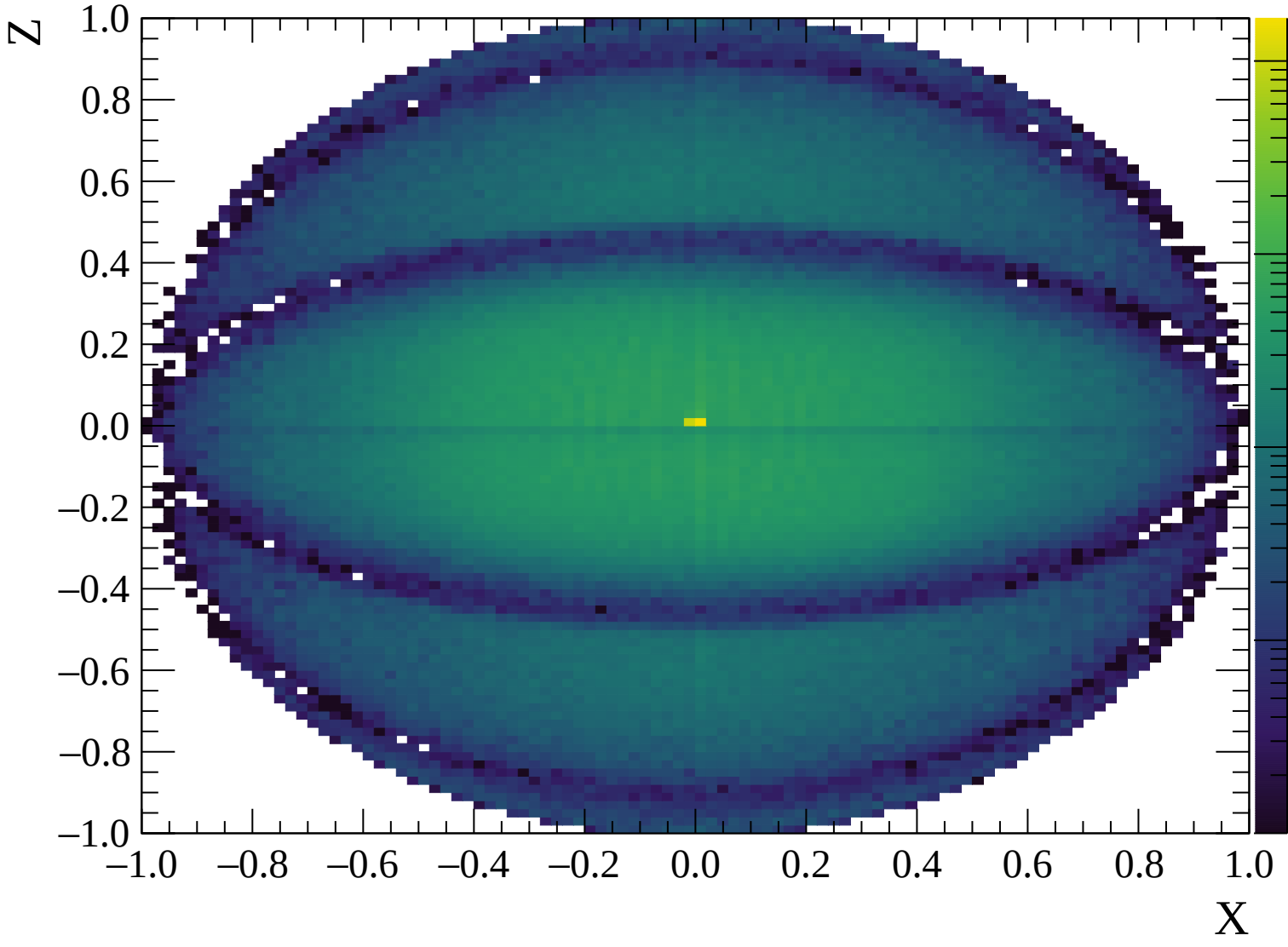


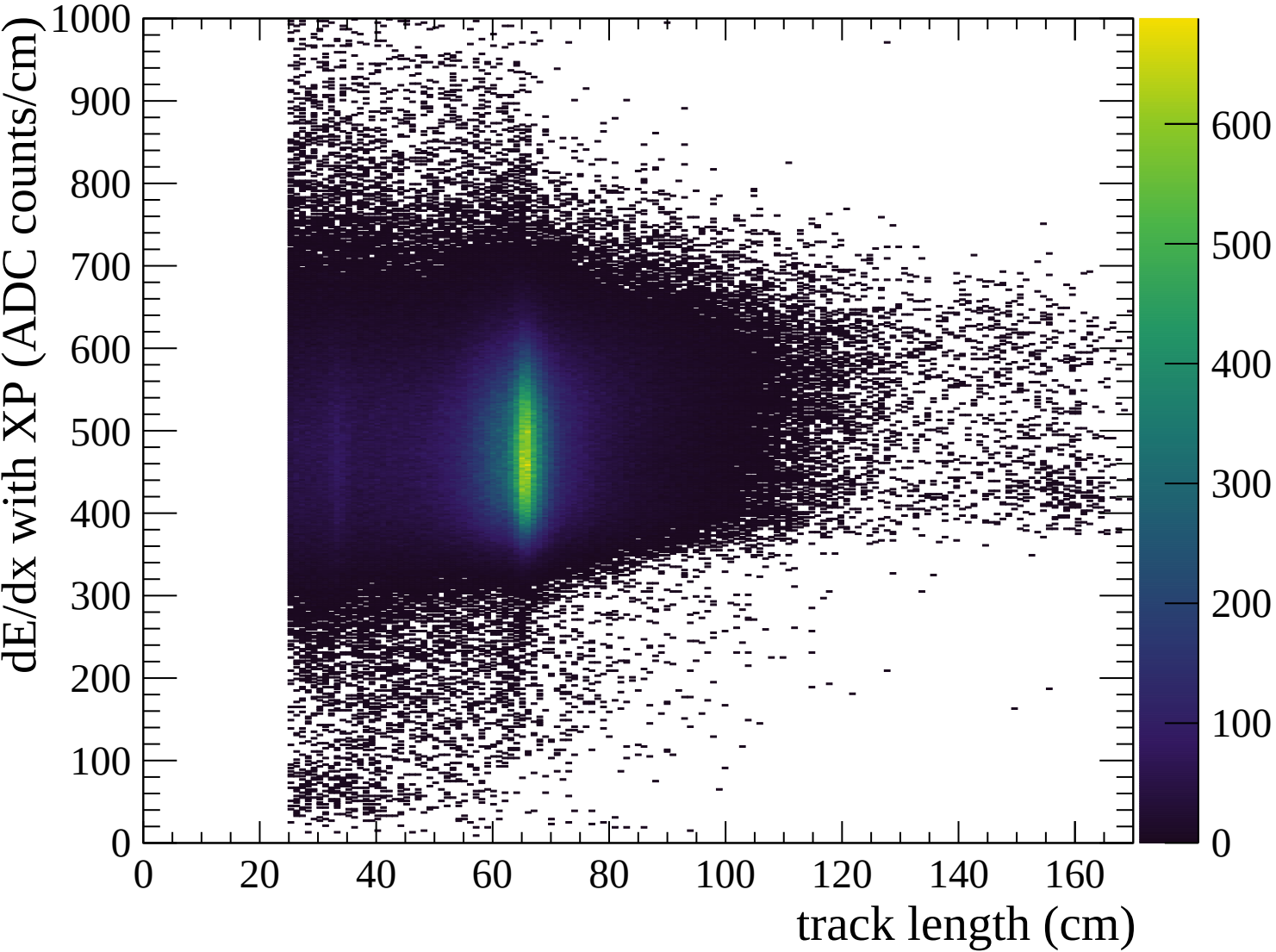


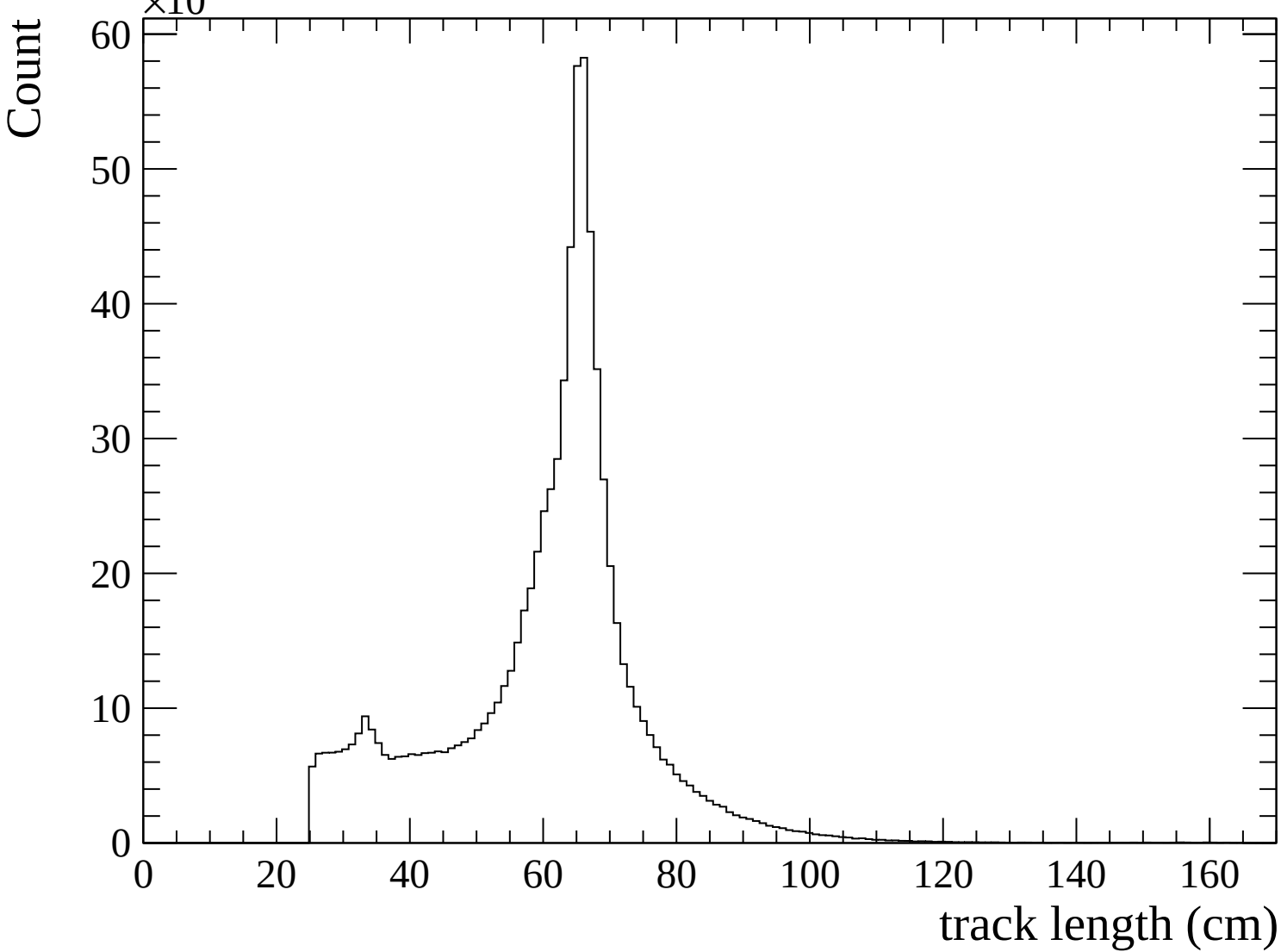


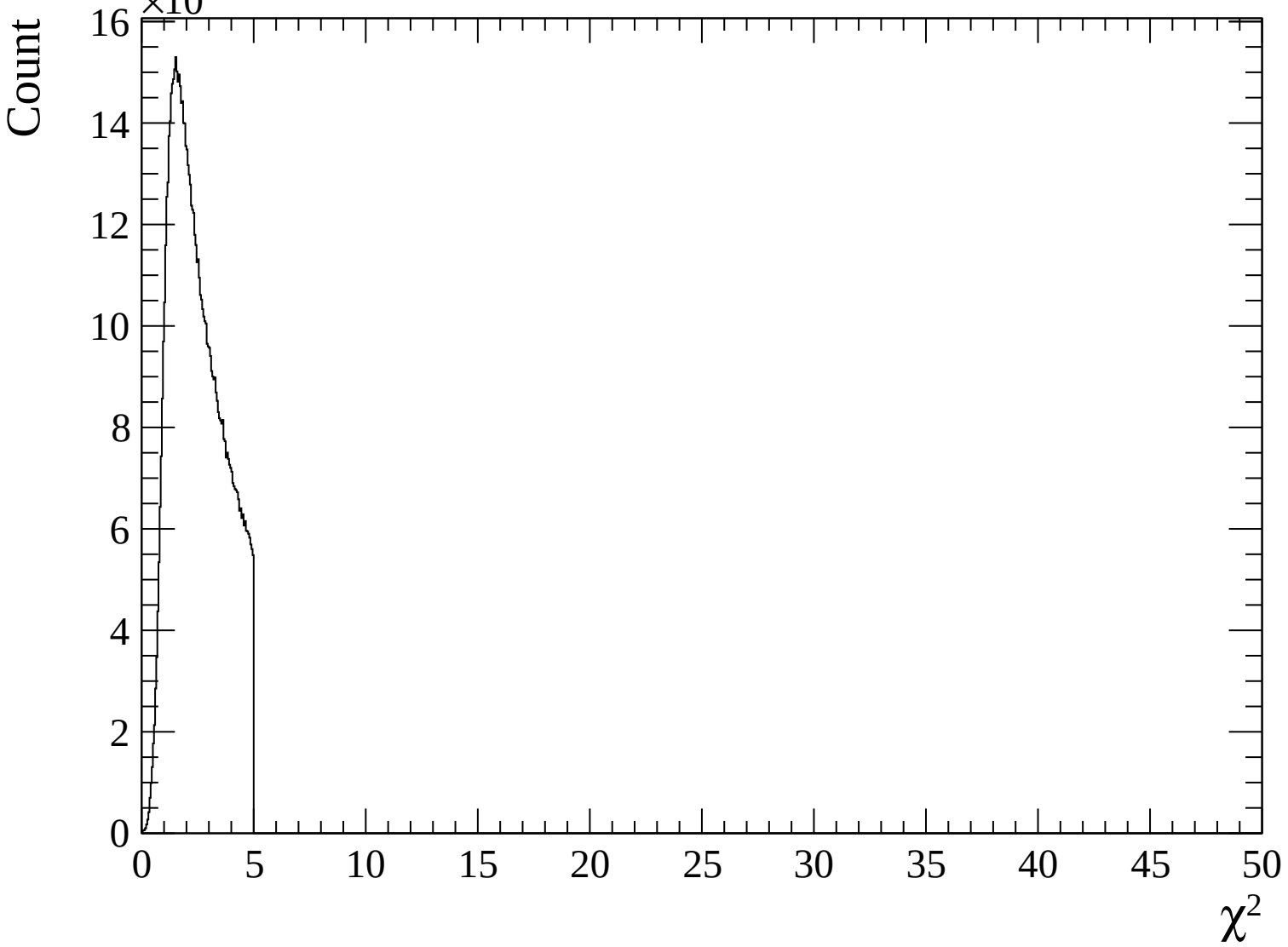


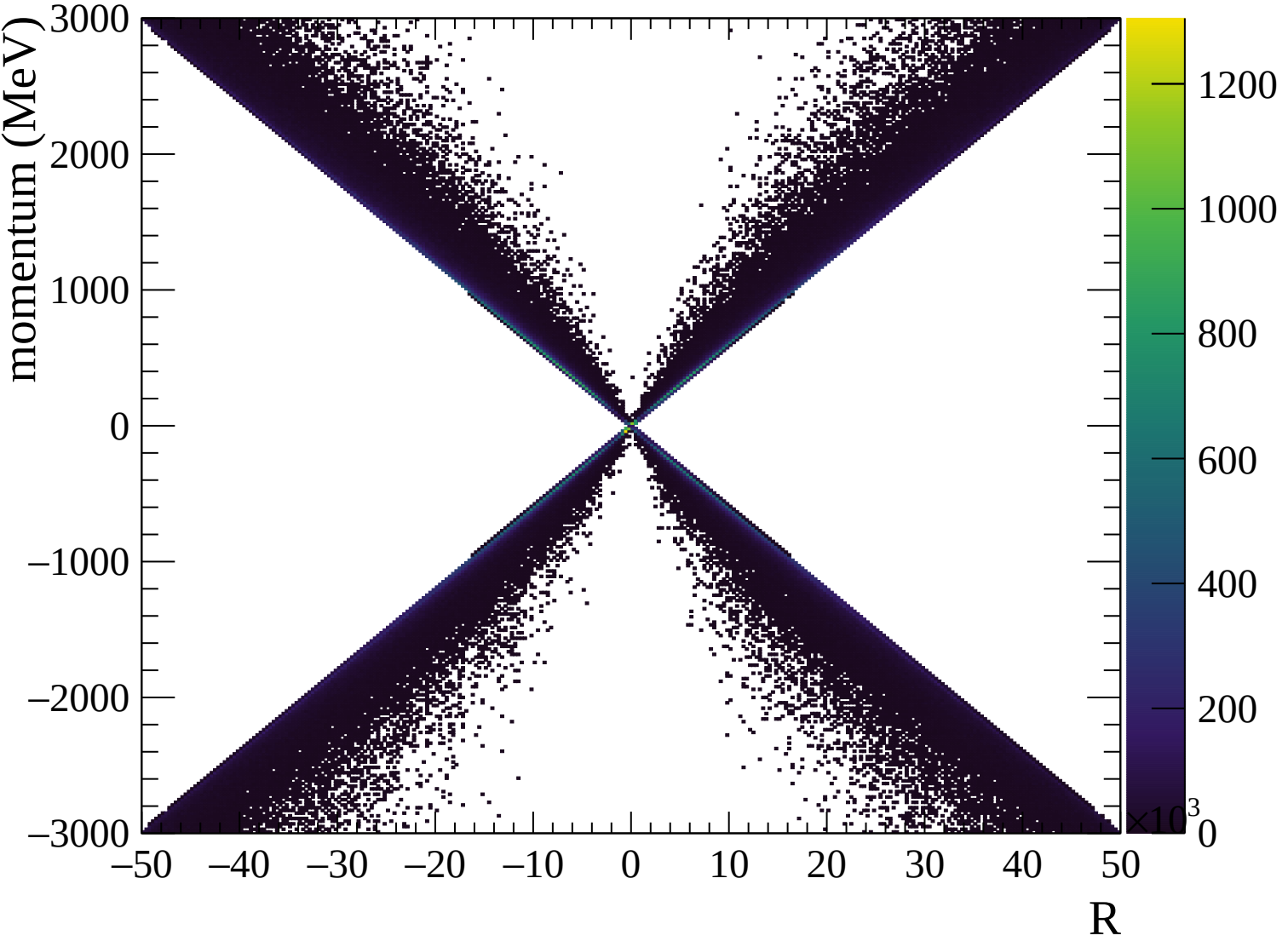


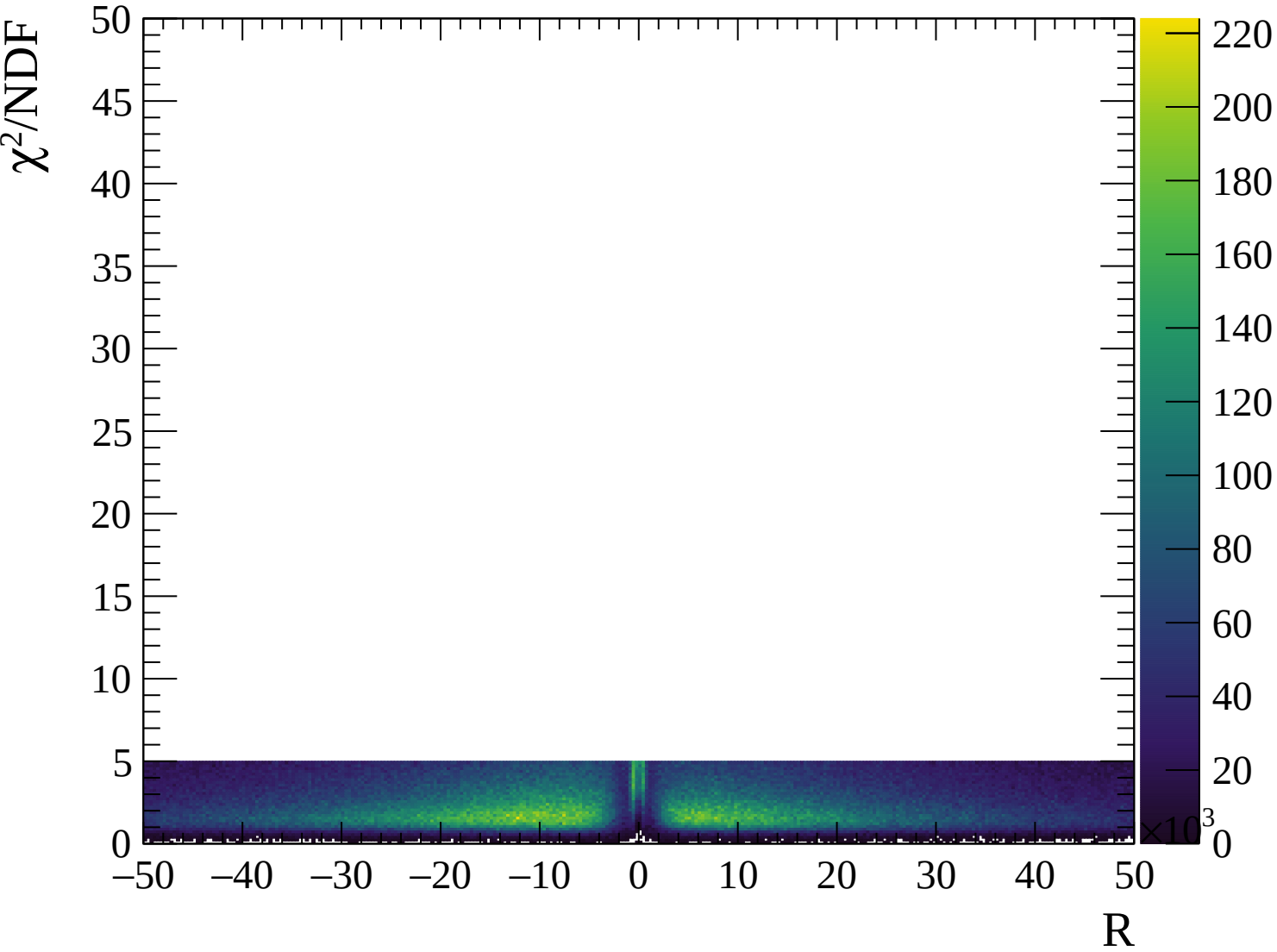






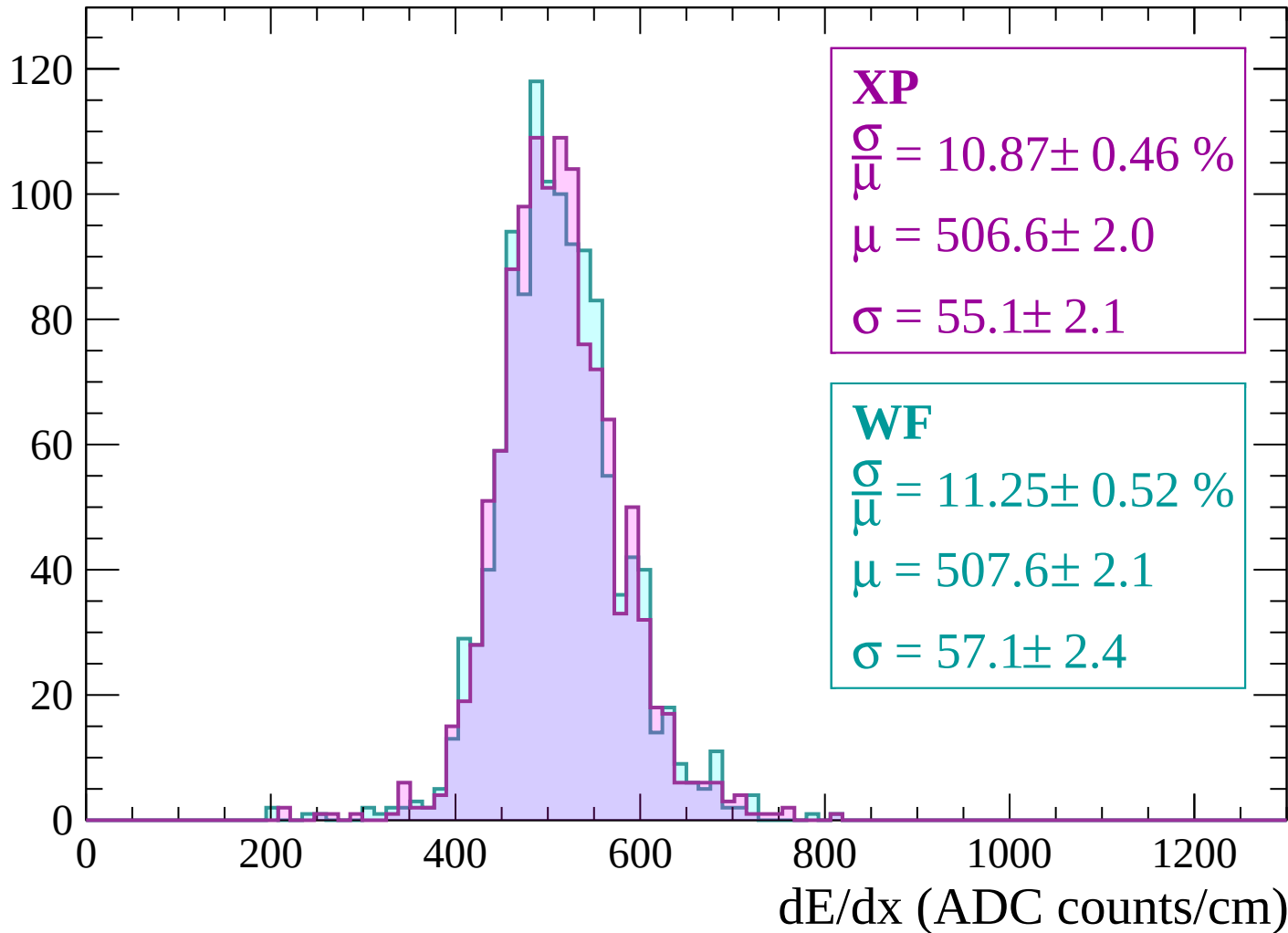






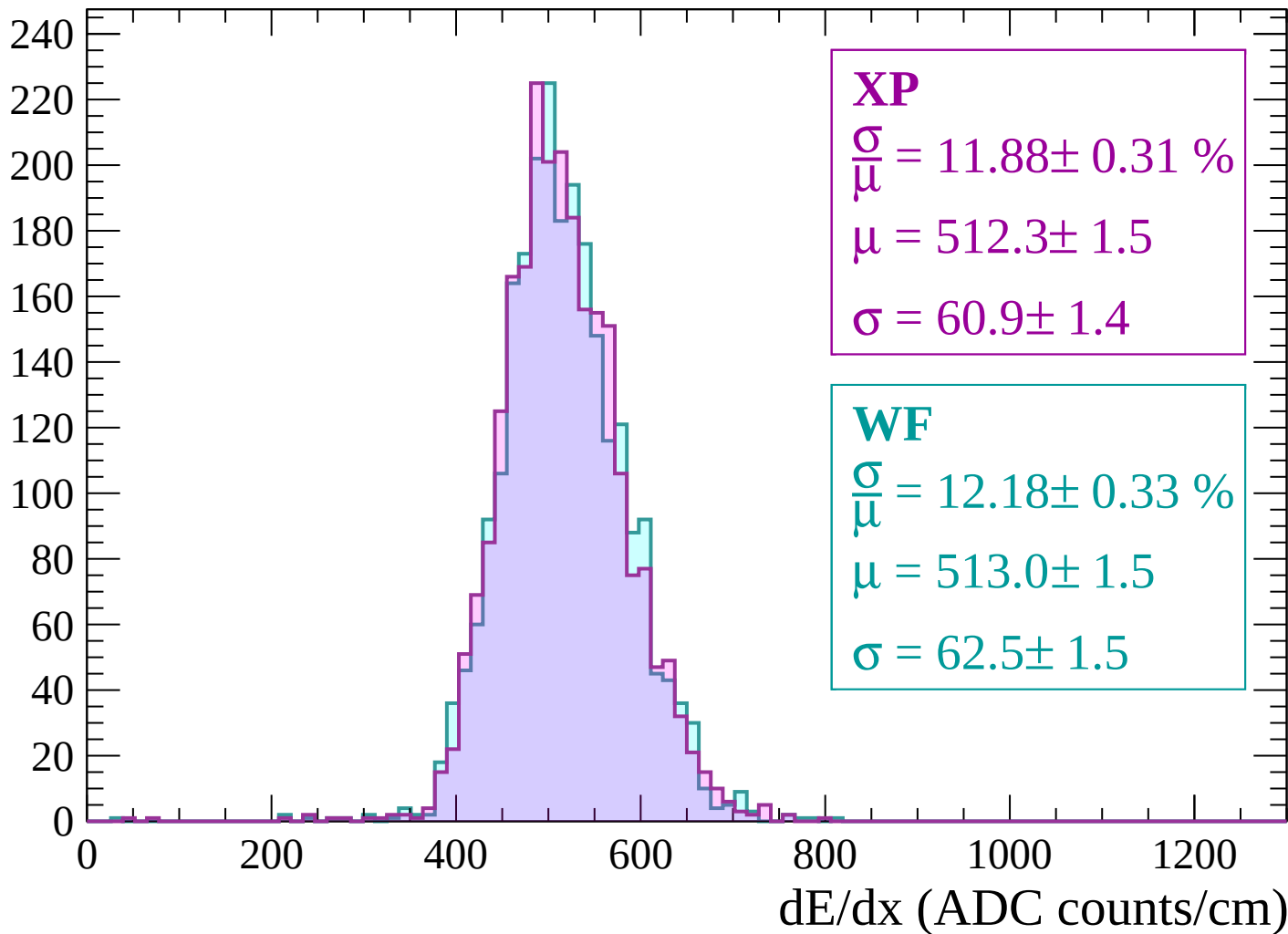
Energy loss | -3000 < p < -2940

Count



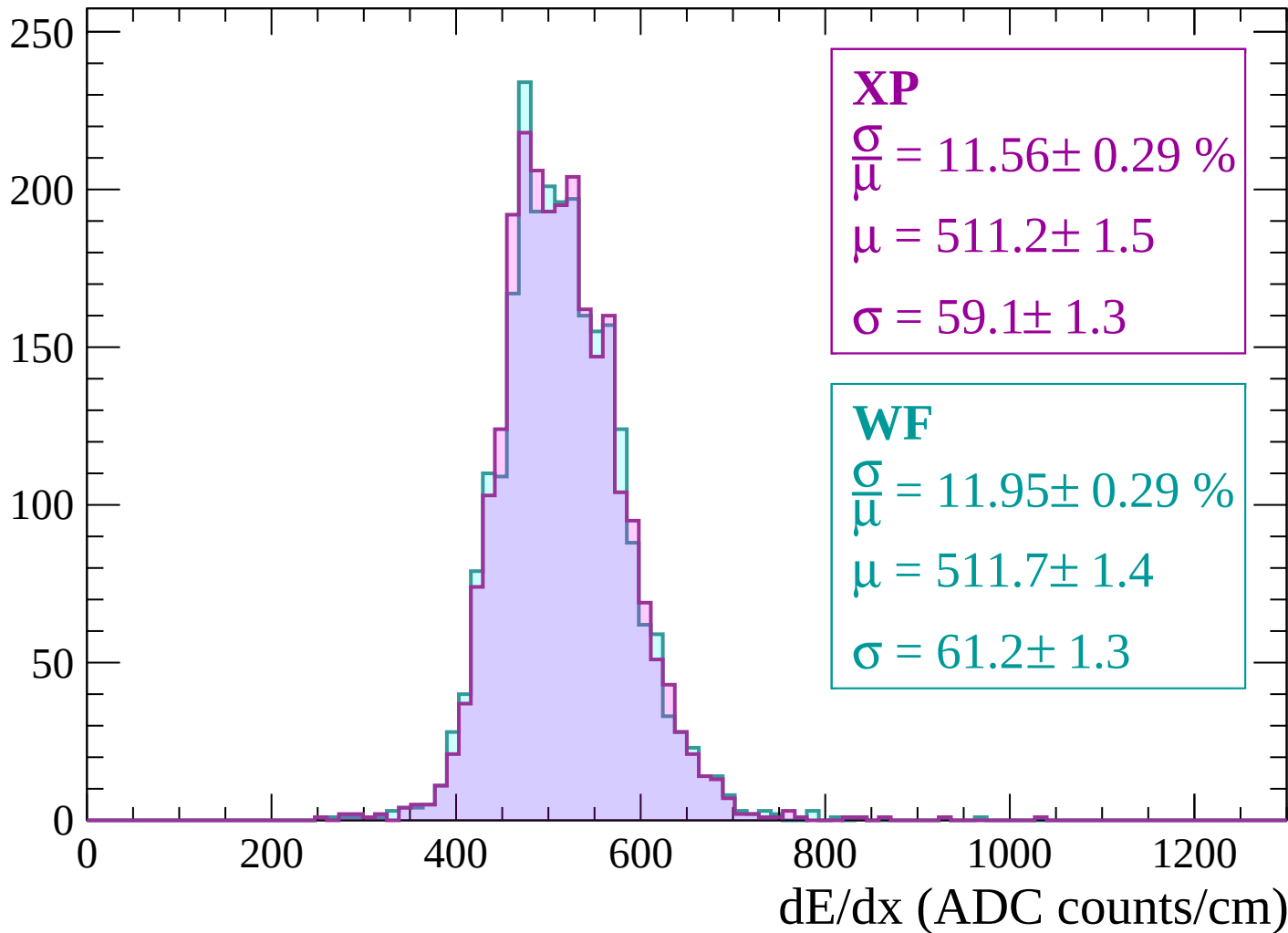
Energy loss | $-2940 < p < -2880$

Count



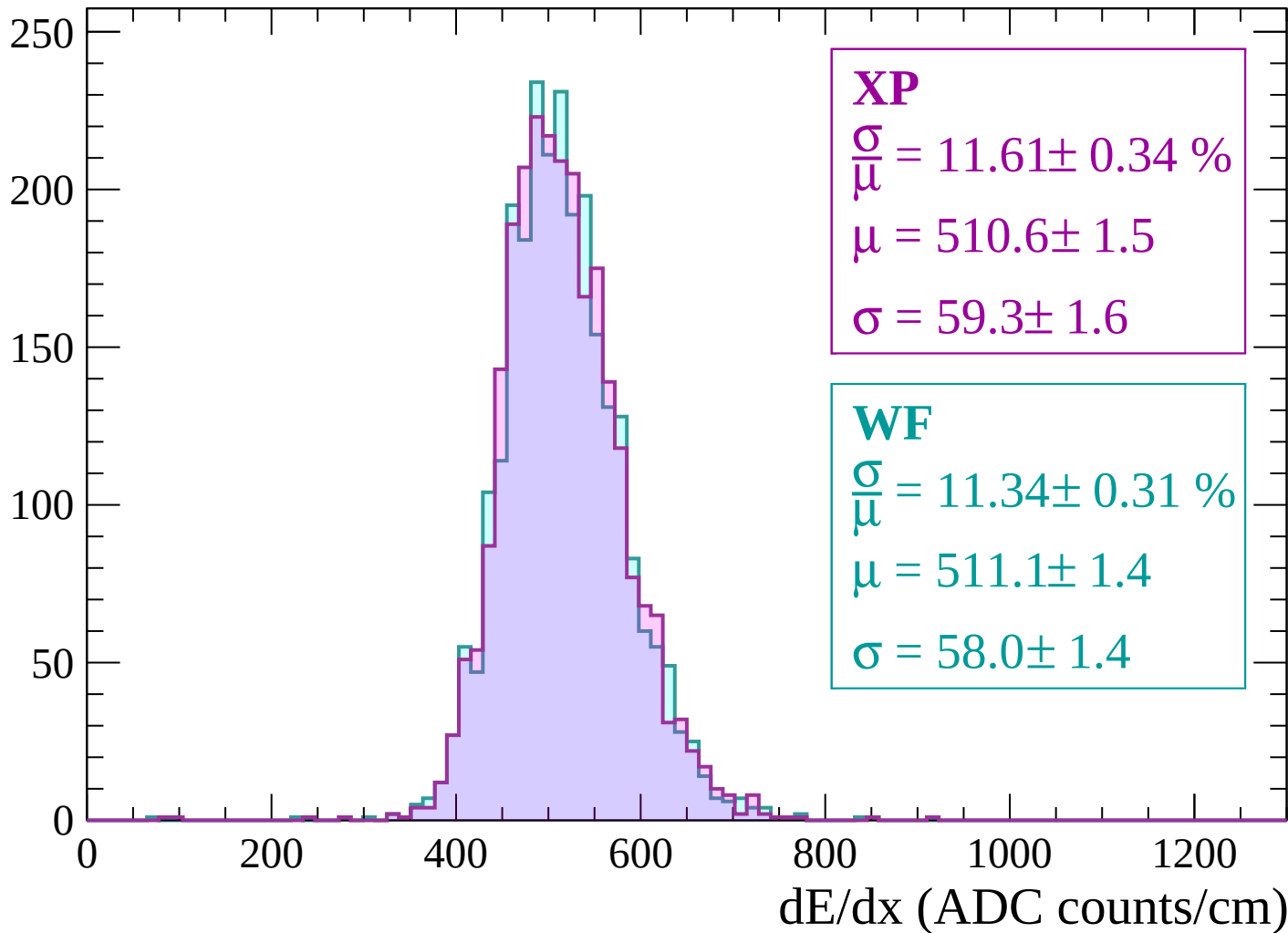
Energy loss | $-2880 < p < -2820$

Count



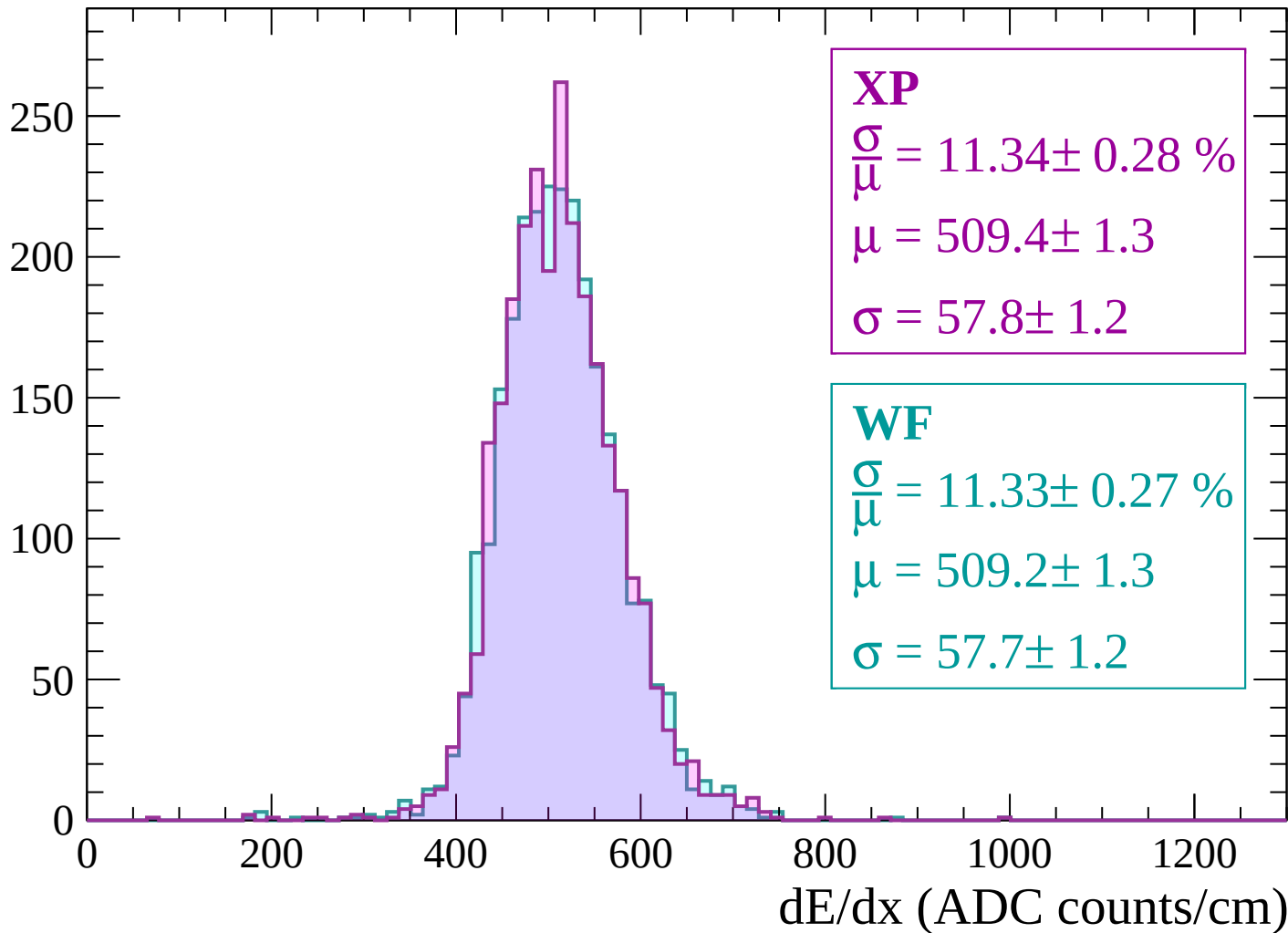
Energy loss | $-2820 < p < -2760$

Count



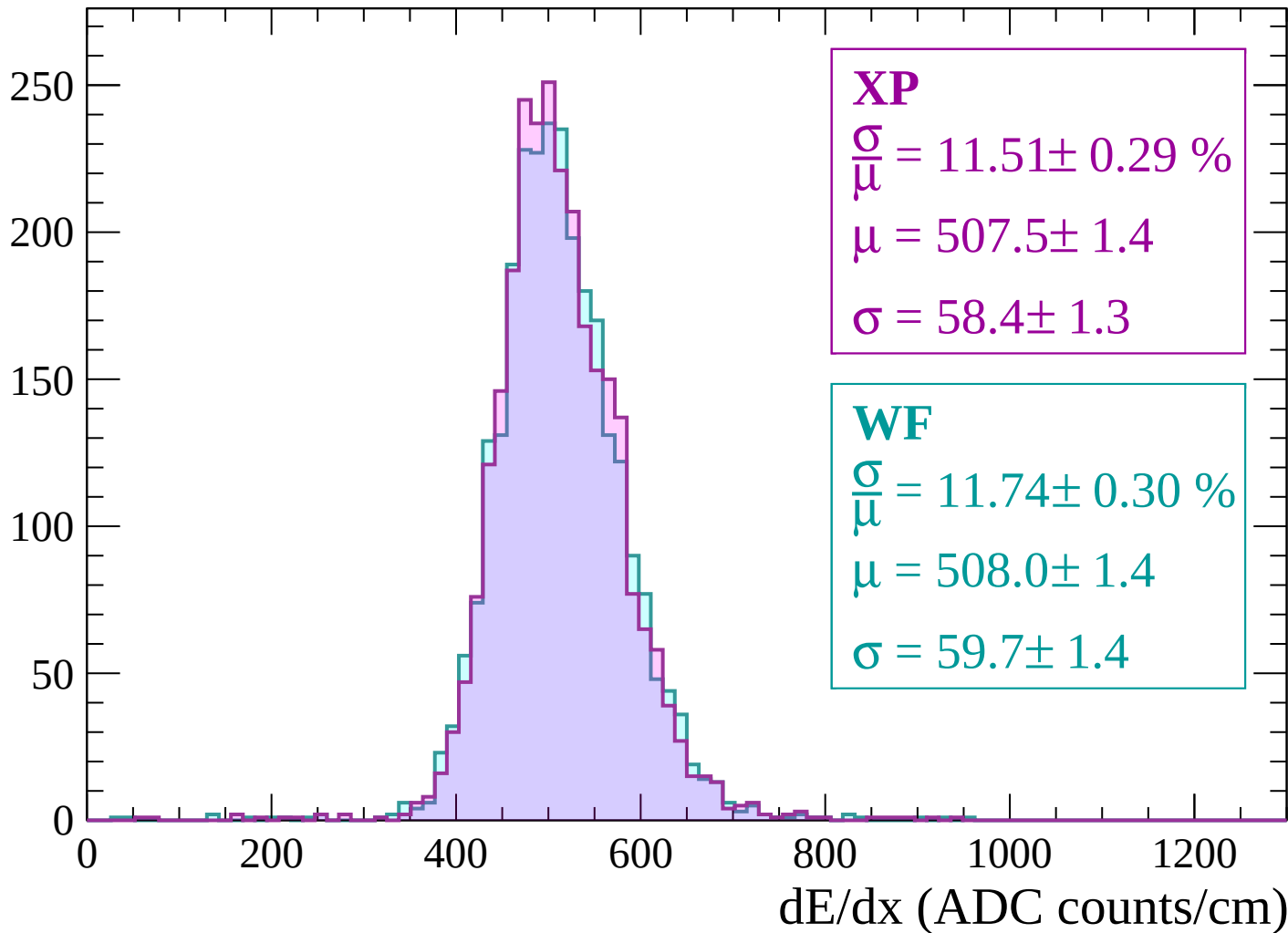
Energy loss | $-2760 < p < -2700$

Count



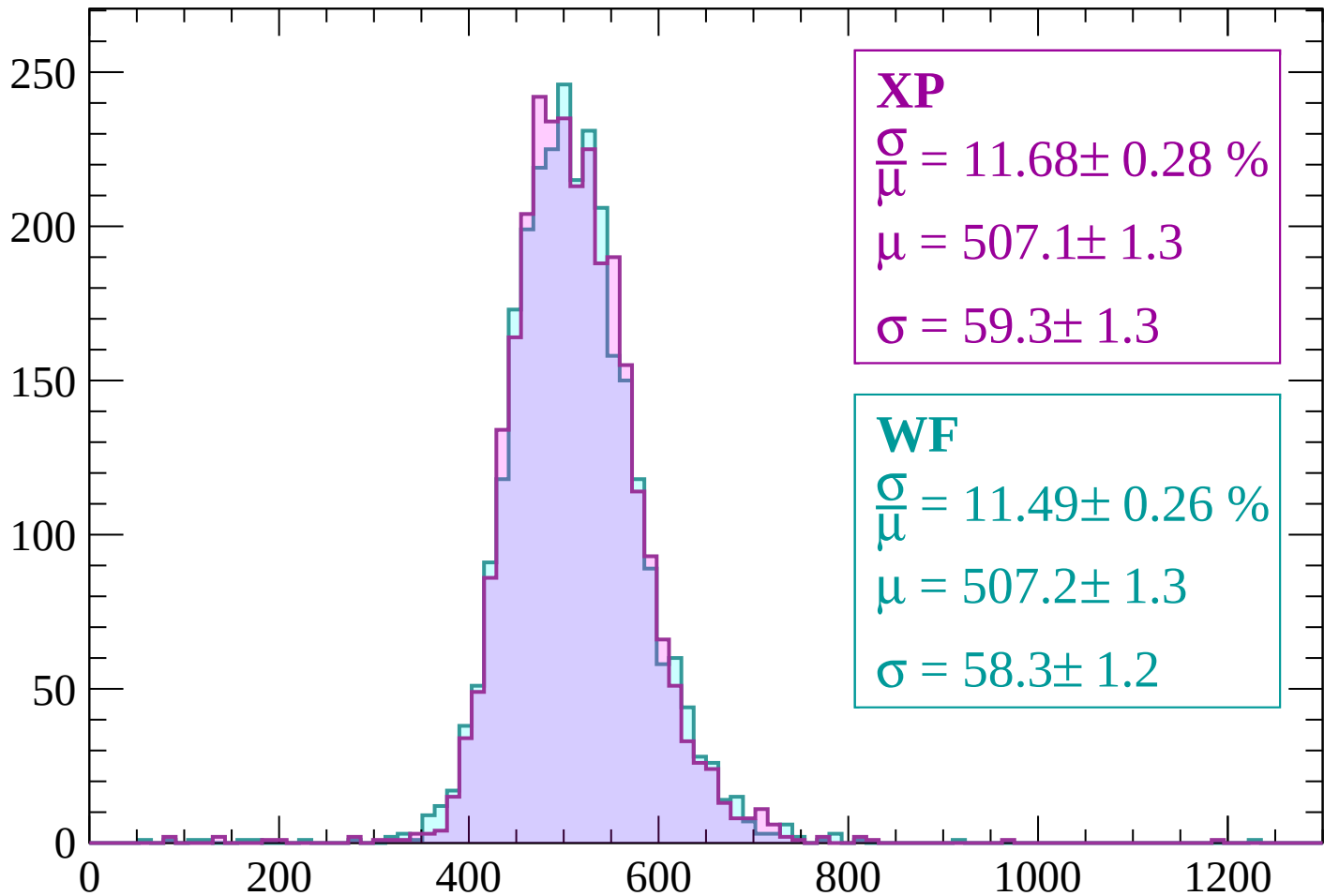
Energy loss | -2700 < p < -2640

Count



Energy loss | $-2640 < p < -2580$

Count



XP

$$\frac{\sigma}{\mu} = 11.68 \pm 0.28 \%$$

$$\mu = 507.1 \pm 1.3$$

$$\sigma = 59.3 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 11.49 \pm 0.26 \%$$

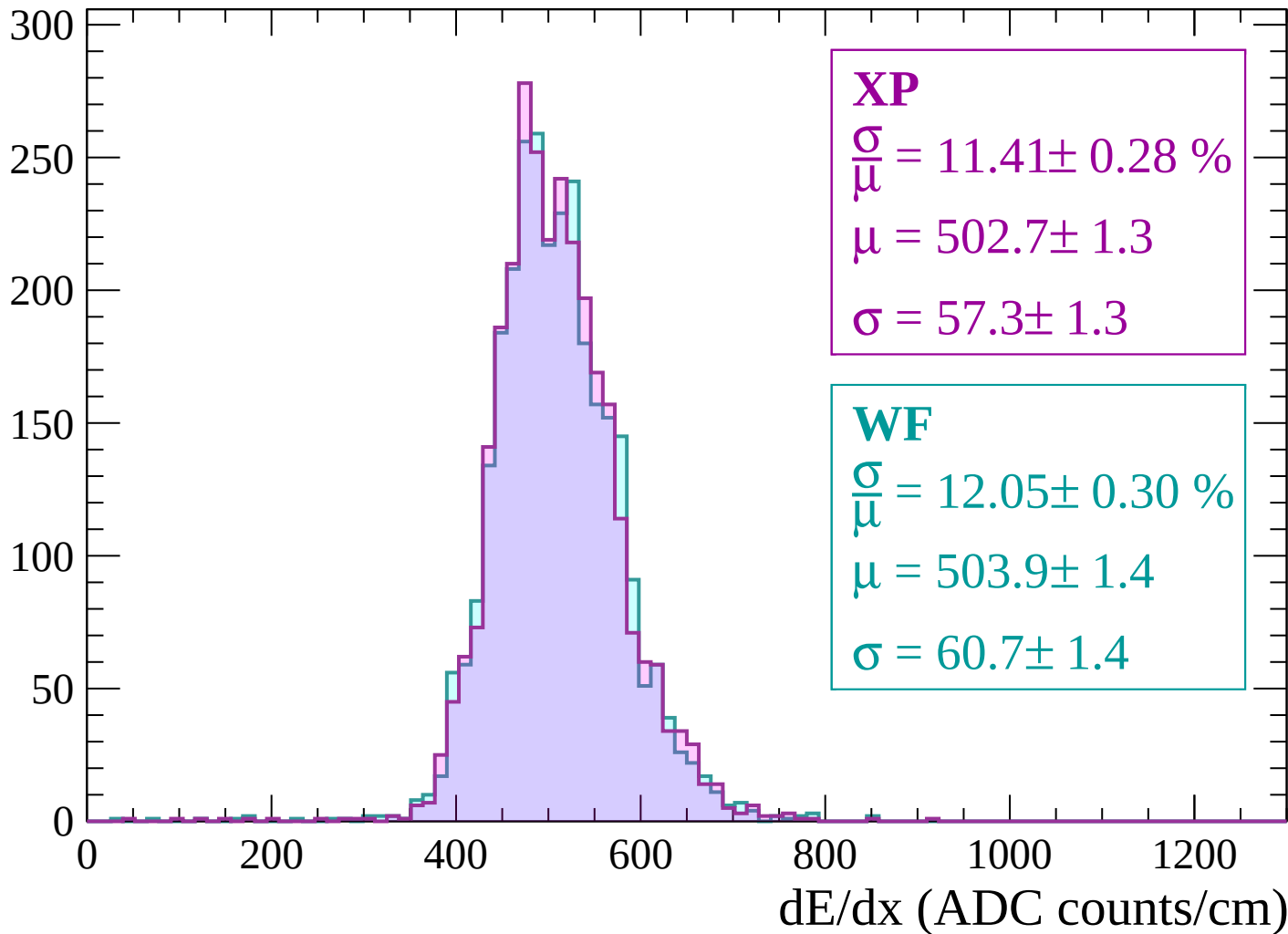
$$\mu = 507.2 \pm 1.3$$

$$\sigma = 58.3 \pm 1.2$$

dE/dx (ADC counts/cm)

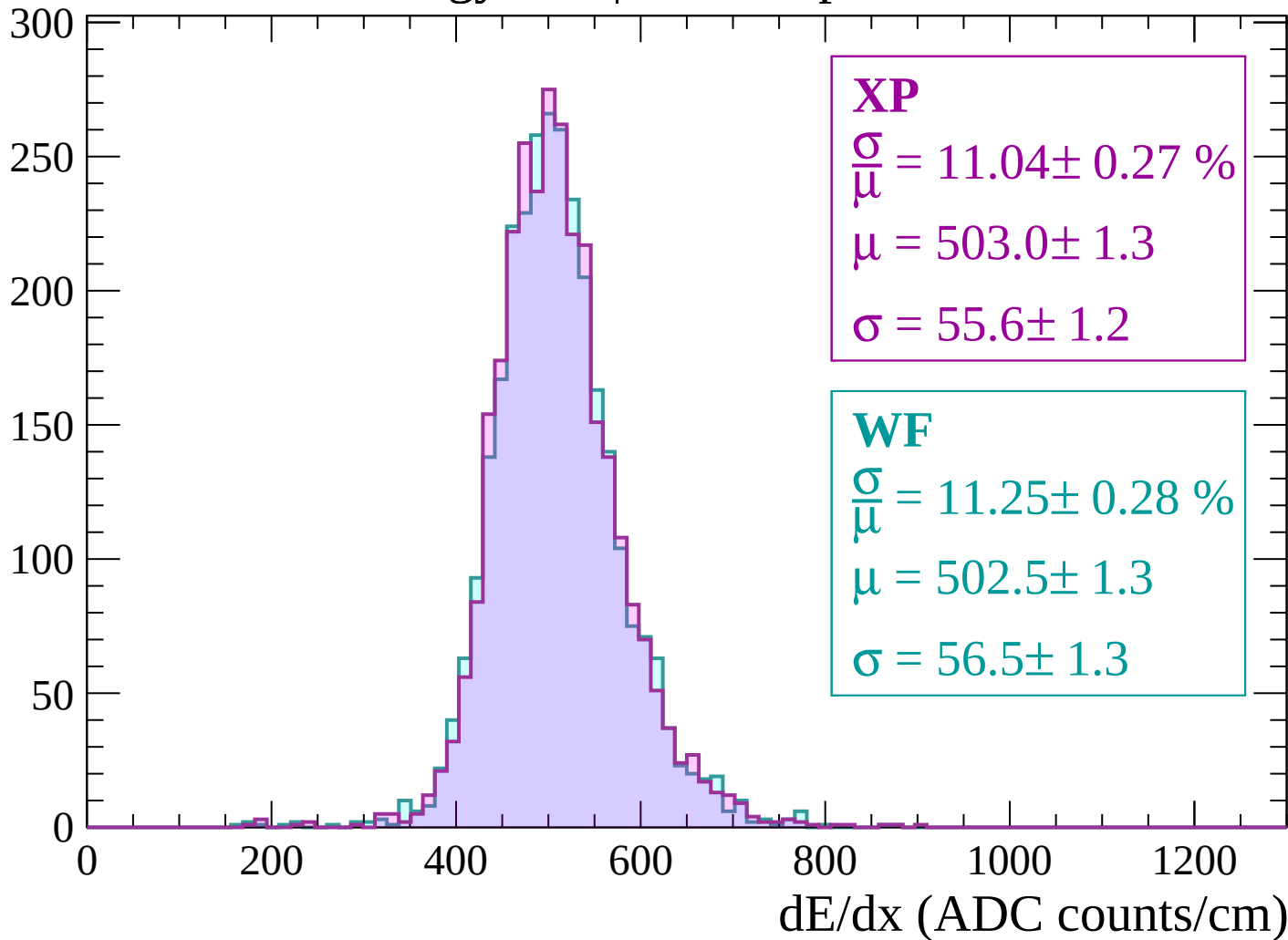
Energy loss | $-2580 < p < -2520$

Count



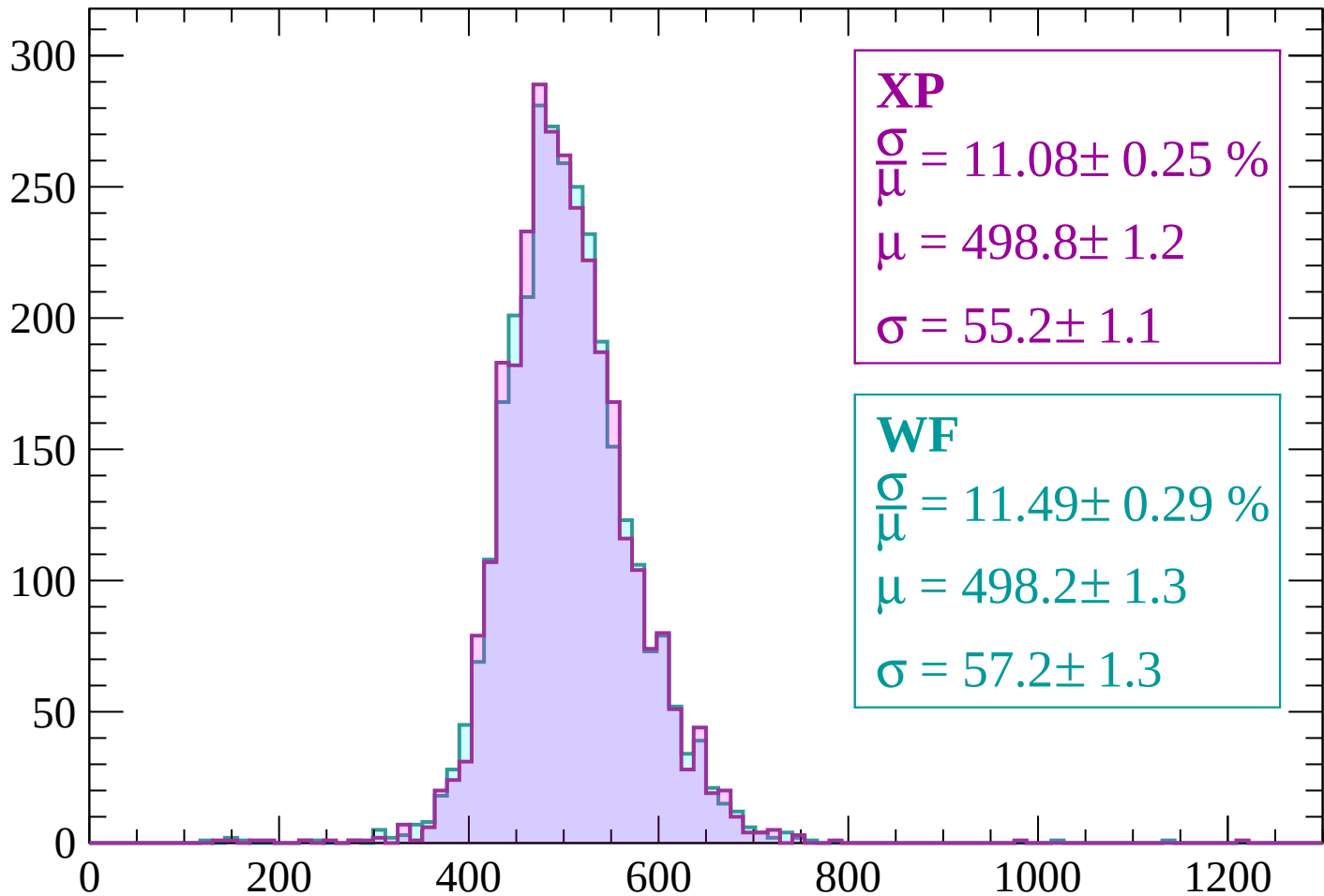
Energy loss | $-2520 < p < -2460$

Count



Energy loss | $-2460 < p < -2400$

Count



XP

$$\frac{\sigma}{\mu} = 11.08 \pm 0.25 \%$$

$$\mu = 498.8 \pm 1.2$$

$$\sigma = 55.2 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 11.49 \pm 0.29 \%$$

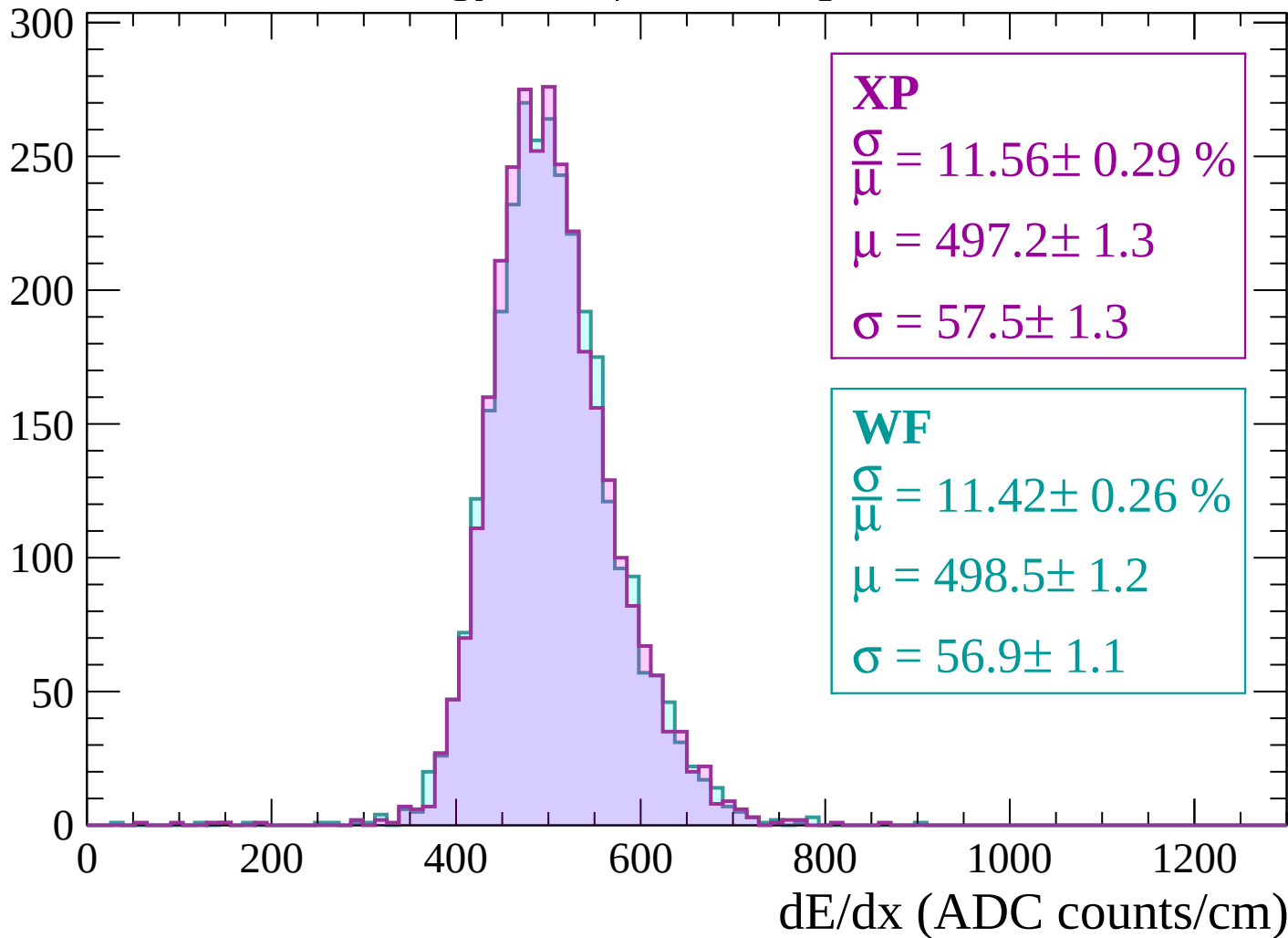
$$\mu = 498.2 \pm 1.3$$

$$\sigma = 57.2 \pm 1.3$$

dE/dx (ADC counts/cm)

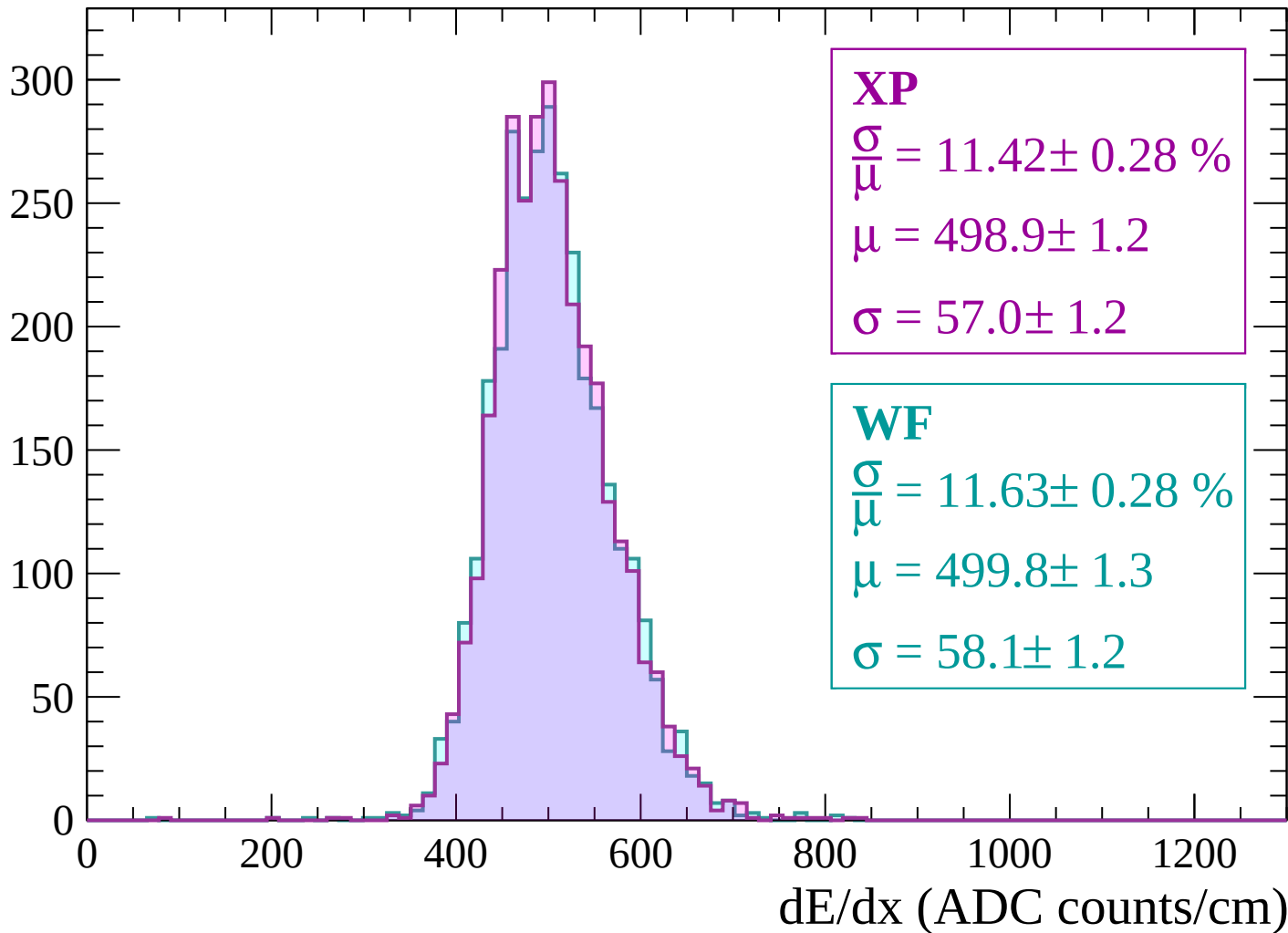
Energy loss | $-2400 < p < -2340$

Count



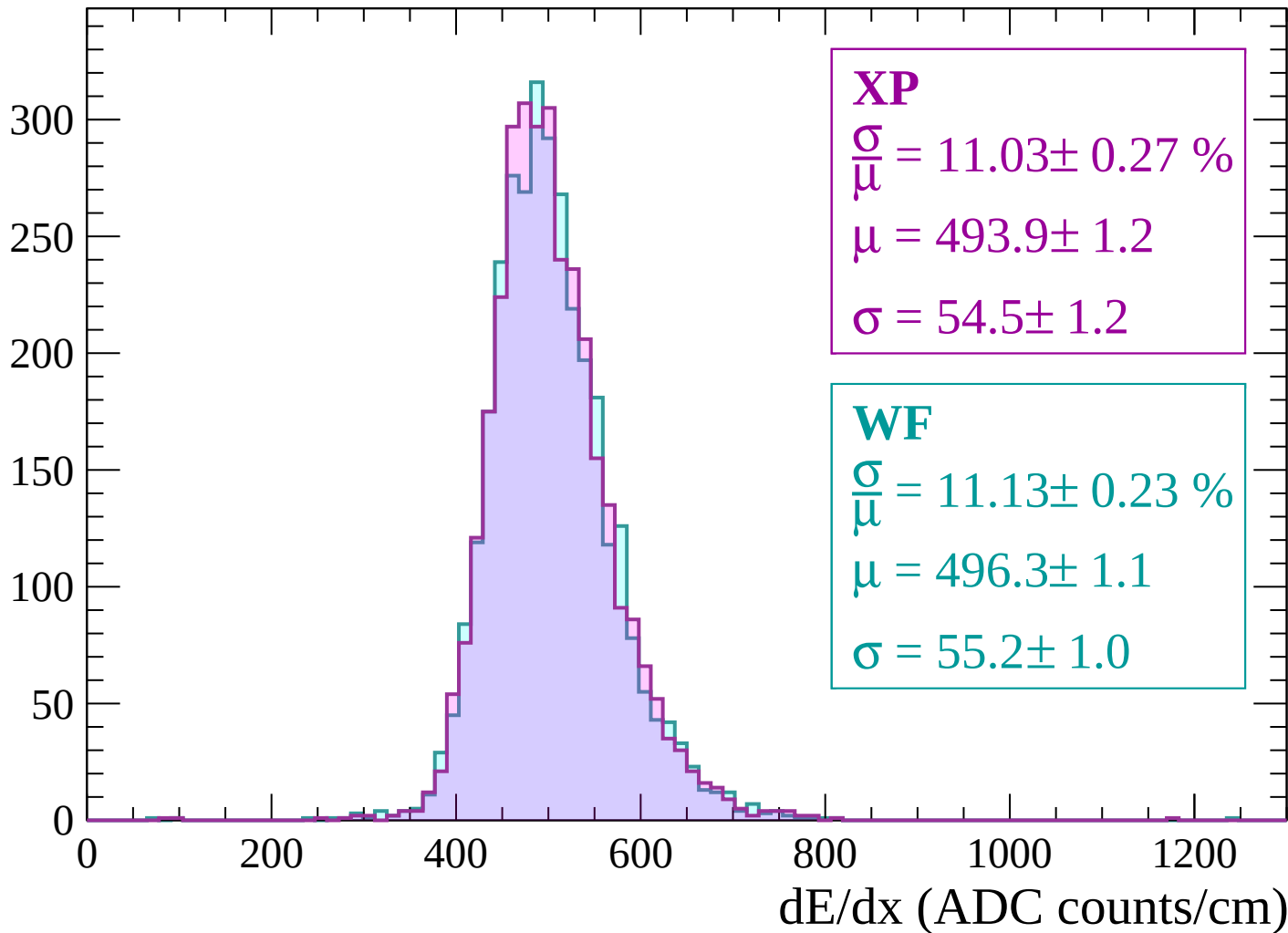
Energy loss | $-2340 < p < -2280$

Count



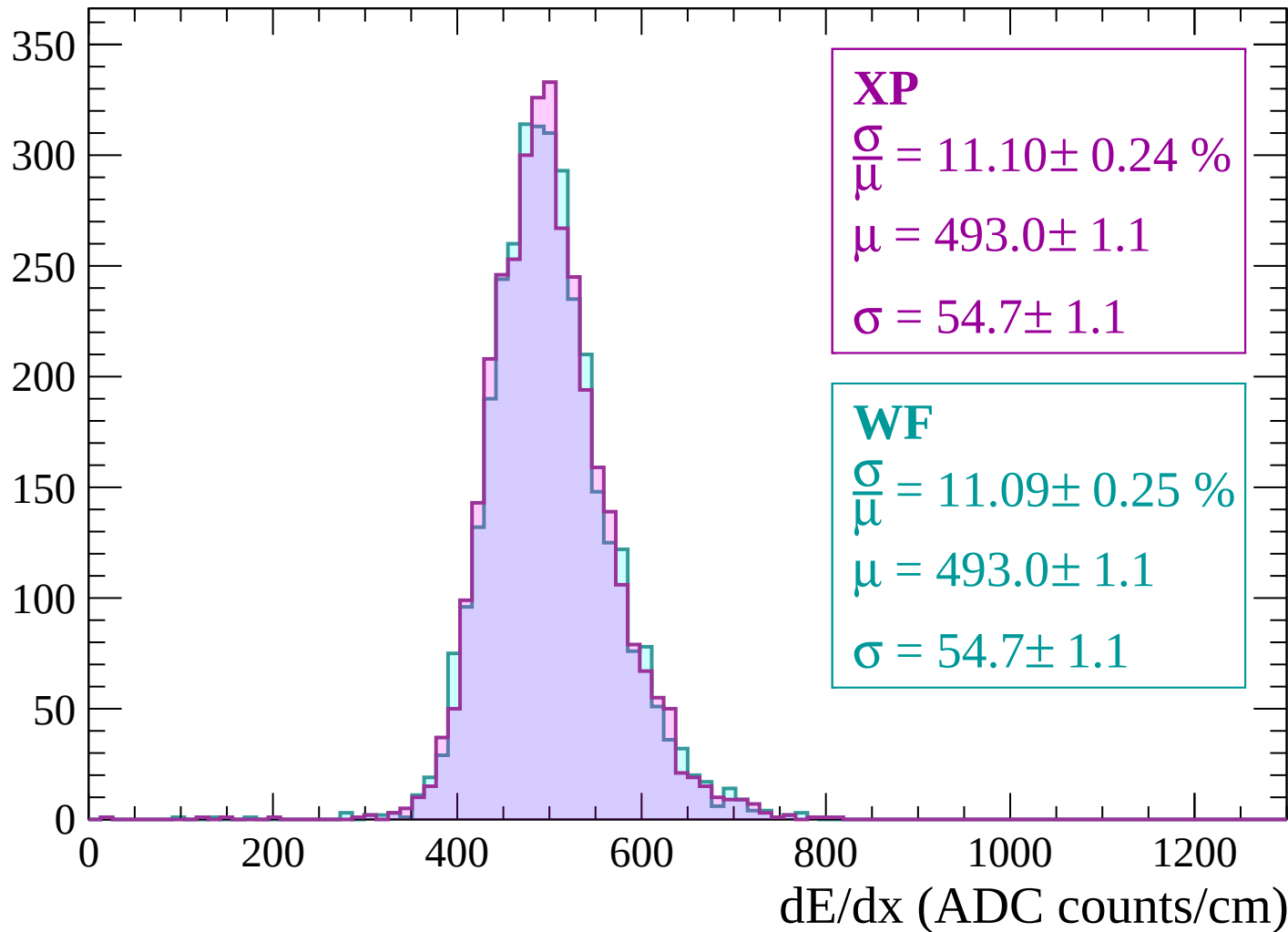
Energy loss | $-2280 < p < -2220$

Count



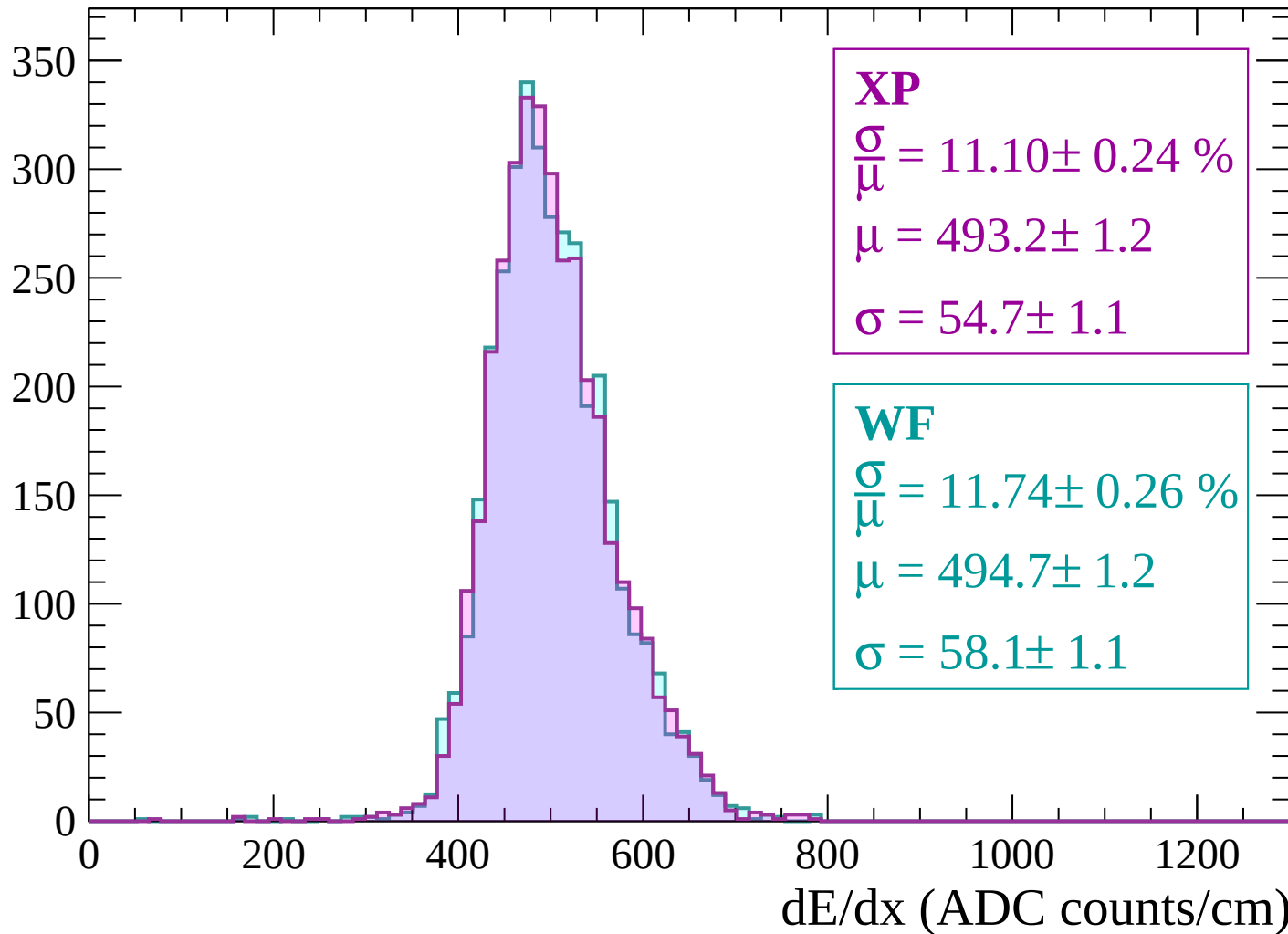
Energy loss | -2220 < p < -2160

Count



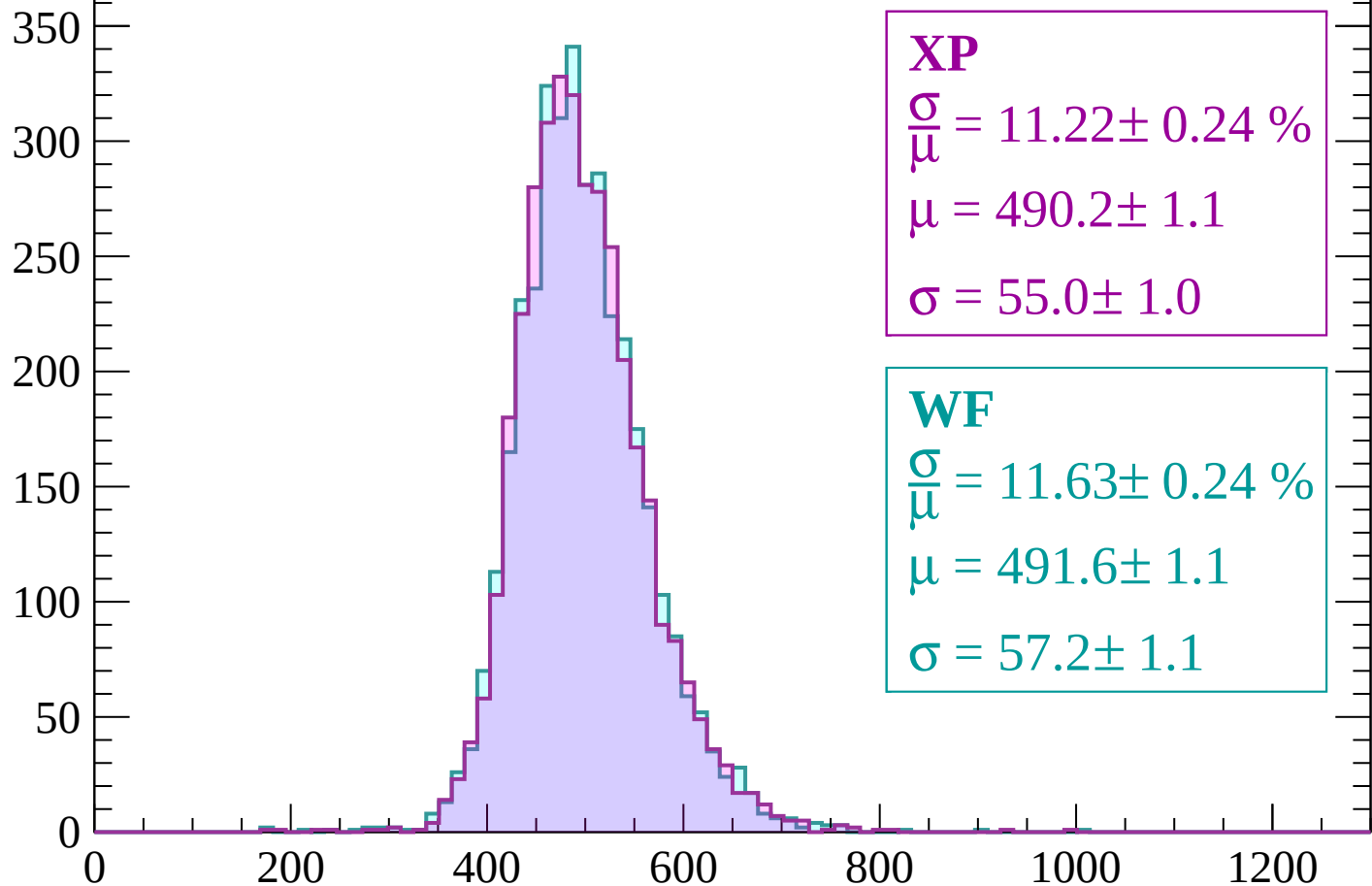
Energy loss | -2160 < p < -2100

Count



Energy loss | $-2100 < p < -2040$

Count



XP

$$\frac{\sigma}{\mu} = 11.22 \pm 0.24 \%$$

$$\mu = 490.2 \pm 1.1$$

$$\sigma = 55.0 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 11.63 \pm 0.24 \%$$

$$\mu = 491.6 \pm 1.1$$

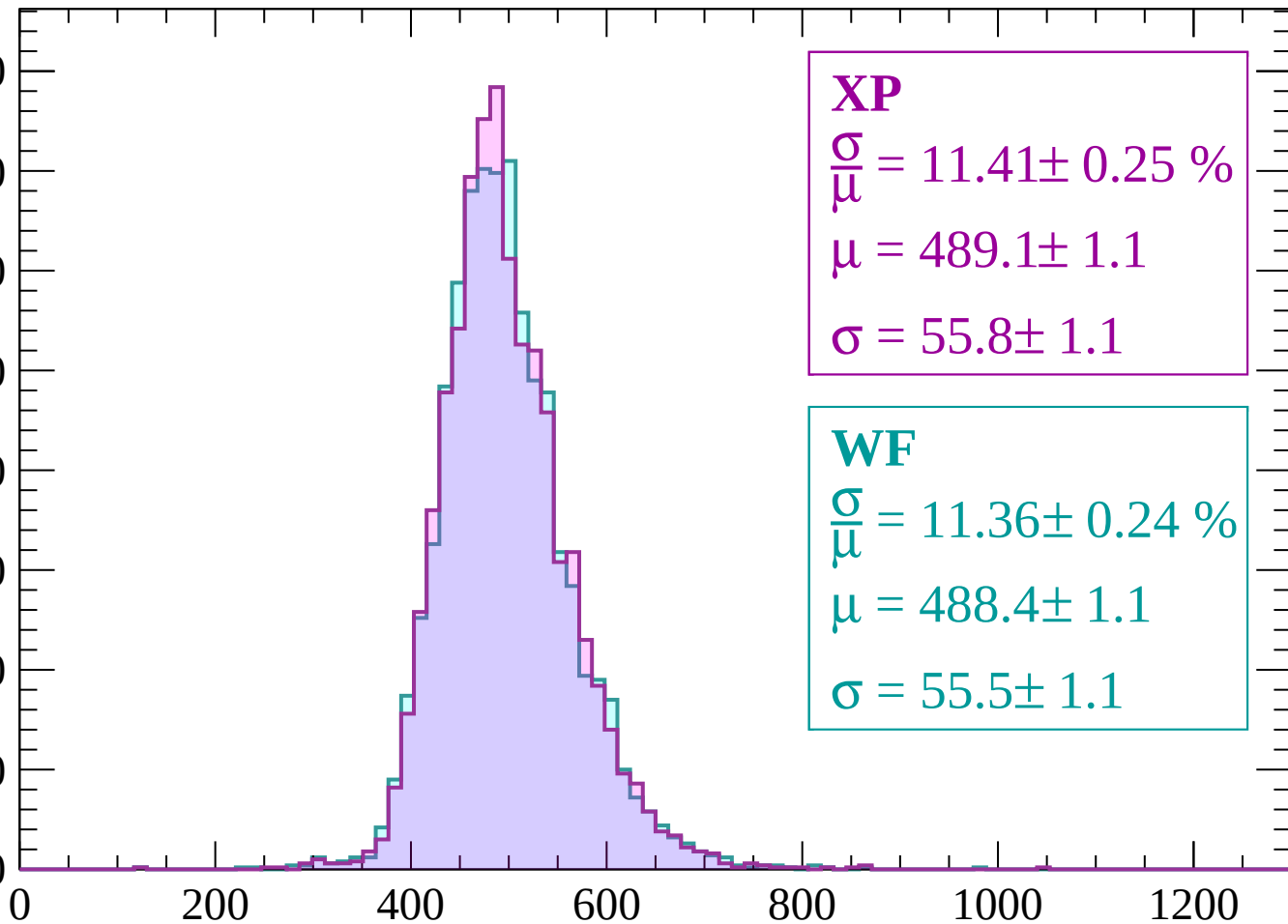
$$\sigma = 57.2 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $-2040 < p < -1980$

Count

400
350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 11.41 \pm 0.25 \%$$

$$\mu = 489.1 \pm 1.1$$

$$\sigma = 55.8 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 11.36 \pm 0.24 \%$$

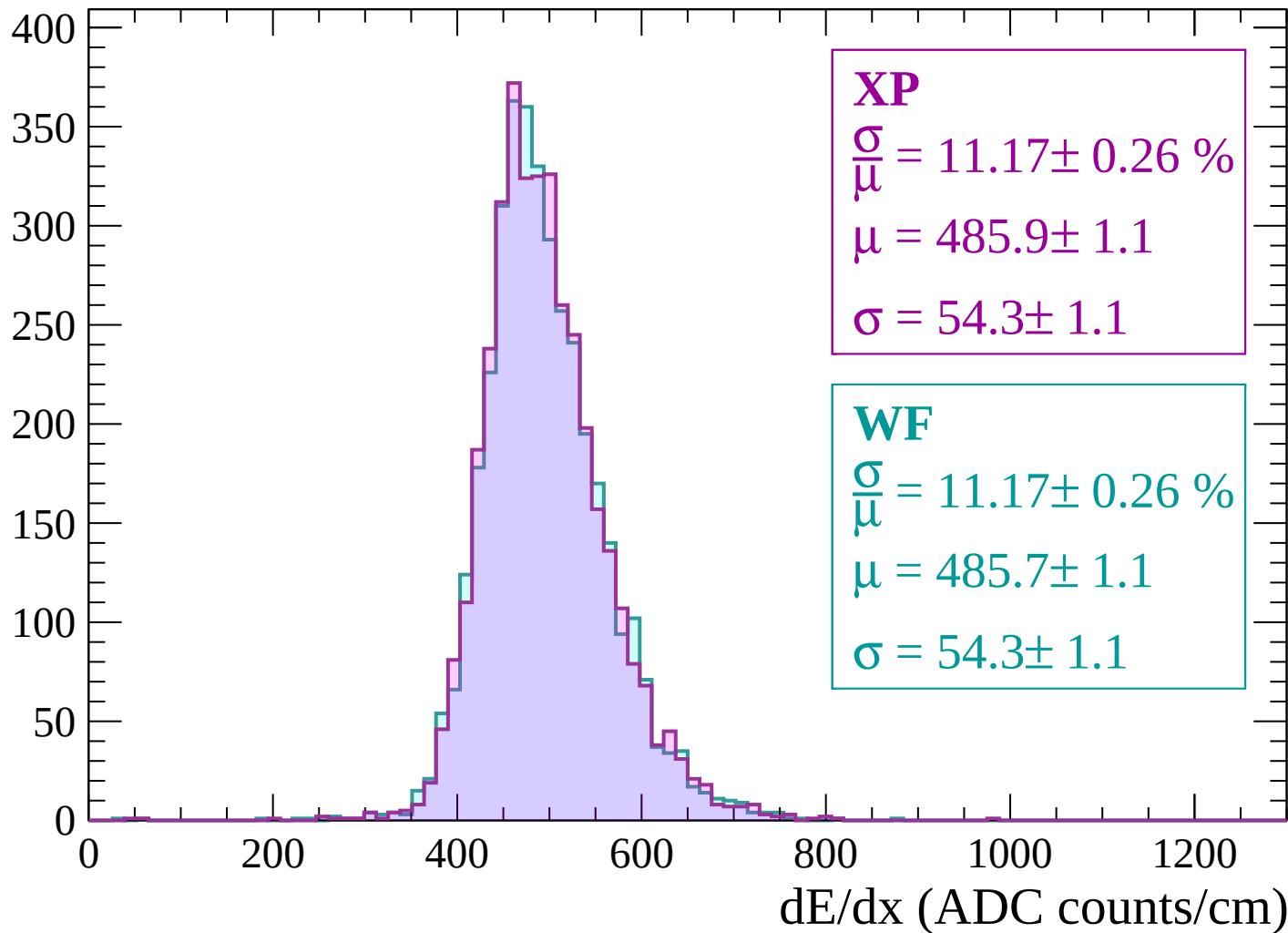
$$\mu = 488.4 \pm 1.1$$

$$\sigma = 55.5 \pm 1.1$$

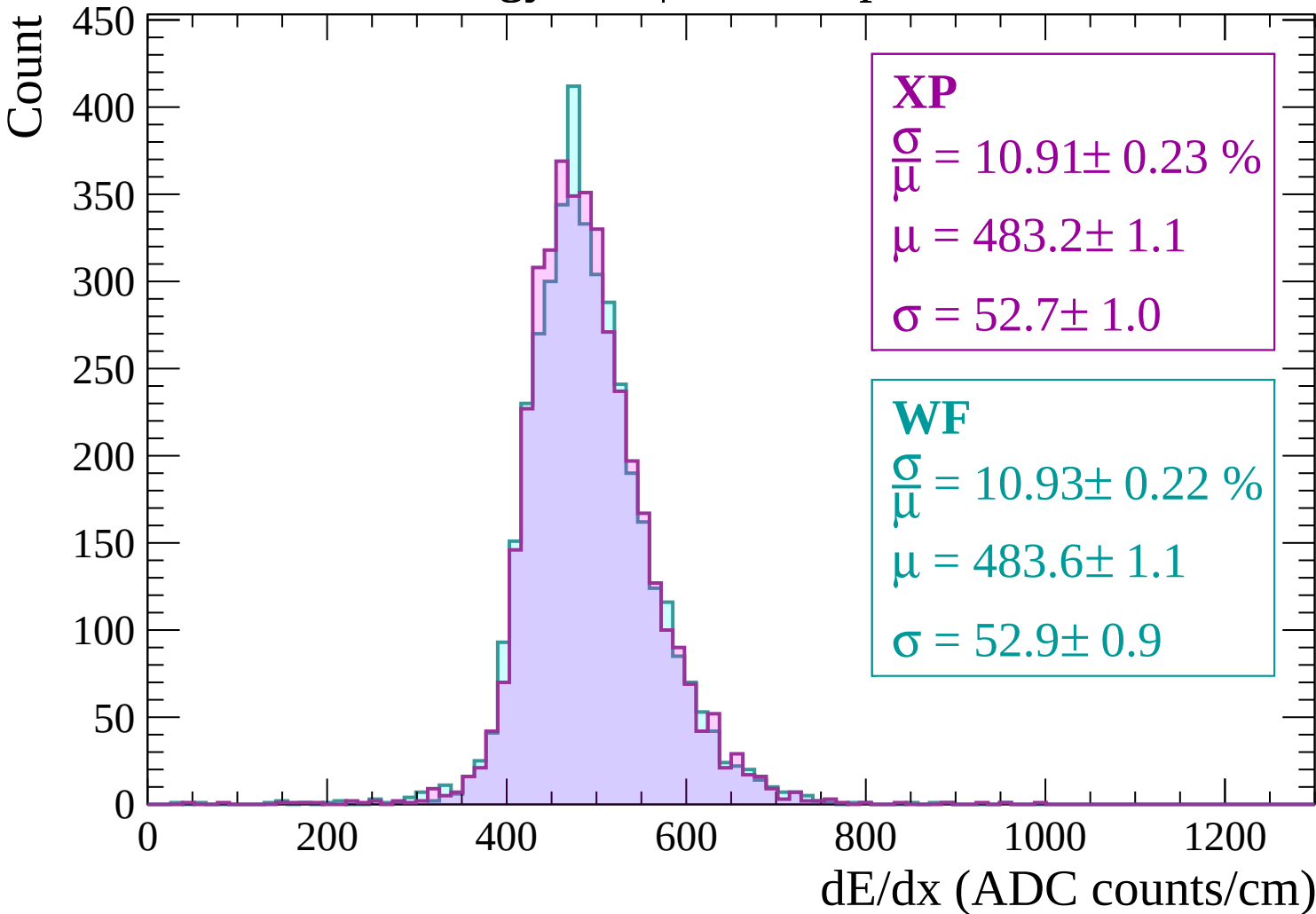
dE/dx (ADC counts/cm)

Energy loss | -1980 < p < -1920

Count

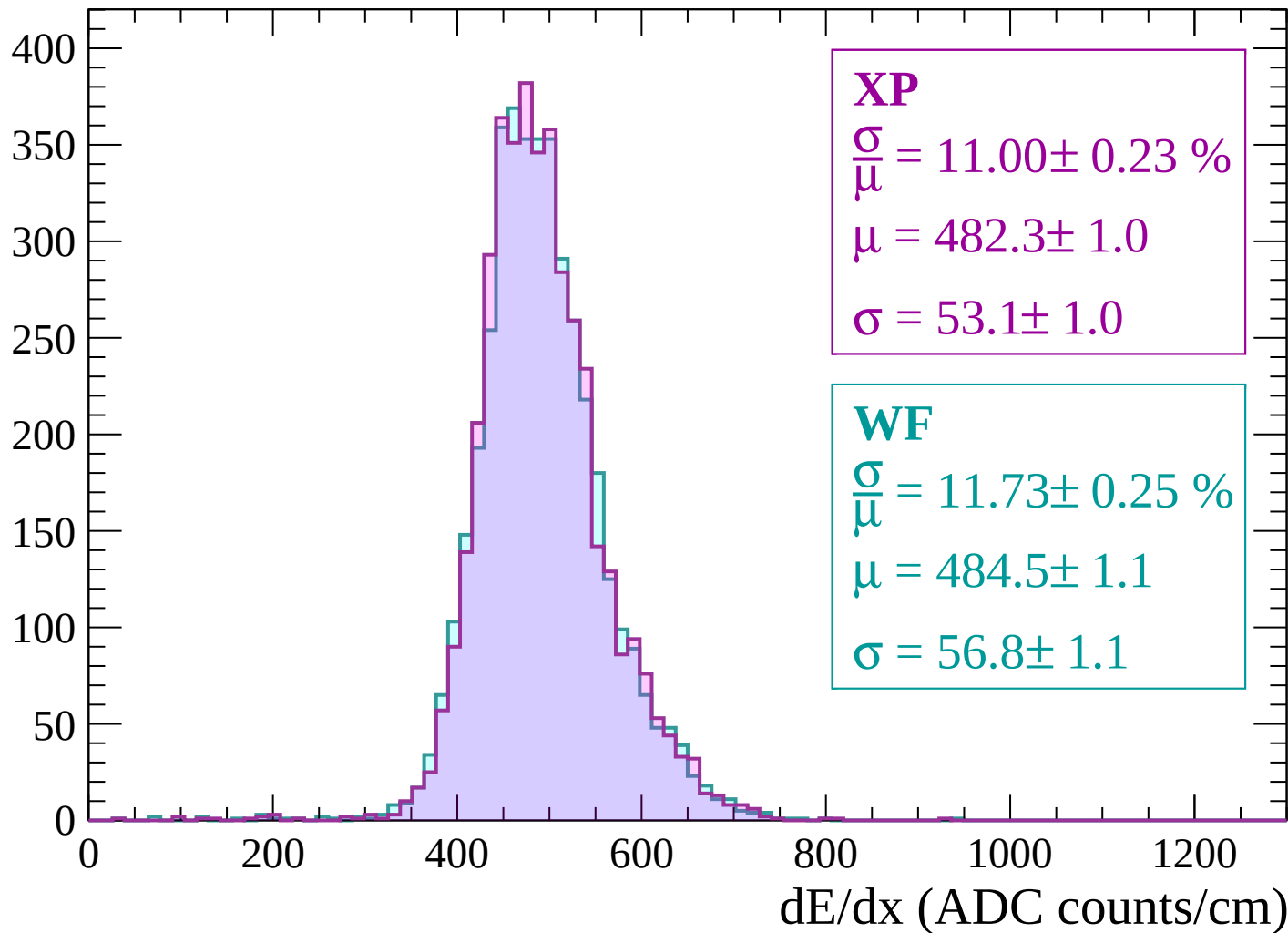


Energy loss | -1920 < p < -1860



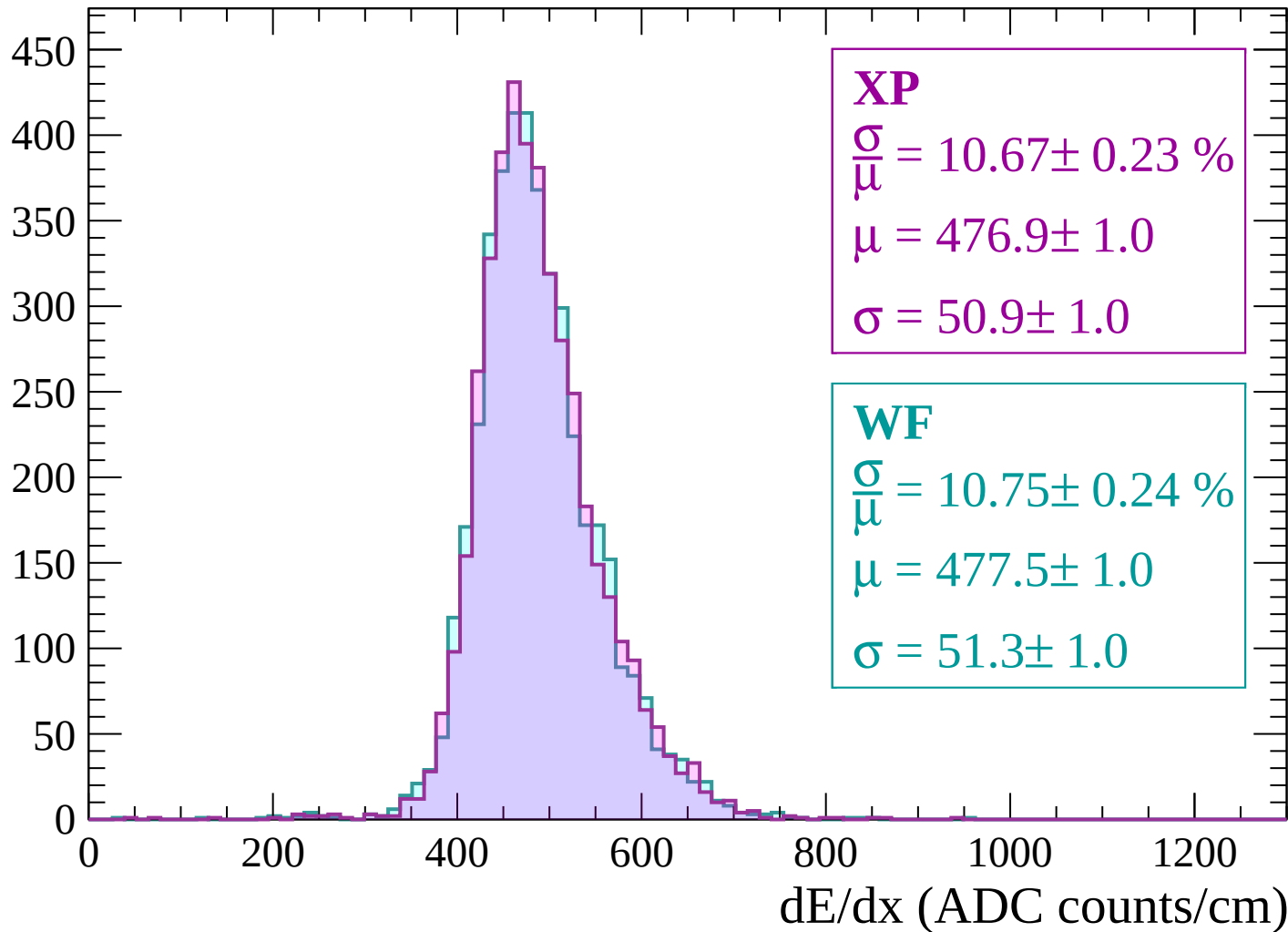
Energy loss | -1860 < p < -1800

Count



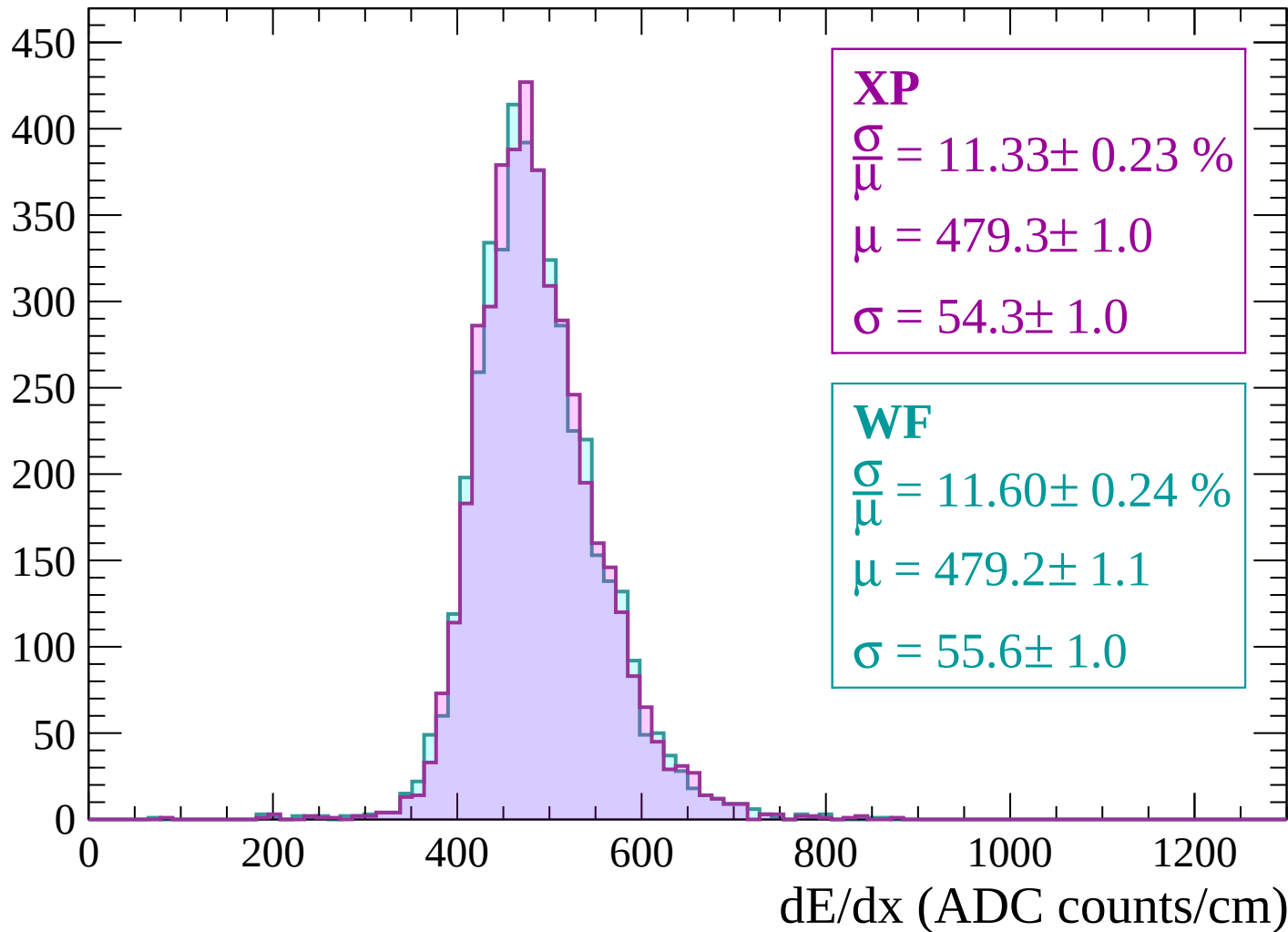
Energy loss | -1800 < p < -1740

Count



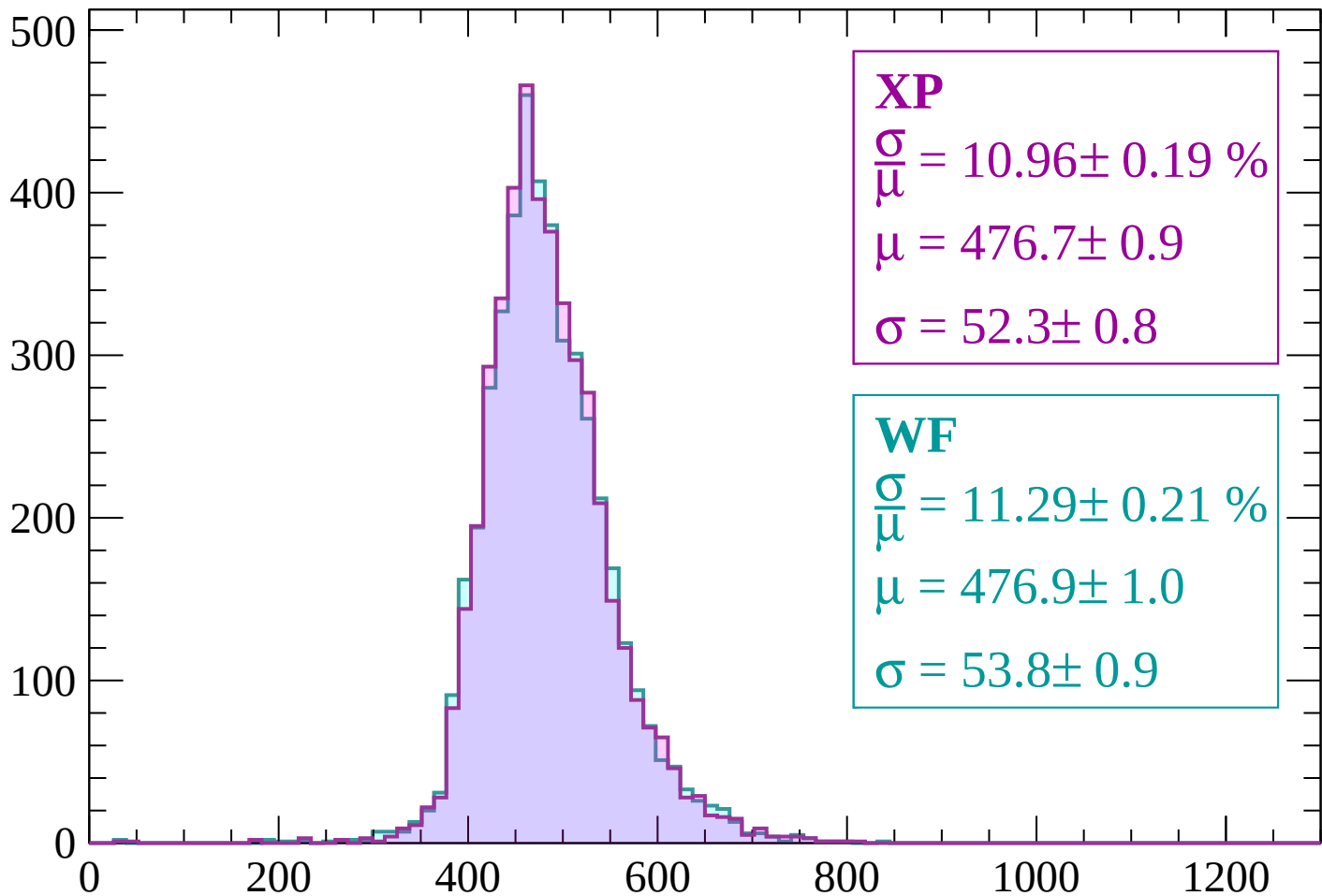
Energy loss | $-1740 < p < -1680$

Count



Energy loss | $-1680 < p < -1620$

Count



XP

$$\frac{\sigma}{\mu} = 10.96 \pm 0.19 \%$$

$$\mu = 476.7 \pm 0.9$$

$$\sigma = 52.3 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 11.29 \pm 0.21 \%$$

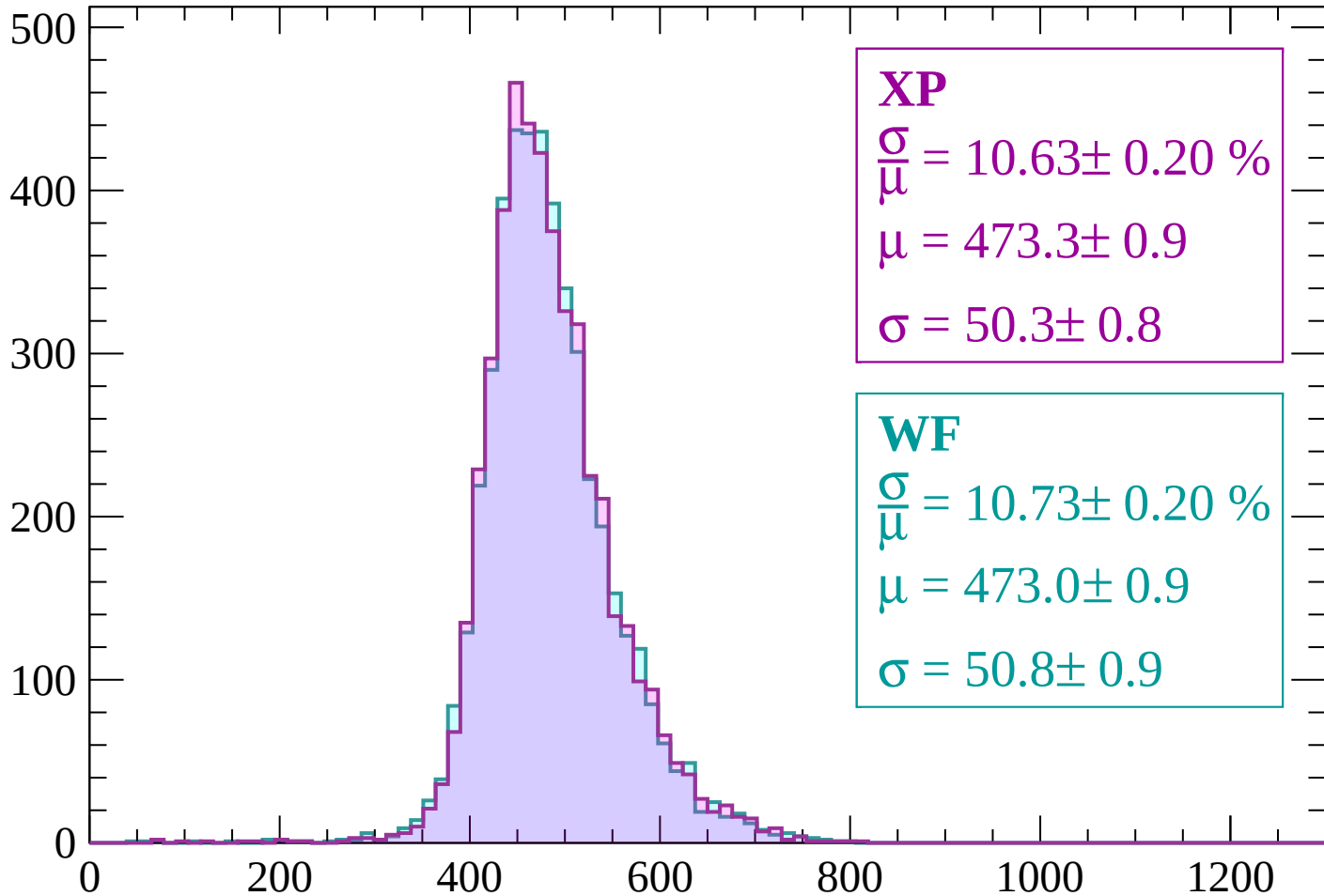
$$\mu = 476.9 \pm 1.0$$

$$\sigma = 53.8 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-1620 < p < -1560$

Count



XP

$$\frac{\sigma}{\mu} = 10.63 \pm 0.20 \%$$

$$\mu = 473.3 \pm 0.9$$

$$\sigma = 50.3 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 10.73 \pm 0.20 \%$$

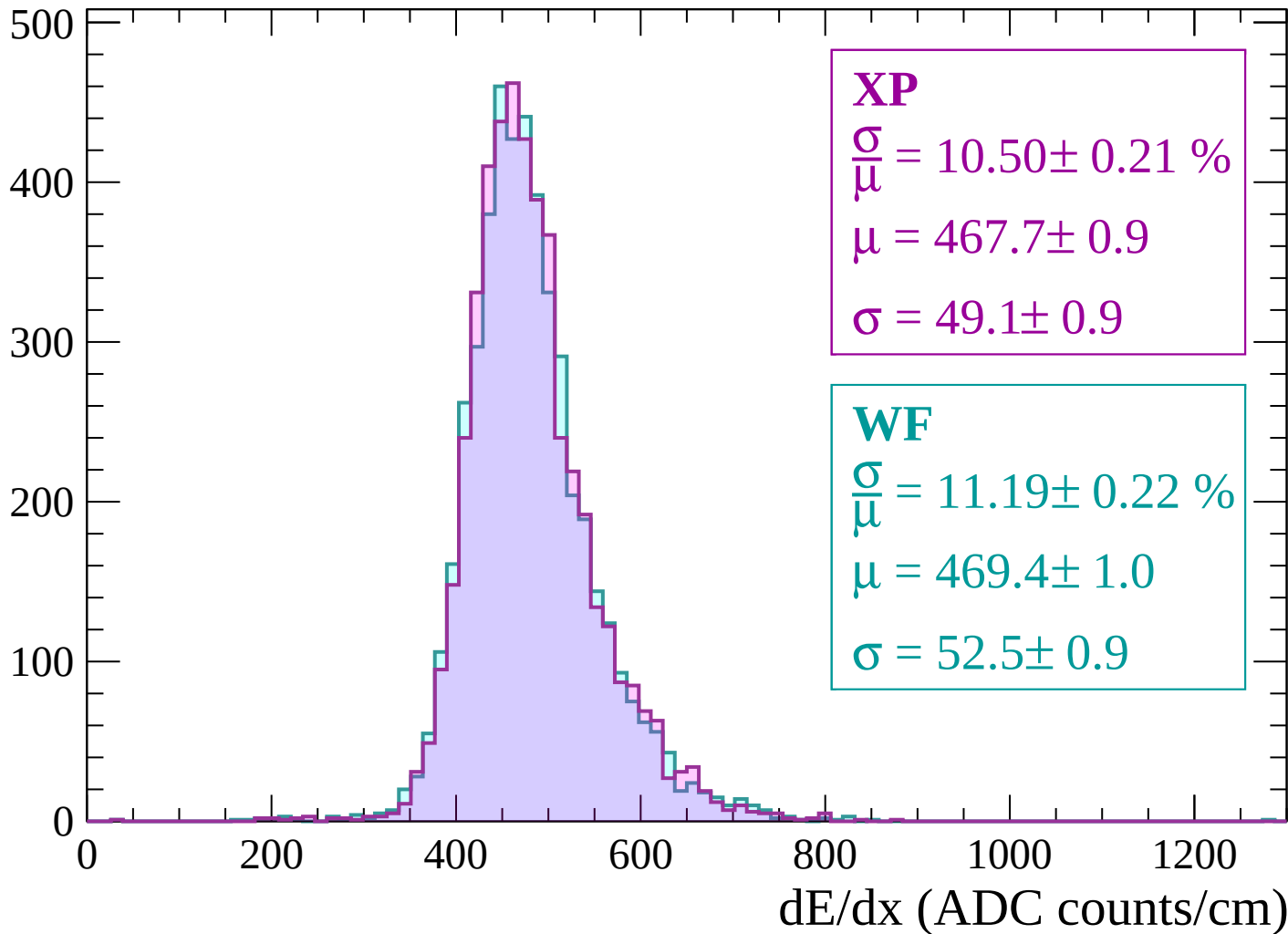
$$\mu = 473.0 \pm 0.9$$

$$\sigma = 50.8 \pm 0.9$$

dE/dx (ADC counts/cm)

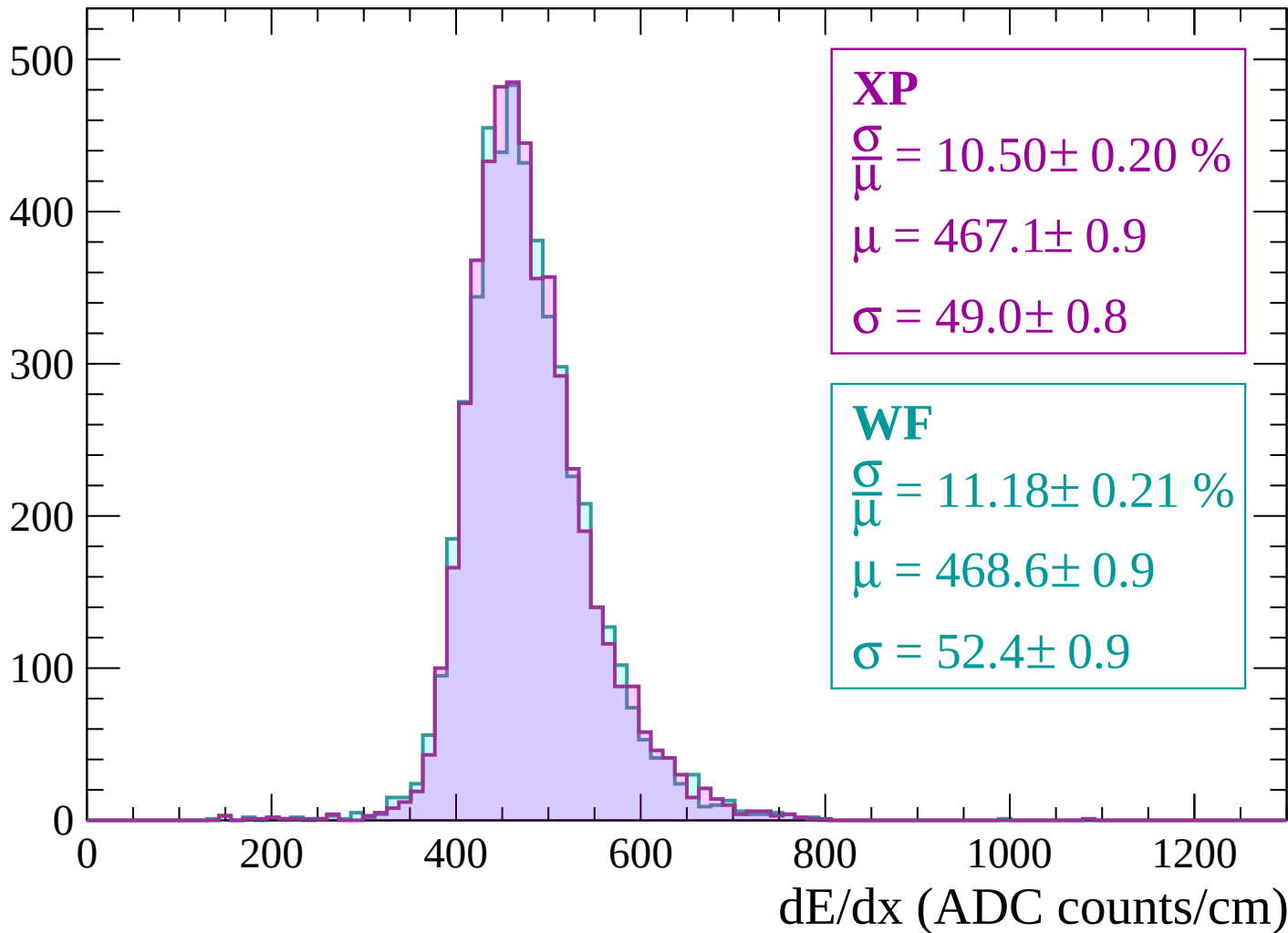
Energy loss | -1560 < p < -1500

Count



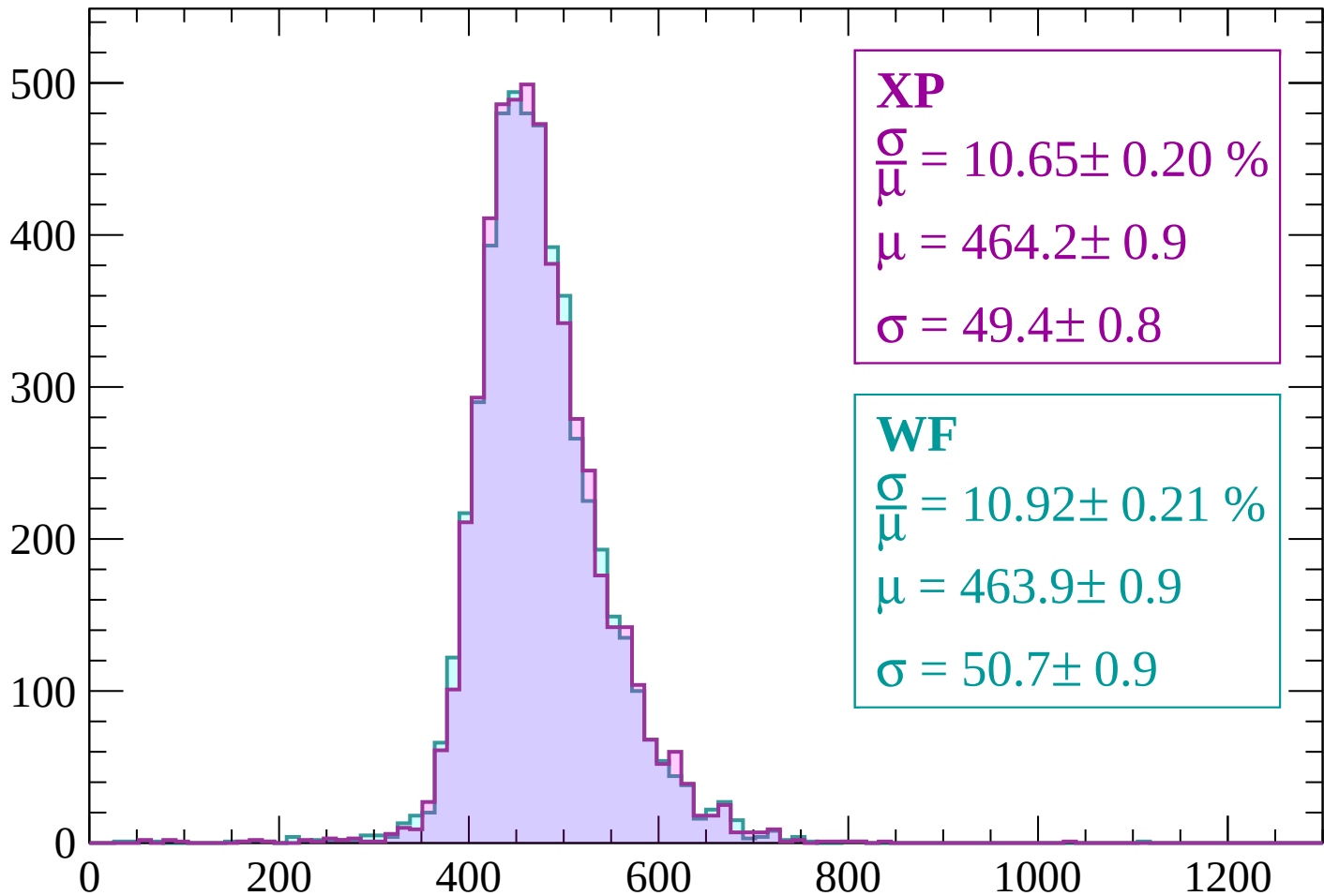
Energy loss | -1500 < p < -1440

Count



Energy loss | $-1440 < p < -1380$

Count



XP

$$\frac{\sigma}{\mu} = 10.65 \pm 0.20 \%$$

$$\mu = 464.2 \pm 0.9$$

$$\sigma = 49.4 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 10.92 \pm 0.21 \%$$

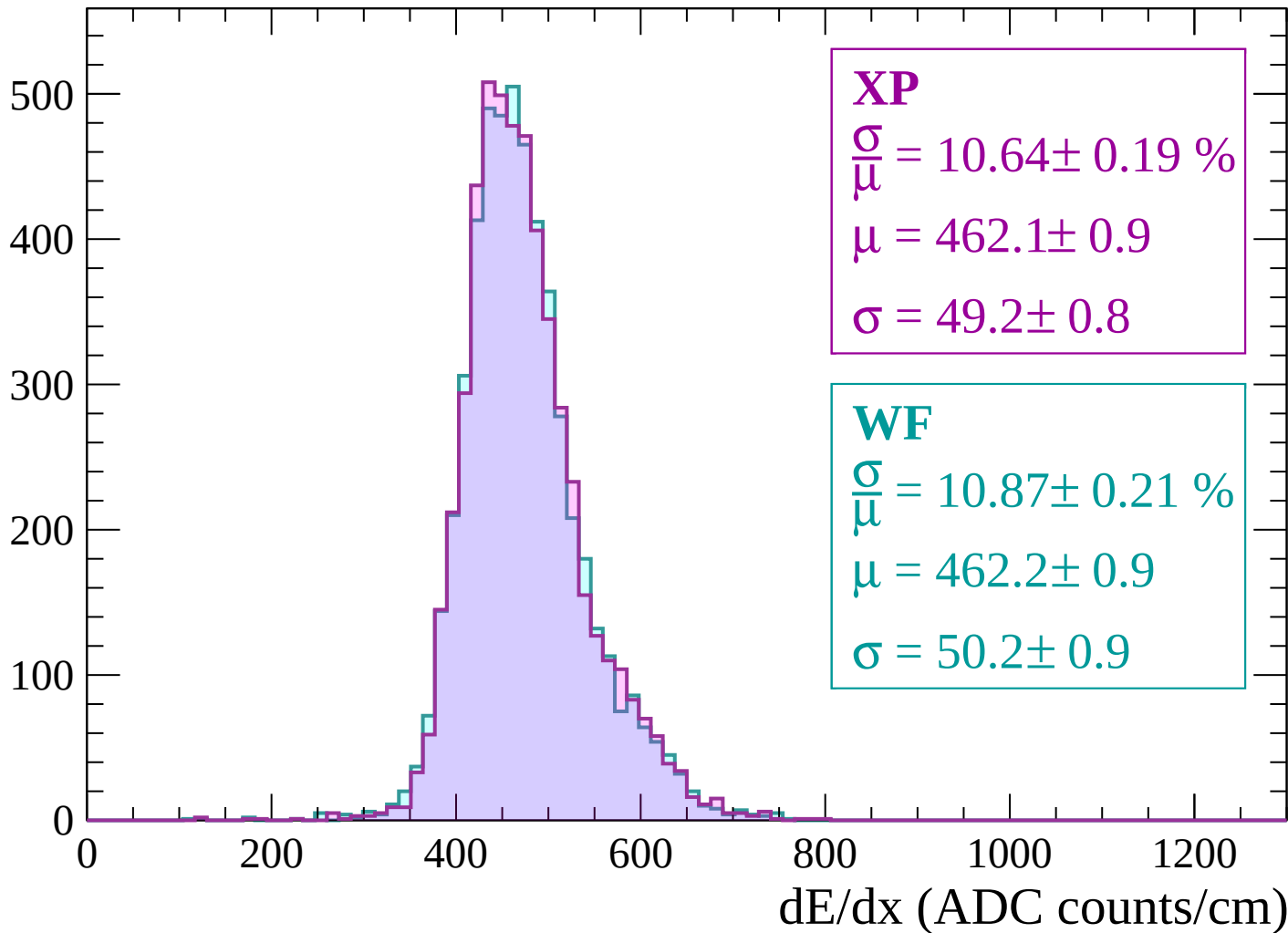
$$\mu = 463.9 \pm 0.9$$

$$\sigma = 50.7 \pm 0.9$$

dE/dx (ADC counts/cm)

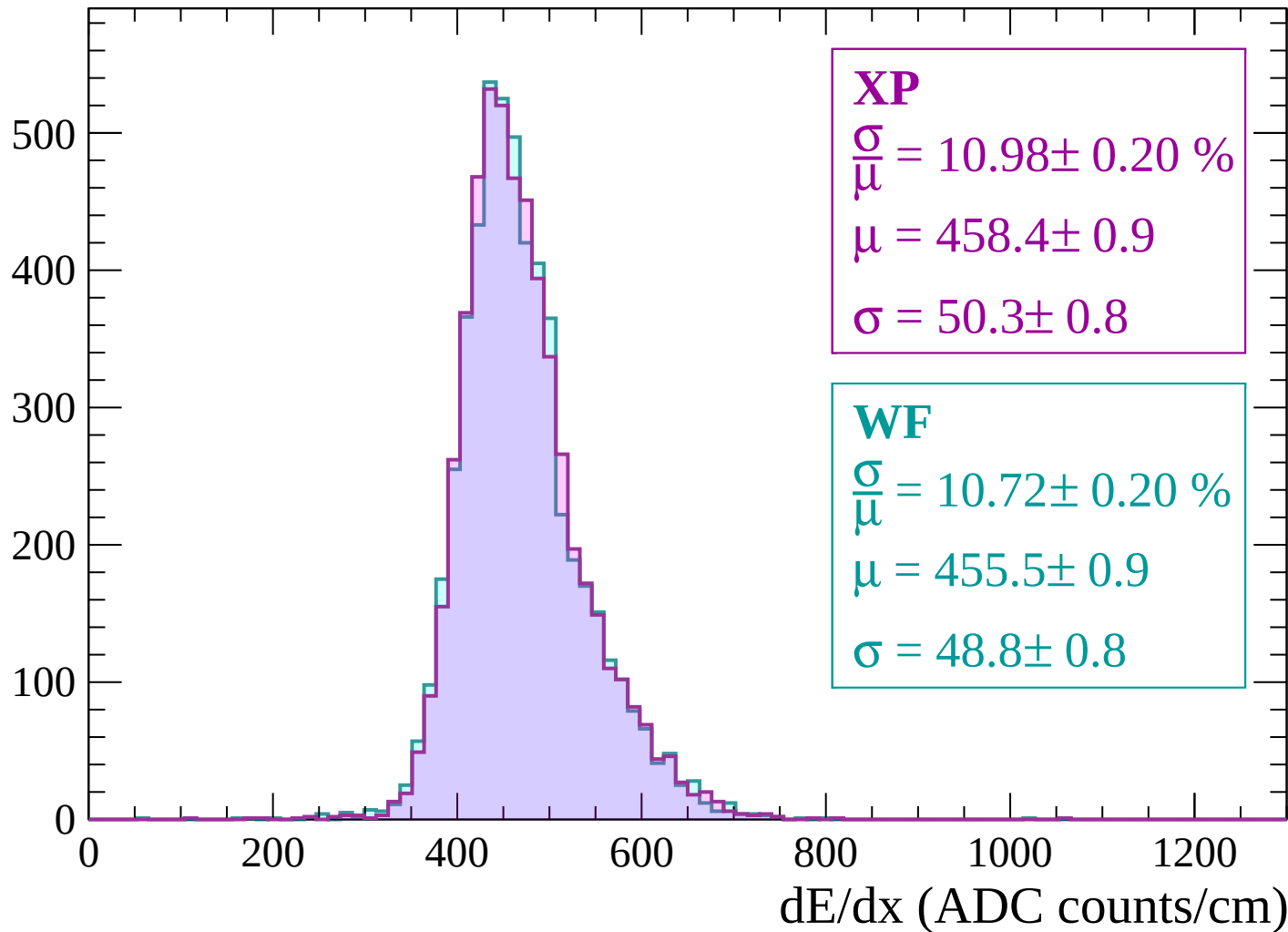
Energy loss | $-1380 < p < -1320$

Count

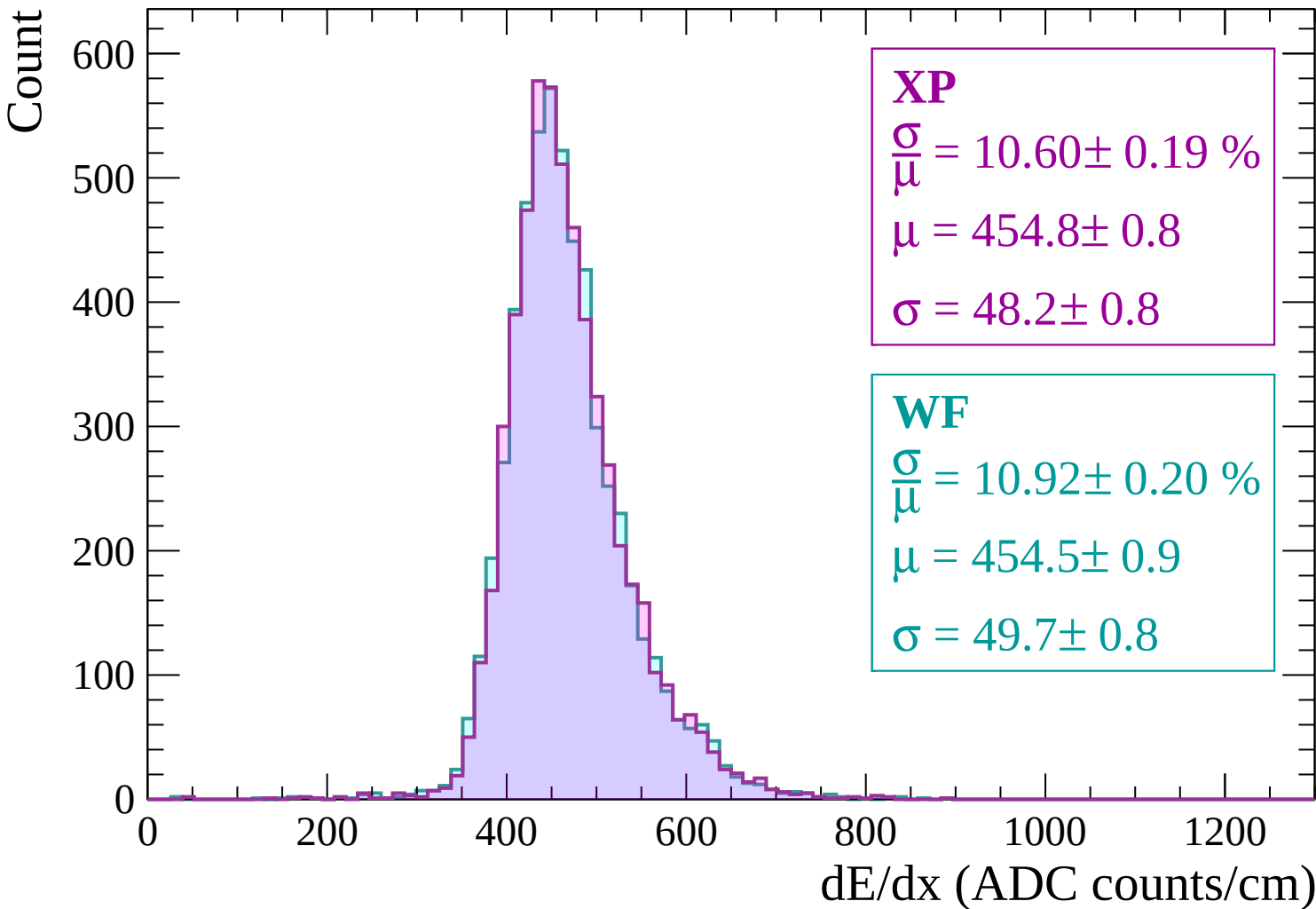


Energy loss | -1320 < p < -1260

Count

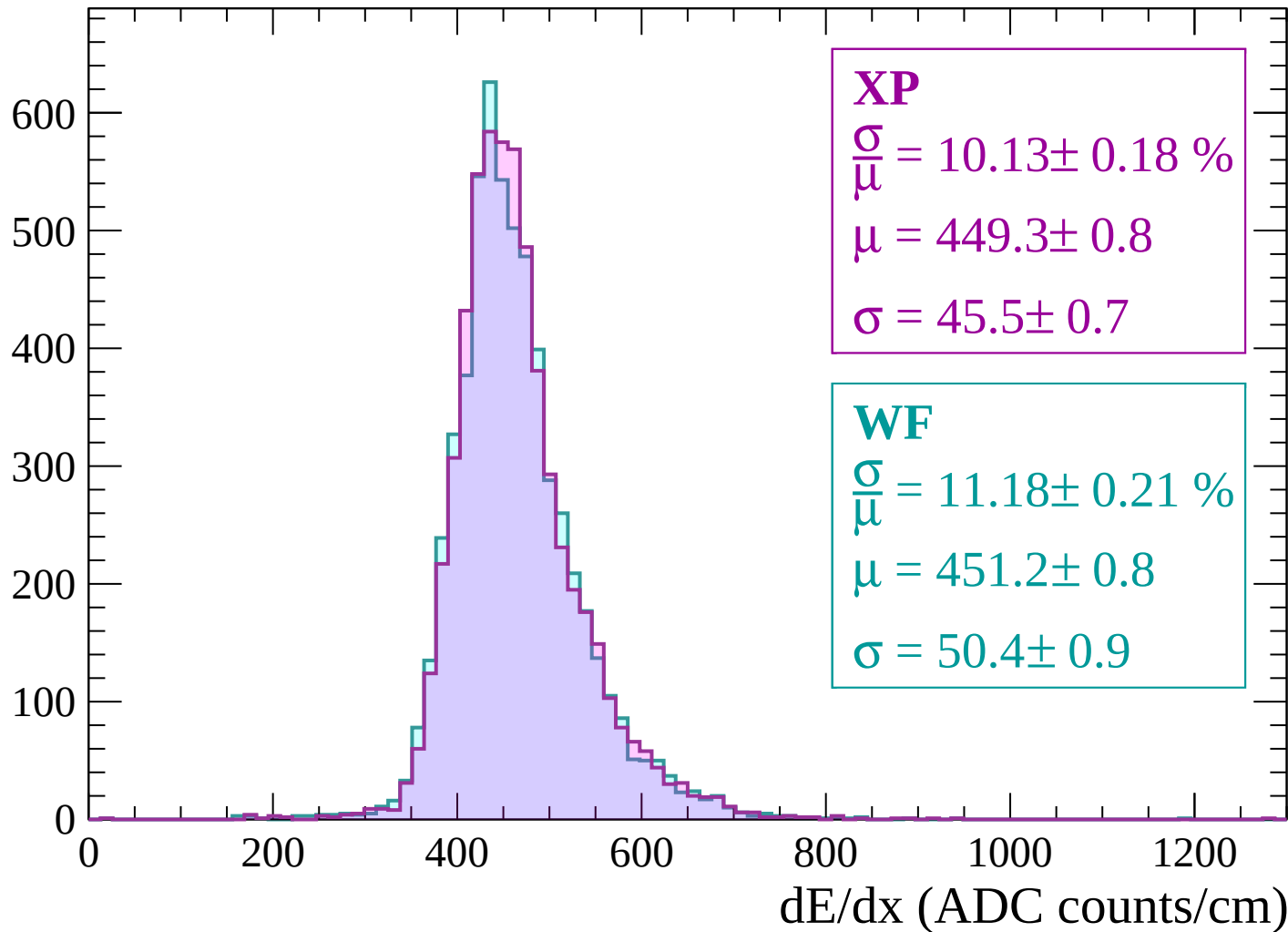


Energy loss | -1260 < p < -1200



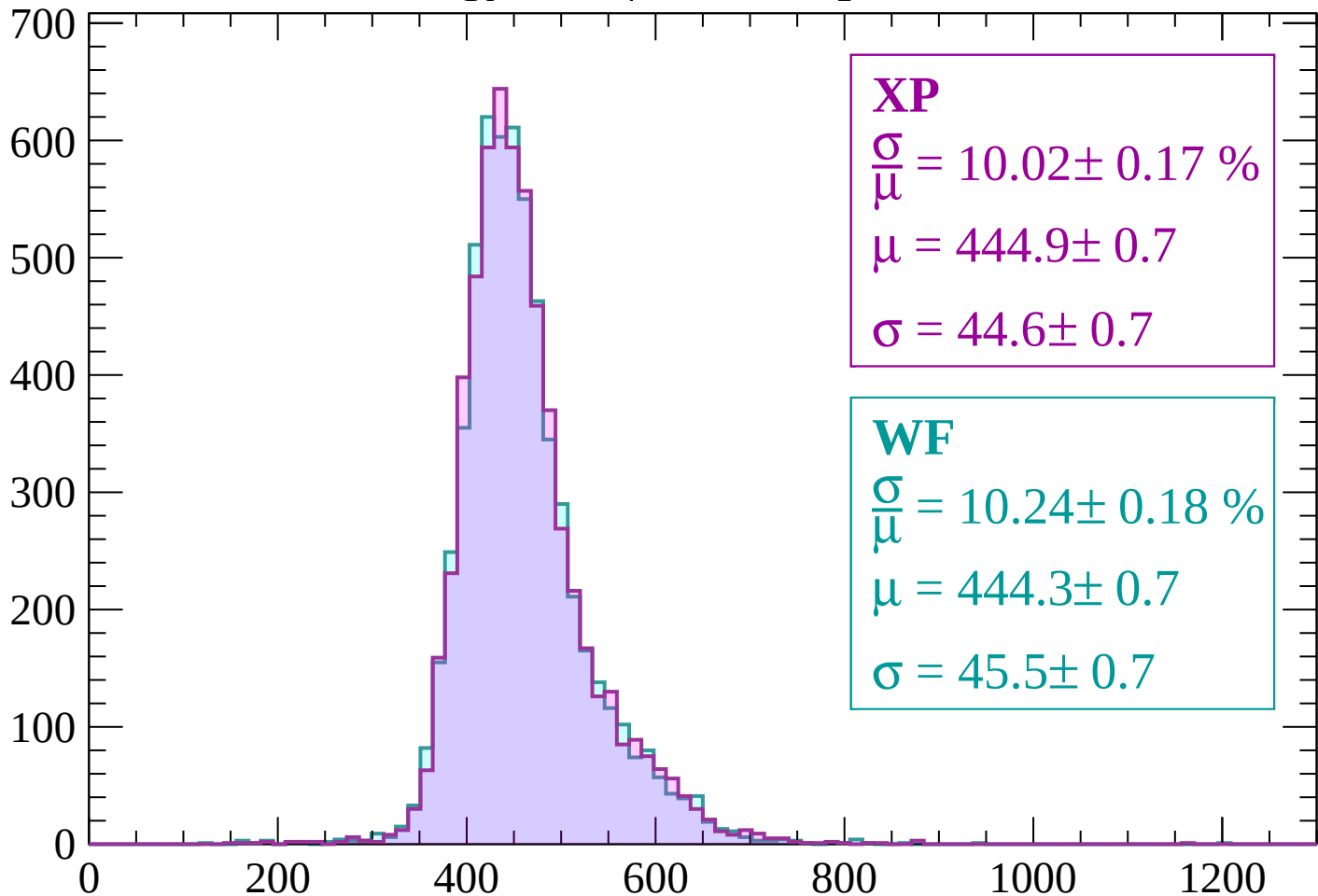
Energy loss | $-1200 < p < -1140$

Count



Energy loss | $-1140 < p < -1080$

Count



XP

$$\frac{\sigma}{\mu} = 10.02 \pm 0.17 \%$$

$$\mu = 444.9 \pm 0.7$$

$$\sigma = 44.6 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 10.24 \pm 0.18 \%$$

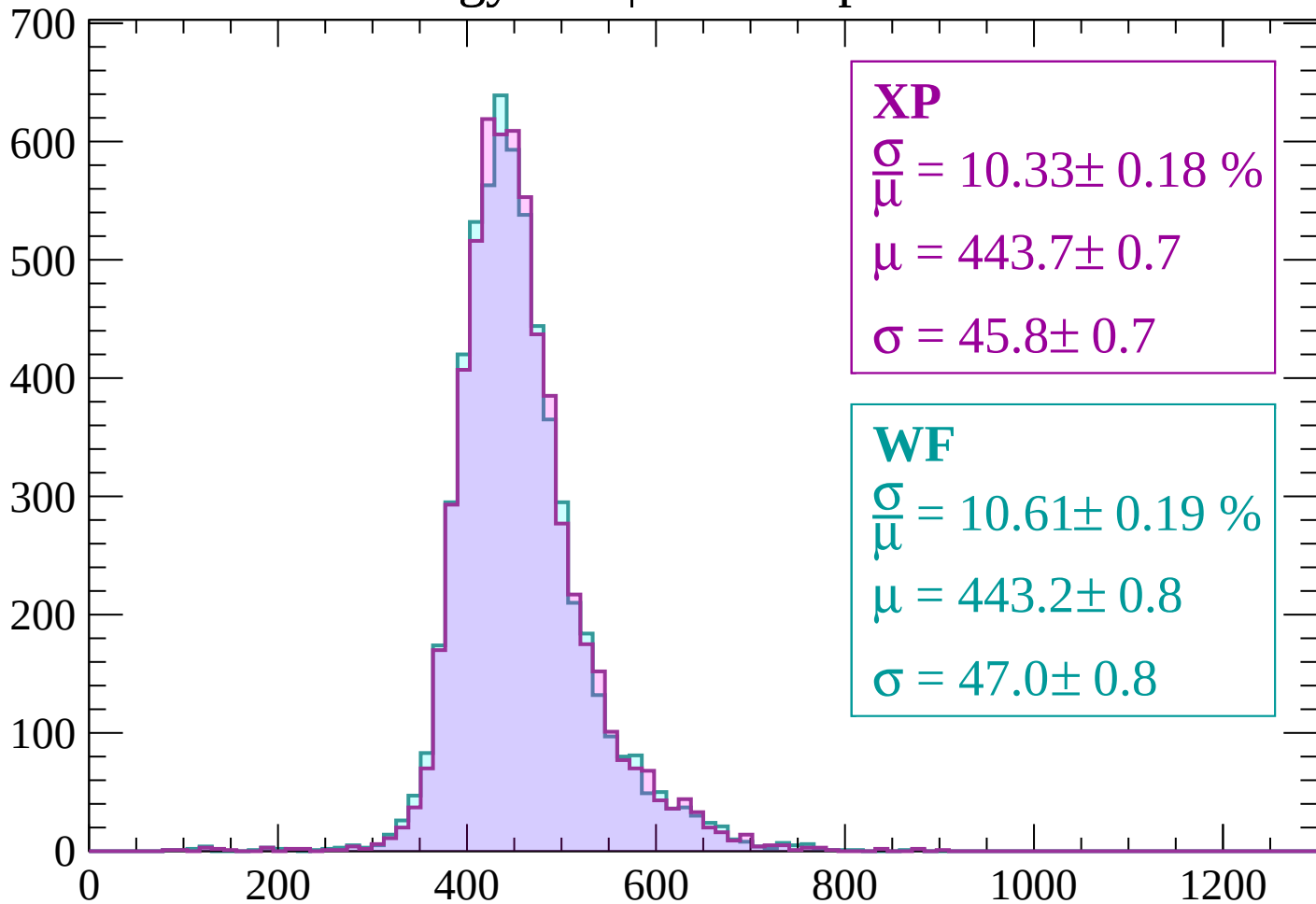
$$\mu = 444.3 \pm 0.7$$

$$\sigma = 45.5 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1080 < p < -1020$

Count



XP

$$\frac{\sigma}{\mu} = 10.33 \pm 0.18 \%$$

$$\mu = 443.7 \pm 0.7$$

$$\sigma = 45.8 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 10.61 \pm 0.19 \%$$

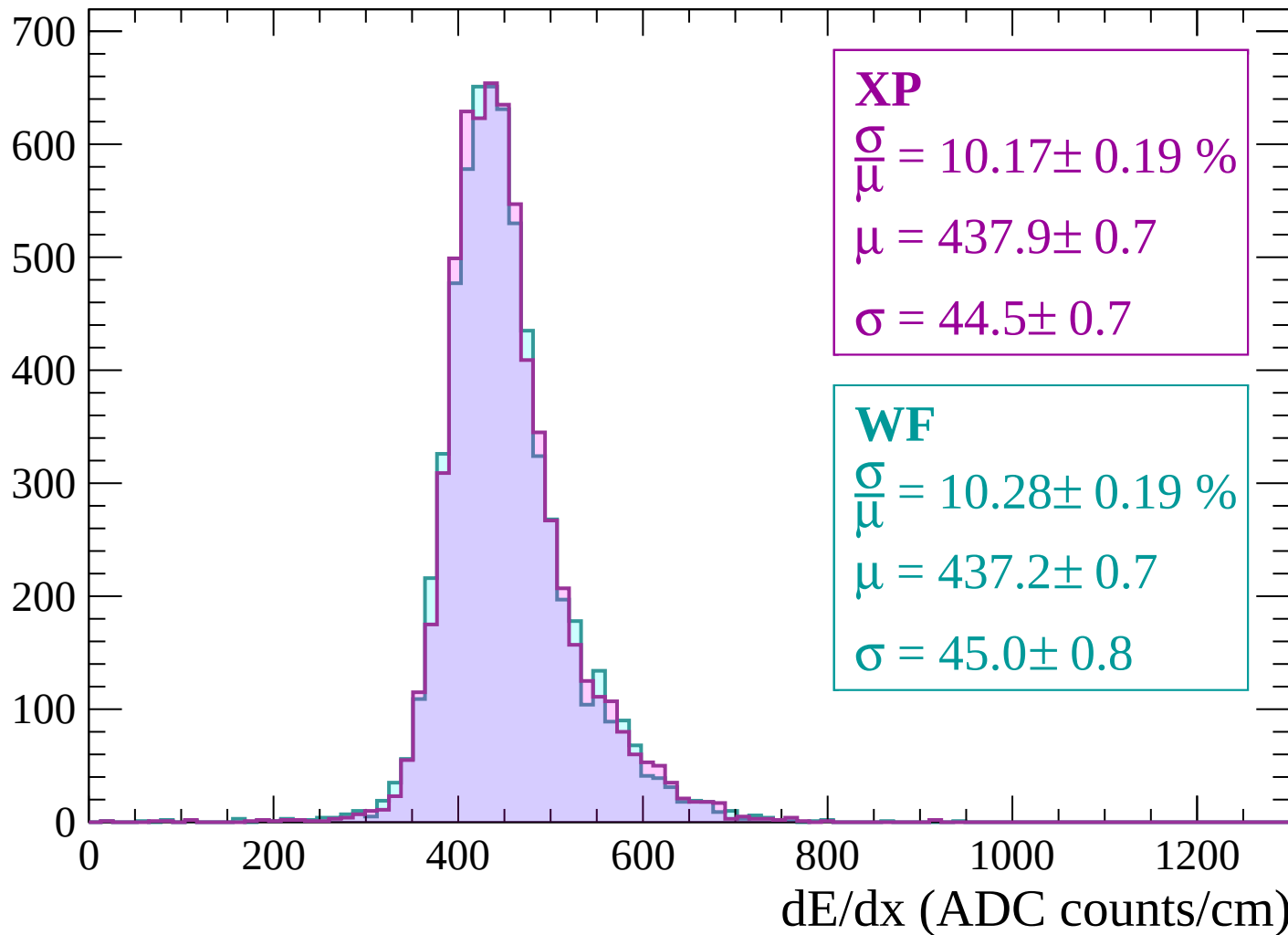
$$\mu = 443.2 \pm 0.8$$

$$\sigma = 47.0 \pm 0.8$$

dE/dx (ADC counts/cm)

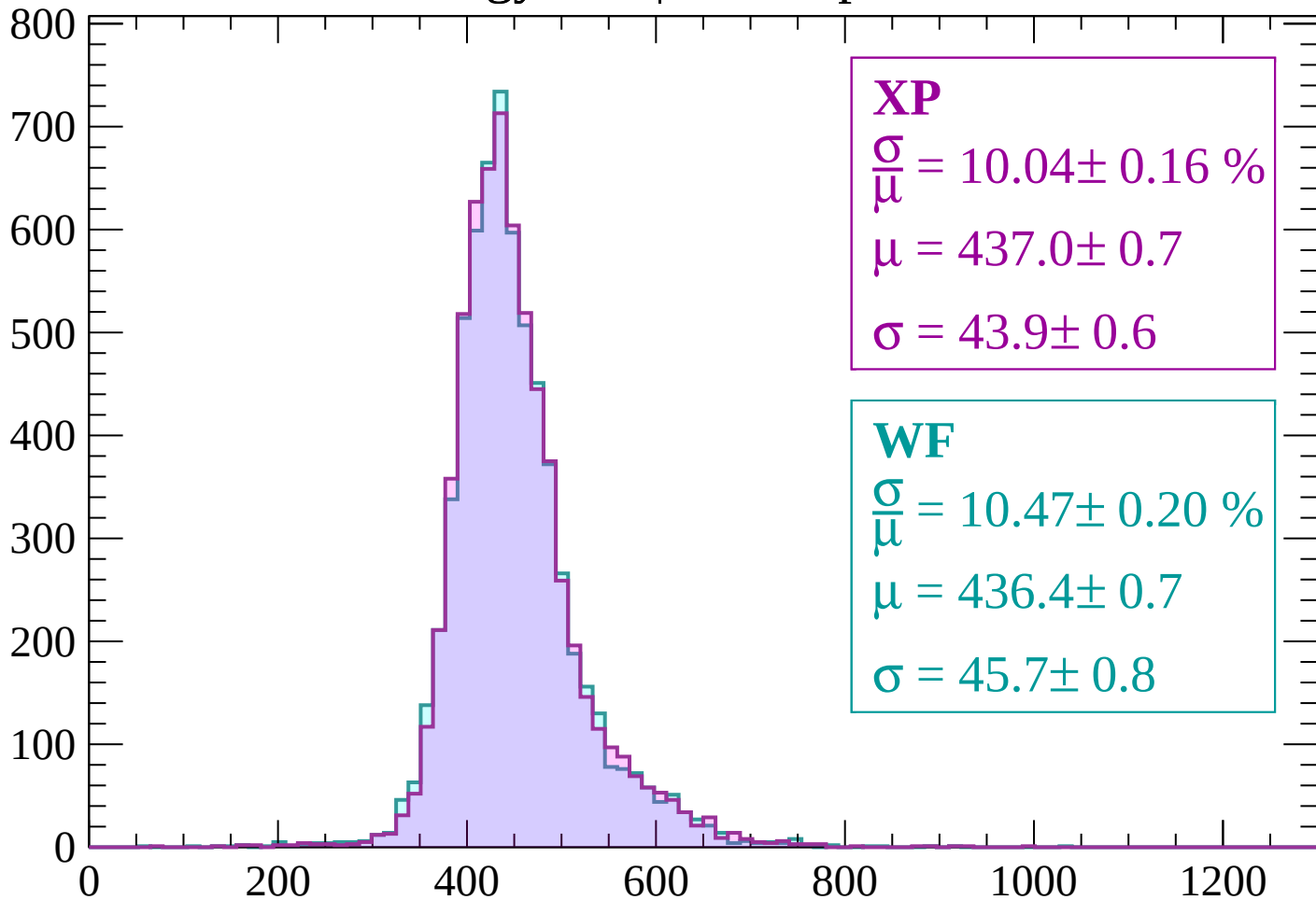
Energy loss | $-1020 < p < -960$

Count



Energy loss | $-960 < p < -900$

Count



XP

$$\frac{\sigma}{\mu} = 10.04 \pm 0.16 \%$$

$$\mu = 437.0 \pm 0.7$$

$$\sigma = 43.9 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 10.47 \pm 0.20 \%$$

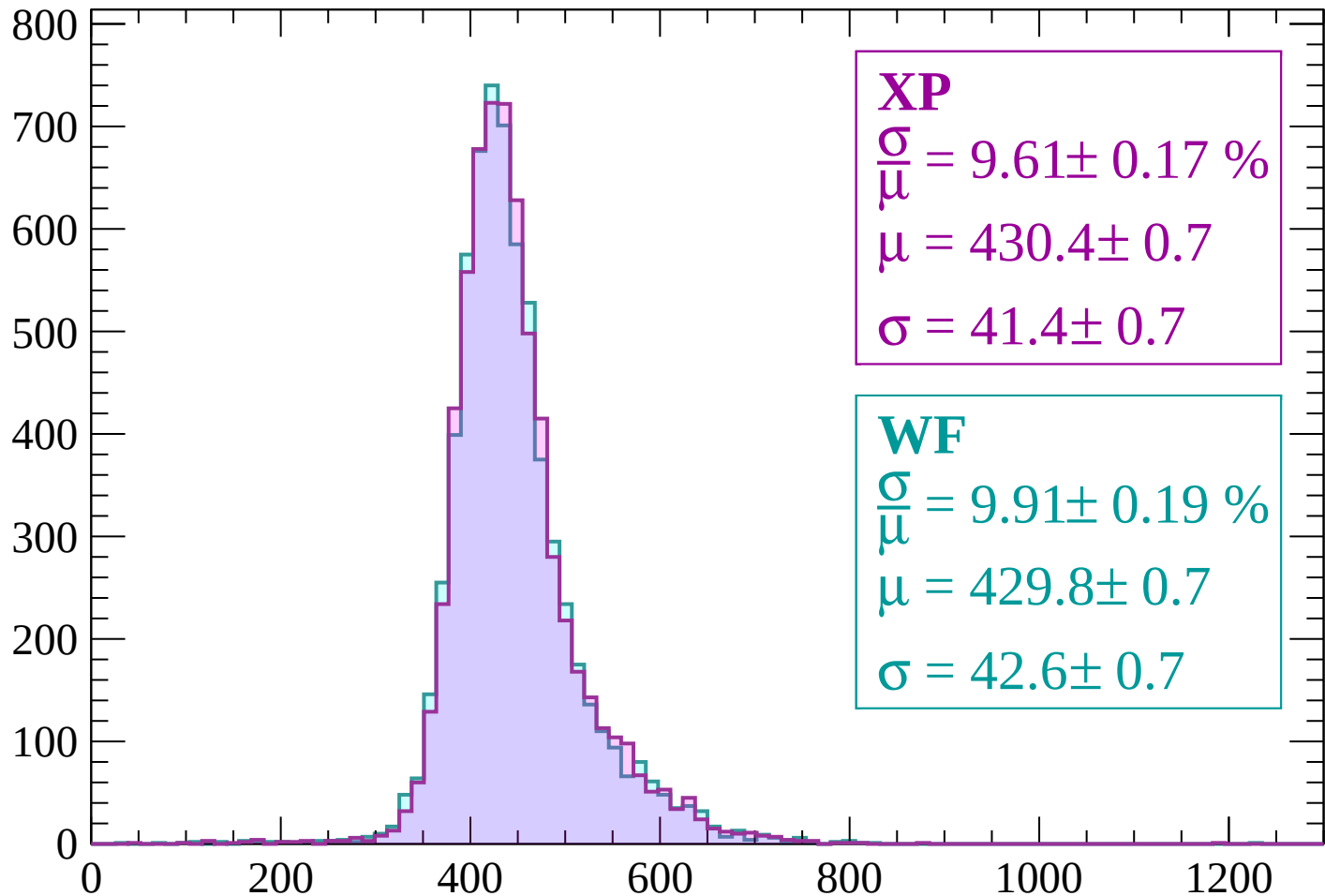
$$\mu = 436.4 \pm 0.7$$

$$\sigma = 45.7 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-900 < p < -840$

Count



XP

$$\frac{\sigma}{\mu} = 9.61 \pm 0.17 \%$$

$$\mu = 430.4 \pm 0.7$$

$$\sigma = 41.4 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.91 \pm 0.19 \%$$

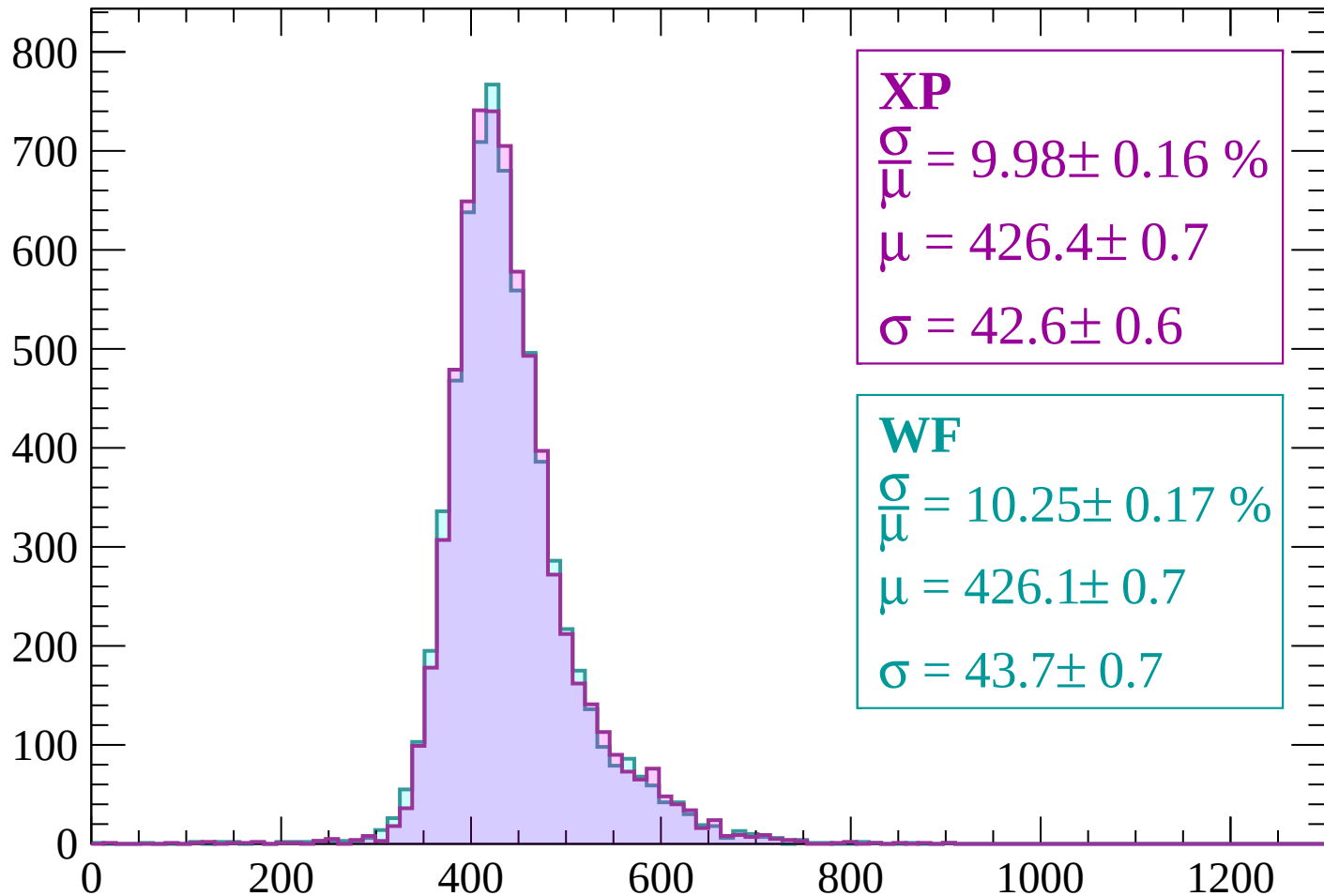
$$\mu = 429.8 \pm 0.7$$

$$\sigma = 42.6 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -780$

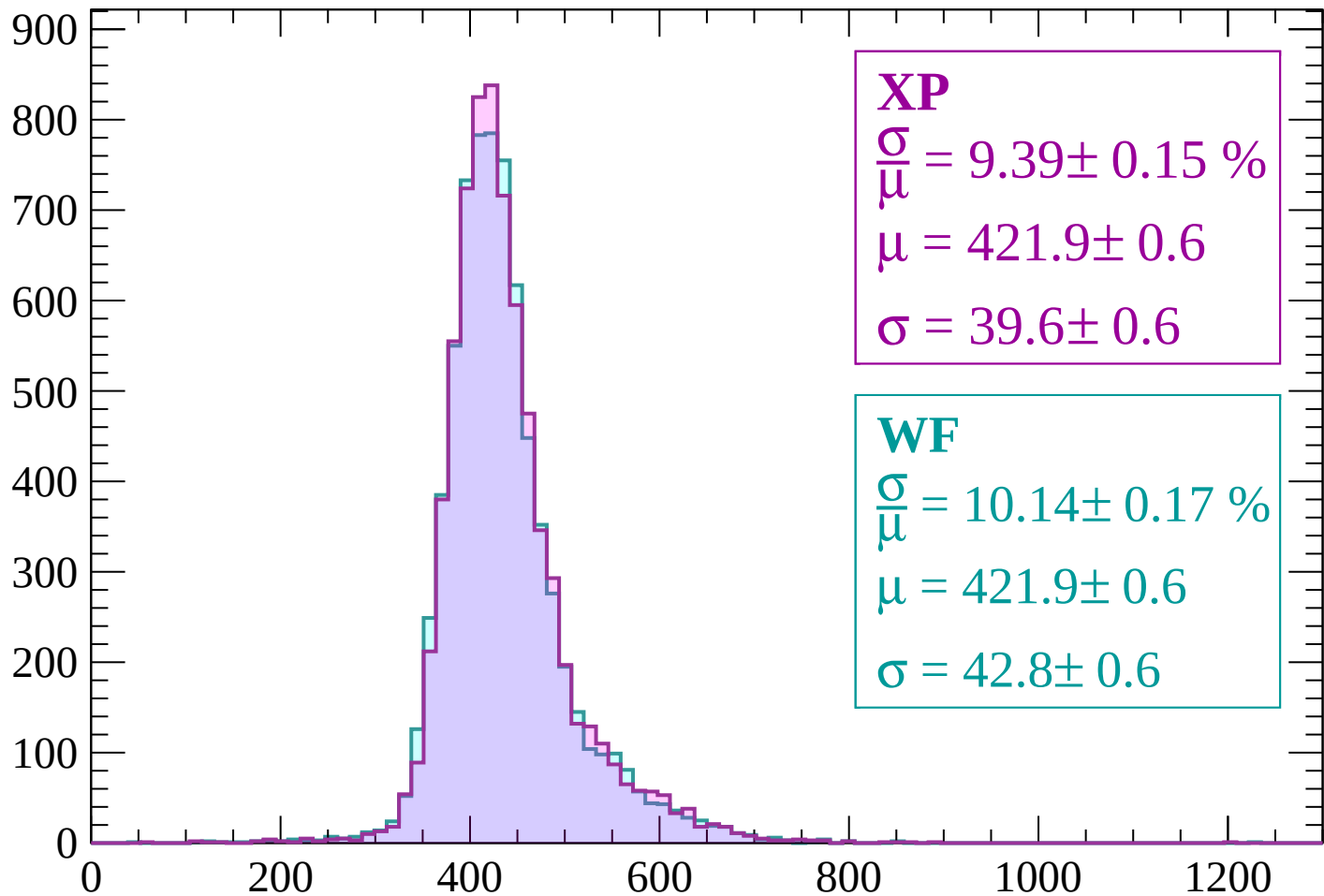
Count



dE/dx (ADC counts/cm)

Energy loss | $-780 < p < -720$

Count



XP

$$\frac{\sigma}{\mu} = 9.39 \pm 0.15 \%$$

$$\mu = 421.9 \pm 0.6$$

$$\sigma = 39.6 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 10.14 \pm 0.17 \%$$

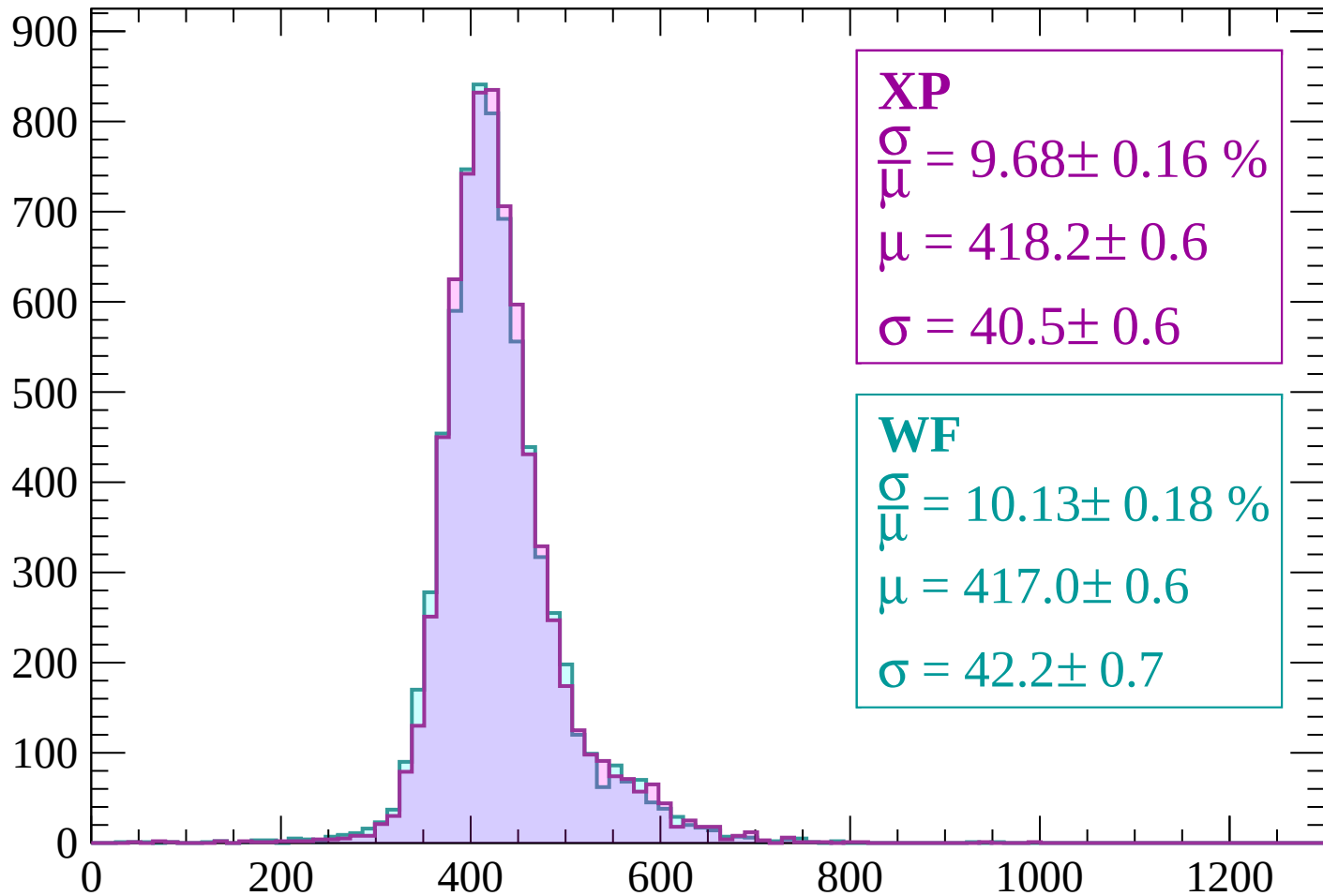
$$\mu = 421.9 \pm 0.6$$

$$\sigma = 42.8 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-720 < p < -660$

Count



XP

$$\frac{\sigma}{\mu} = 9.68 \pm 0.16 \%$$

$$\mu = 418.2 \pm 0.6$$

$$\sigma = 40.5 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 10.13 \pm 0.18 \%$$

$$\mu = 417.0 \pm 0.6$$

$$\sigma = 42.2 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-660 < p < -600$

Count

1000
800
600
400
200
0

XP

$$\frac{\sigma}{\mu} = 9.60 \pm 0.16 \%$$

$$\mu = 414.1 \pm 0.6$$

$$\sigma = 39.8 \pm 0.6$$

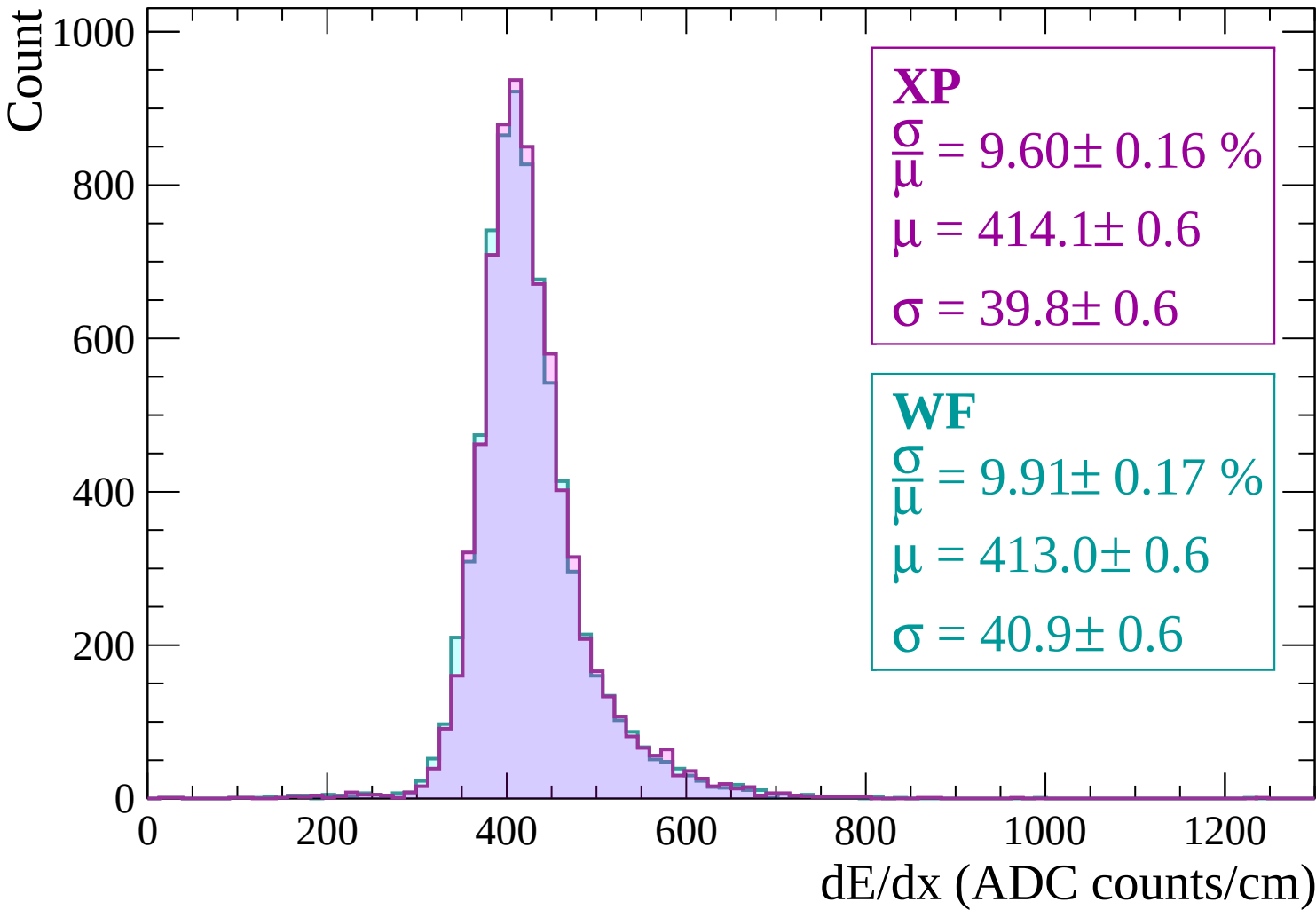
WF

$$\frac{\sigma}{\mu} = 9.91 \pm 0.17 \%$$

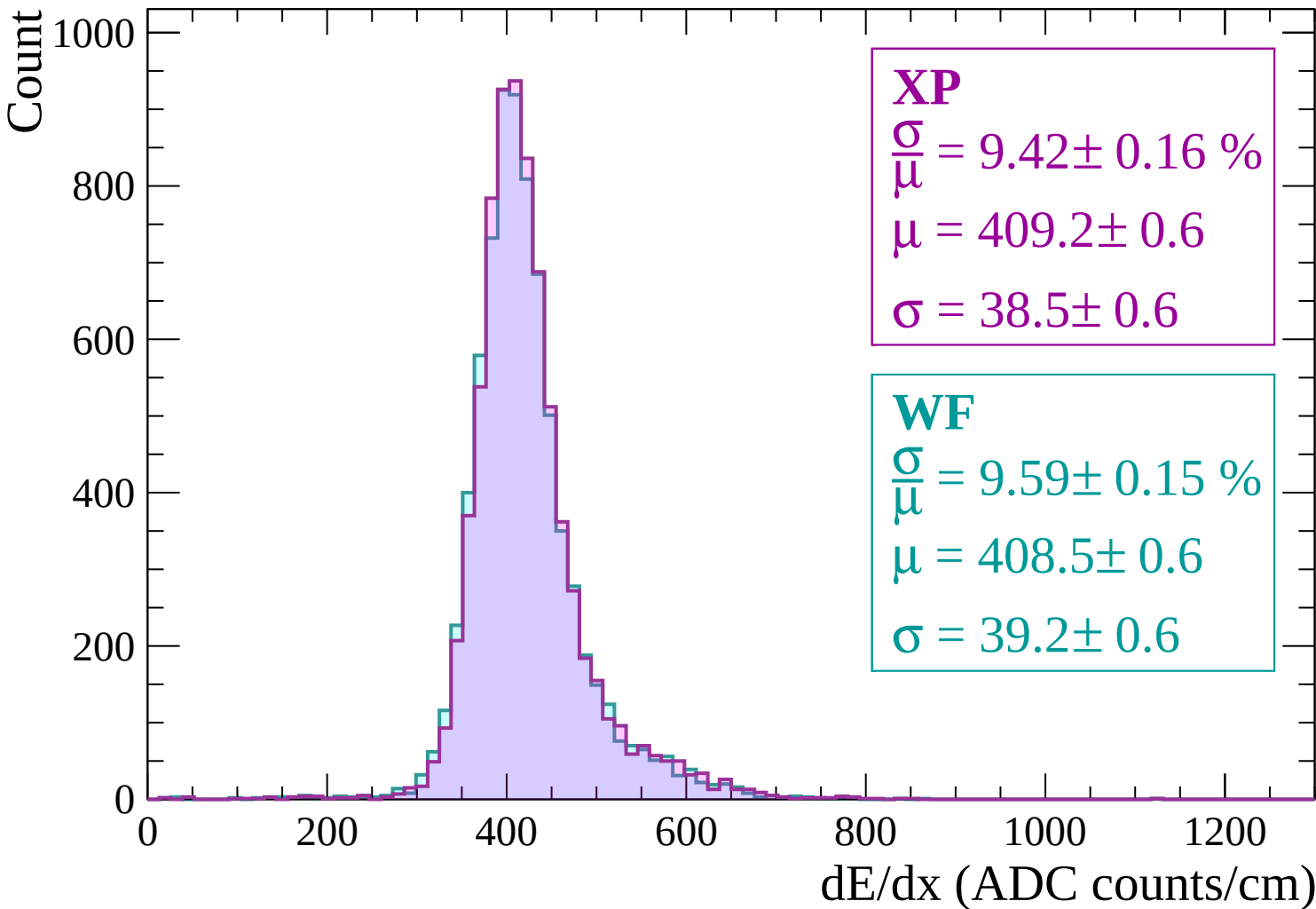
$$\mu = 413.0 \pm 0.6$$

$$\sigma = 40.9 \pm 0.6$$

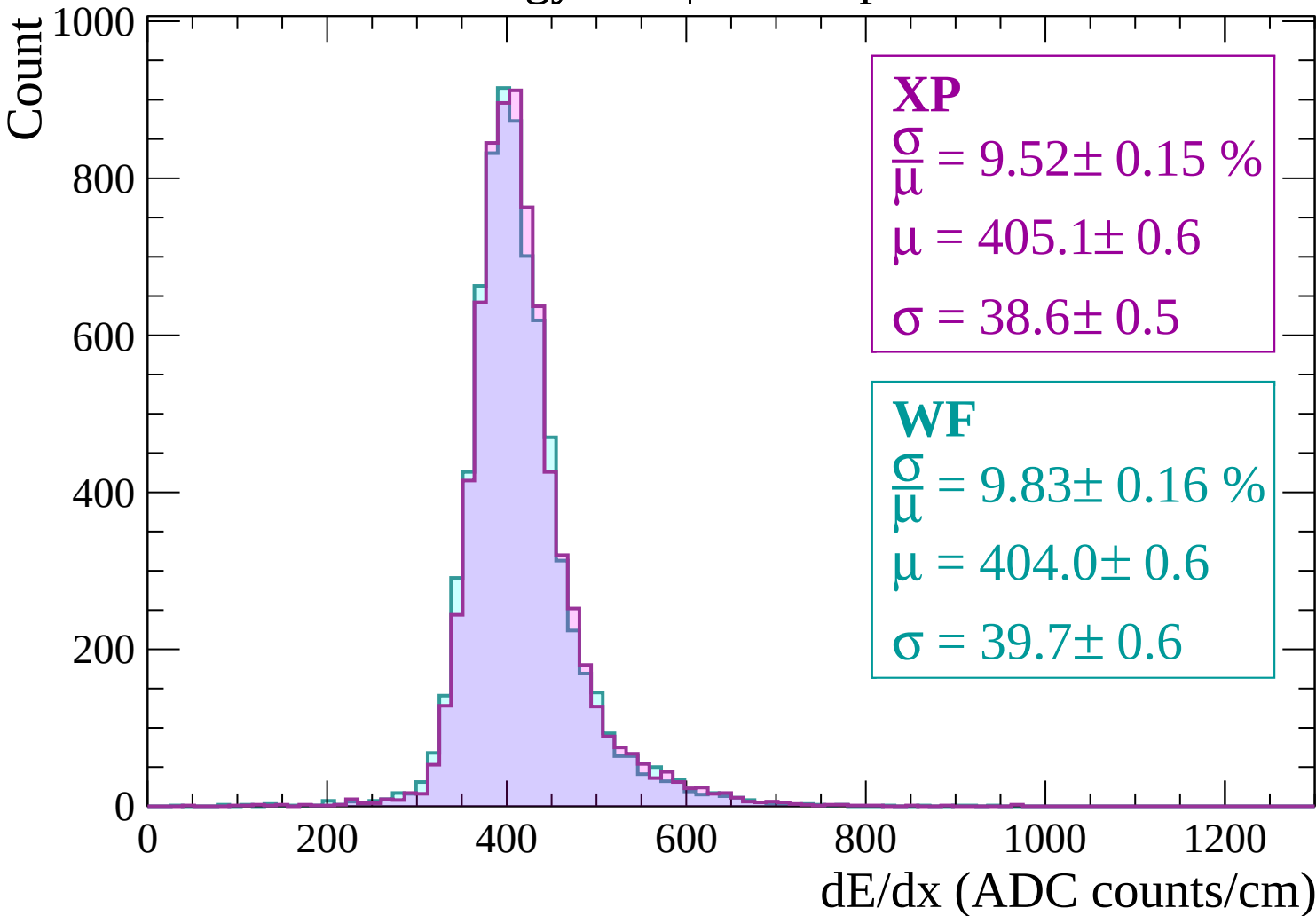
dE/dx (ADC counts/cm)



Energy loss | $-600 < p < -540$



Energy loss | $-540 < p < -480$



Energy loss | $-480 < p < -420$

Count

1000

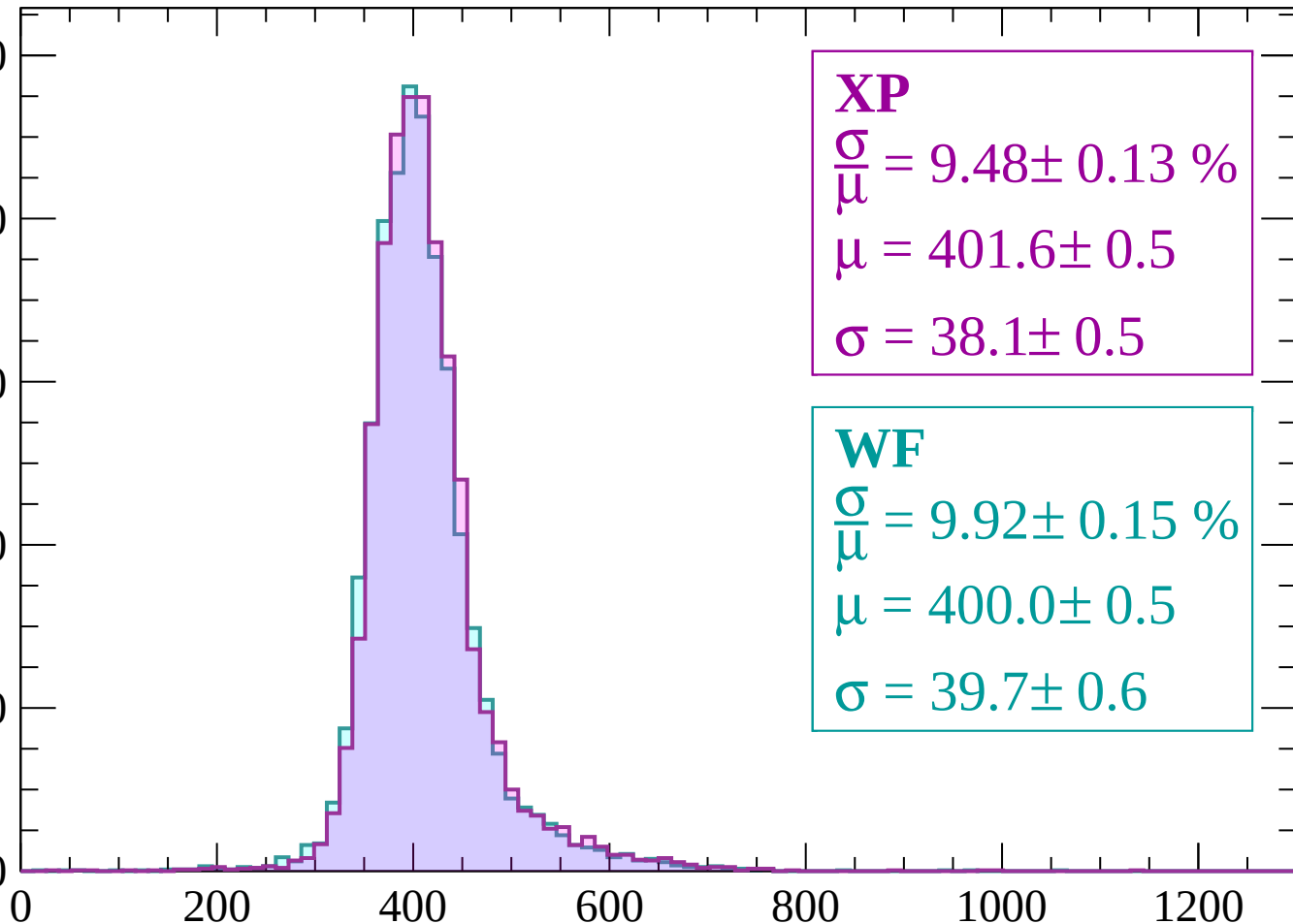
800

600

400

200

0



XP

$$\frac{\sigma}{\mu} = 9.48 \pm 0.13 \%$$

$$\mu = 401.6 \pm 0.5$$

$$\sigma = 38.1 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 9.92 \pm 0.15 \%$$

$$\mu = 400.0 \pm 0.5$$

$$\sigma = 39.7 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-420 < p < -360$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 9.21 \pm 0.14 \%$$

$$\mu = 398.0 \pm 0.5$$

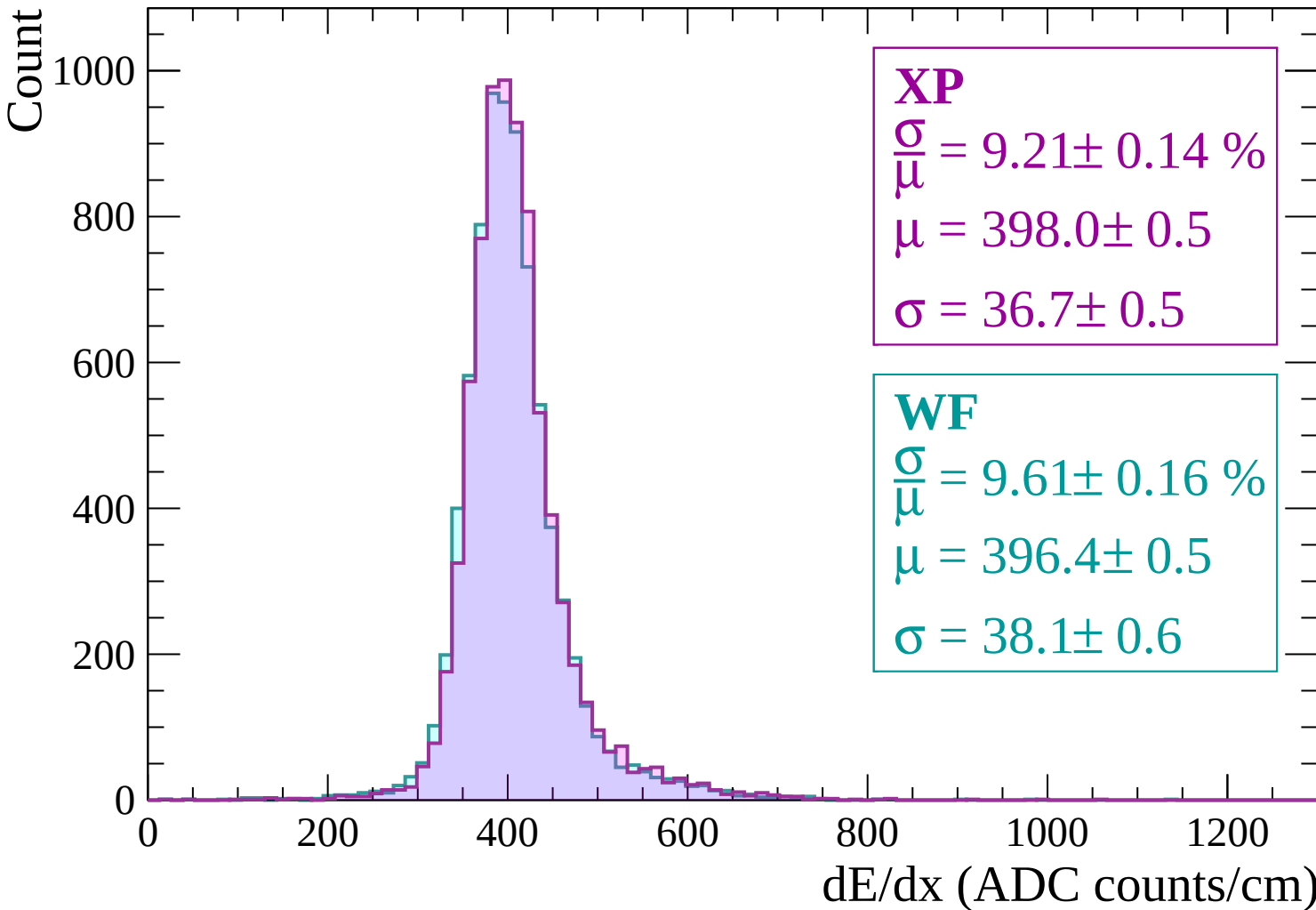
$$\sigma = 36.7 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 9.61 \pm 0.16 \%$$

$$\mu = 396.4 \pm 0.5$$

$$\sigma = 38.1 \pm 0.6$$



Energy loss | $-360 < p < -300$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 9.40 \pm 0.15 \%$$

$$\mu = 396.9 \pm 0.5$$

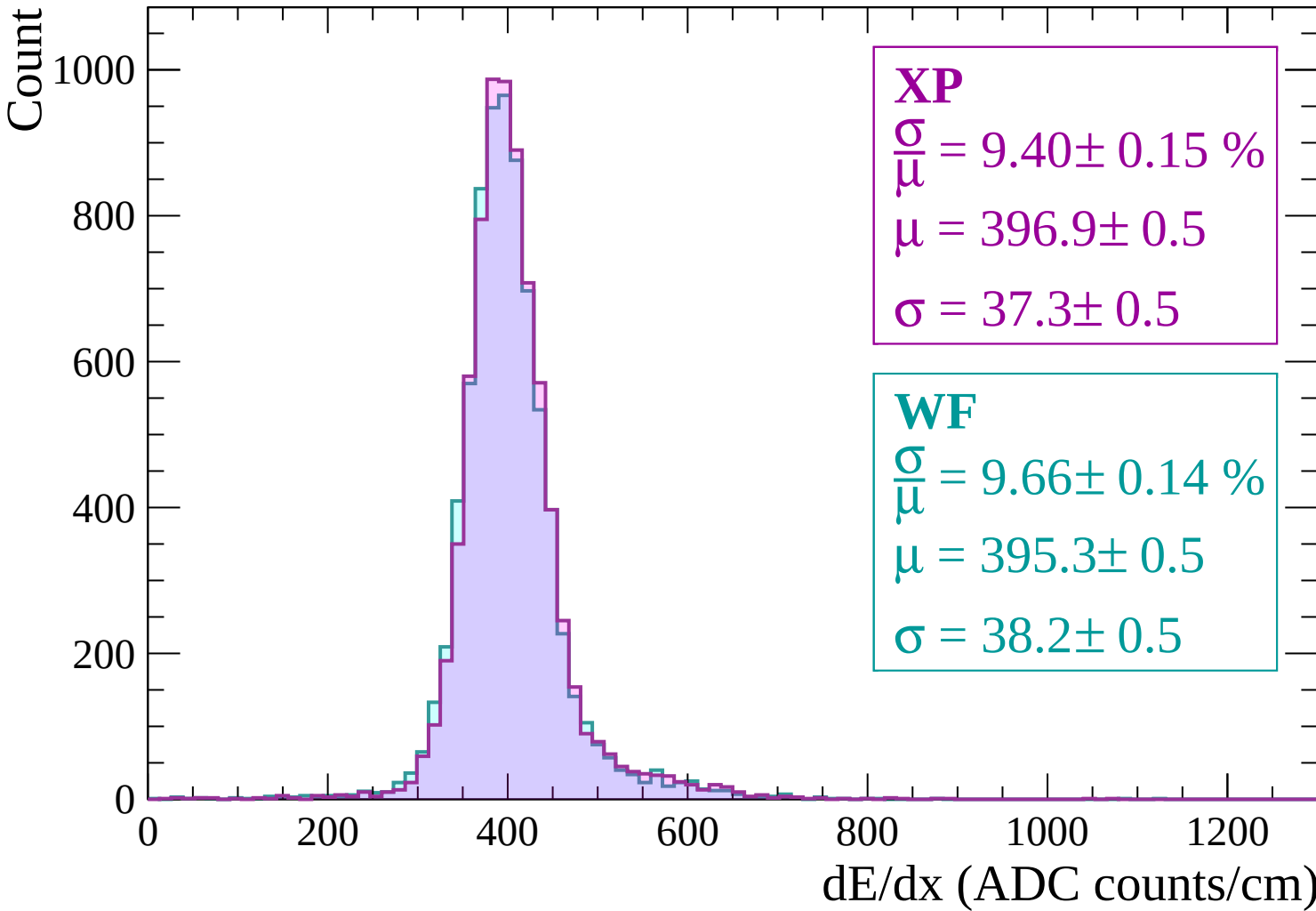
$$\sigma = 37.3 \pm 0.5$$

WF

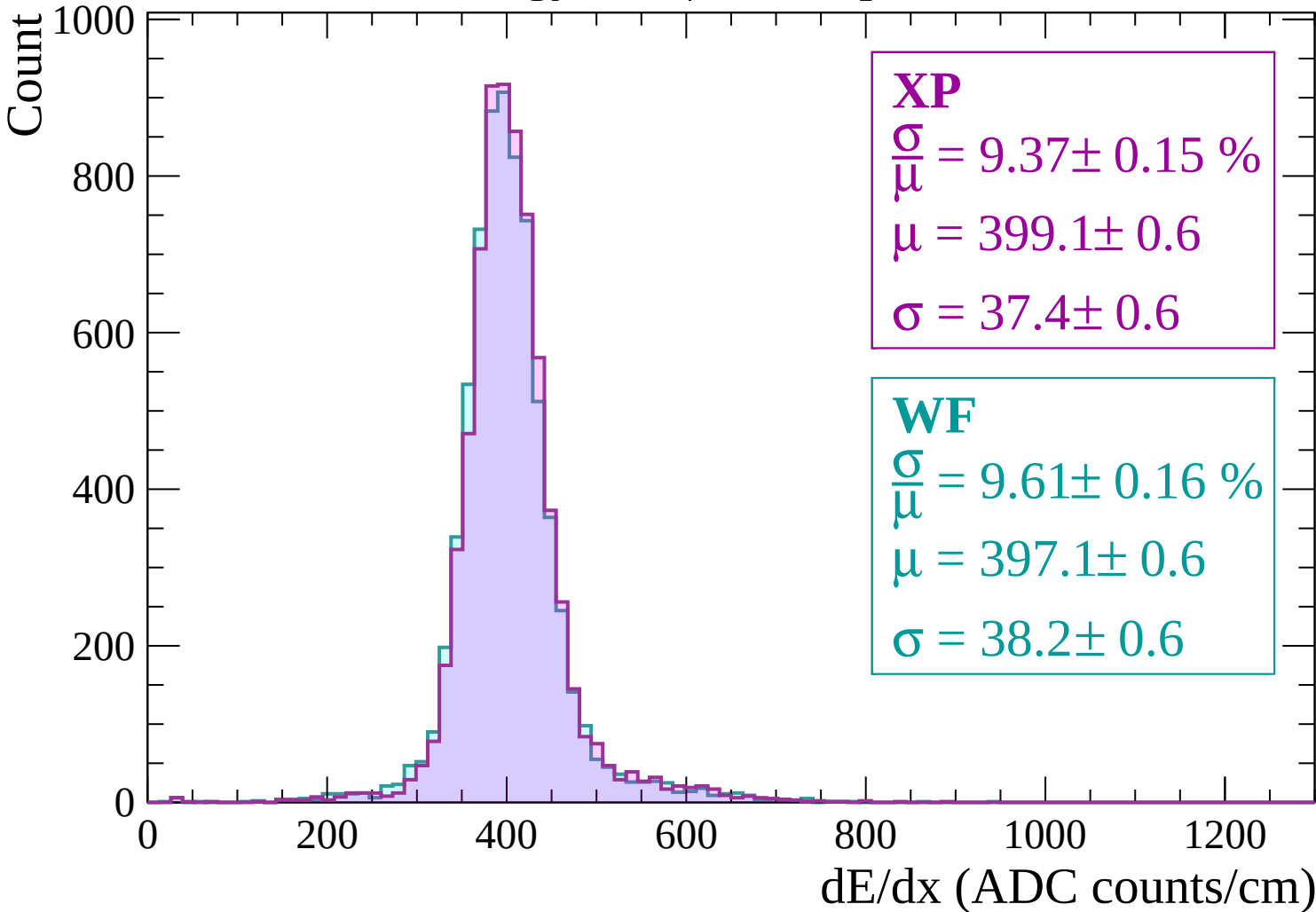
$$\frac{\sigma}{\mu} = 9.66 \pm 0.14 \%$$

$$\mu = 395.3 \pm 0.5$$

$$\sigma = 38.2 \pm 0.5$$

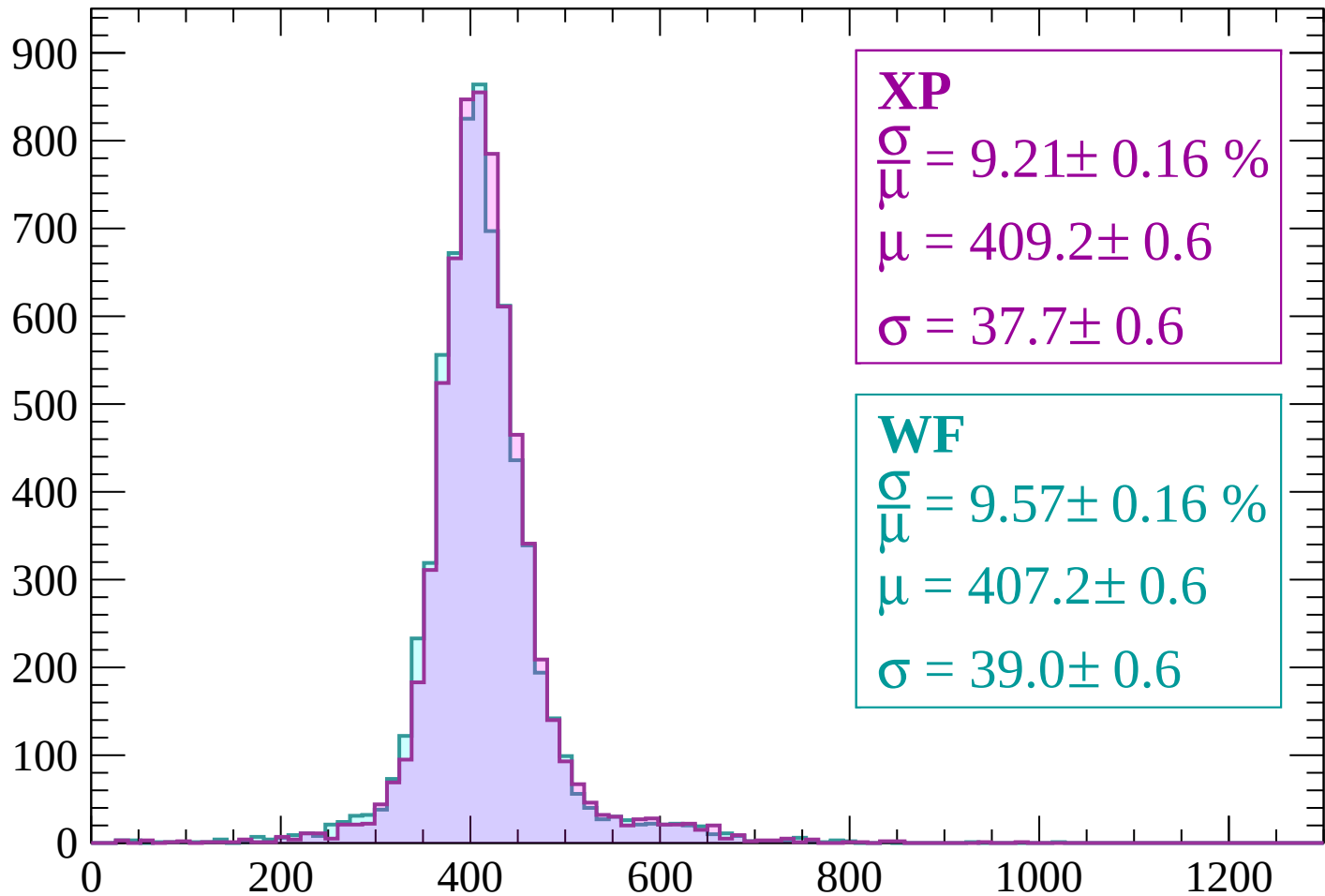


Energy loss | $-300 < p < -240$



Energy loss | $-240 < p < -180$

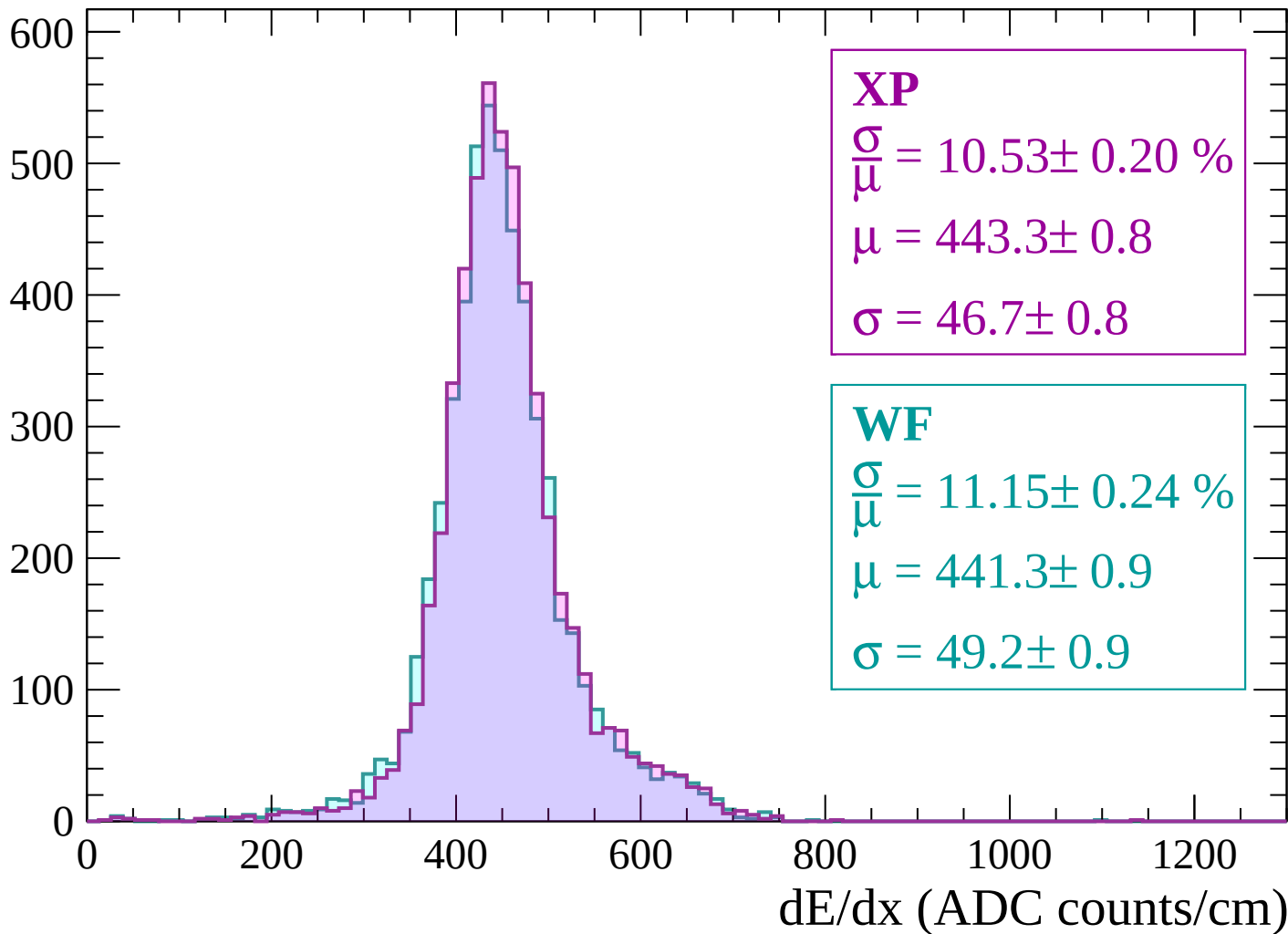
Count



dE/dx (ADC counts/cm)

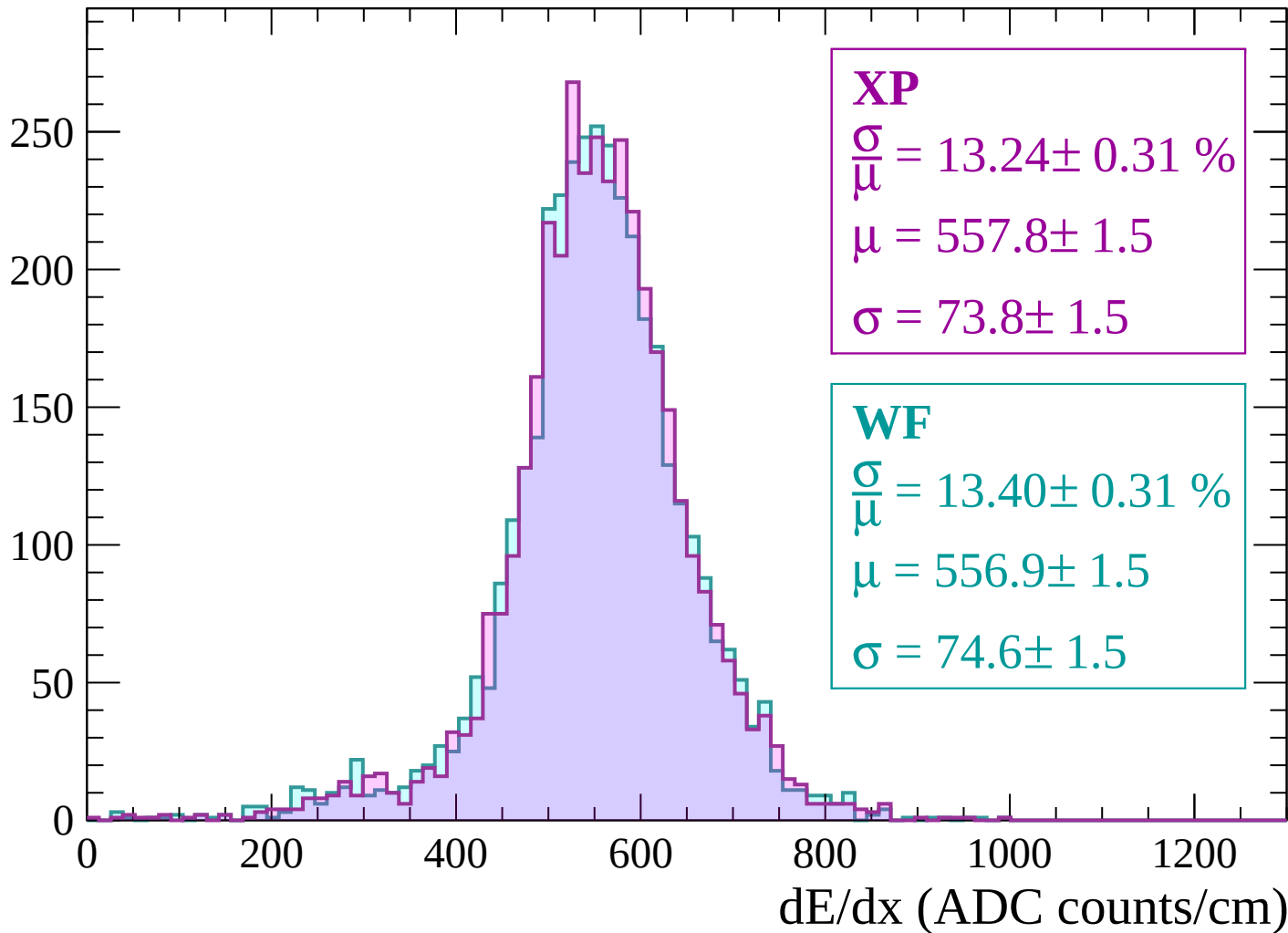
Energy loss | -180 < p < -120

Count



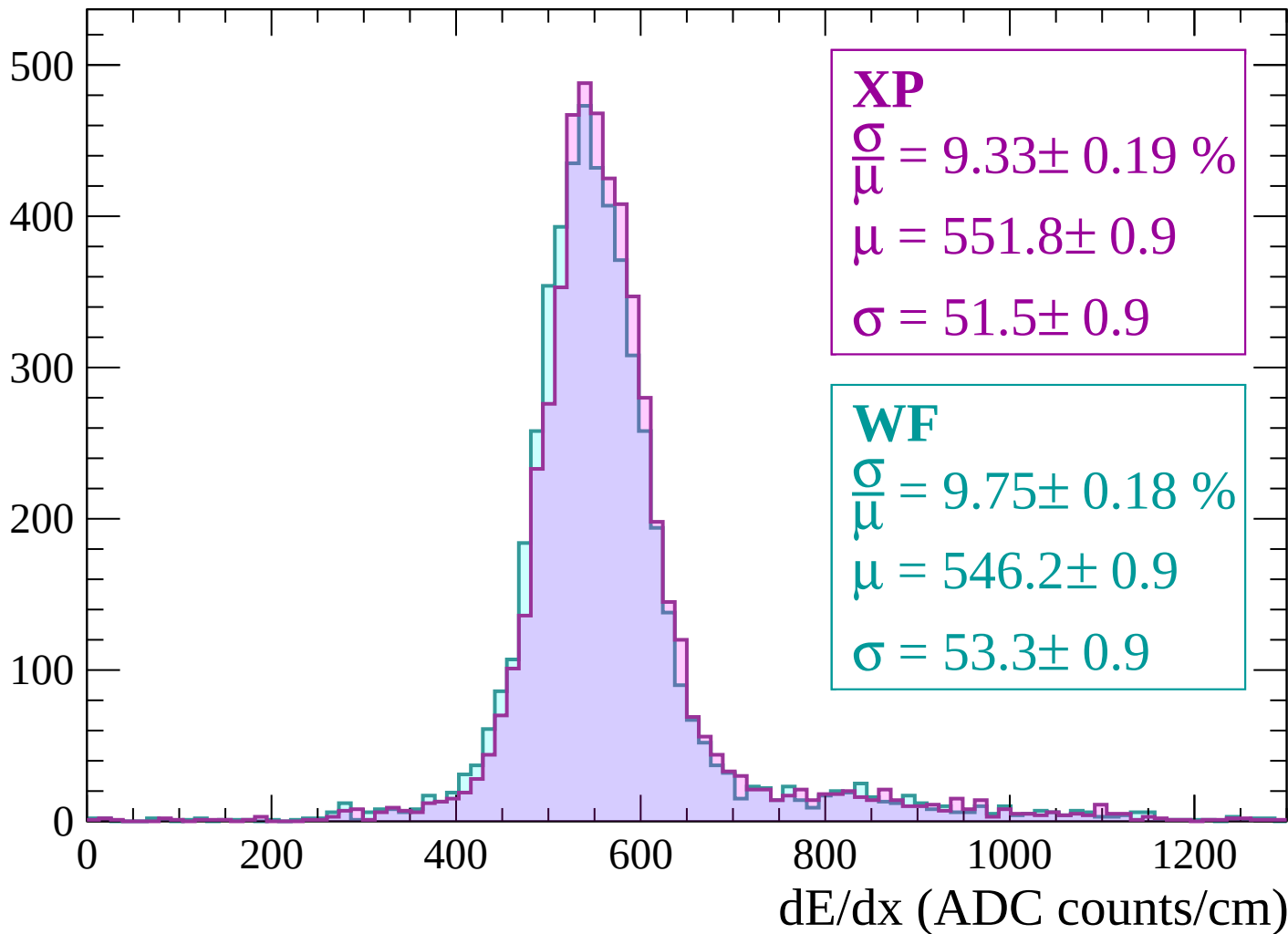
Energy loss | $-120 < p < -60$

Count



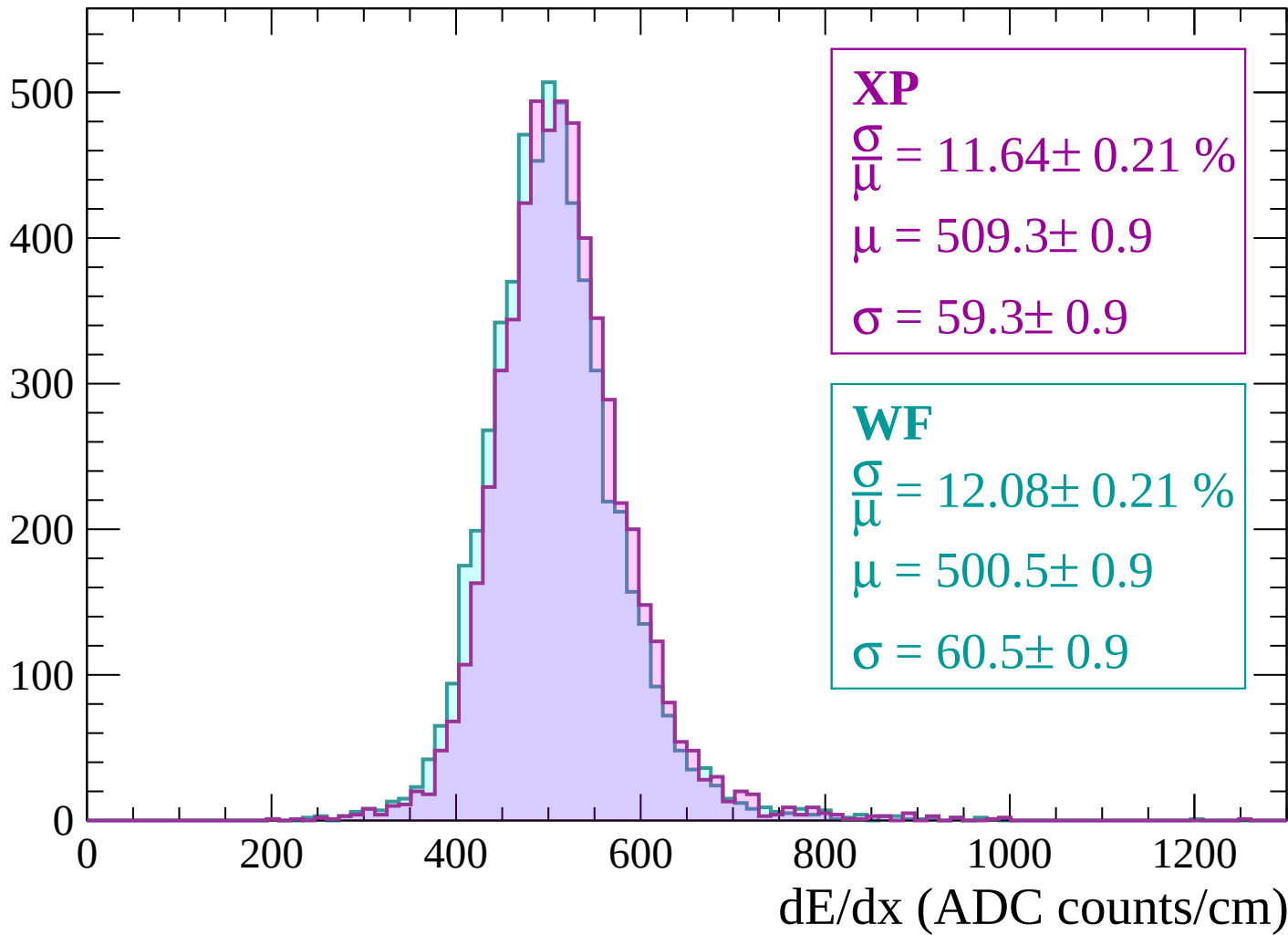
Energy loss | $-60 < p < 0$

Count



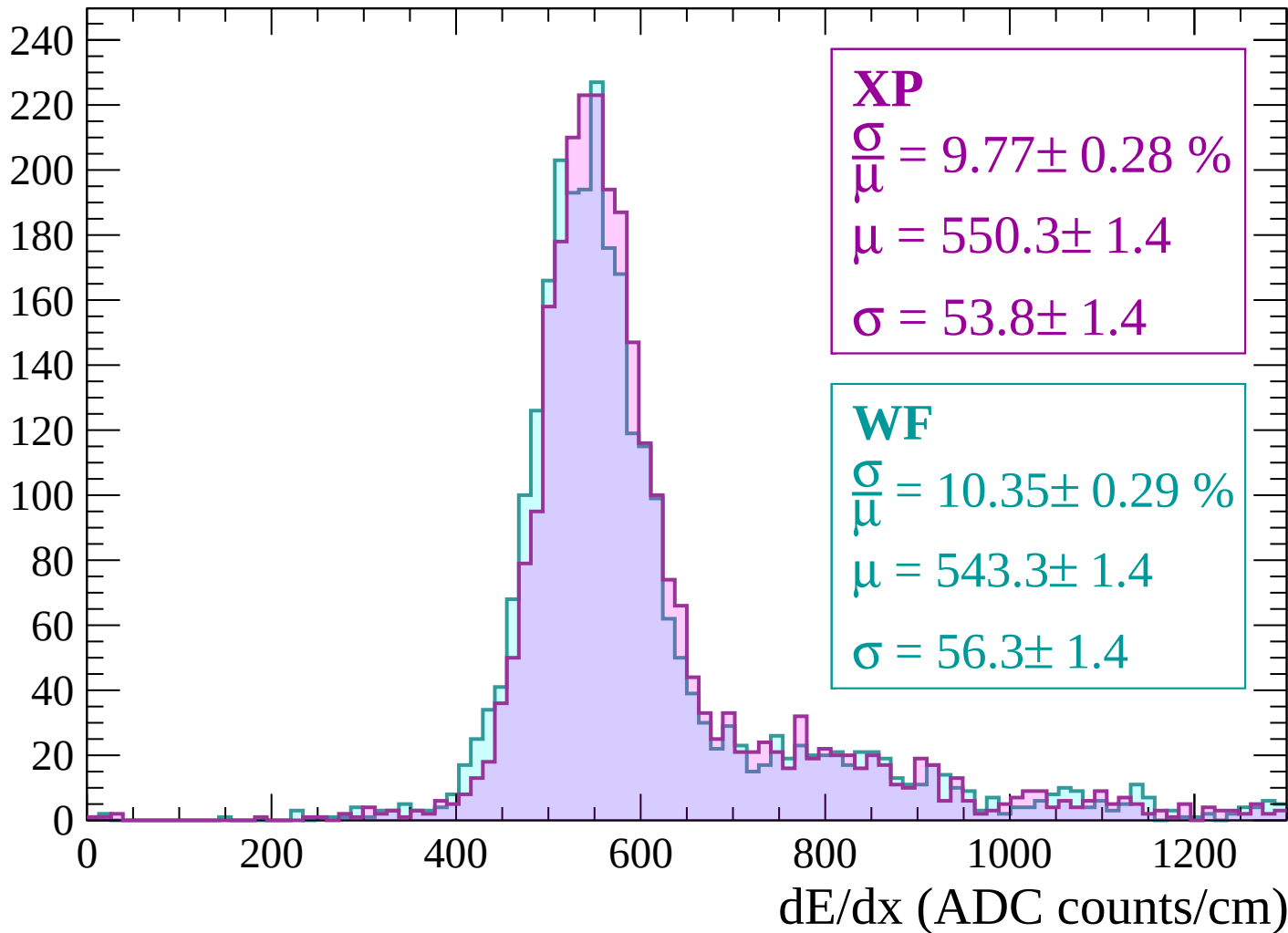
Energy loss | $0 < p < 60$

Count



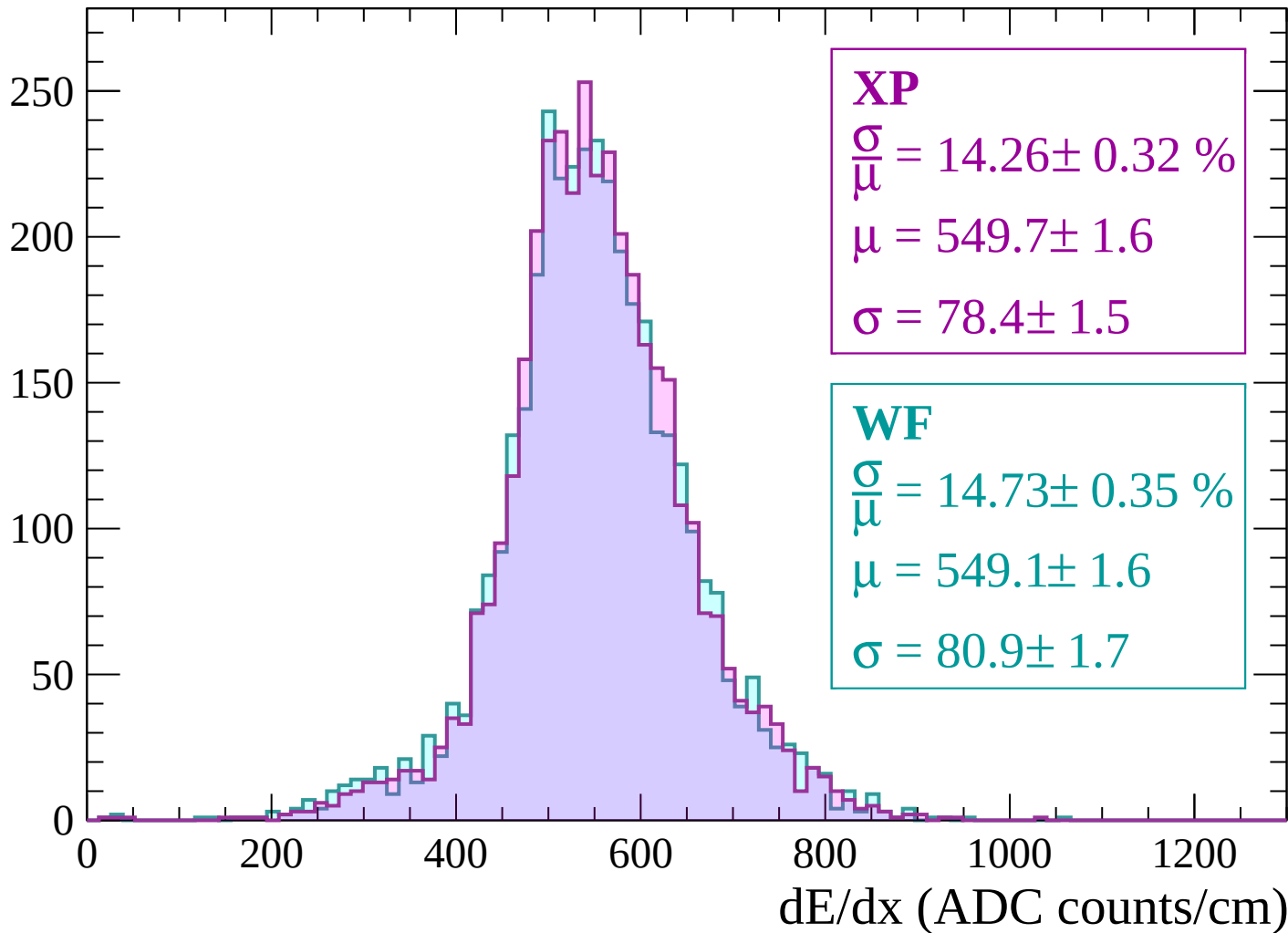
Energy loss | $60 < p < 120$

Count



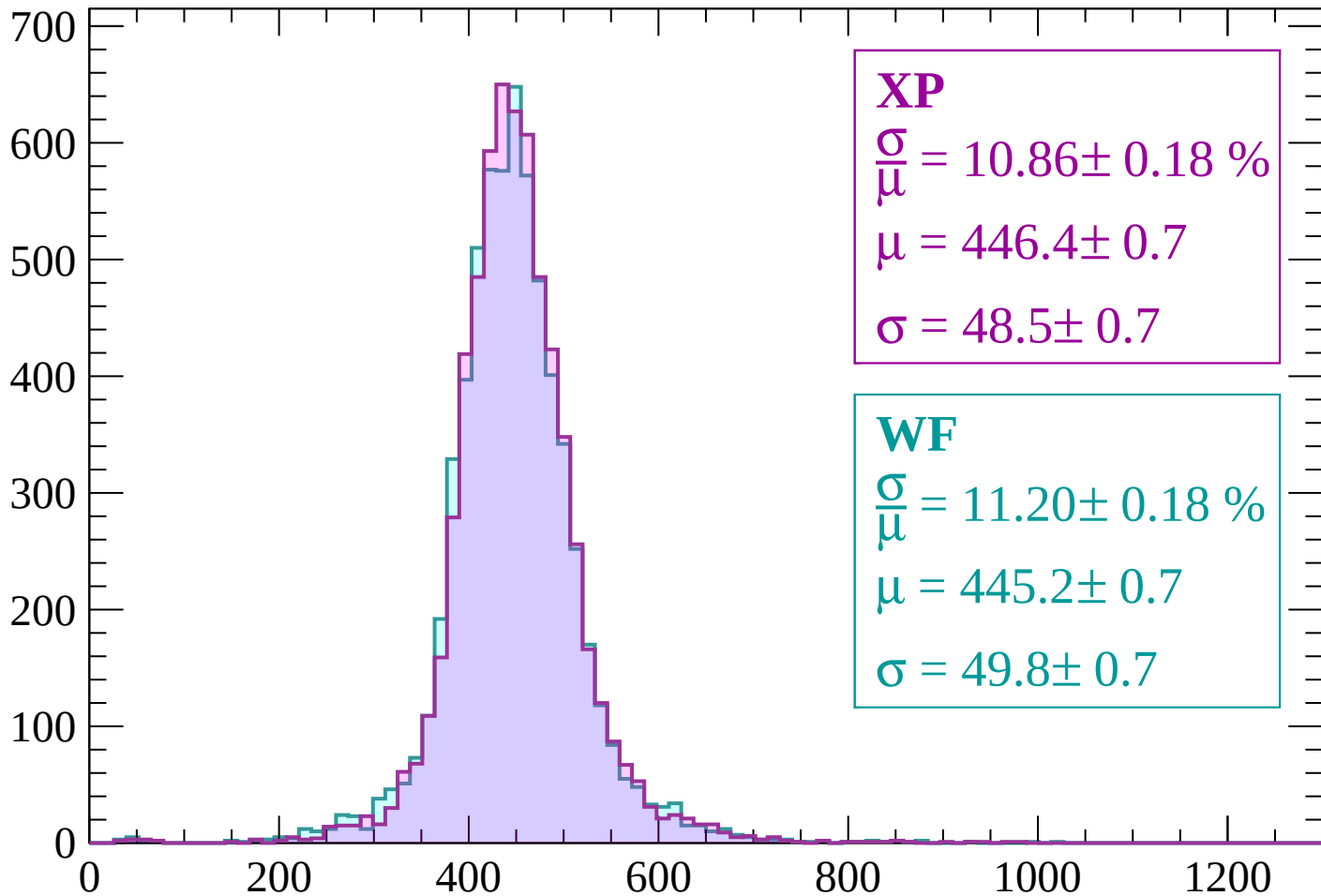
Energy loss | $120 < p < 180$

Count



Energy loss | $180 < p < 240$

Count



XP

$$\frac{\sigma}{\mu} = 10.86 \pm 0.18 \%$$

$$\mu = 446.4 \pm 0.7$$

$$\sigma = 48.5 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 11.20 \pm 0.18 \%$$

$$\mu = 445.2 \pm 0.7$$

$$\sigma = 49.8 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $240 < p < 300$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 9.64 \pm 0.15 \%$$

$$\mu = 411.6 \pm 0.5$$

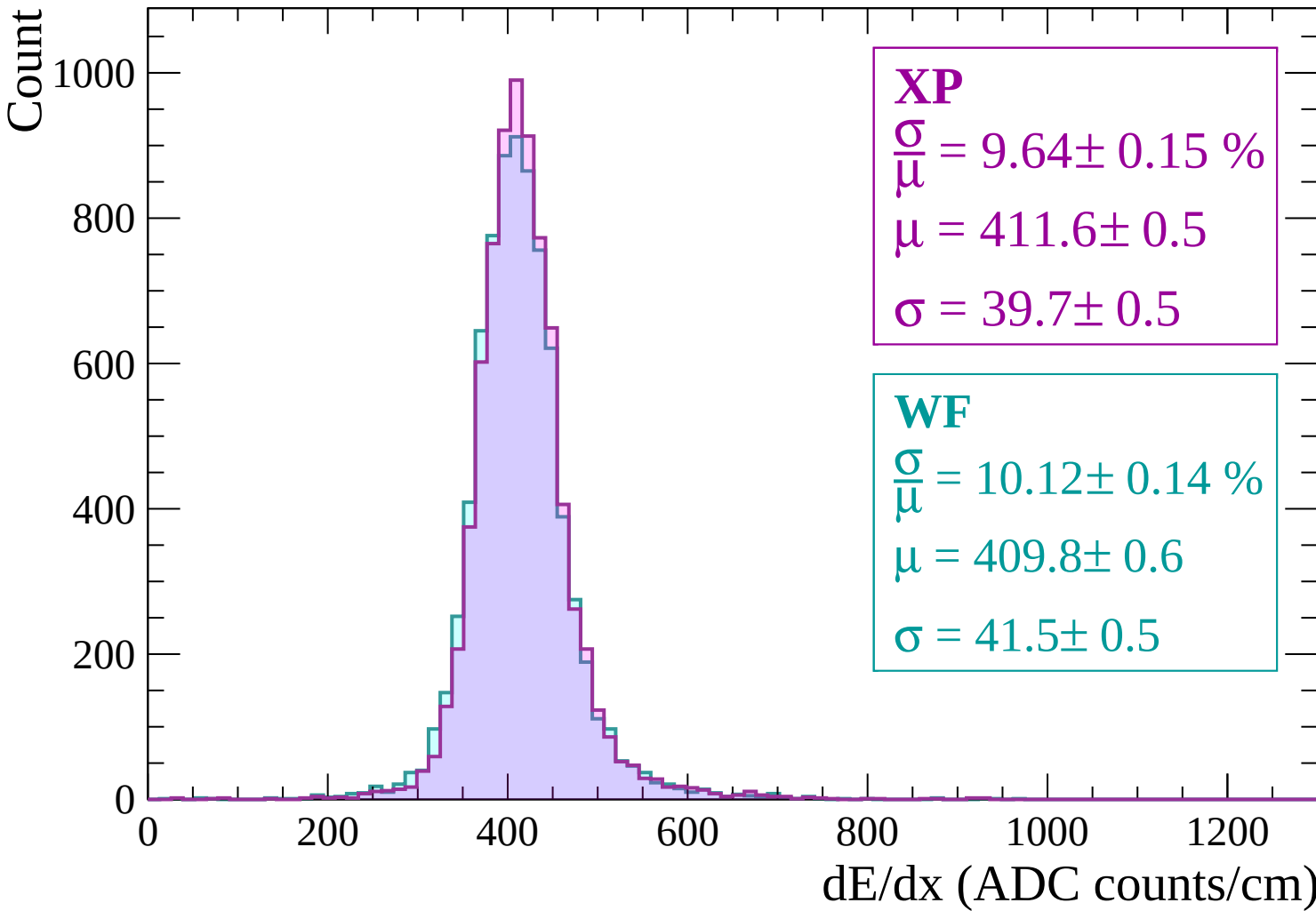
$$\sigma = 39.7 \pm 0.5$$

WF

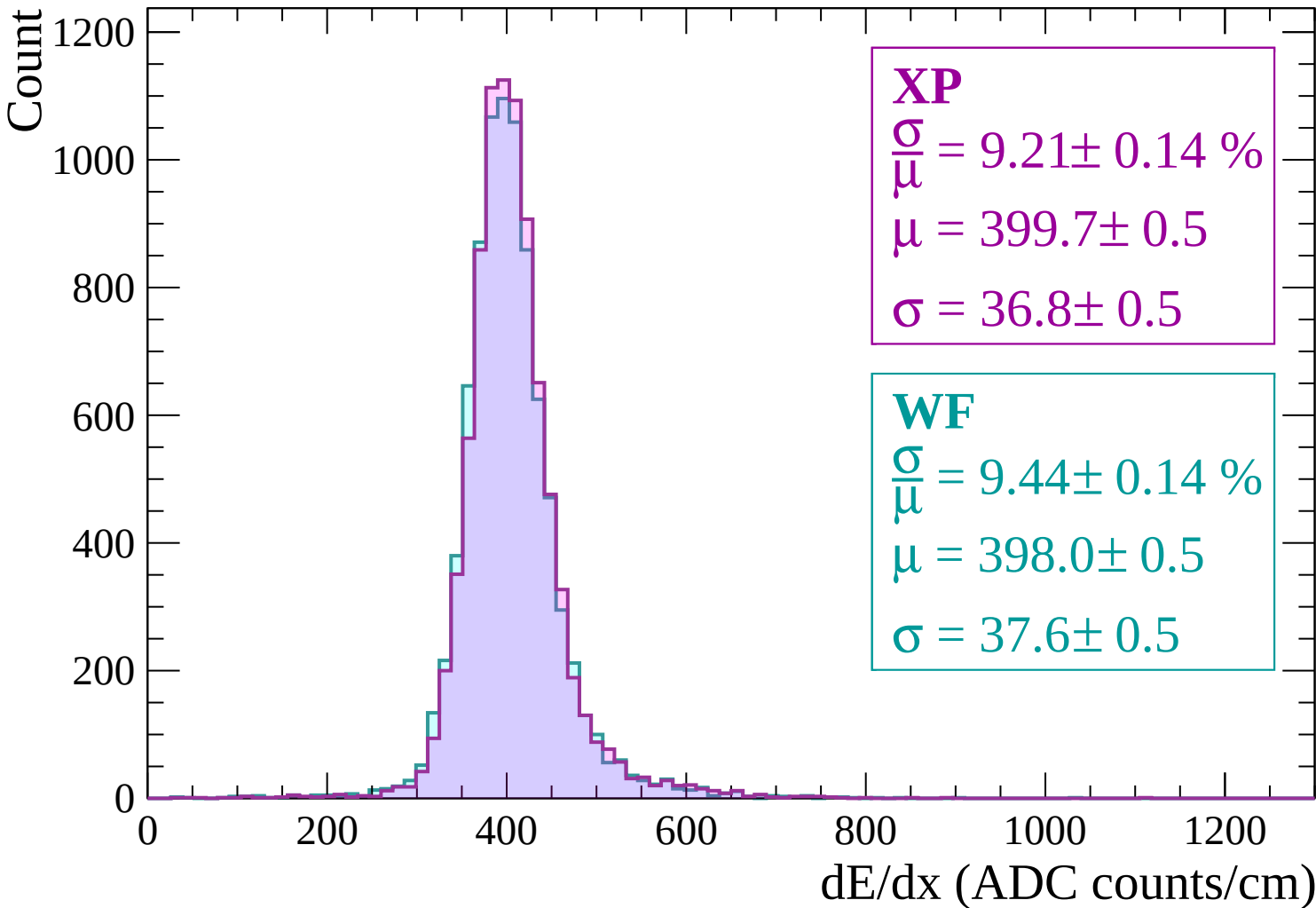
$$\frac{\sigma}{\mu} = 10.12 \pm 0.14 \%$$

$$\mu = 409.8 \pm 0.6$$

$$\sigma = 41.5 \pm 0.5$$



Energy loss | $300 < p < 360$



Energy loss | $360 < p < 420$

Count

1200

1000

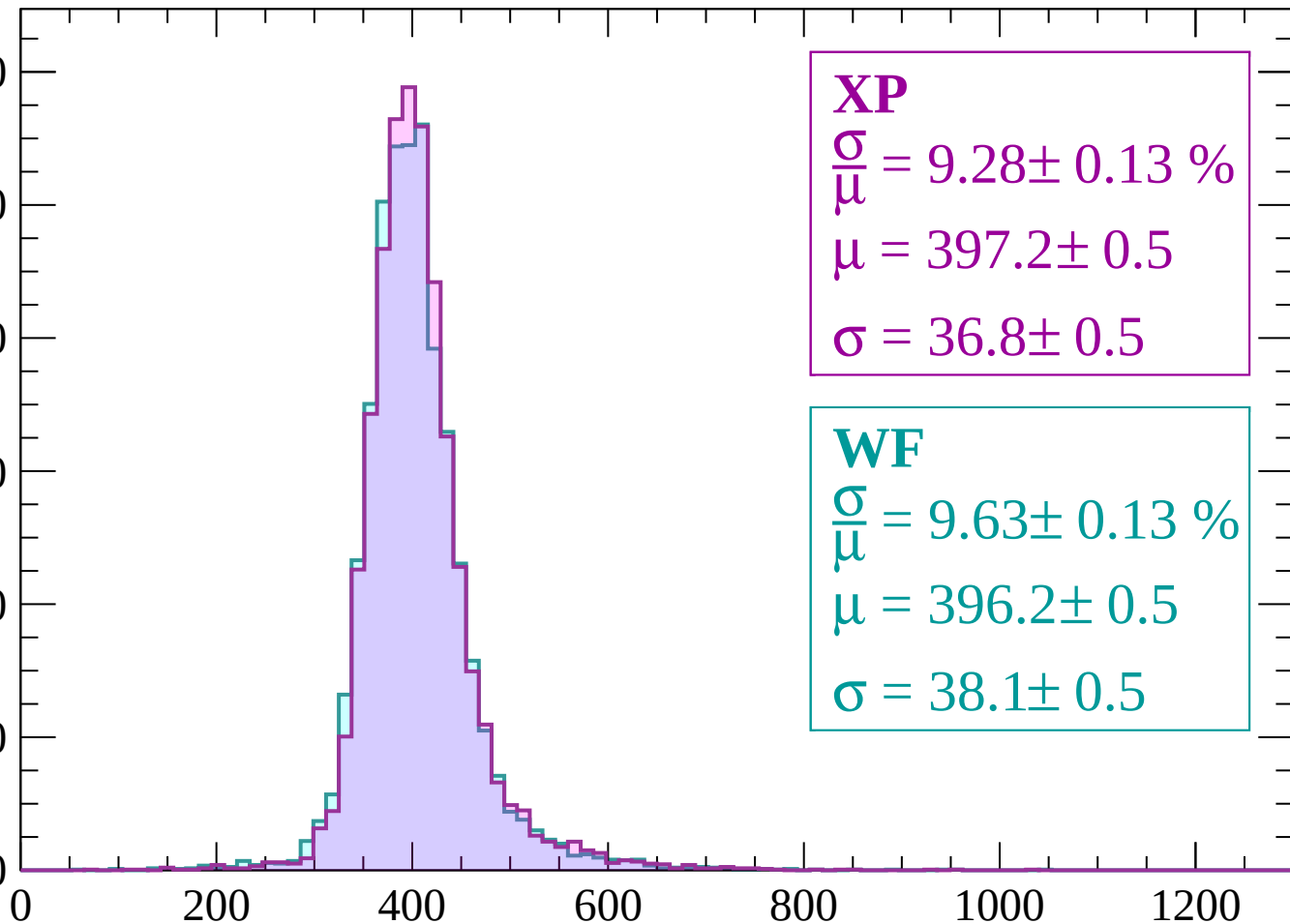
800

600

400

200

0



dE/dx (ADC counts/cm)

Energy loss | $420 < p < 480$

Count

1200
1000
800
600
400
200
0

XP

$$\frac{\sigma}{\mu} = 9.33 \pm 0.13 \%$$

$$\mu = 399.1 \pm 0.5$$

$$\sigma = 37.2 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 9.71 \pm 0.15 \%$$

$$\mu = 397.6 \pm 0.5$$

$$\sigma = 38.6 \pm 0.5$$

0

200

400

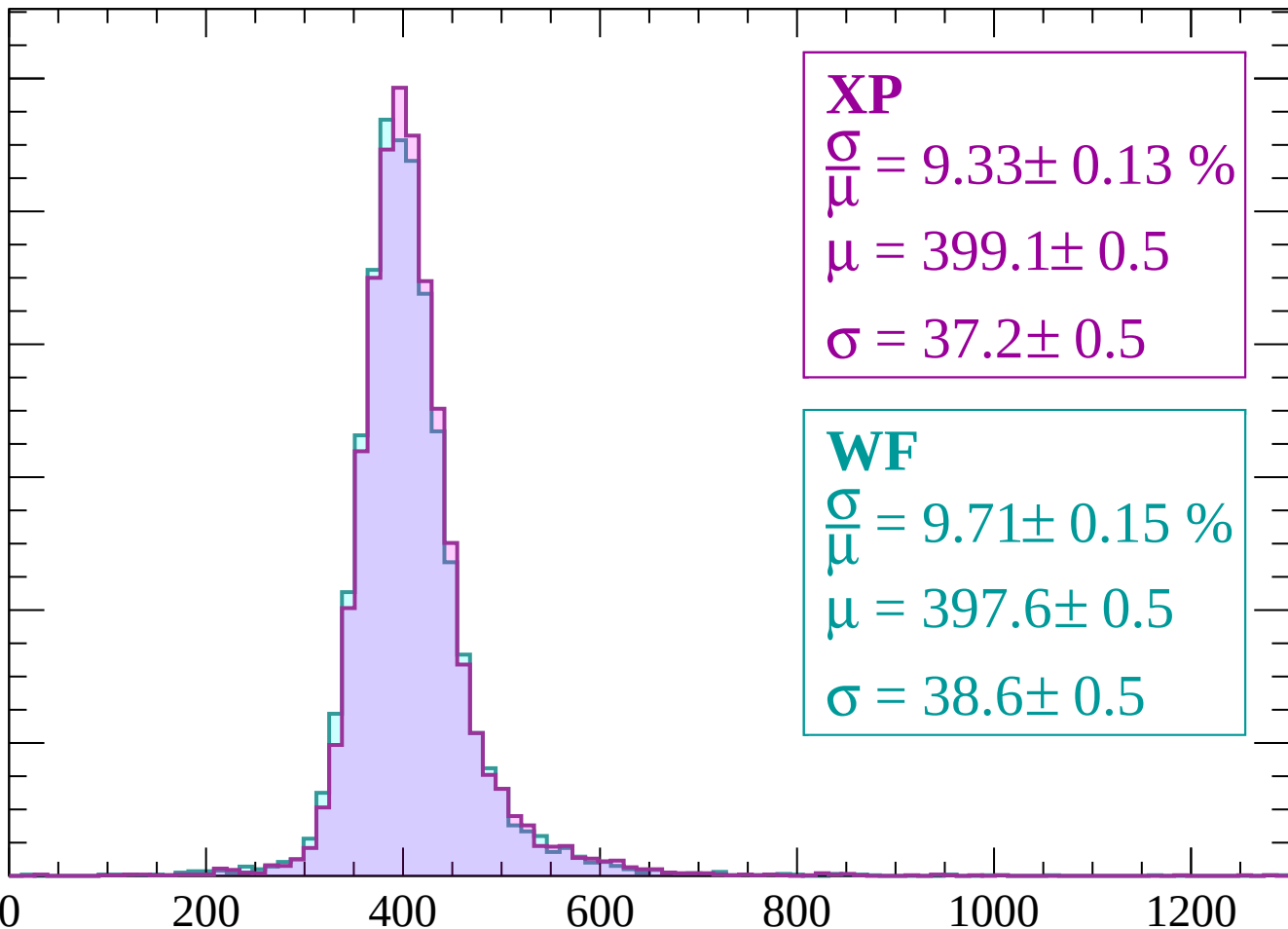
600

800

1000

1200

dE/dx (ADC counts/cm)



Energy loss | $480 < p < 540$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 9.17 \pm 0.14 \%$$

$$\mu = 401.0 \pm 0.5$$

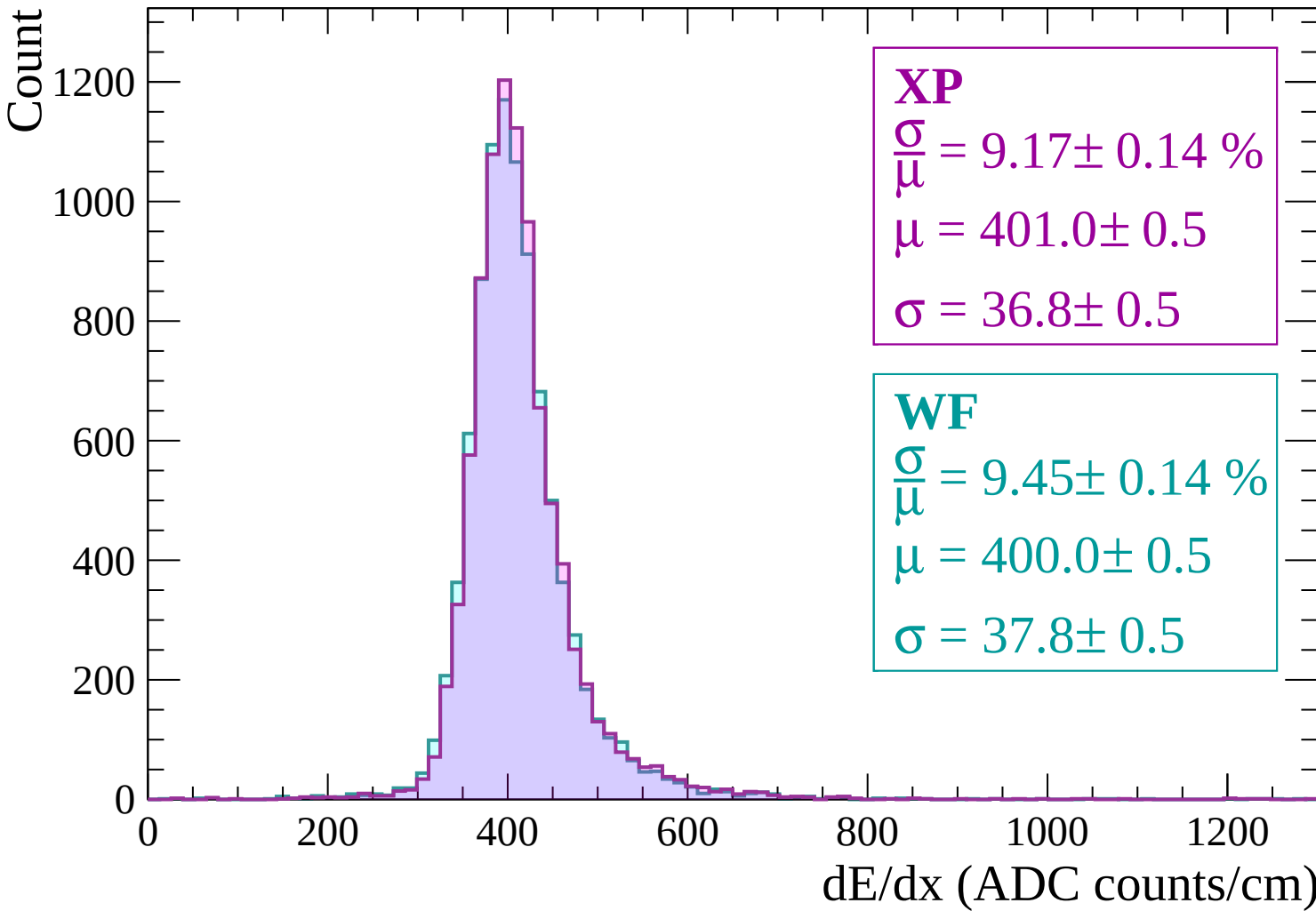
$$\sigma = 36.8 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 9.45 \pm 0.14 \%$$

$$\mu = 400.0 \pm 0.5$$

$$\sigma = 37.8 \pm 0.5$$



Energy loss | $540 < p < 600$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 9.33 \pm 0.13 \%$$

$$\mu = 406.2 \pm 0.5$$

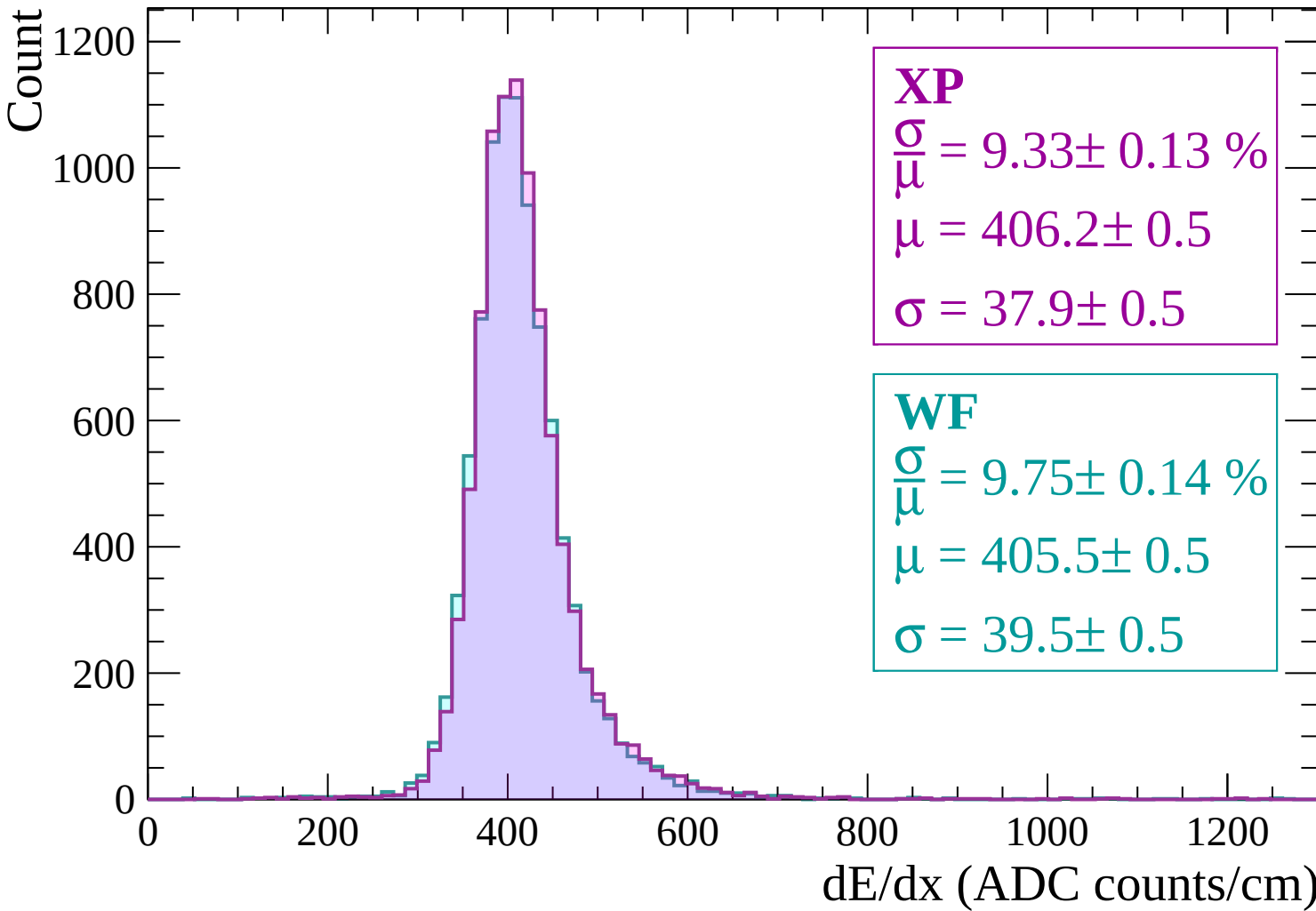
$$\sigma = 37.9 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 9.75 \pm 0.14 \%$$

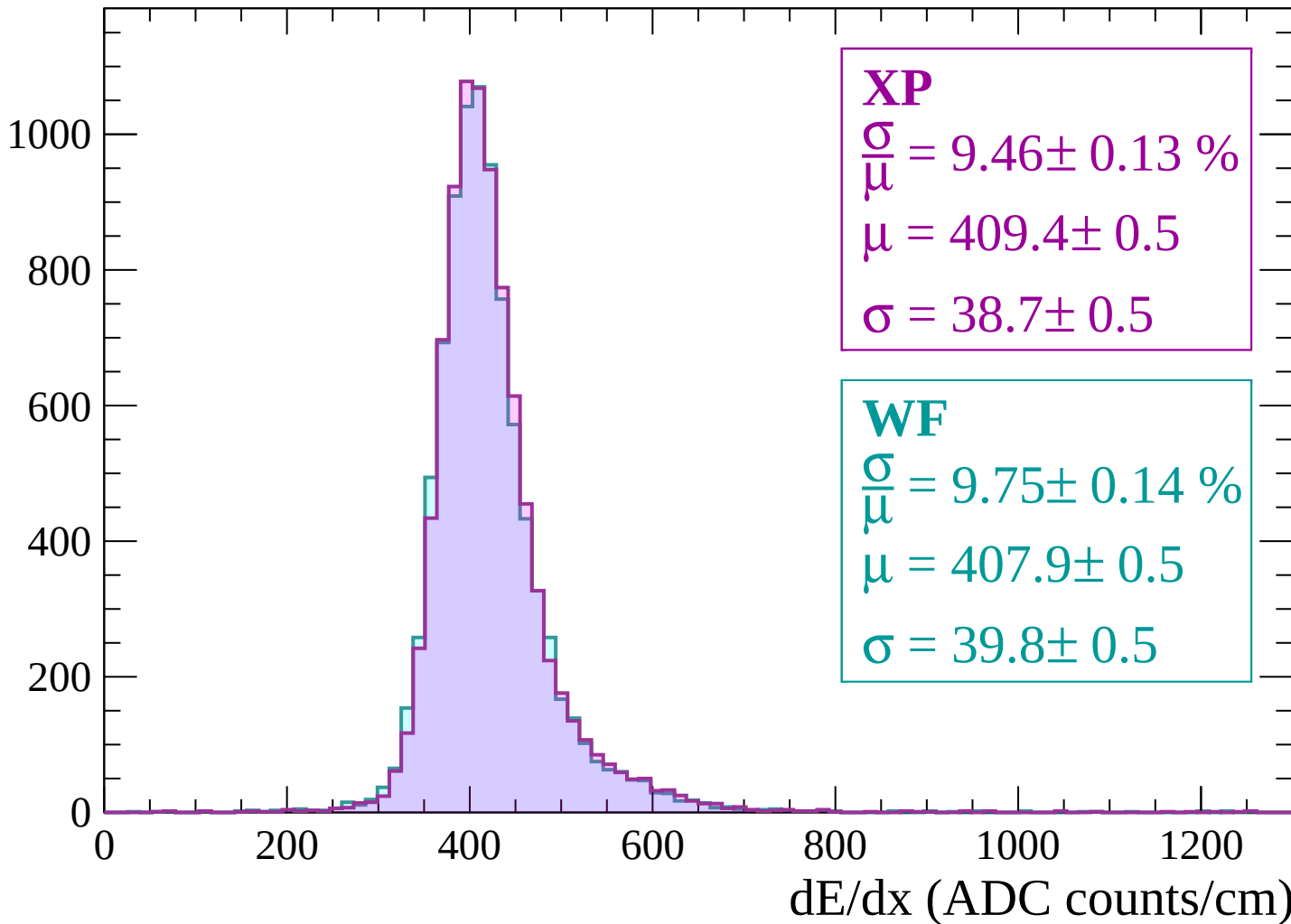
$$\mu = 405.5 \pm 0.5$$

$$\sigma = 39.5 \pm 0.5$$



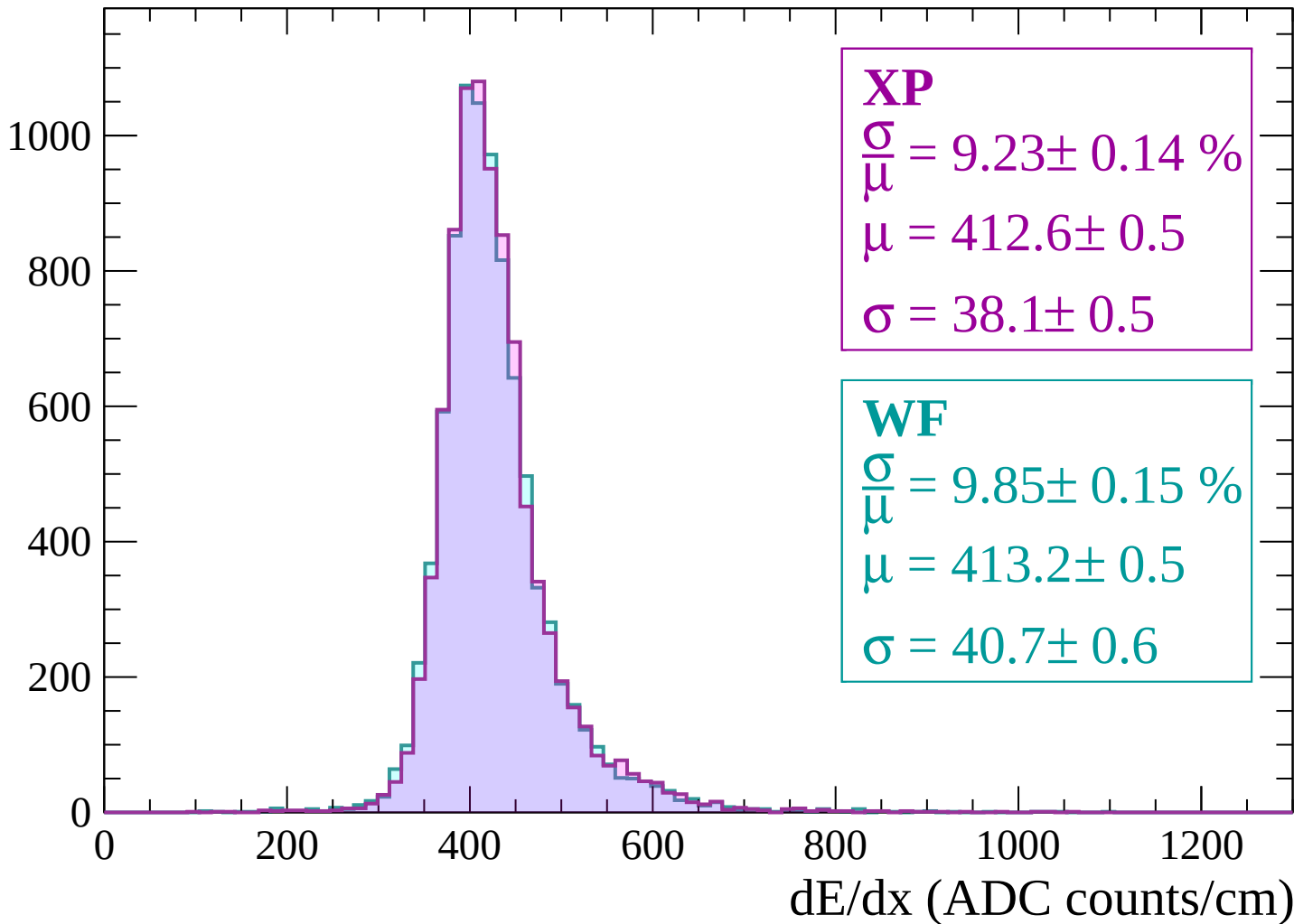
Energy loss | $600 < p < 660$

Count



Energy loss | $660 < p < 720$

Count



Energy loss | $720 < p < 780$

Count

1000

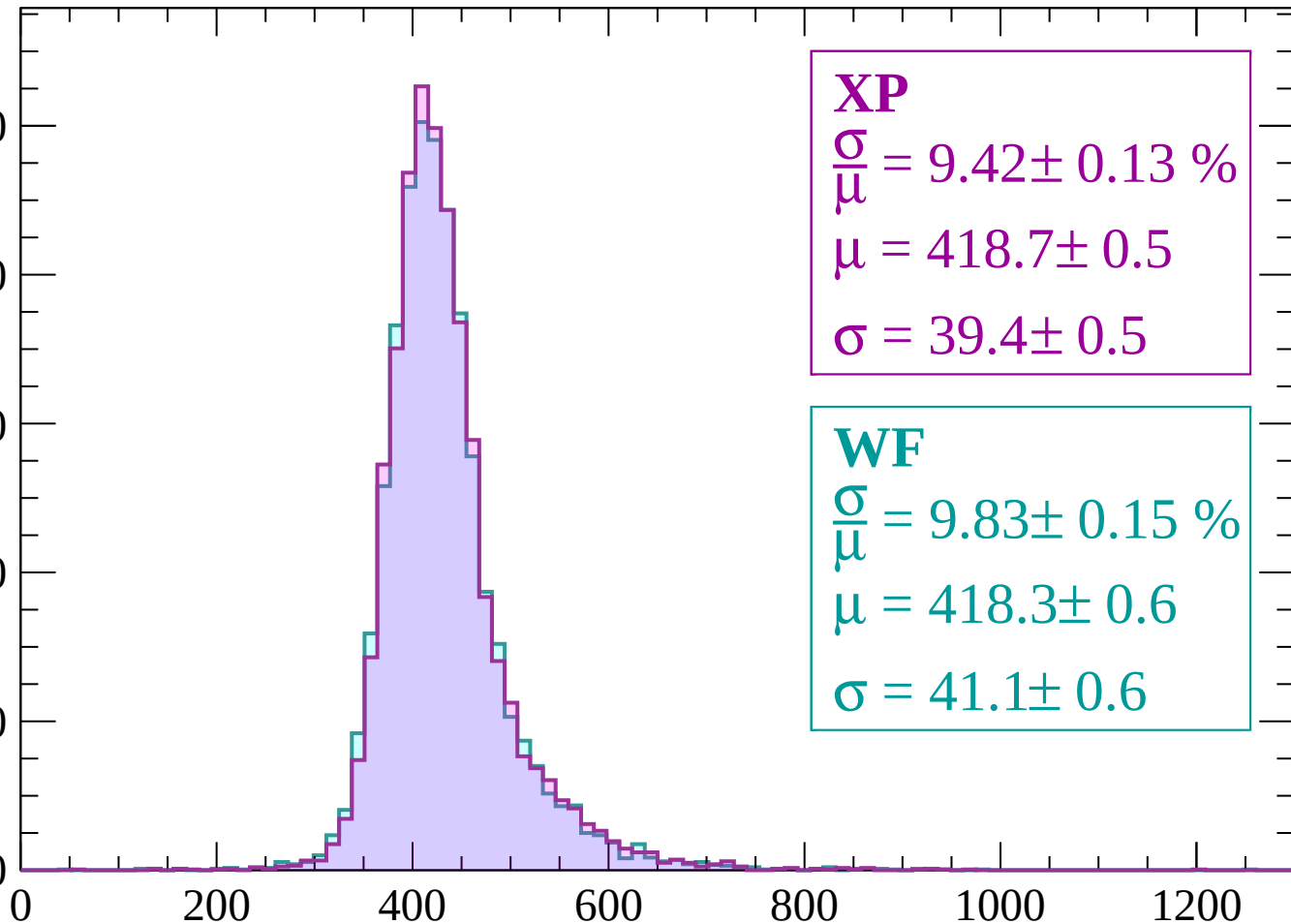
800

600

400

200

0



XP

$$\frac{\sigma}{\mu} = 9.42 \pm 0.13 \%$$

$$\mu = 418.7 \pm 0.5$$

$$\sigma = 39.4 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 9.83 \pm 0.15 \%$$

$$\mu = 418.3 \pm 0.6$$

$$\sigma = 41.1 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $780 < p < 840$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 9.52 \pm 0.14 \%$$

$$\mu = 421.6 \pm 0.6$$

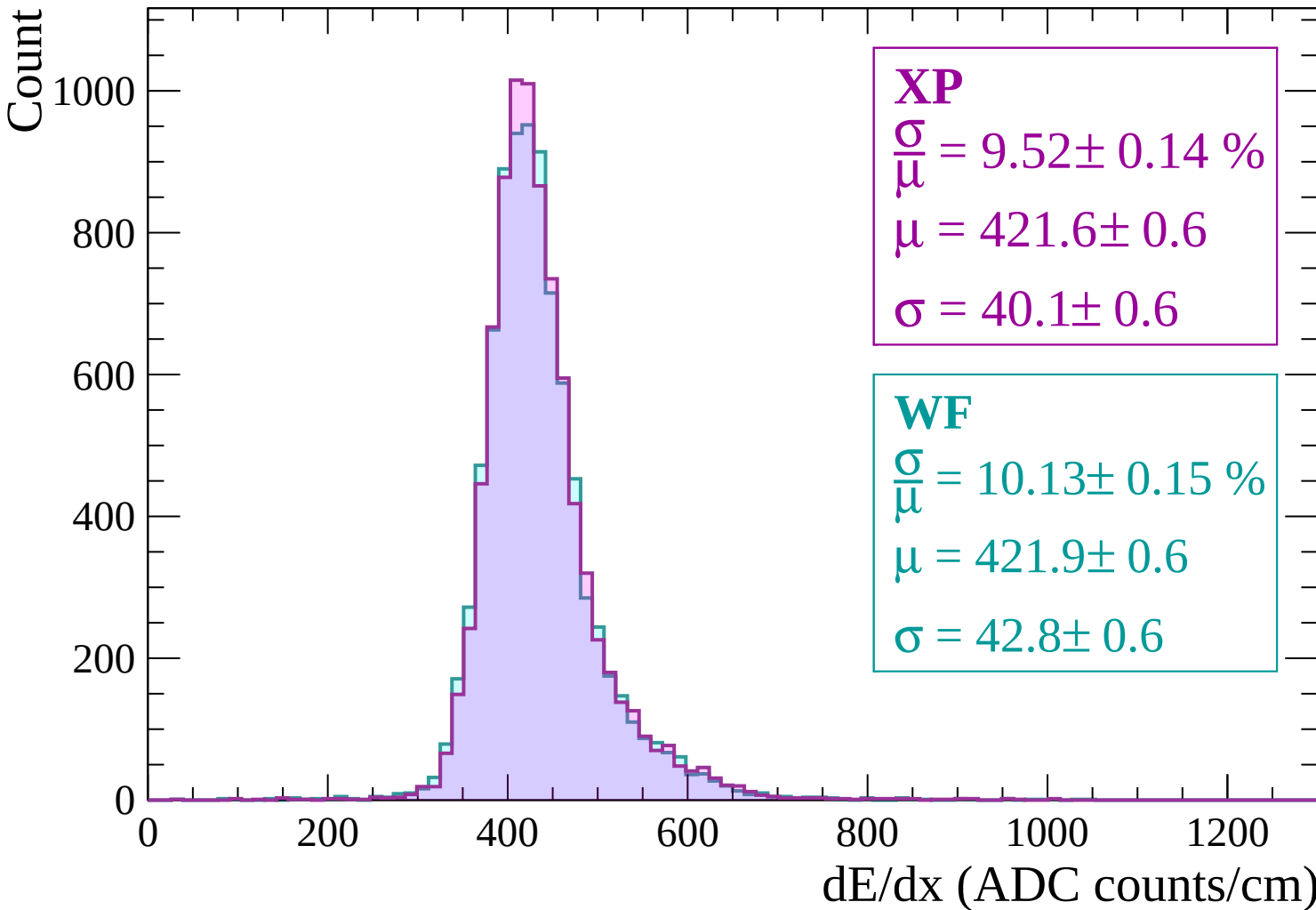
$$\sigma = 40.1 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 10.13 \pm 0.15 \%$$

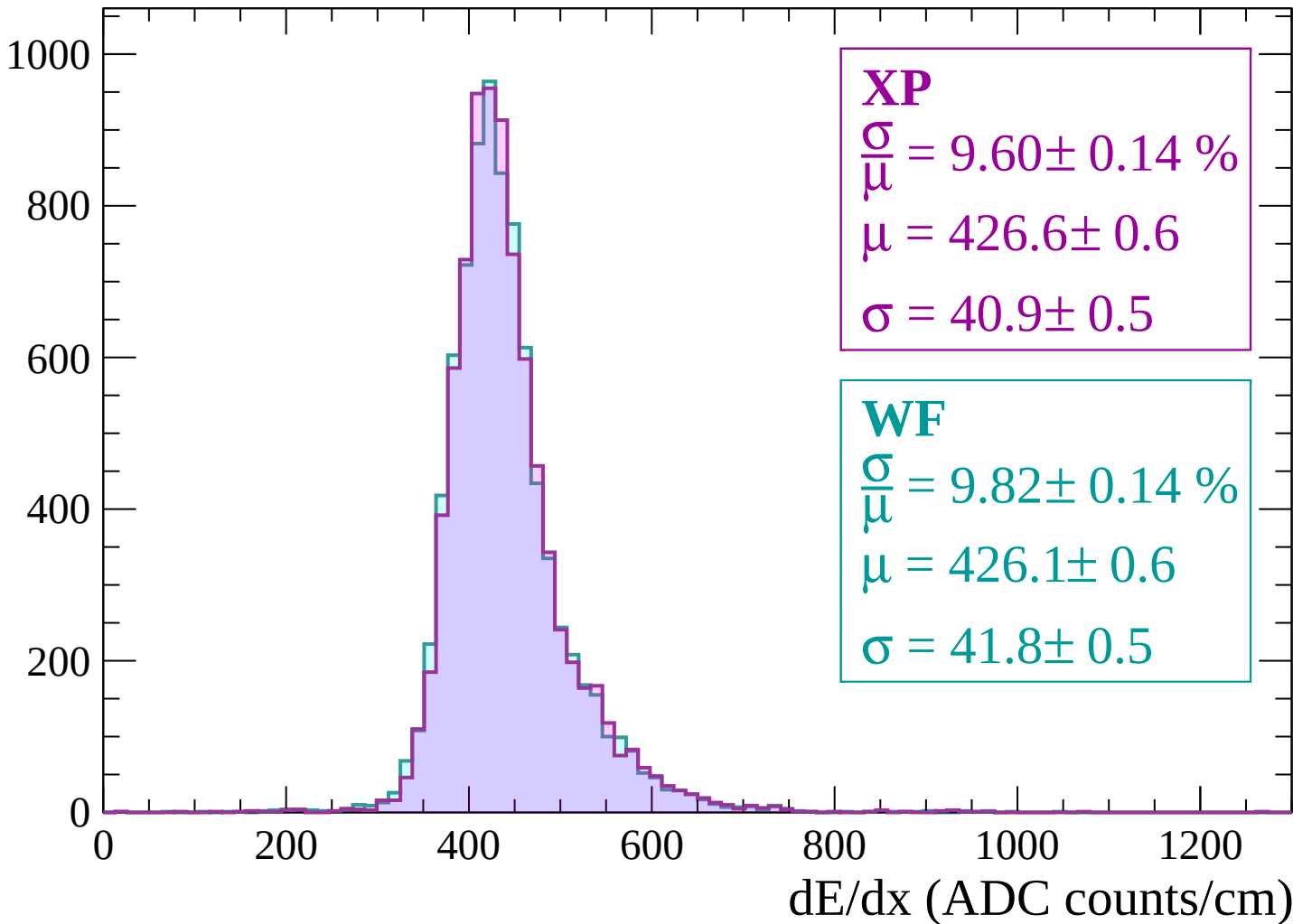
$$\mu = 421.9 \pm 0.6$$

$$\sigma = 42.8 \pm 0.6$$

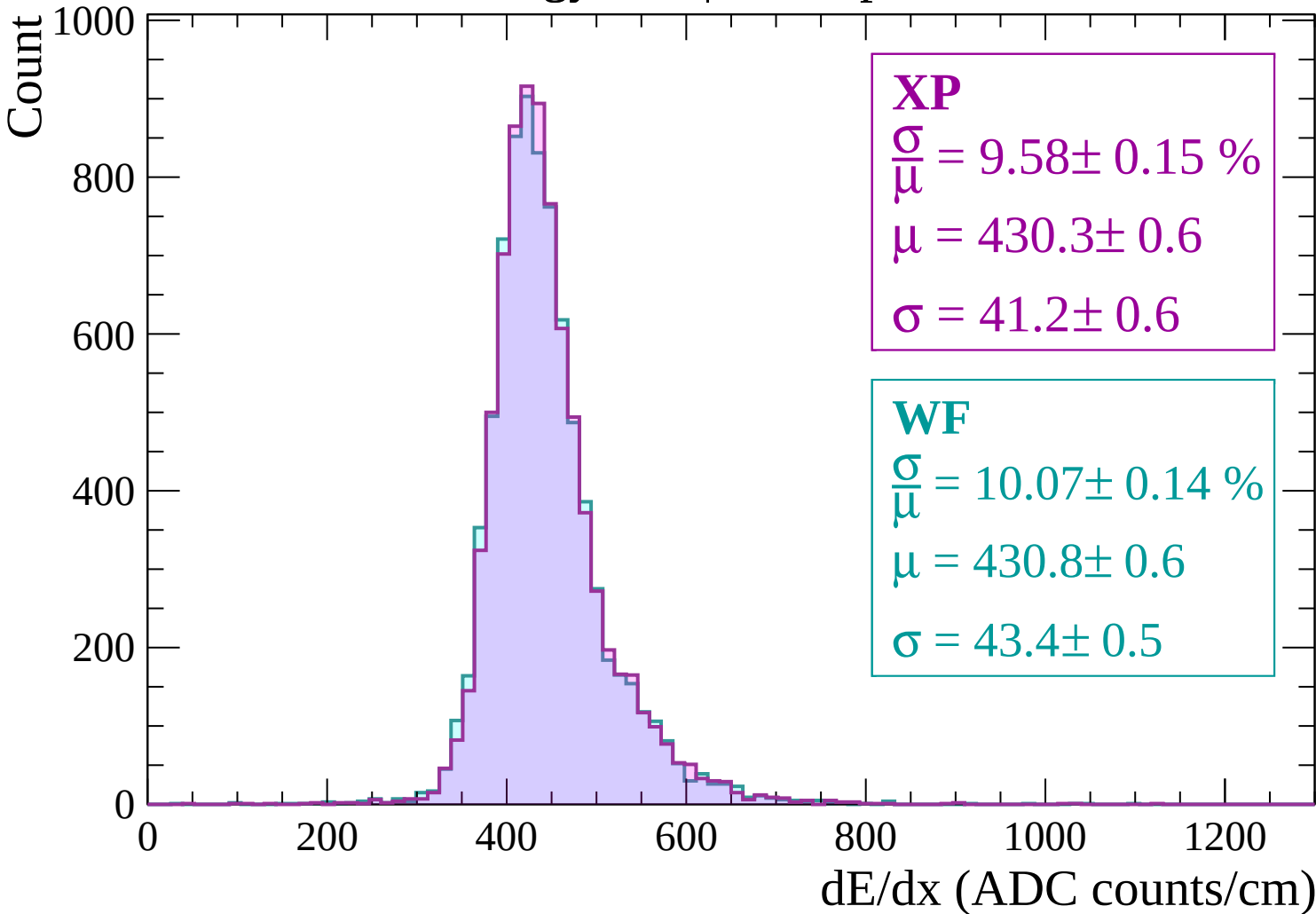


Energy loss | $840 < p < 900$

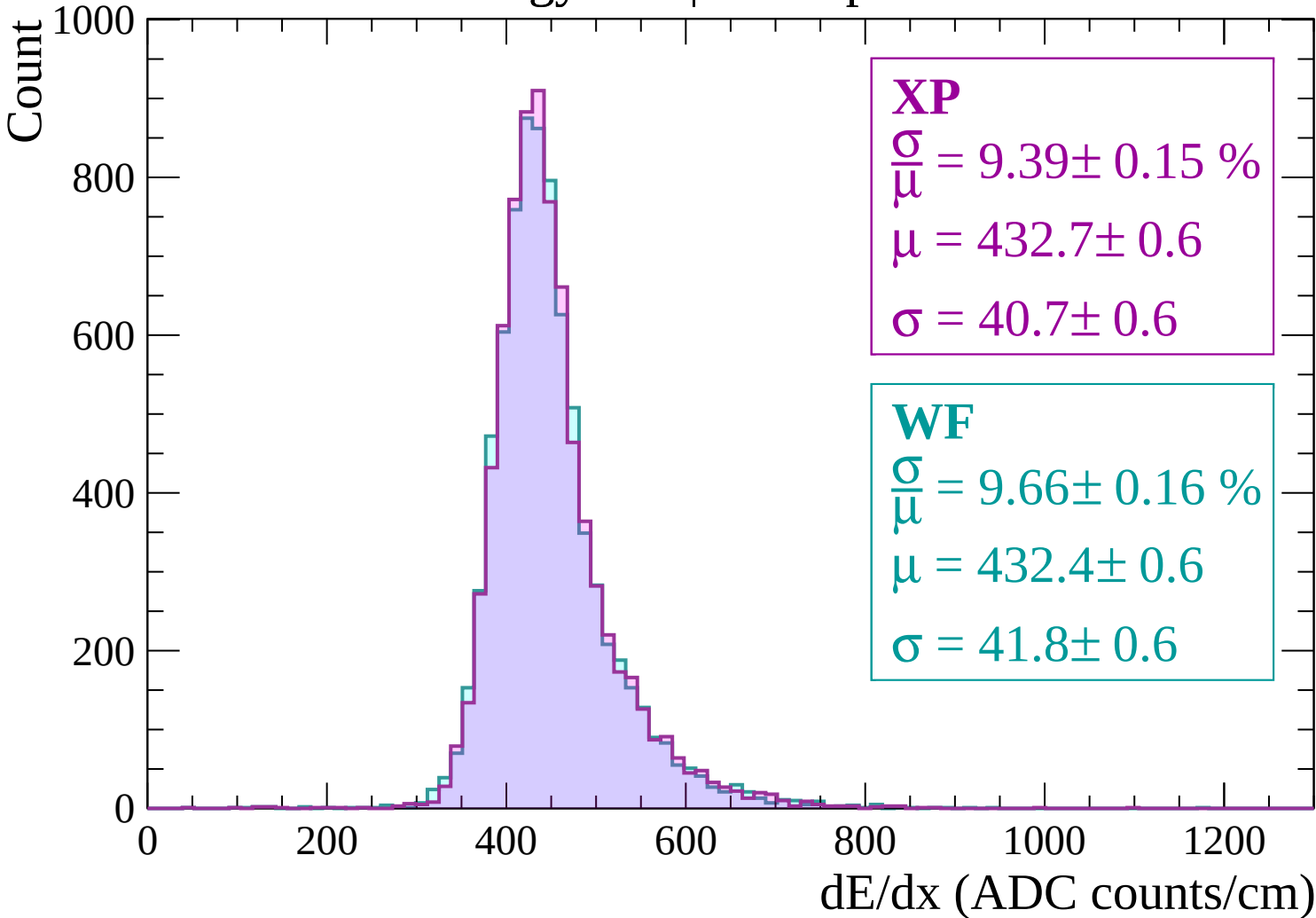
Count



Energy loss | $900 < p < 960$

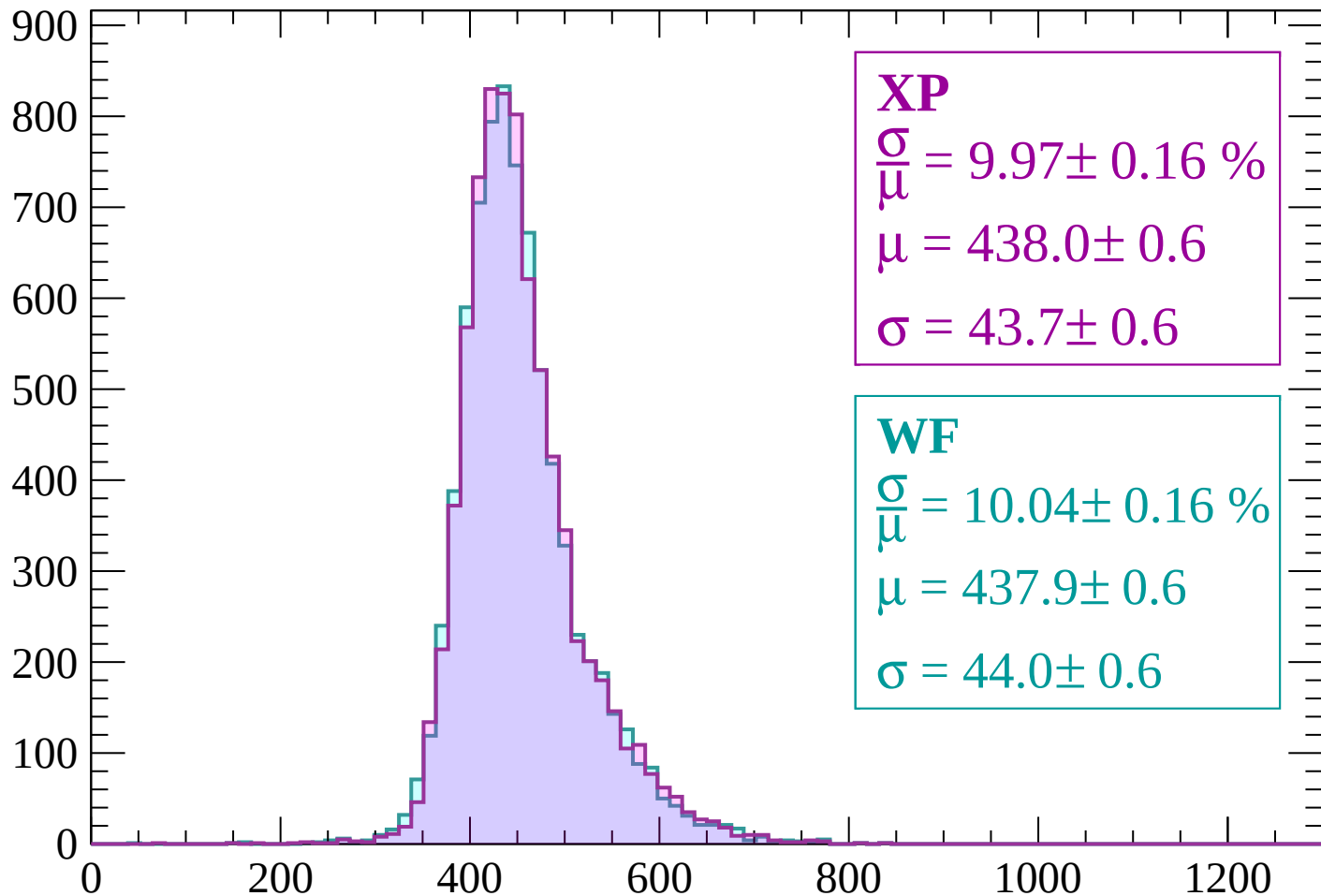


Energy loss | $960 < p < 1020$



Energy loss | $1020 < p < 1080$

Count



XP

$$\frac{\sigma}{\mu} = 9.97 \pm 0.16 \%$$

$$\mu = 438.0 \pm 0.6$$

$$\sigma = 43.7 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 10.04 \pm 0.16 \%$$

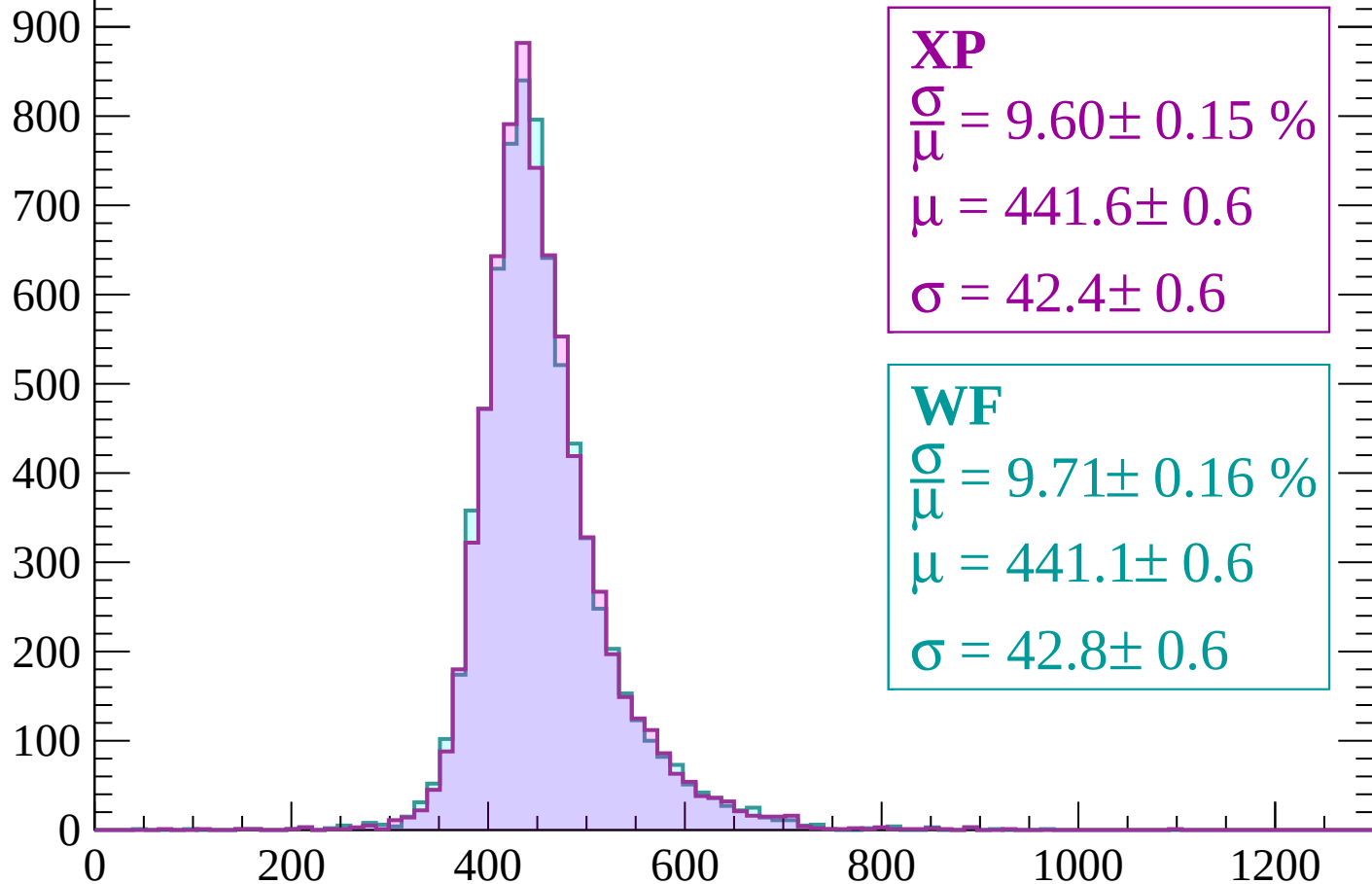
$$\mu = 437.9 \pm 0.6$$

$$\sigma = 44.0 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $1080 < p < 1140$

Count



XP

$$\frac{\sigma}{\mu} = 9.60 \pm 0.15 \%$$

$$\mu = 441.6 \pm 0.6$$

$$\sigma = 42.4 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.71 \pm 0.16 \%$$

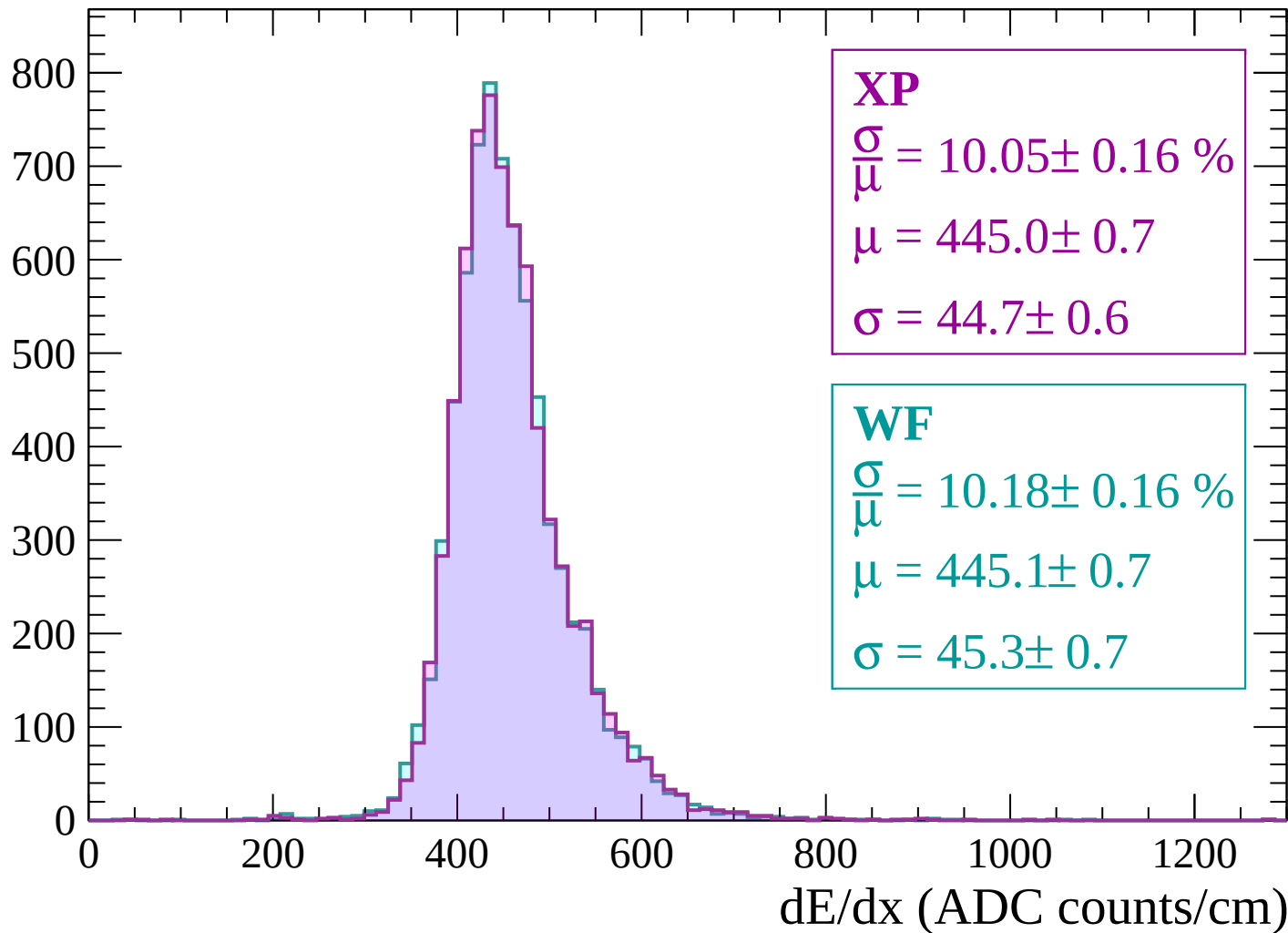
$$\mu = 441.1 \pm 0.6$$

$$\sigma = 42.8 \pm 0.6$$

dE/dx (ADC counts/cm)

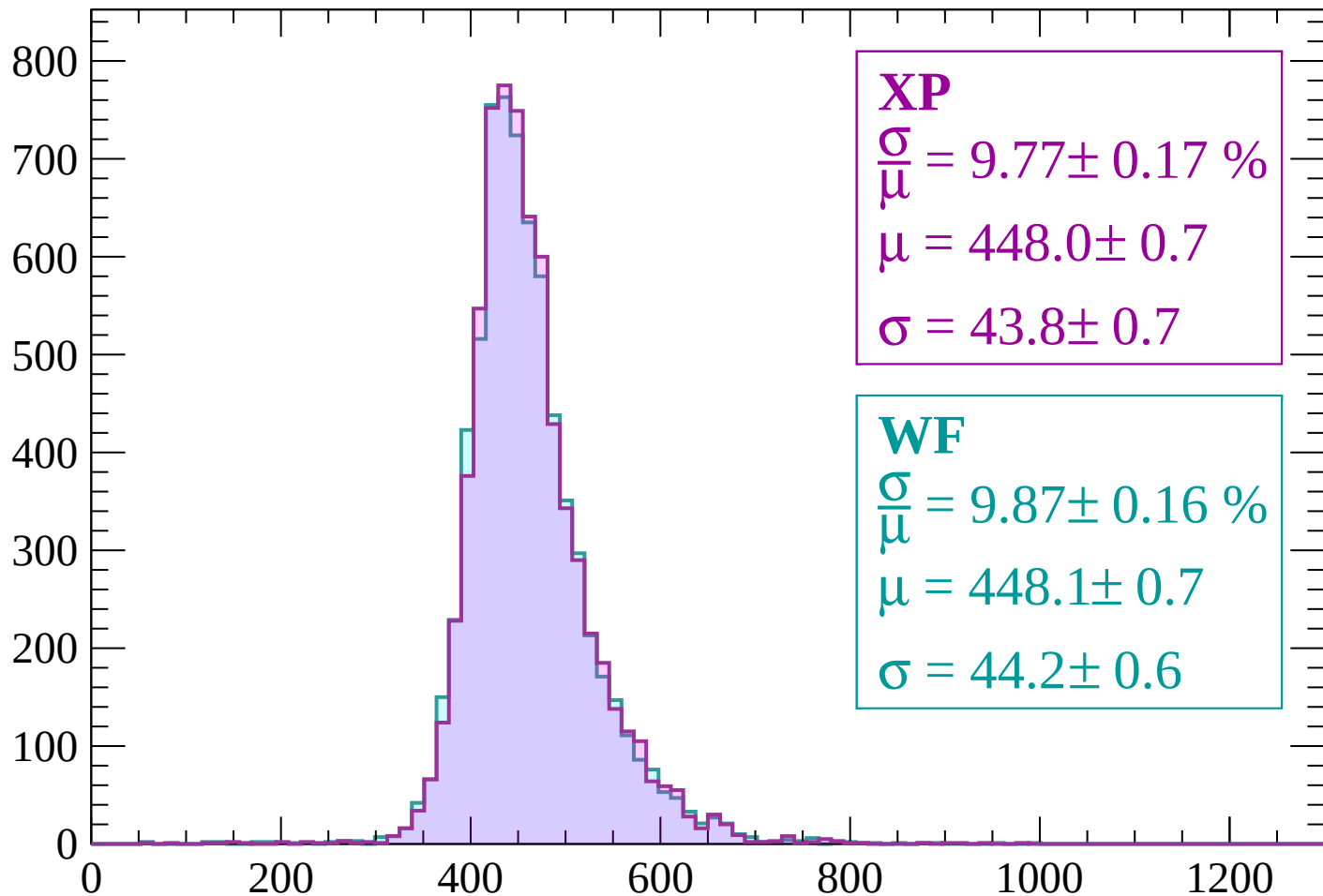
Energy loss | $1140 < p < 1200$

Count



Energy loss | $1200 < p < 1260$

Count



XP

$$\frac{\sigma}{\mu} = 9.77 \pm 0.17 \%$$

$$\mu = 448.0 \pm 0.7$$

$$\sigma = 43.8 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.87 \pm 0.16 \%$$

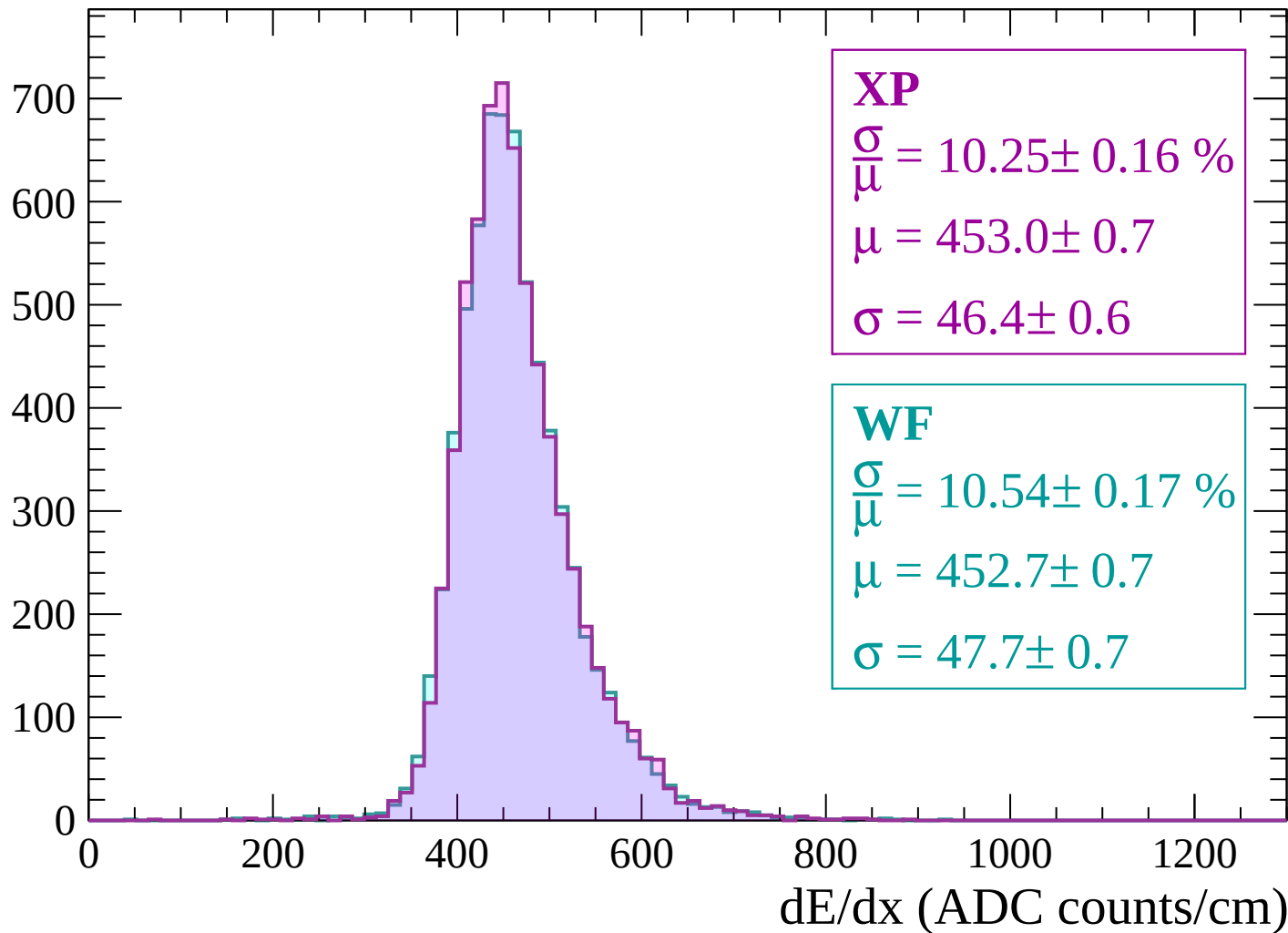
$$\mu = 448.1 \pm 0.7$$

$$\sigma = 44.2 \pm 0.6$$

dE/dx (ADC counts/cm)

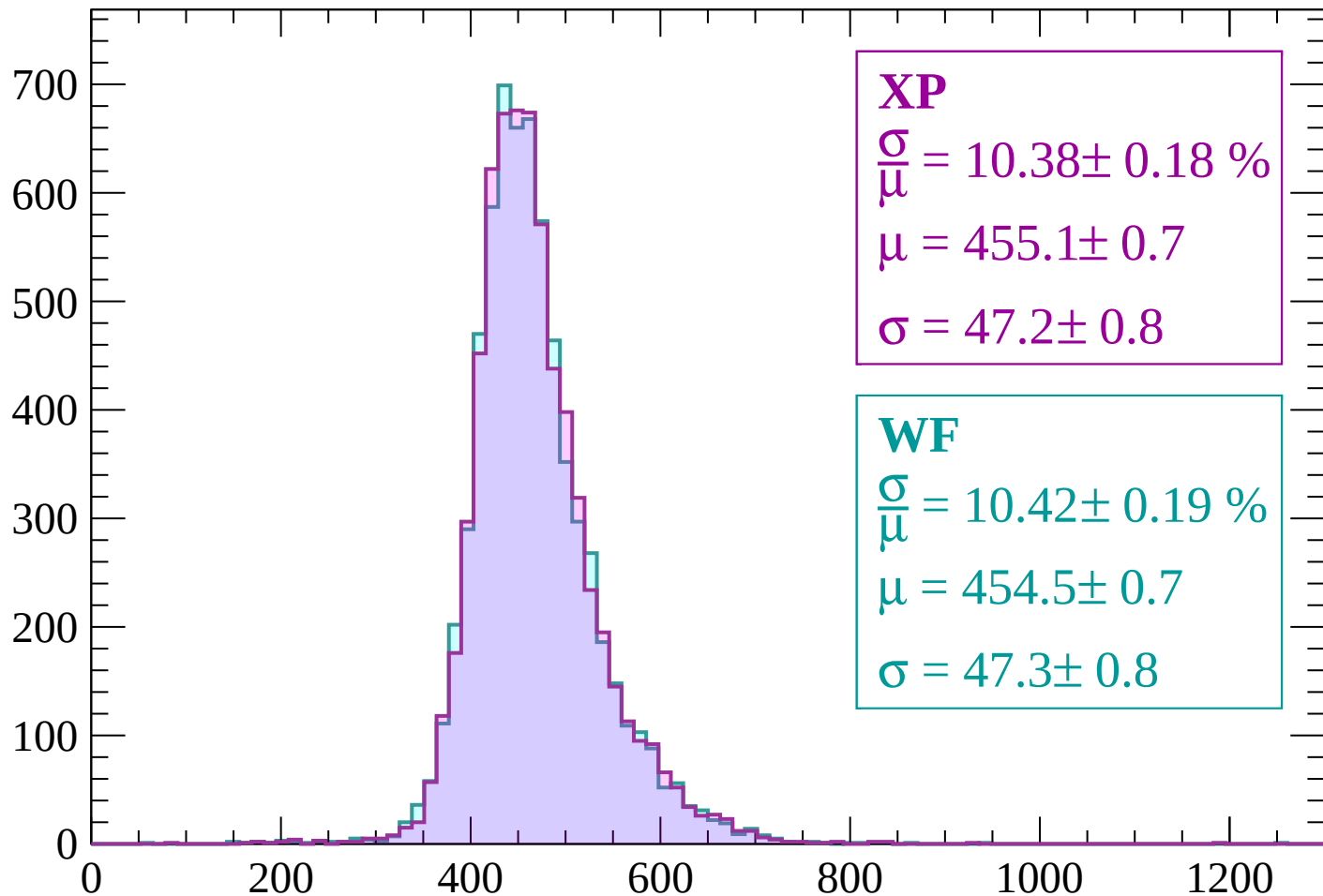
Energy loss | $1260 < p < 1320$

Count



Energy loss | $1320 < p < 1380$

Count



XP

$$\frac{\sigma}{\mu} = 10.38 \pm 0.18 \%$$

$$\mu = 455.1 \pm 0.7$$

$$\sigma = 47.2 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 10.42 \pm 0.19 \%$$

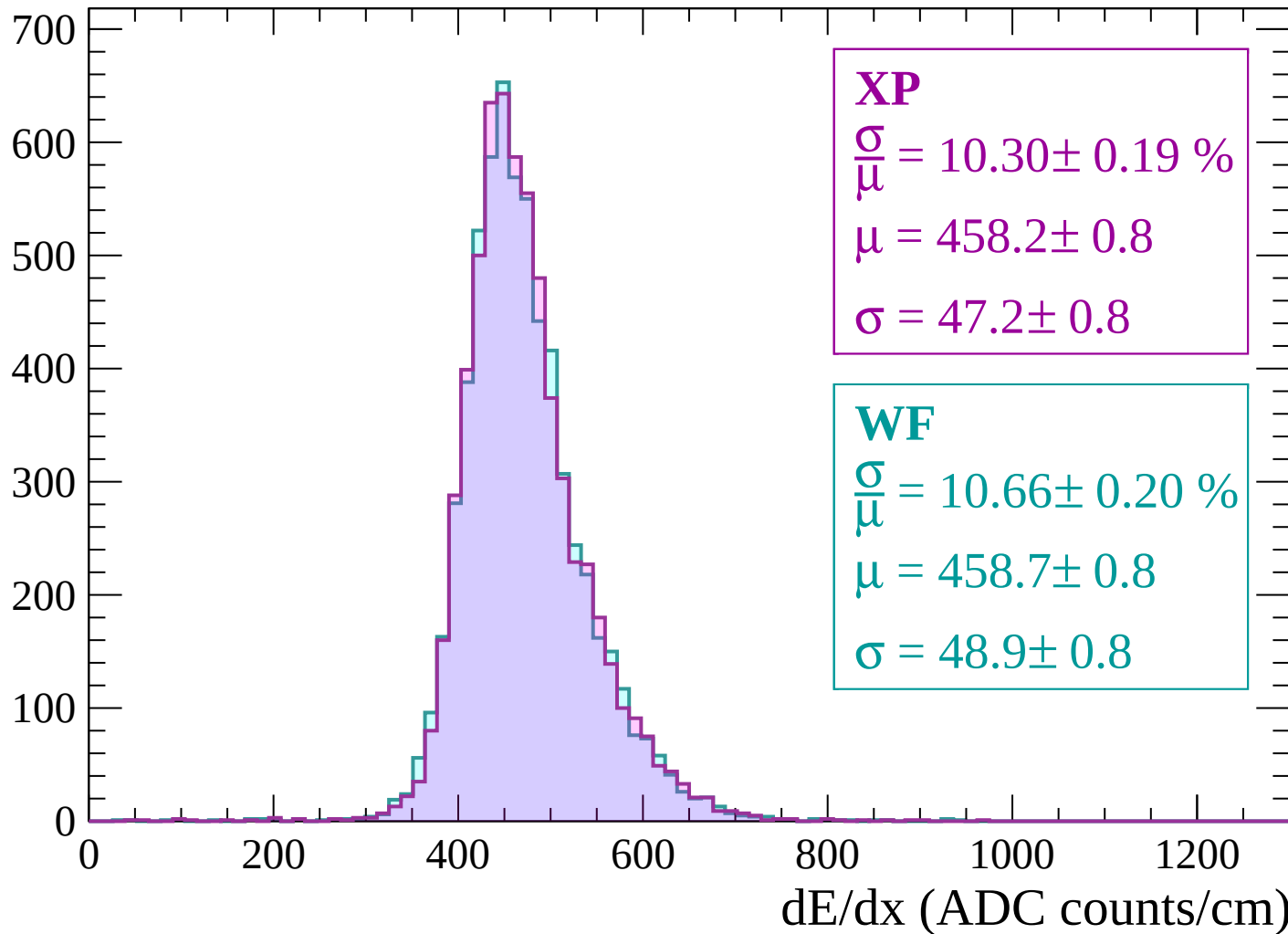
$$\mu = 454.5 \pm 0.7$$

$$\sigma = 47.3 \pm 0.8$$

dE/dx (ADC counts/cm)

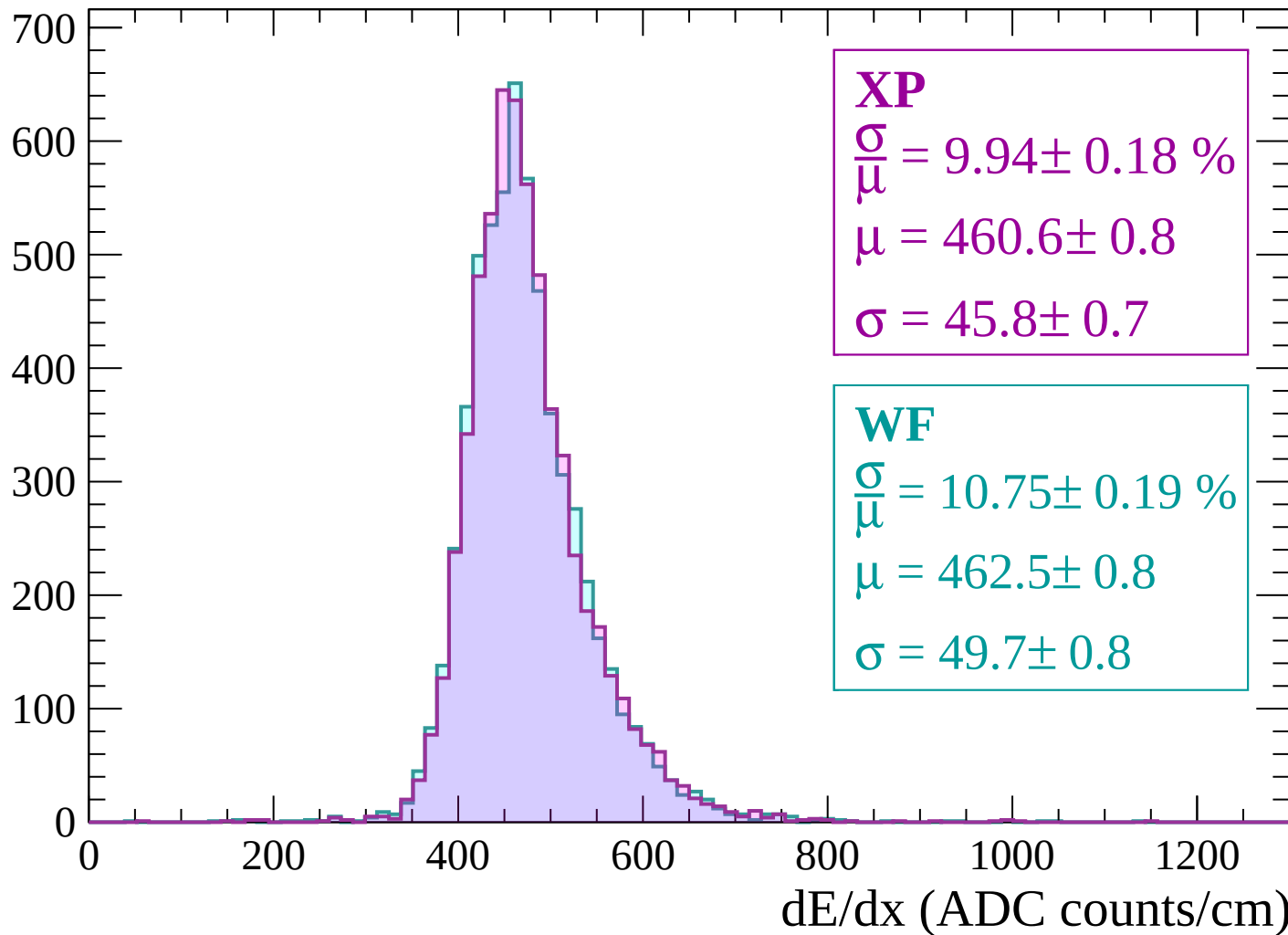
Energy loss | $1380 < p < 1440$

Count



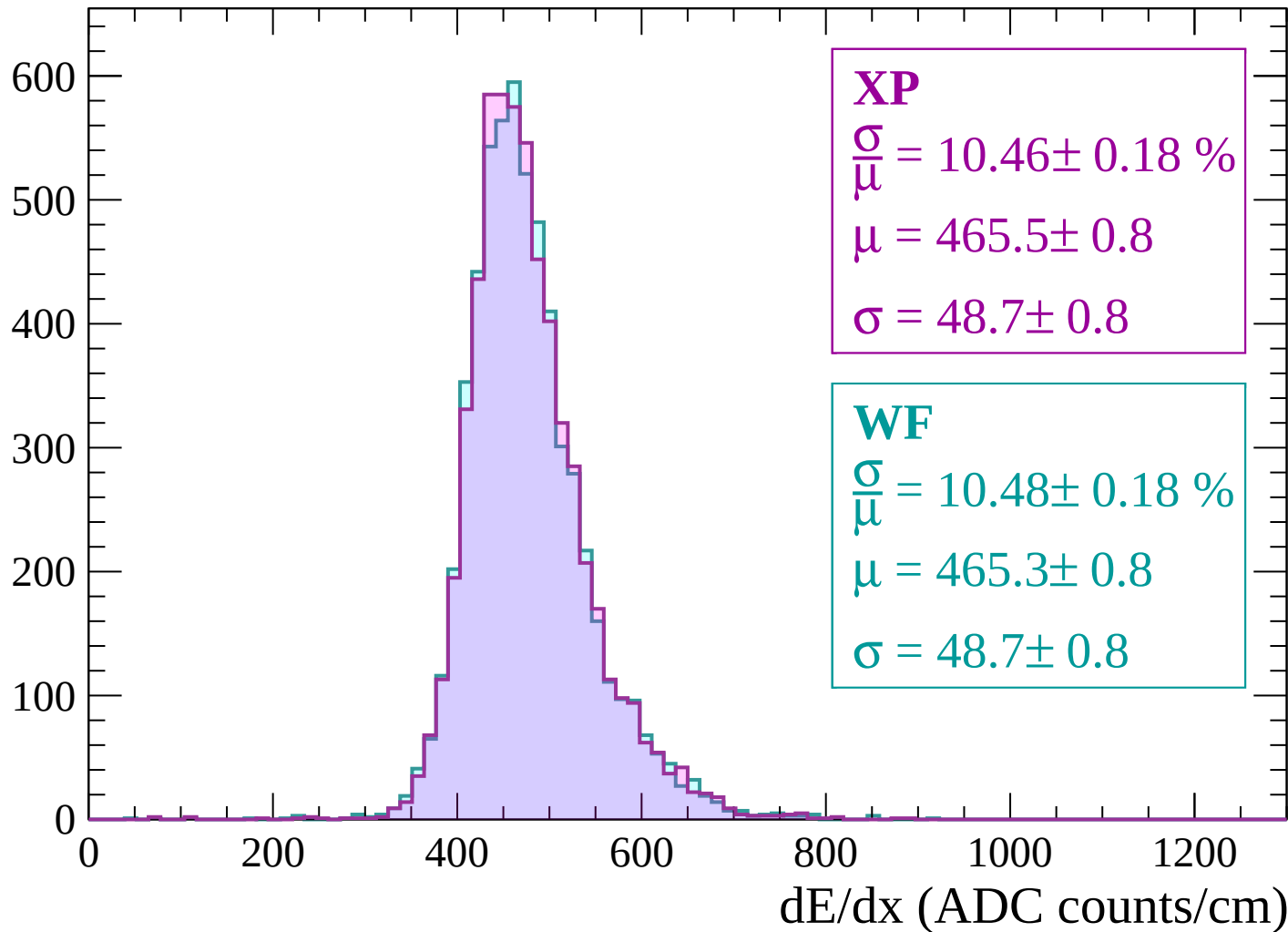
Energy loss | $1440 < p < 1500$

Count



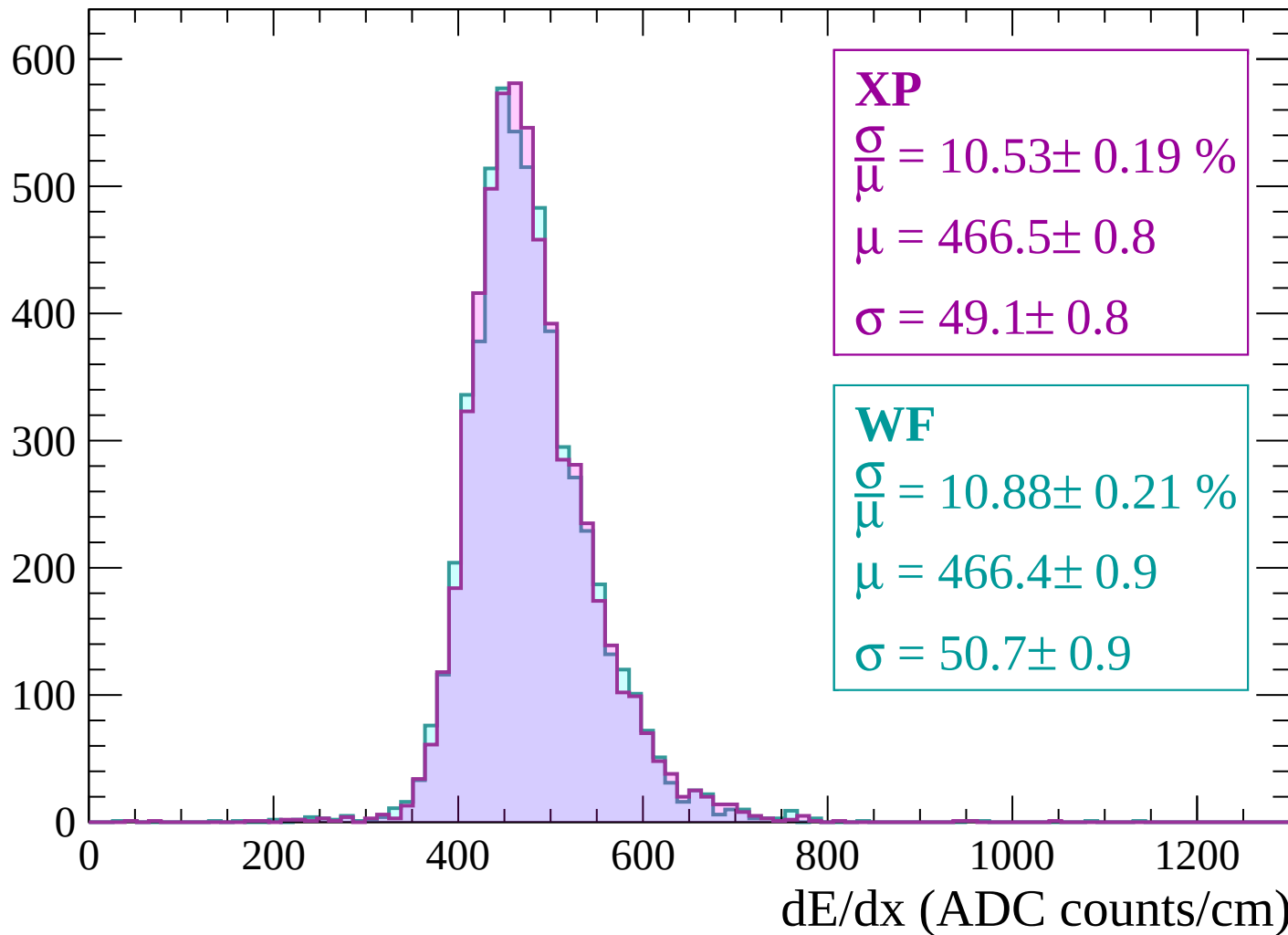
Energy loss | $1500 < p < 1560$

Count



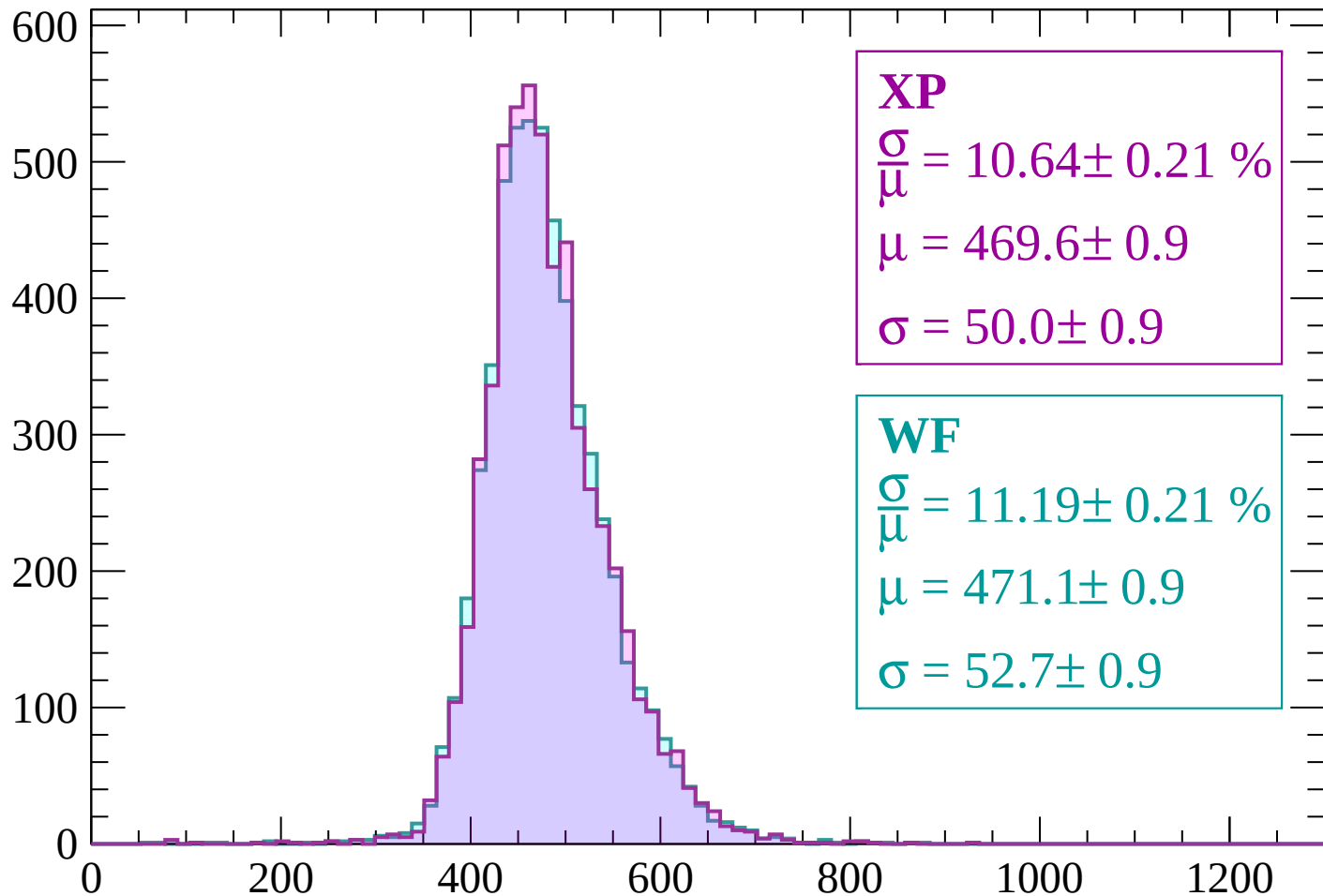
Energy loss | $1560 < p < 1620$

Count



Energy loss | $1620 < p < 1680$

Count



XP

$$\frac{\sigma}{\mu} = 10.64 \pm 0.21 \%$$

$$\mu = 469.6 \pm 0.9$$

$$\sigma = 50.0 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 11.19 \pm 0.21 \%$$

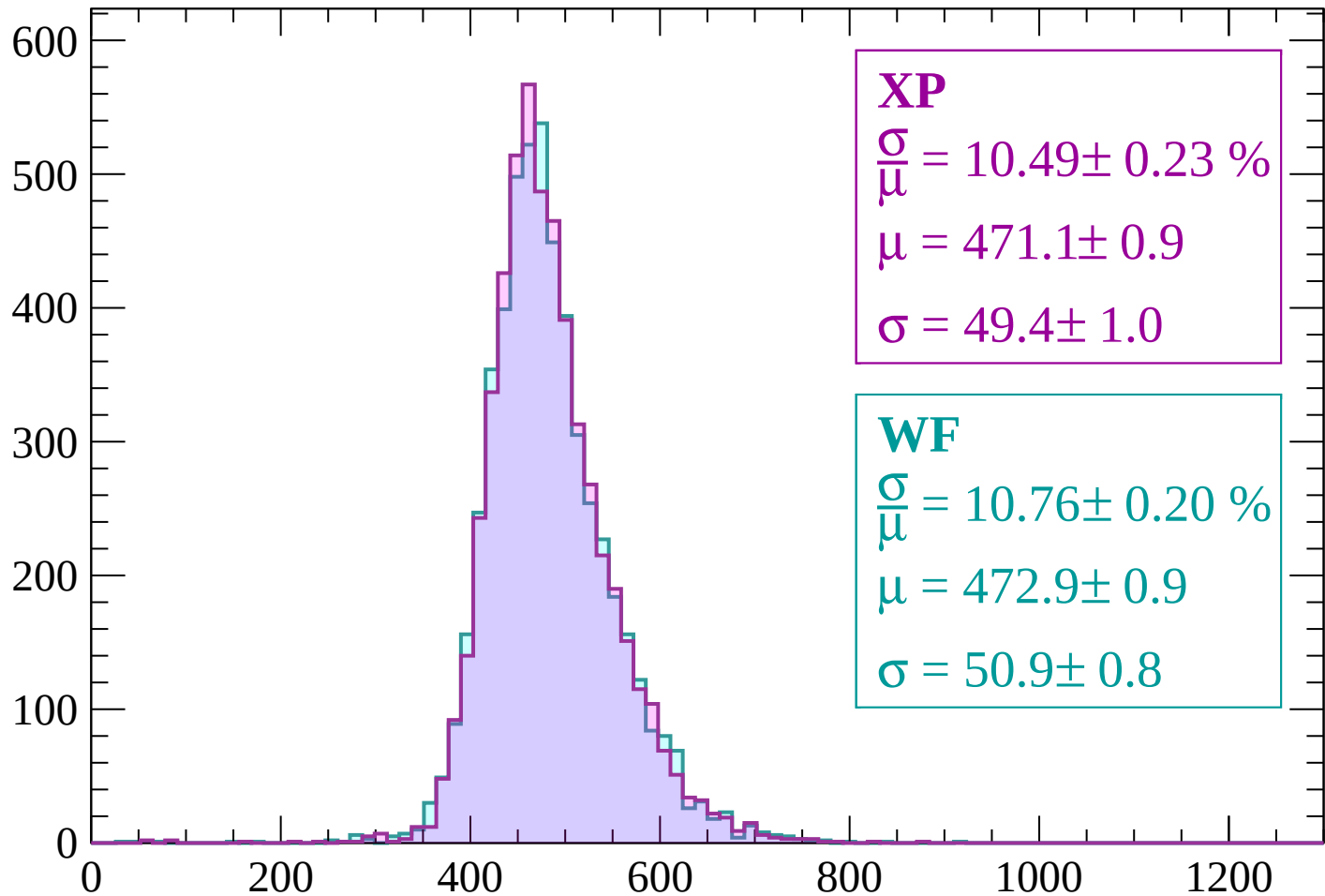
$$\mu = 471.1 \pm 0.9$$

$$\sigma = 52.7 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1680 < p < 1740$

Count



XP

$$\frac{\sigma}{\mu} = 10.49 \pm 0.23 \%$$

$$\mu = 471.1 \pm 0.9$$

$$\sigma = 49.4 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 10.76 \pm 0.20 \%$$

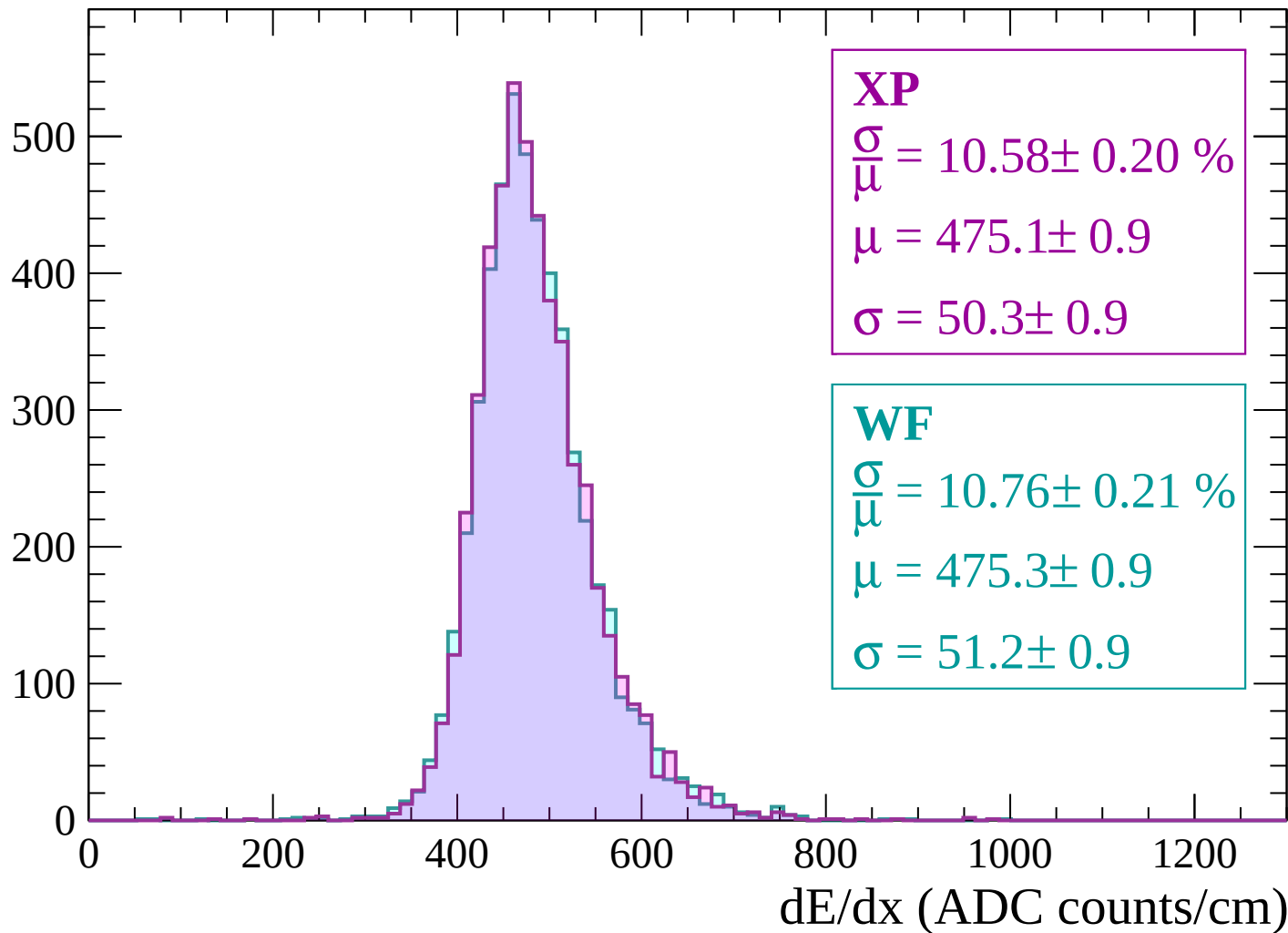
$$\mu = 472.9 \pm 0.9$$

$$\sigma = 50.9 \pm 0.8$$

dE/dx (ADC counts/cm)

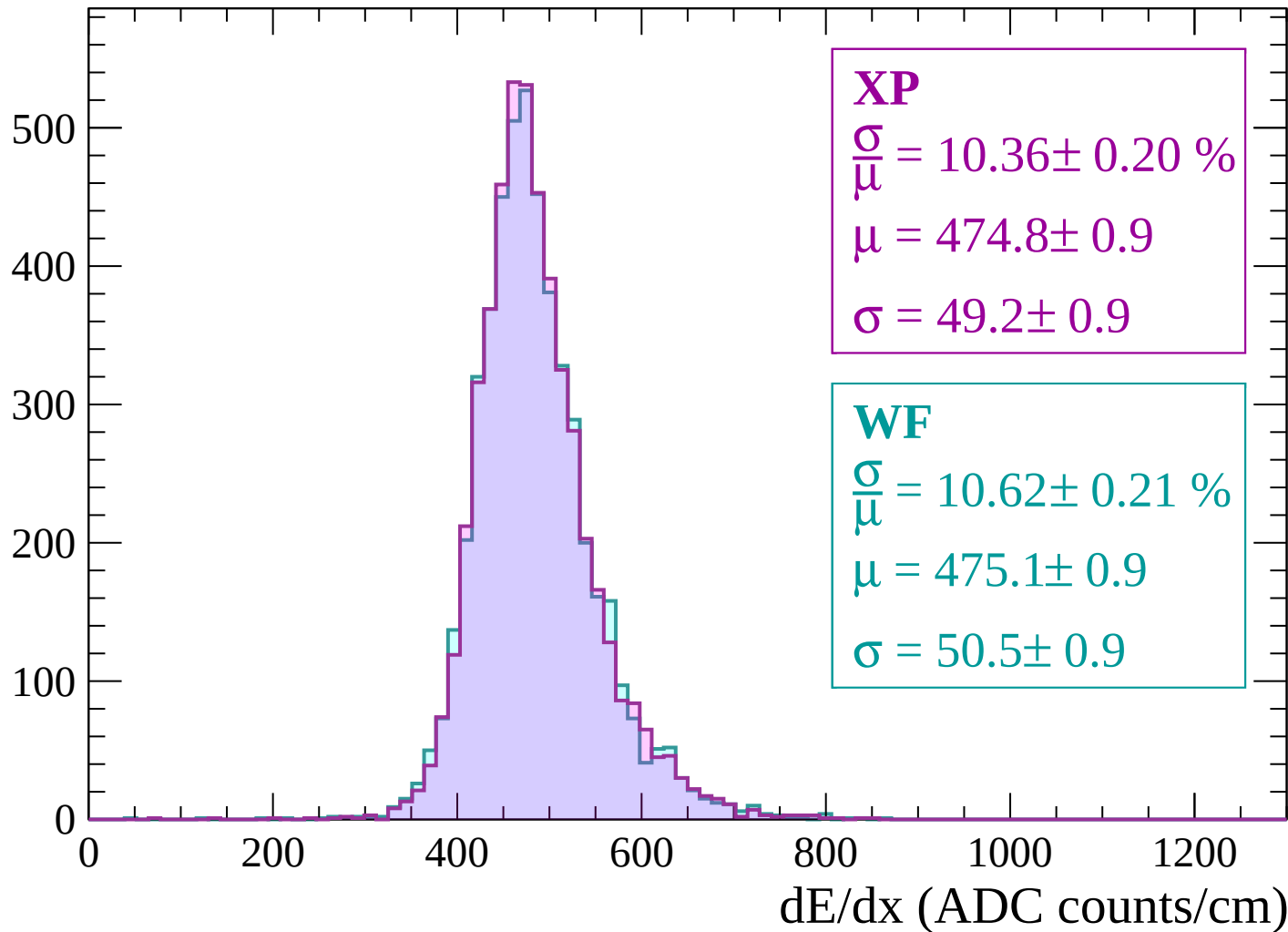
Energy loss | $1740 < p < 1800$

Count



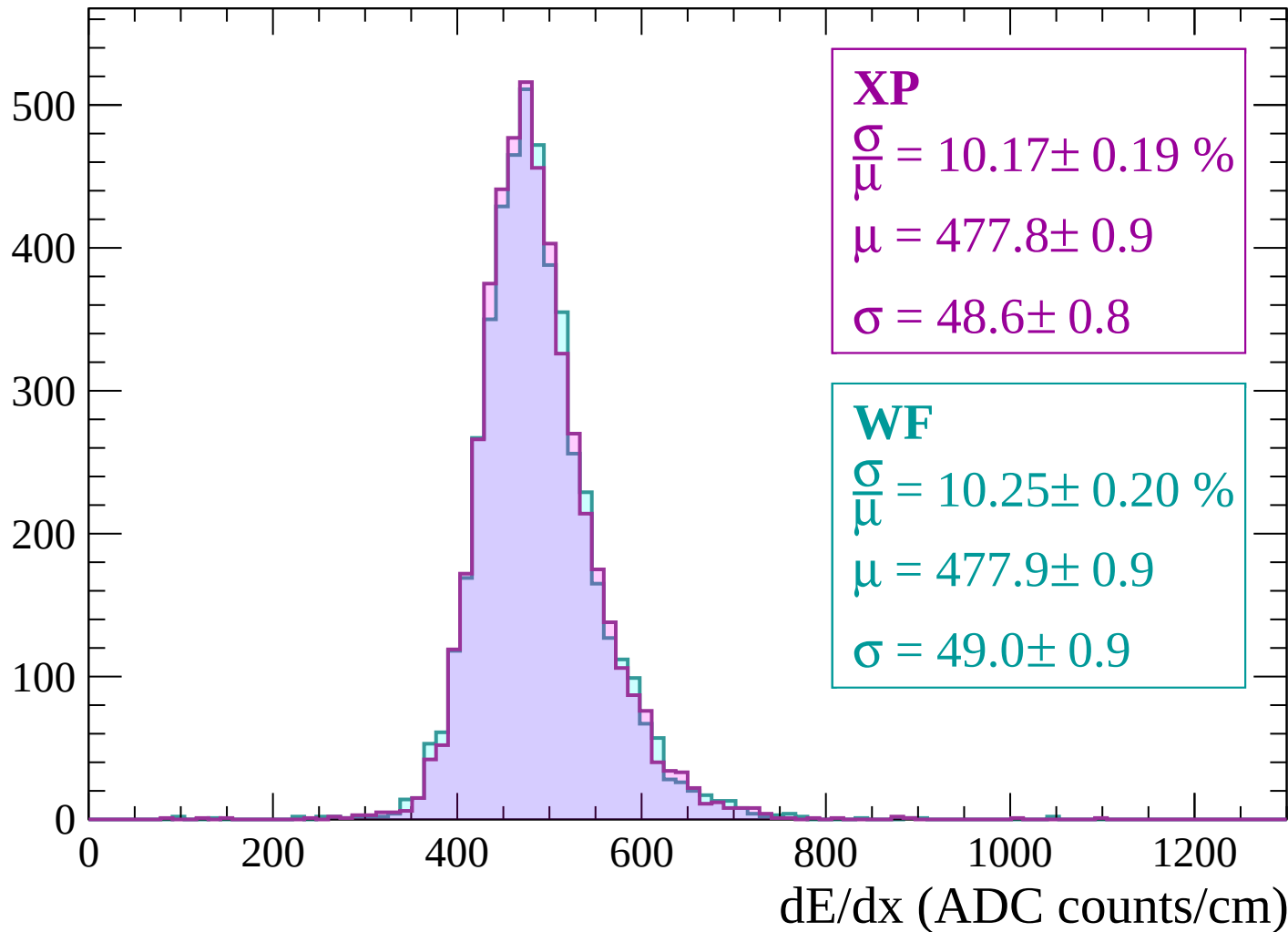
Energy loss | $1800 < p < 1860$

Count



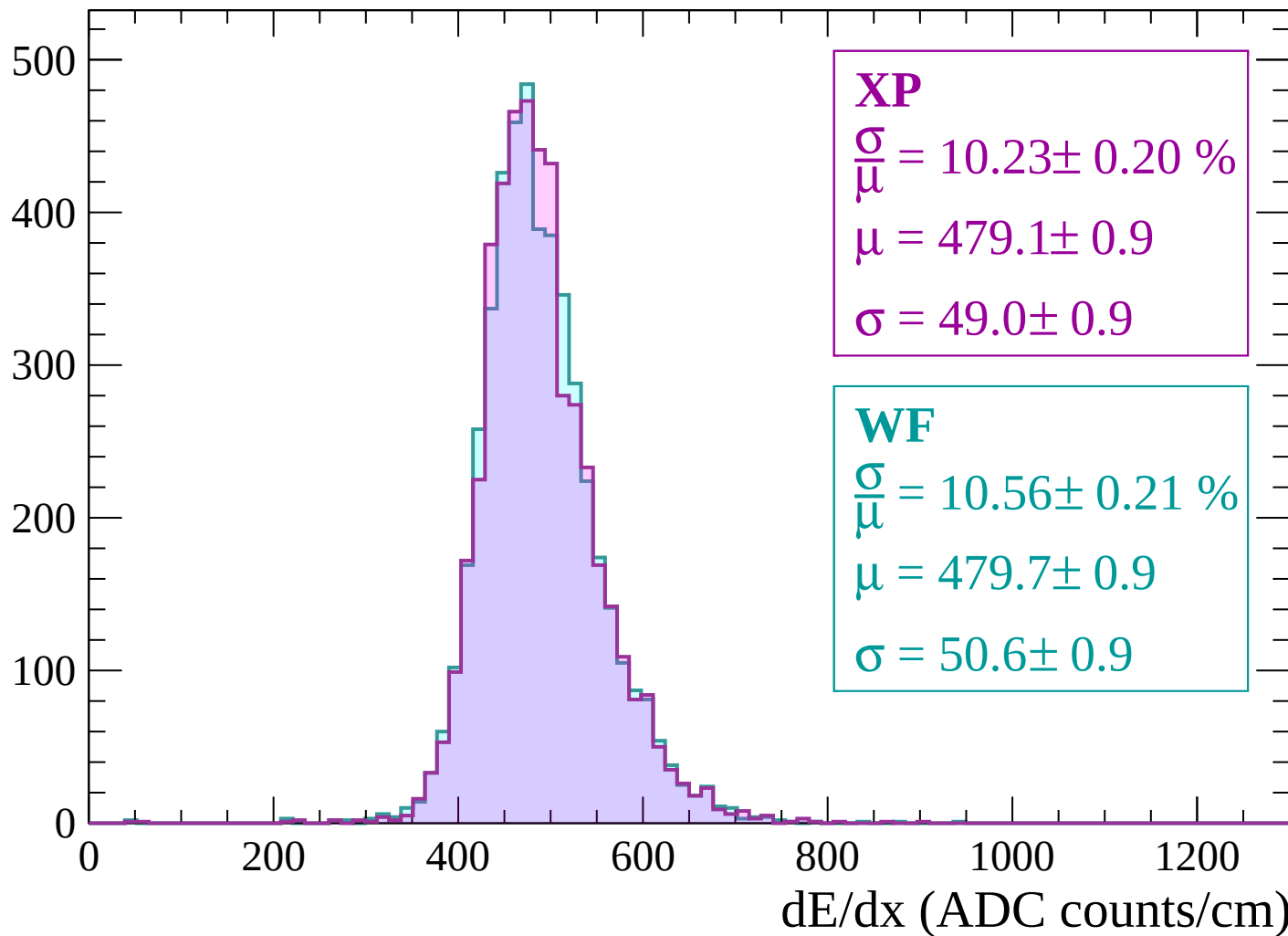
Energy loss | $1860 < p < 1920$

Count



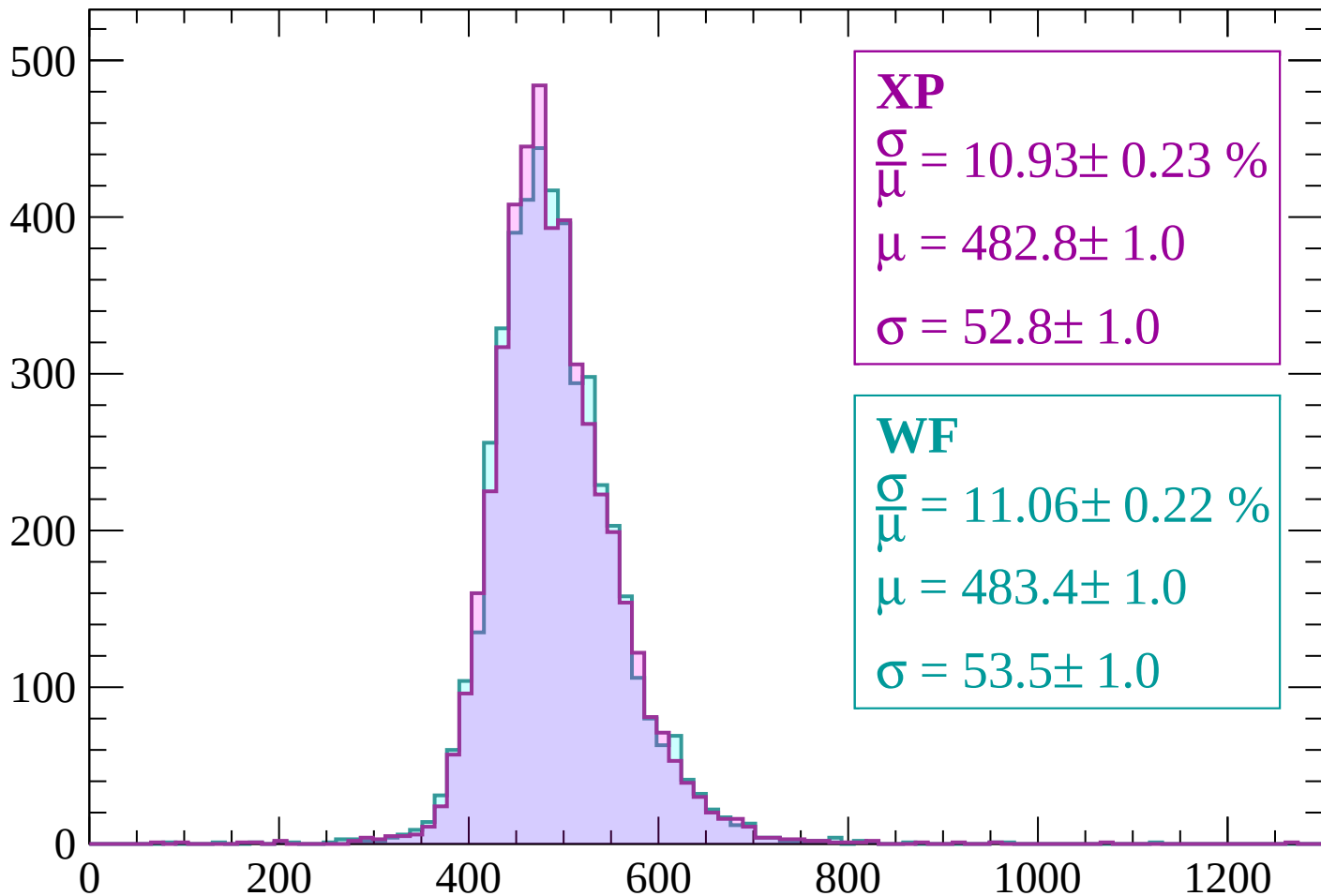
Energy loss | $1920 < p < 1980$

Count



Energy loss | $1980 < p < 2040$

Count



XP

$$\frac{\sigma}{\mu} = 10.93 \pm 0.23 \%$$

$$\mu = 482.8 \pm 1.0$$

$$\sigma = 52.8 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 11.06 \pm 0.22 \%$$

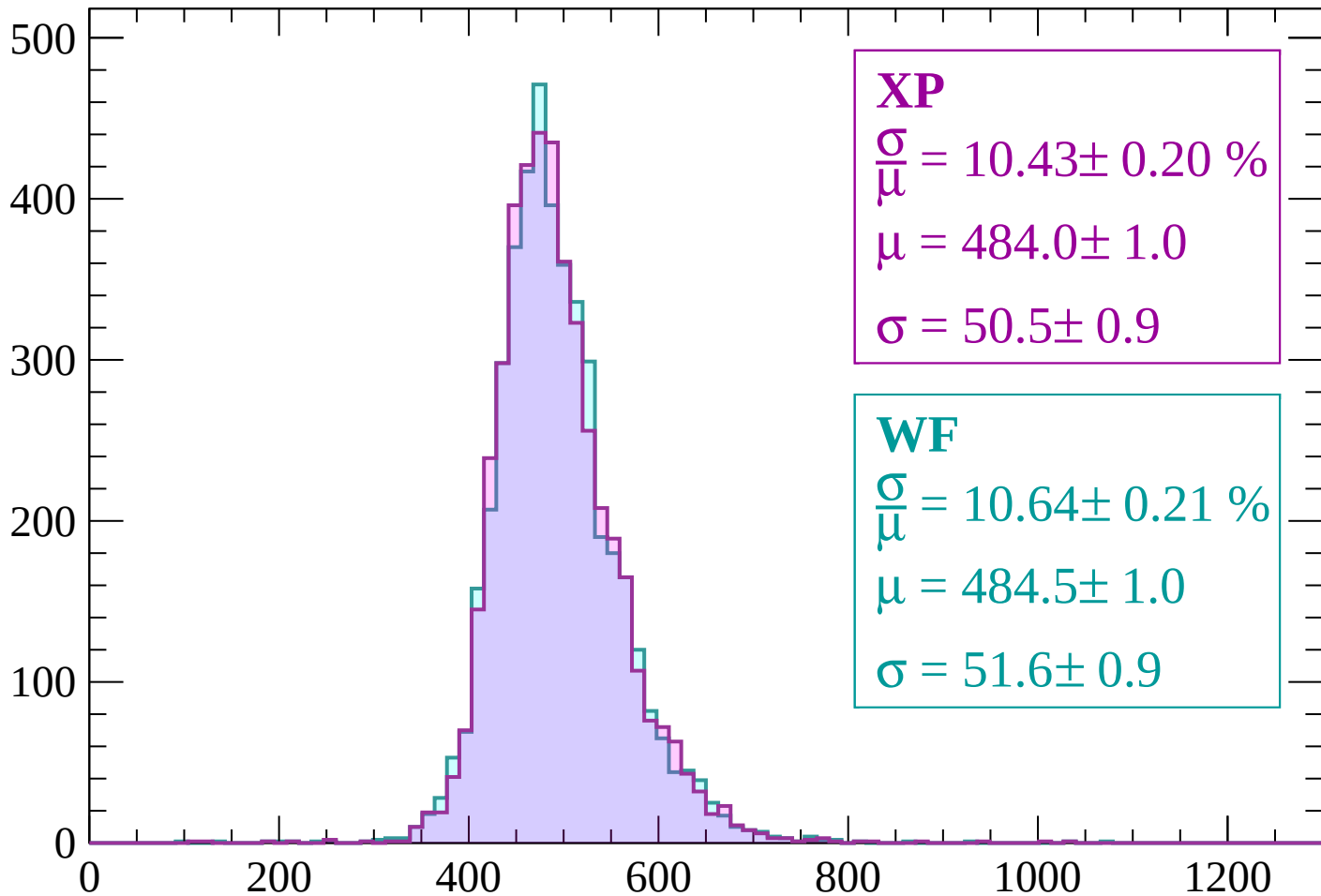
$$\mu = 483.4 \pm 1.0$$

$$\sigma = 53.5 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $2040 < p < 2100$

Count



XP

$$\frac{\sigma}{\mu} = 10.43 \pm 0.20 \%$$

$$\mu = 484.0 \pm 1.0$$

$$\sigma = 50.5 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 10.64 \pm 0.21 \%$$

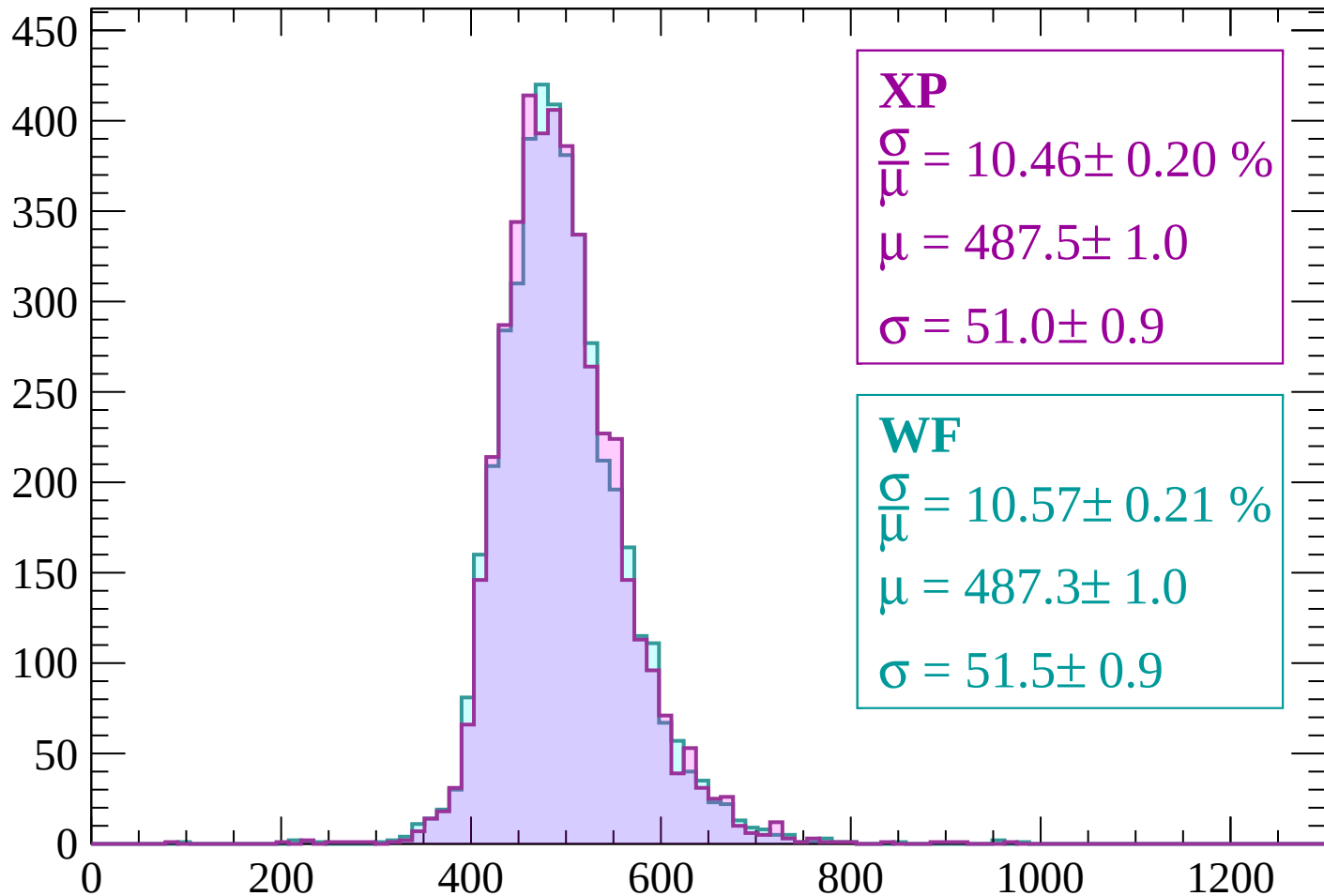
$$\mu = 484.5 \pm 1.0$$

$$\sigma = 51.6 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $2100 < p < 2160$

Count



XP

$$\frac{\sigma}{\mu} = 10.46 \pm 0.20 \%$$

$$\mu = 487.5 \pm 1.0$$

$$\sigma = 51.0 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 10.57 \pm 0.21 \%$$

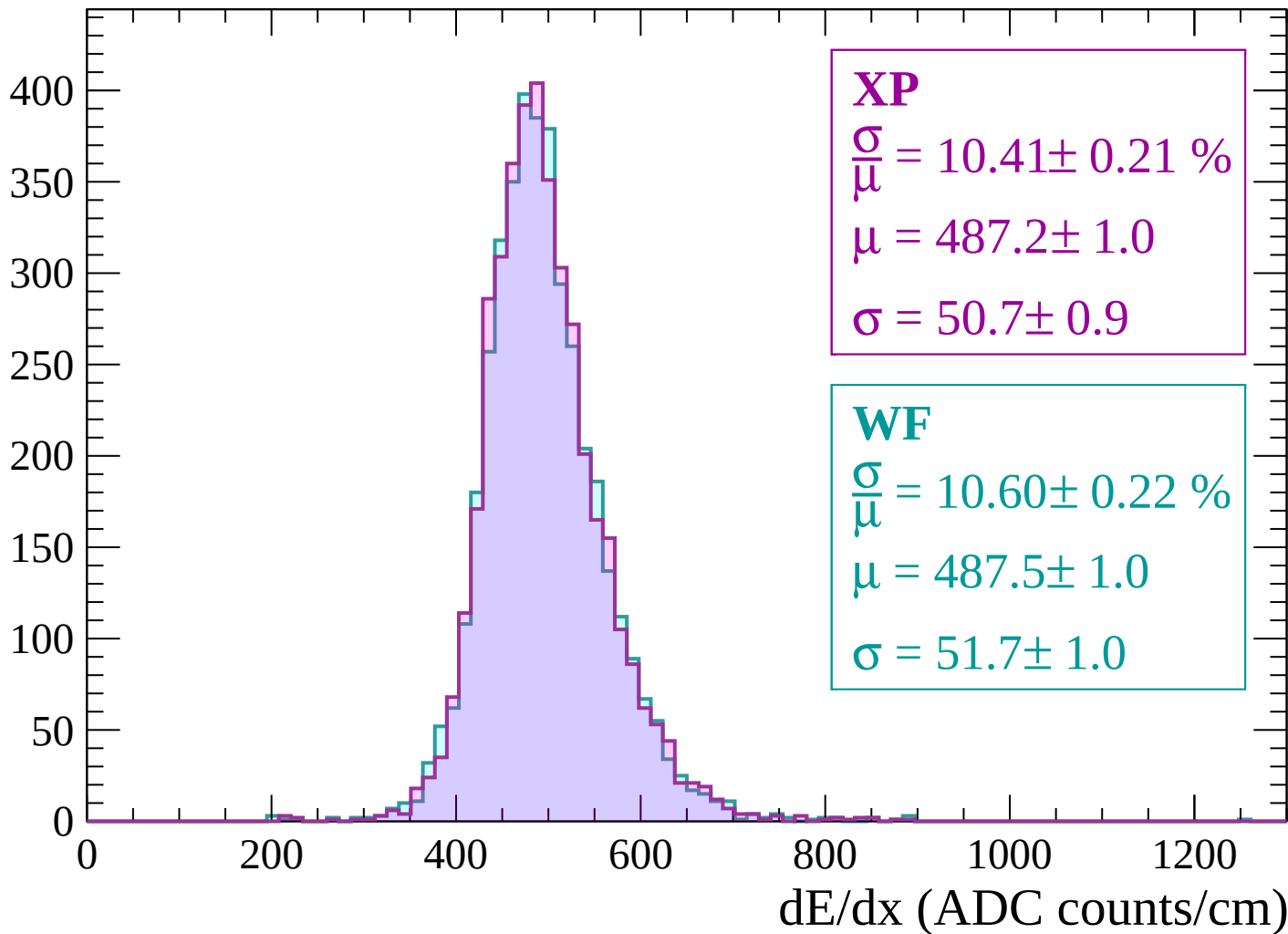
$$\mu = 487.3 \pm 1.0$$

$$\sigma = 51.5 \pm 0.9$$

dE/dx (ADC counts/cm)

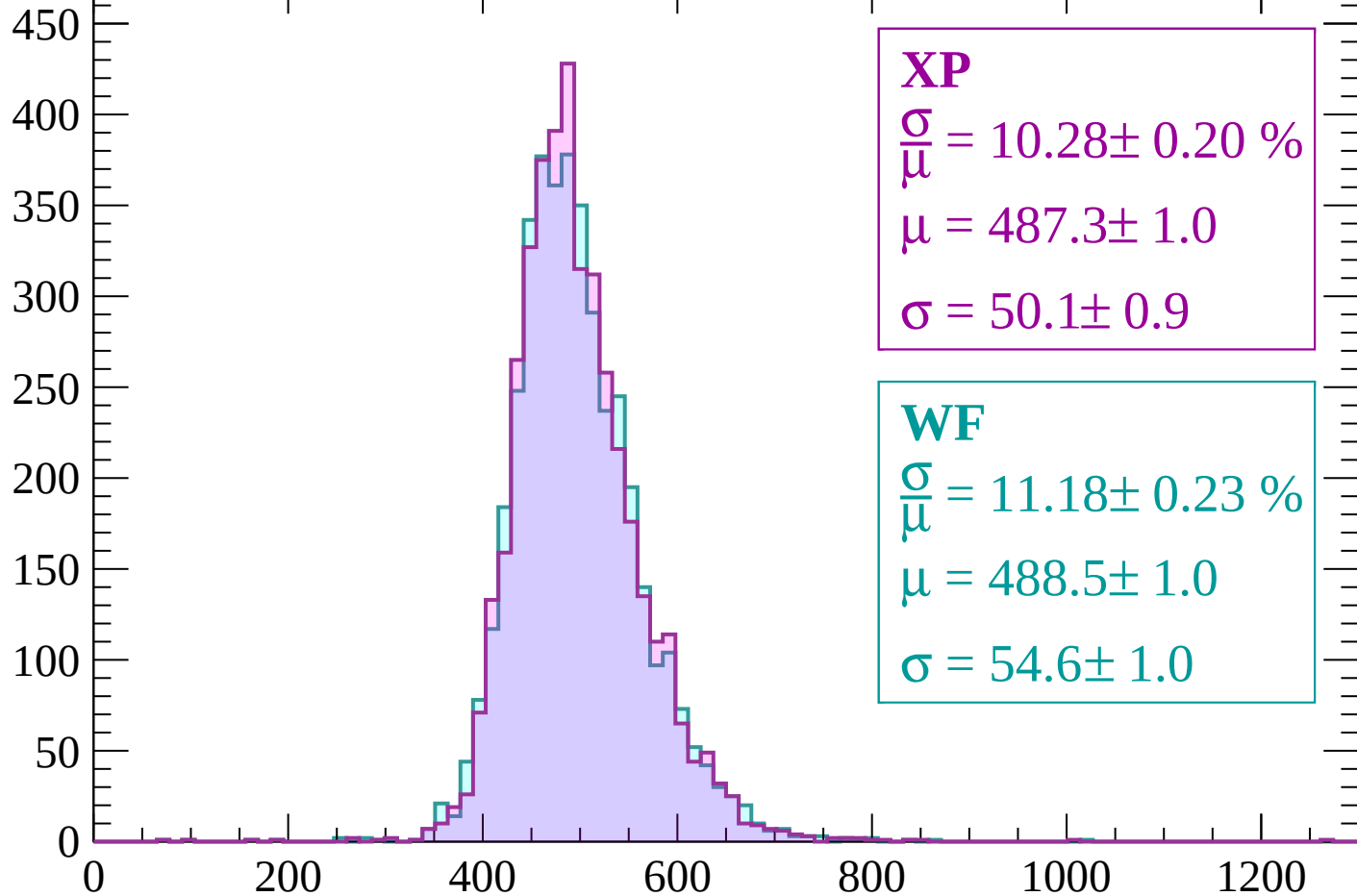
Energy loss | $2160 < p < 2220$

Count



Energy loss | $2220 < p < 2280$

Count



XP

$$\frac{\sigma}{\mu} = 10.28 \pm 0.20 \%$$

$$\mu = 487.3 \pm 1.0$$

$$\sigma = 50.1 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 11.18 \pm 0.23 \%$$

$$\mu = 488.5 \pm 1.0$$

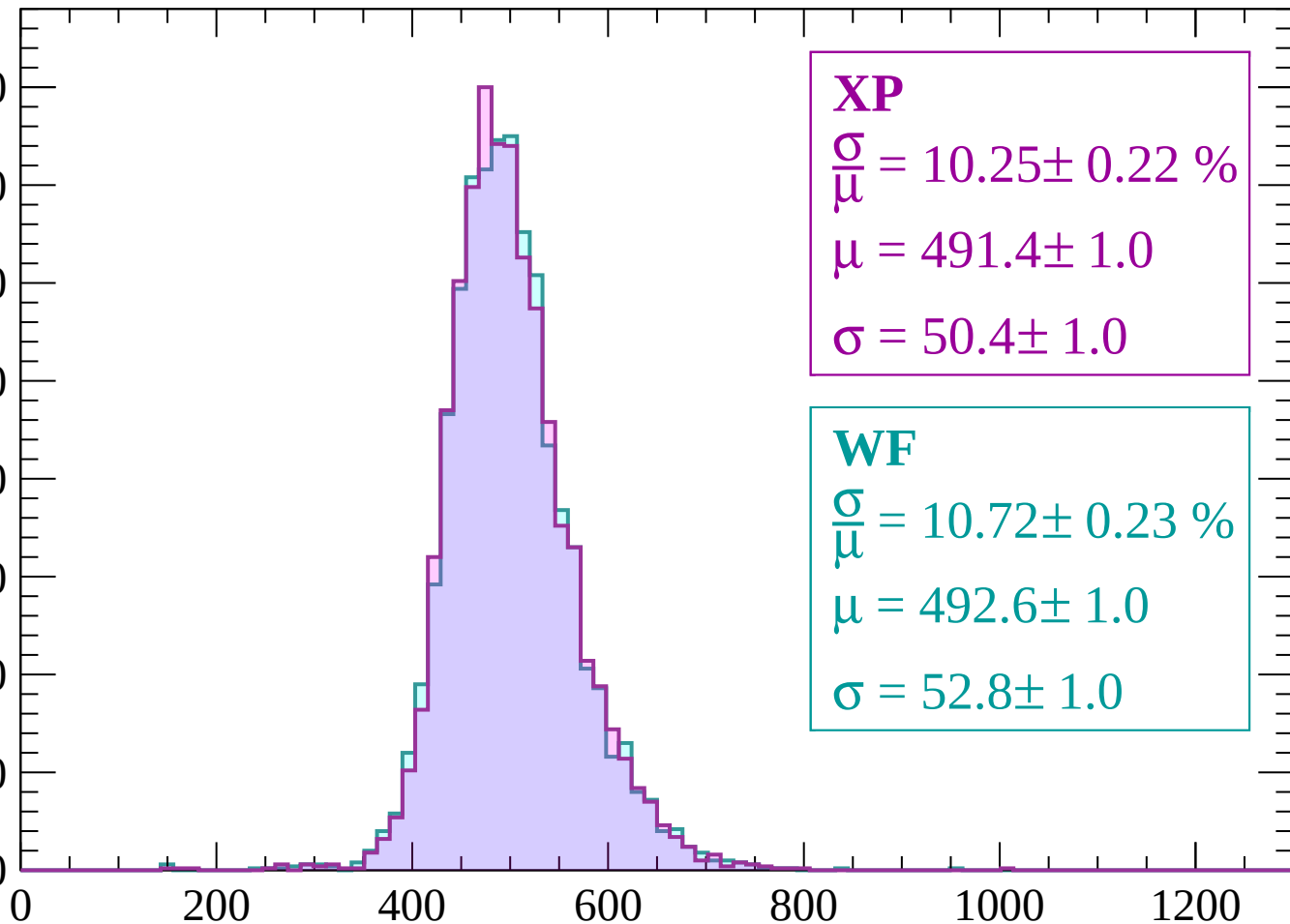
$$\sigma = 54.6 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $2280 < p < 2340$

Count

400
350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 10.25 \pm 0.22 \%$$

$$\mu = 491.4 \pm 1.0$$

$$\sigma = 50.4 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 10.72 \pm 0.23 \%$$

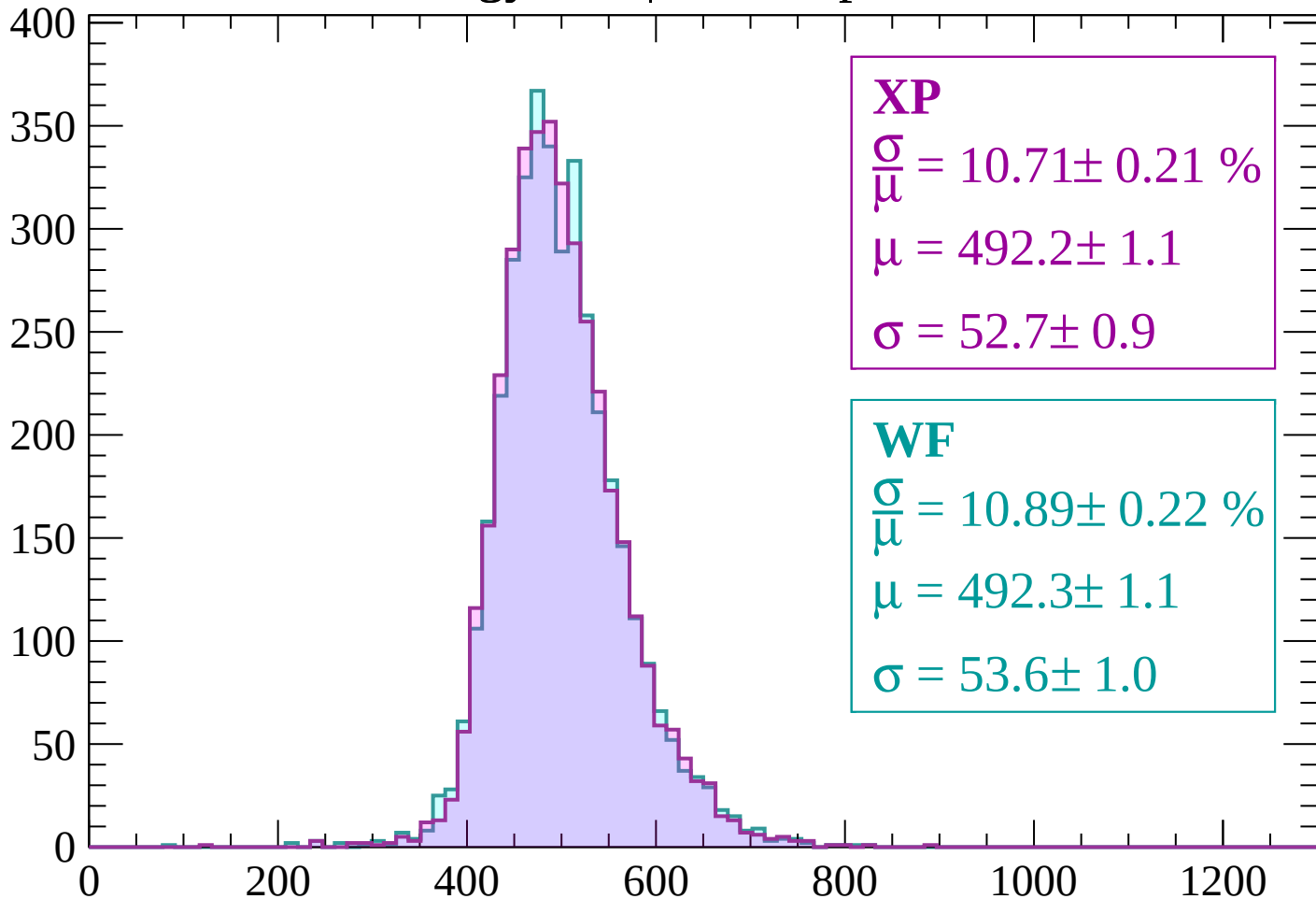
$$\mu = 492.6 \pm 1.0$$

$$\sigma = 52.8 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $2340 < p < 2400$

Count



XP

$$\frac{\sigma}{\mu} = 10.71 \pm 0.21 \%$$

$$\mu = 492.2 \pm 1.1$$

$$\sigma = 52.7 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 10.89 \pm 0.22 \%$$

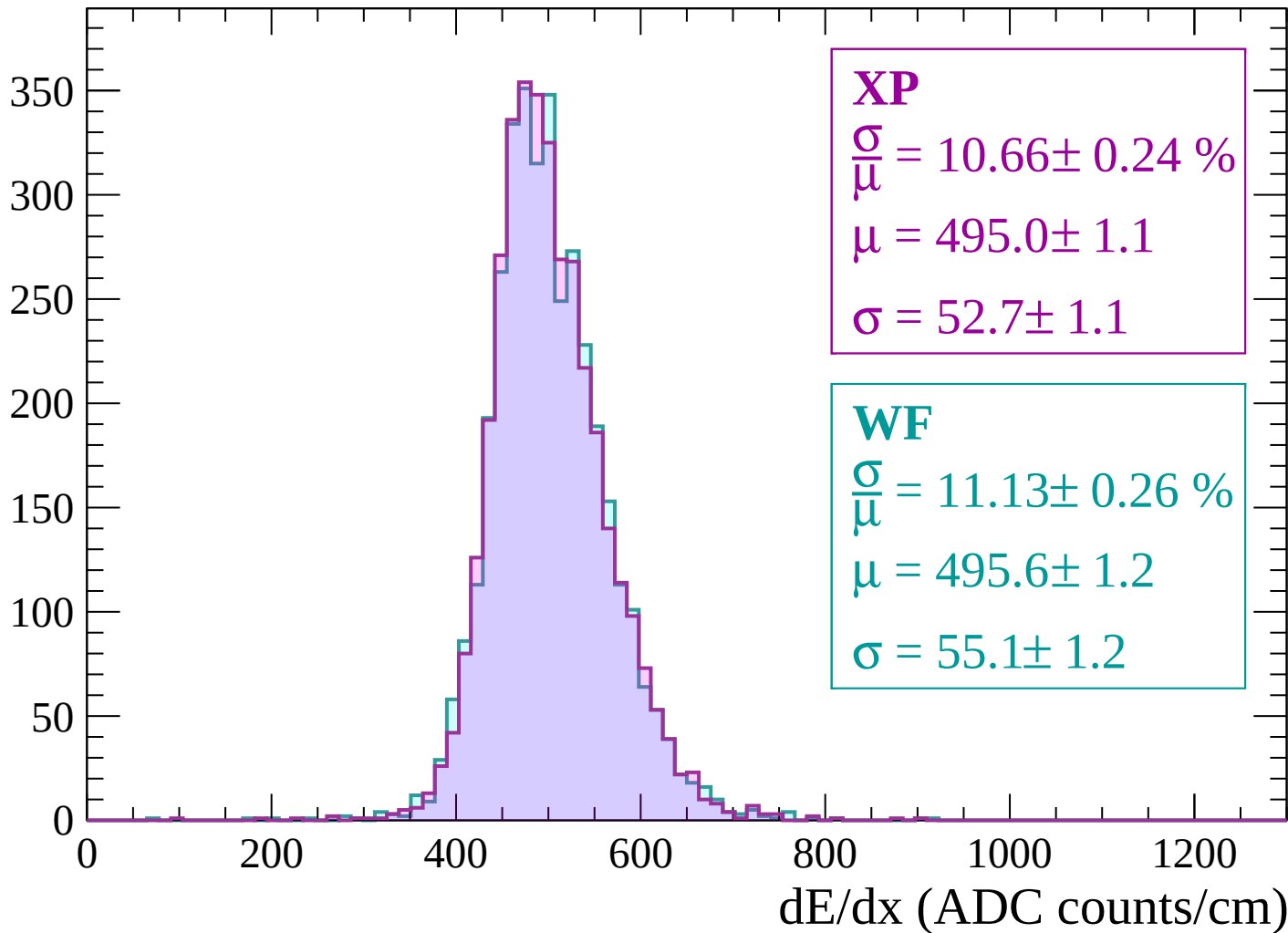
$$\mu = 492.3 \pm 1.1$$

$$\sigma = 53.6 \pm 1.0$$

dE/dx (ADC counts/cm)

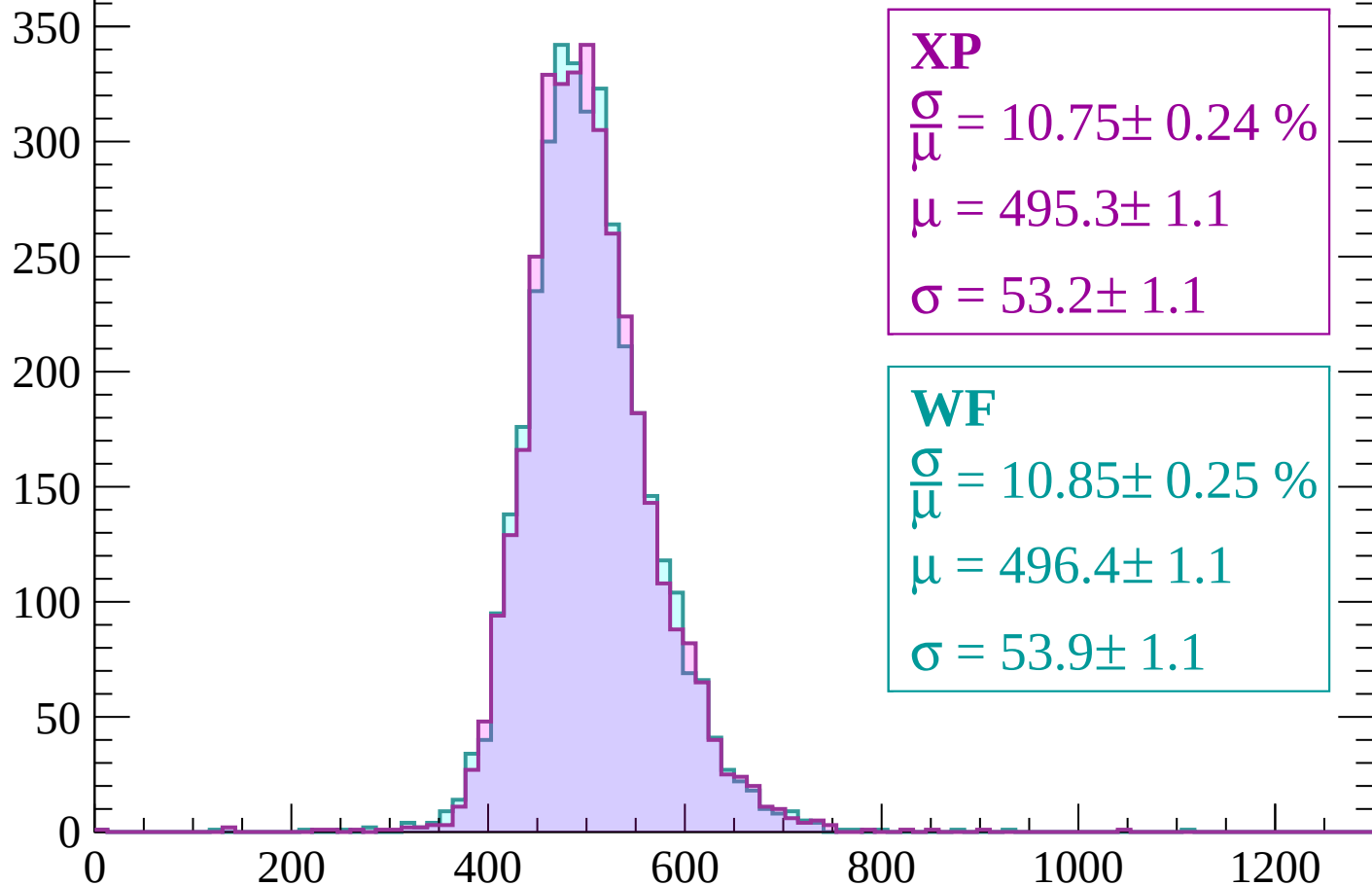
Energy loss | $2400 < p < 2460$

Count



Energy loss | $2460 < p < 2520$

Count



XP

$$\frac{\sigma}{\mu} = 10.75 \pm 0.24 \%$$

$$\mu = 495.3 \pm 1.1$$

$$\sigma = 53.2 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 10.85 \pm 0.25 \%$$

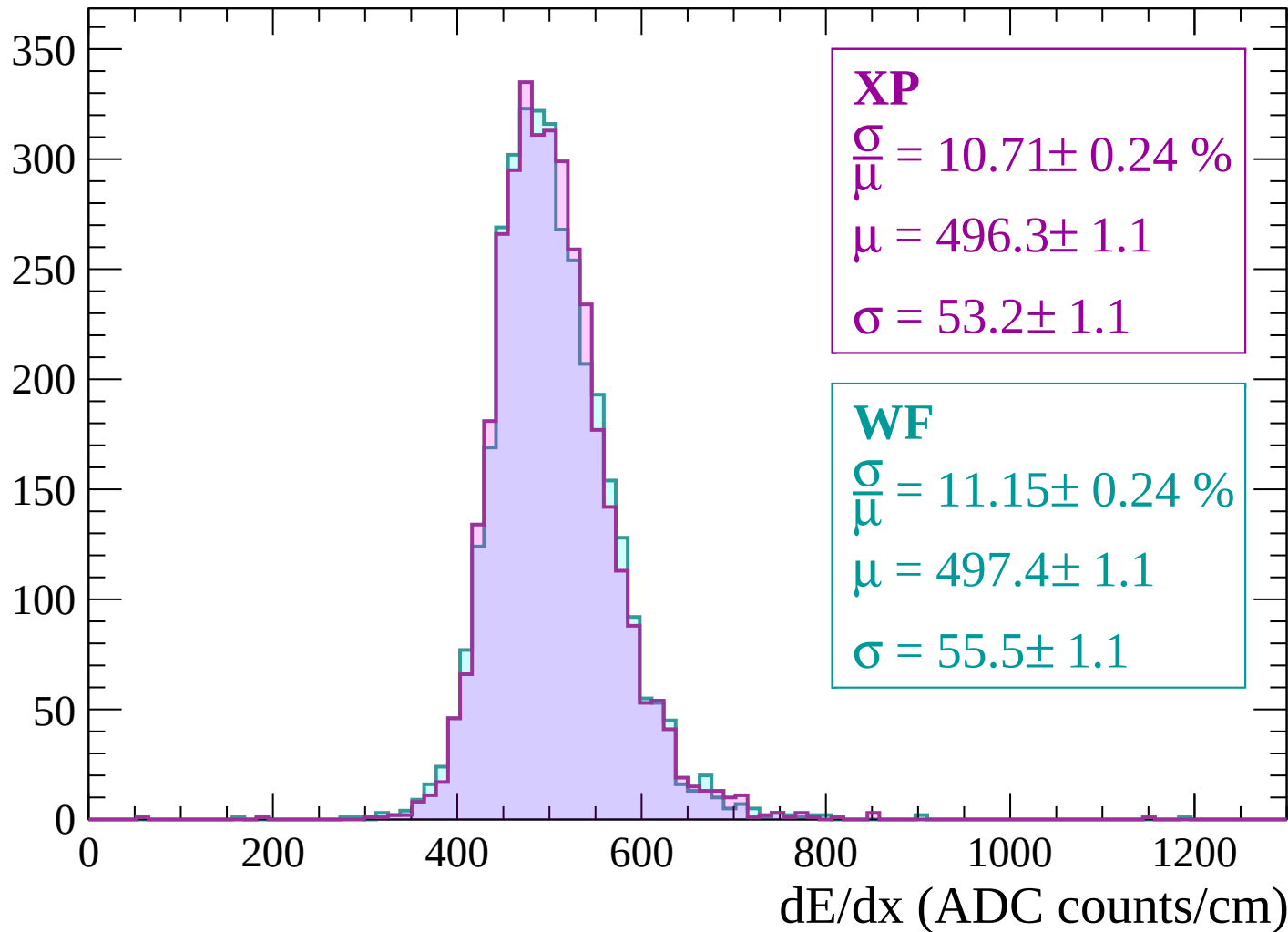
$$\mu = 496.4 \pm 1.1$$

$$\sigma = 53.9 \pm 1.1$$

dE/dx (ADC counts/cm)

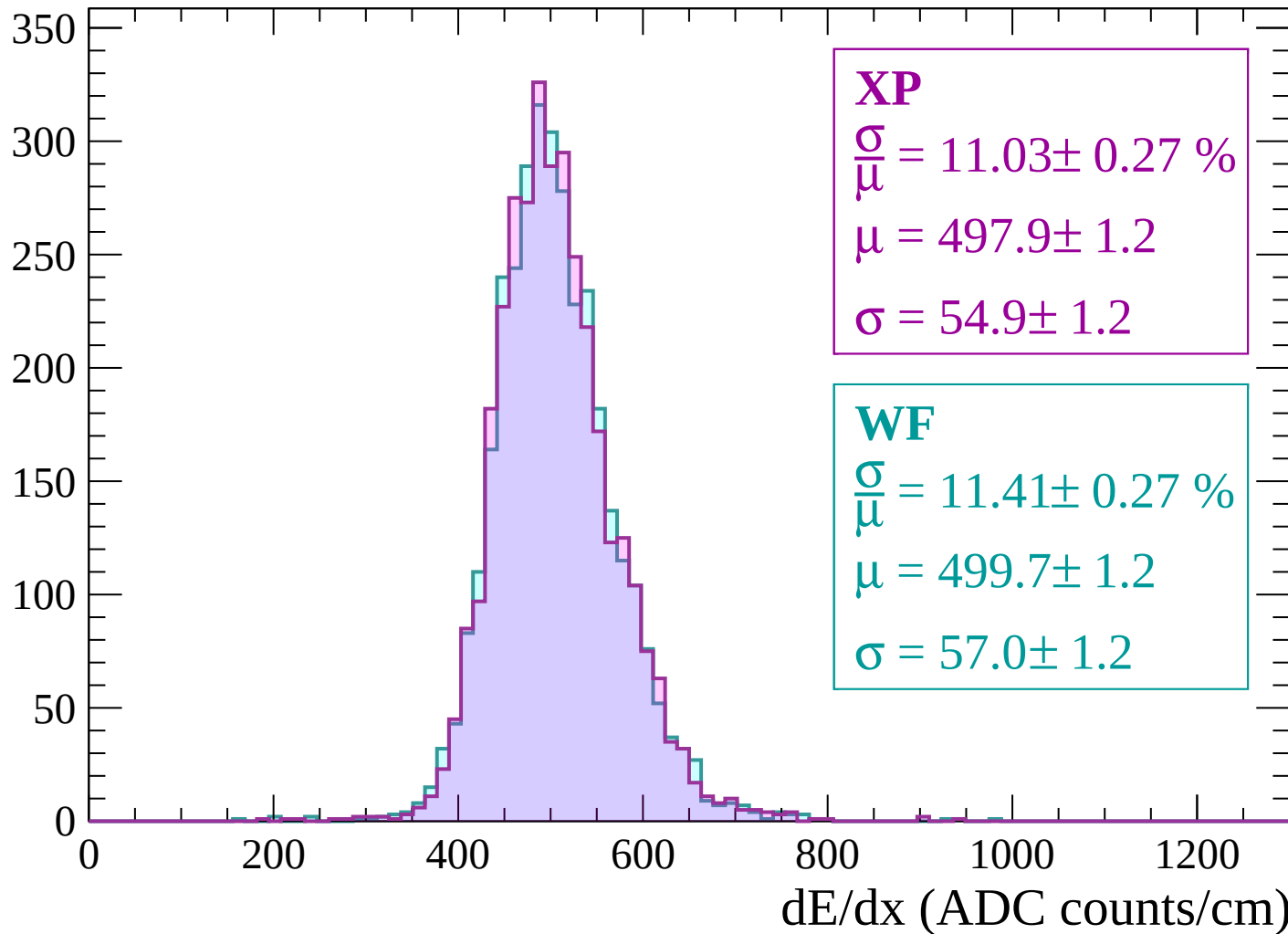
Energy loss | $2520 < p < 2580$

Count



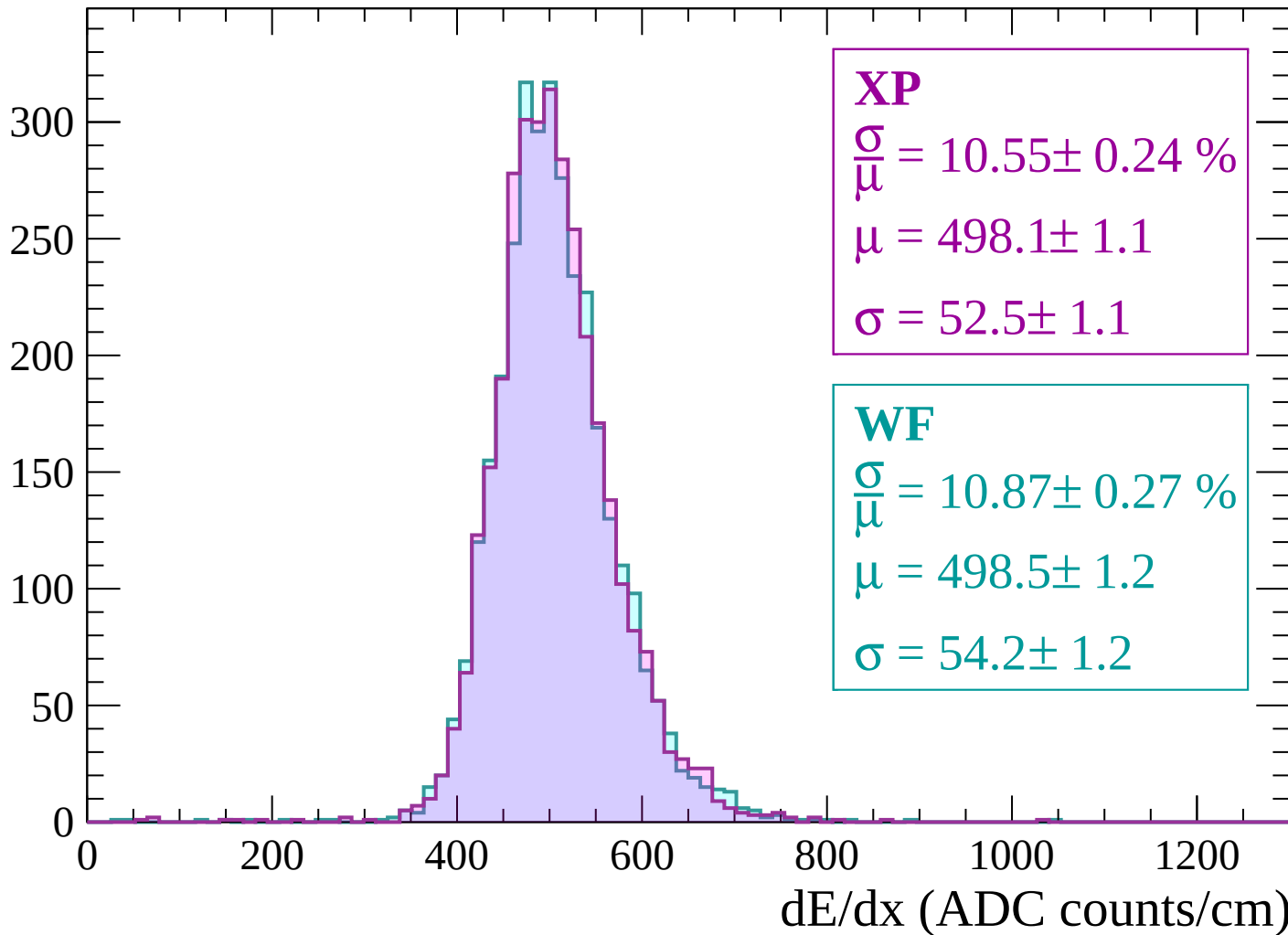
Energy loss | $2580 < p < 2640$

Count



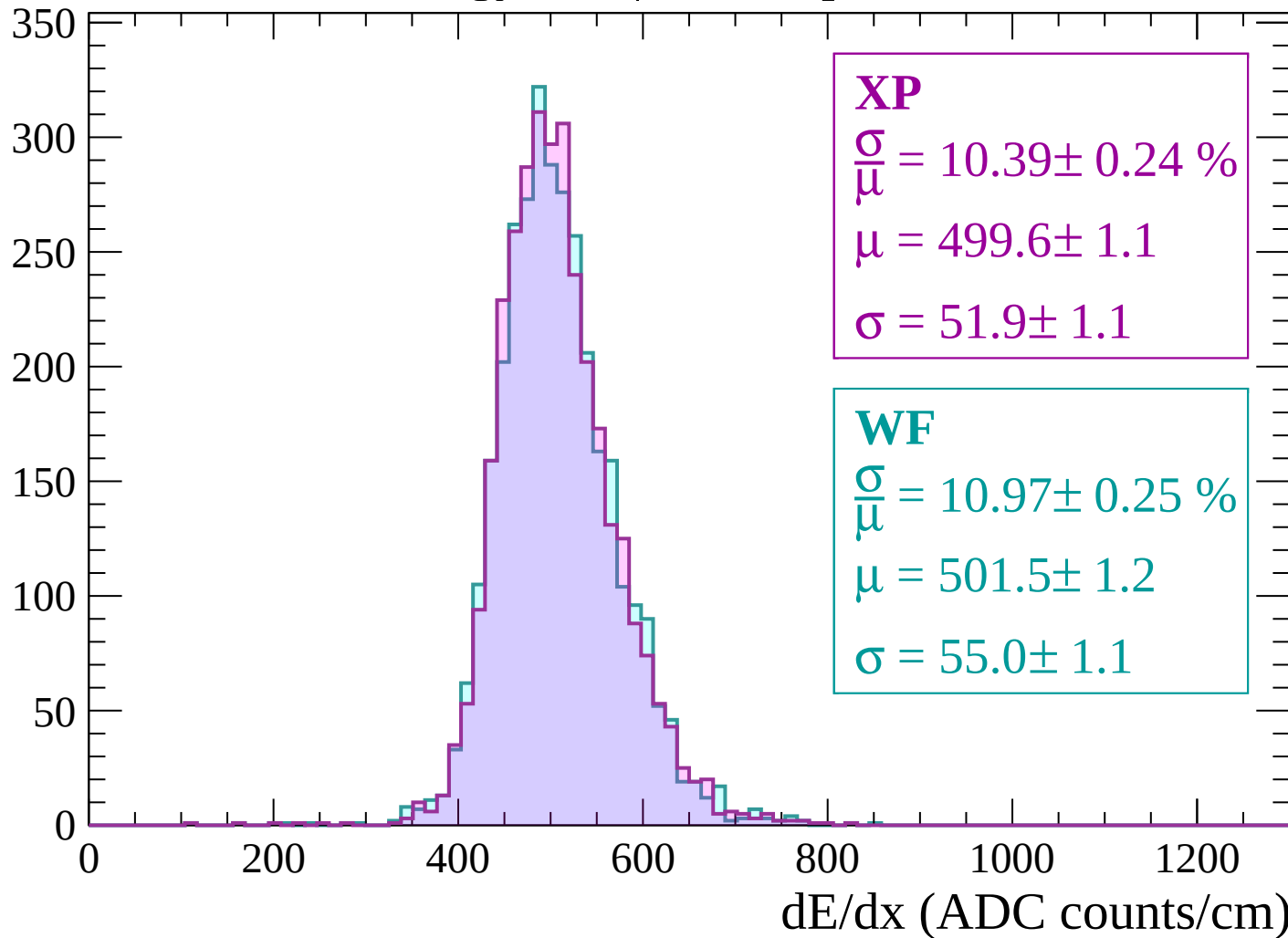
Energy loss | $2640 < p < 2700$

Count



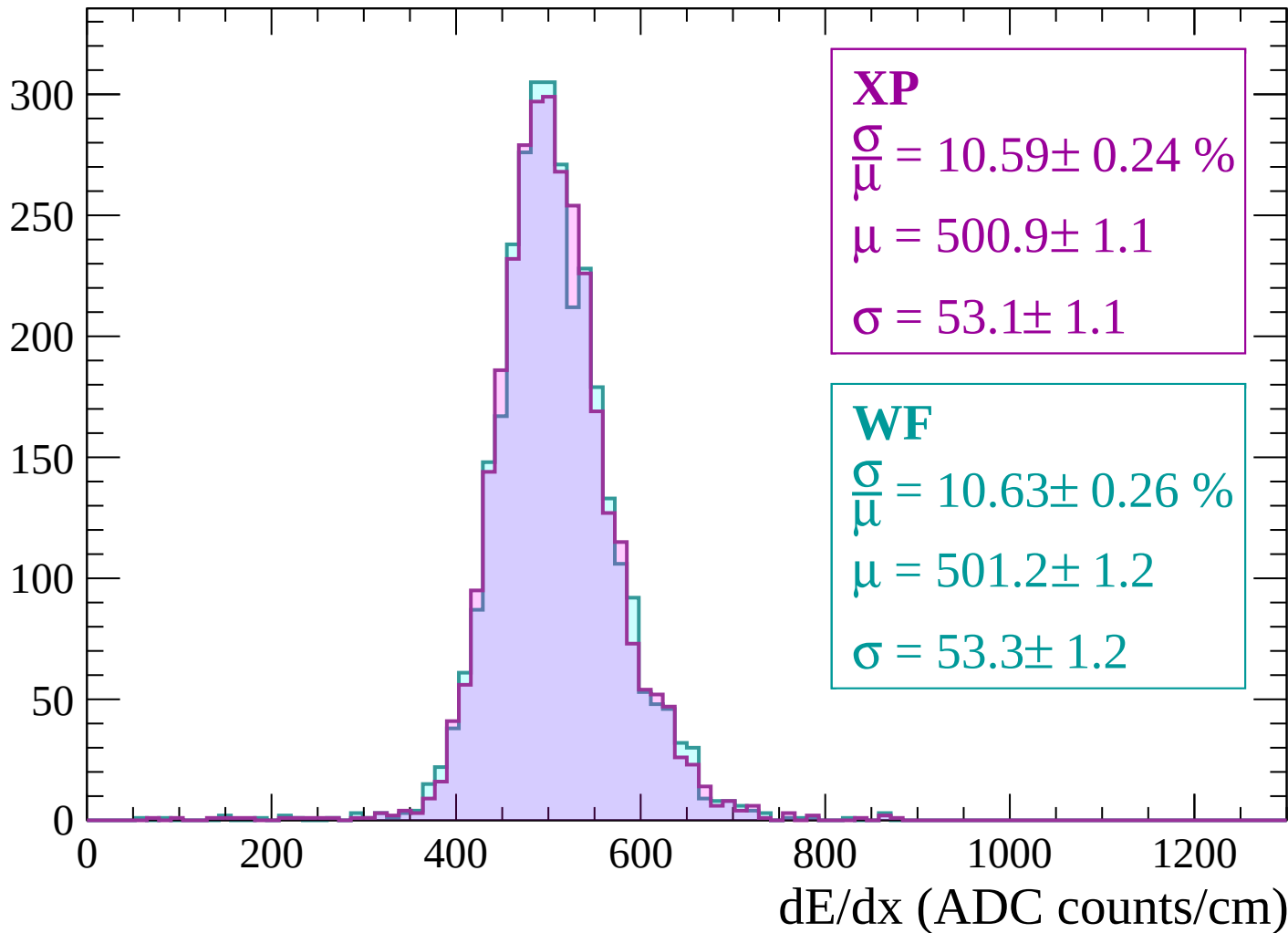
Energy loss | $2700 < p < 2760$

Count



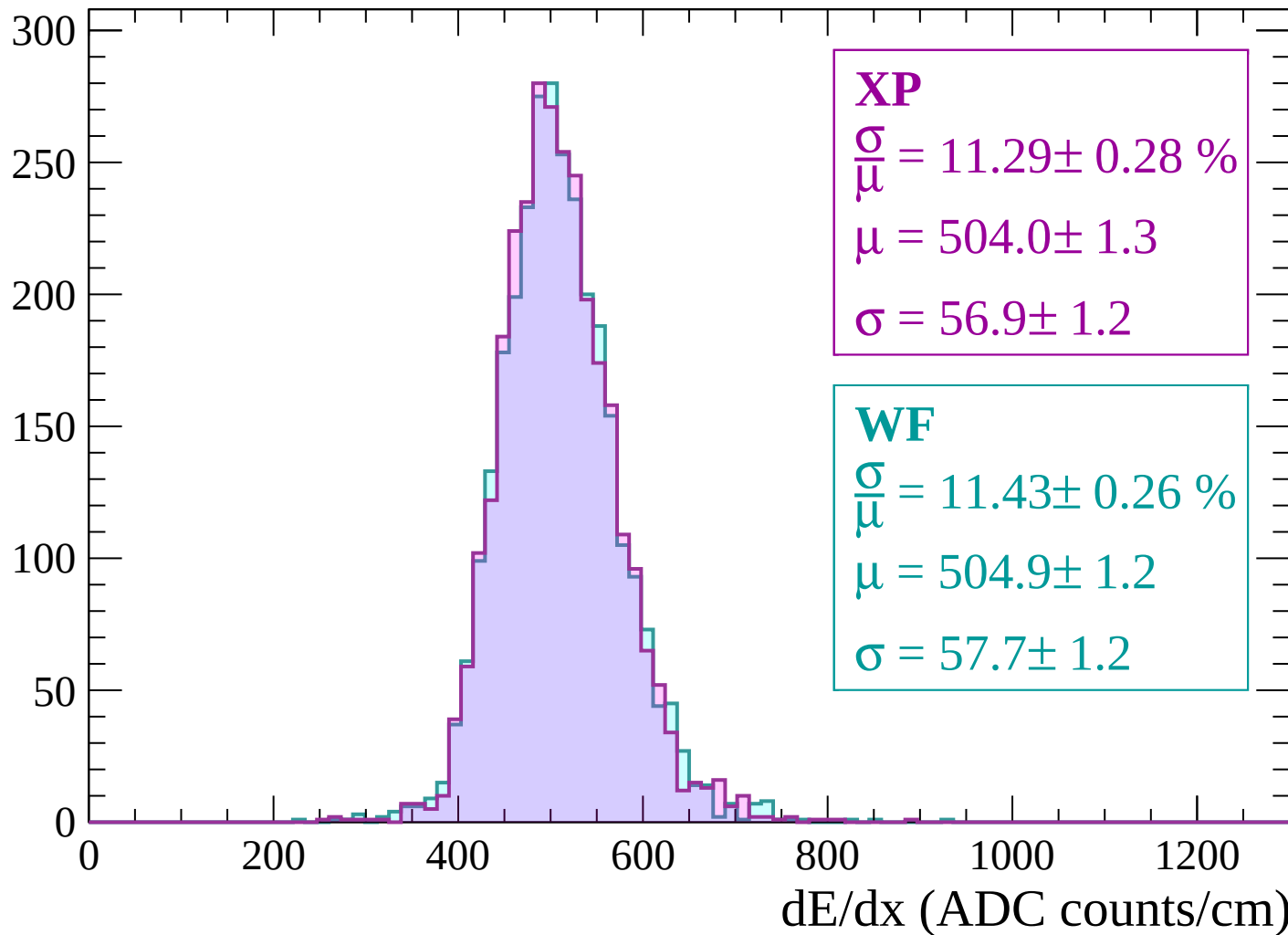
Energy loss | $2760 < p < 2820$

Count



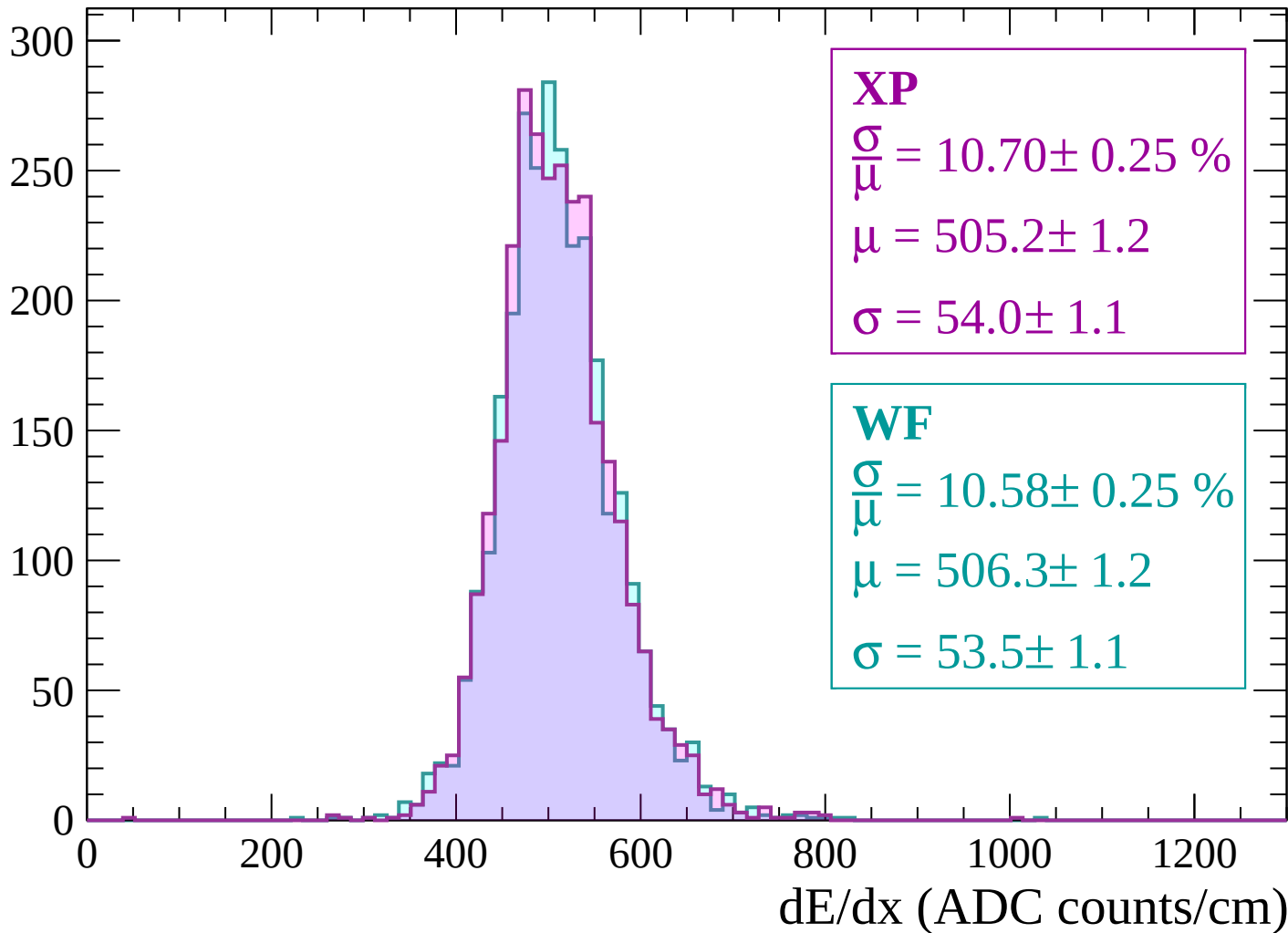
Energy loss | $2820 < p < 2880$

Count



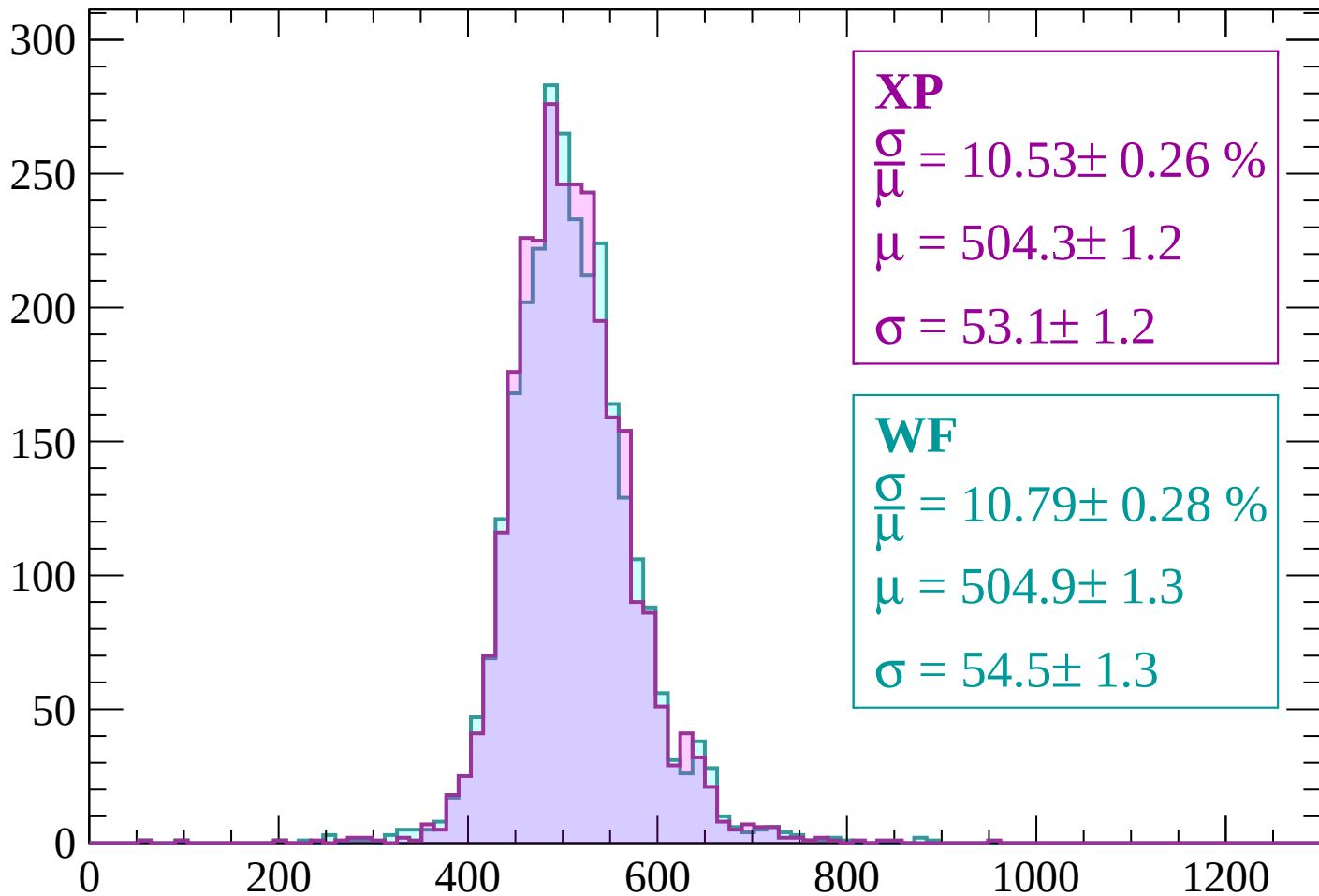
Energy loss | $2880 < p < 2940$

Count



Energy loss | $2940 < p < 3000$

Count



XP

$$\frac{\sigma}{\mu} = 10.53 \pm 0.26 \%$$

$$\mu = 504.3 \pm 1.2$$

$$\sigma = 53.1 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 10.79 \pm 0.28 \%$$

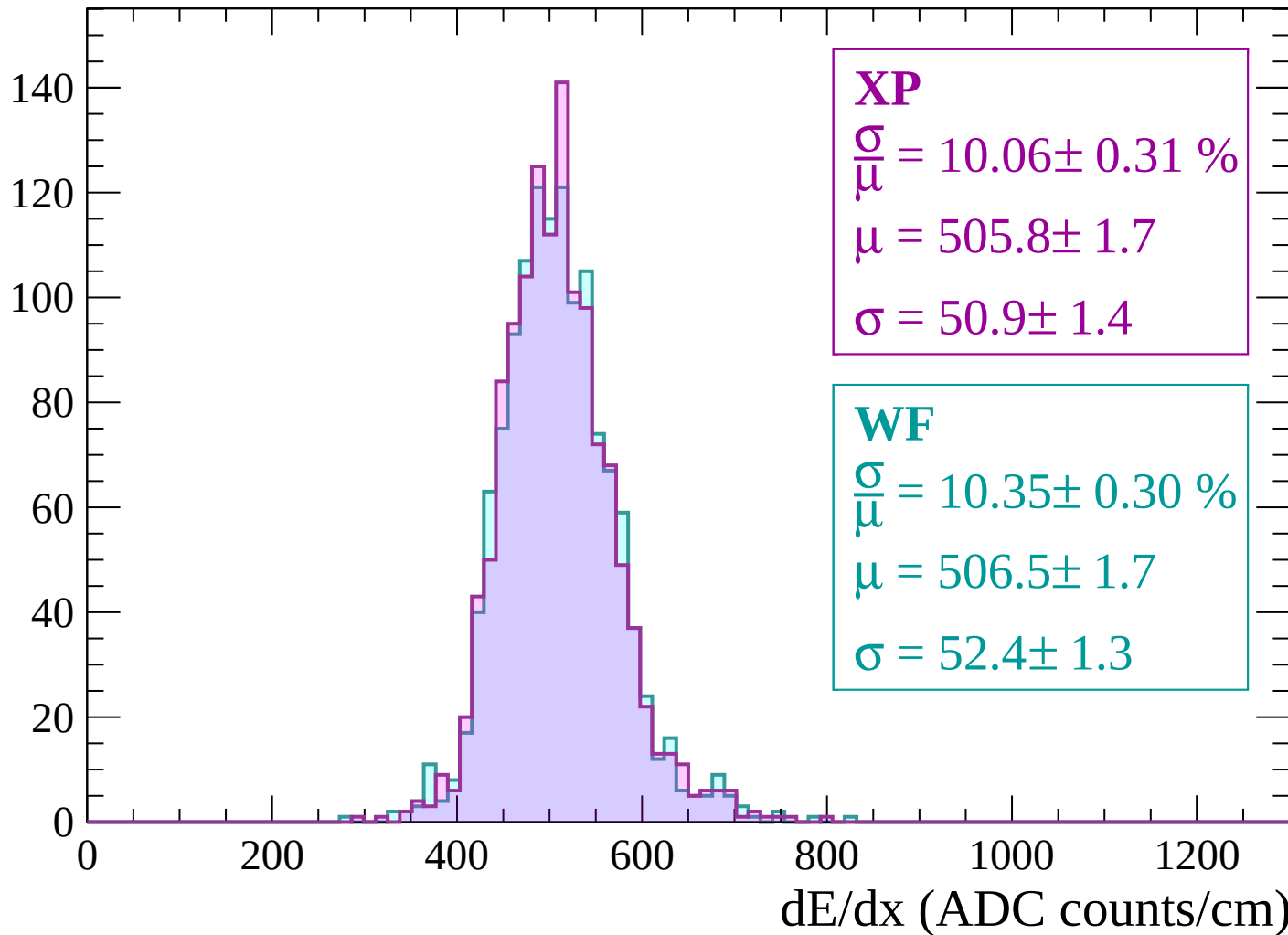
$$\mu = 504.9 \pm 1.3$$

$$\sigma = 54.5 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $3000 < p < 3060$

Count



Energy loss | $-90 < \theta < -88$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 4.16 \pm 21.99 \%$$

$$\mu = 6.2 \pm 2.4$$

$$\sigma = 0.3 \pm 1.3$$

WF

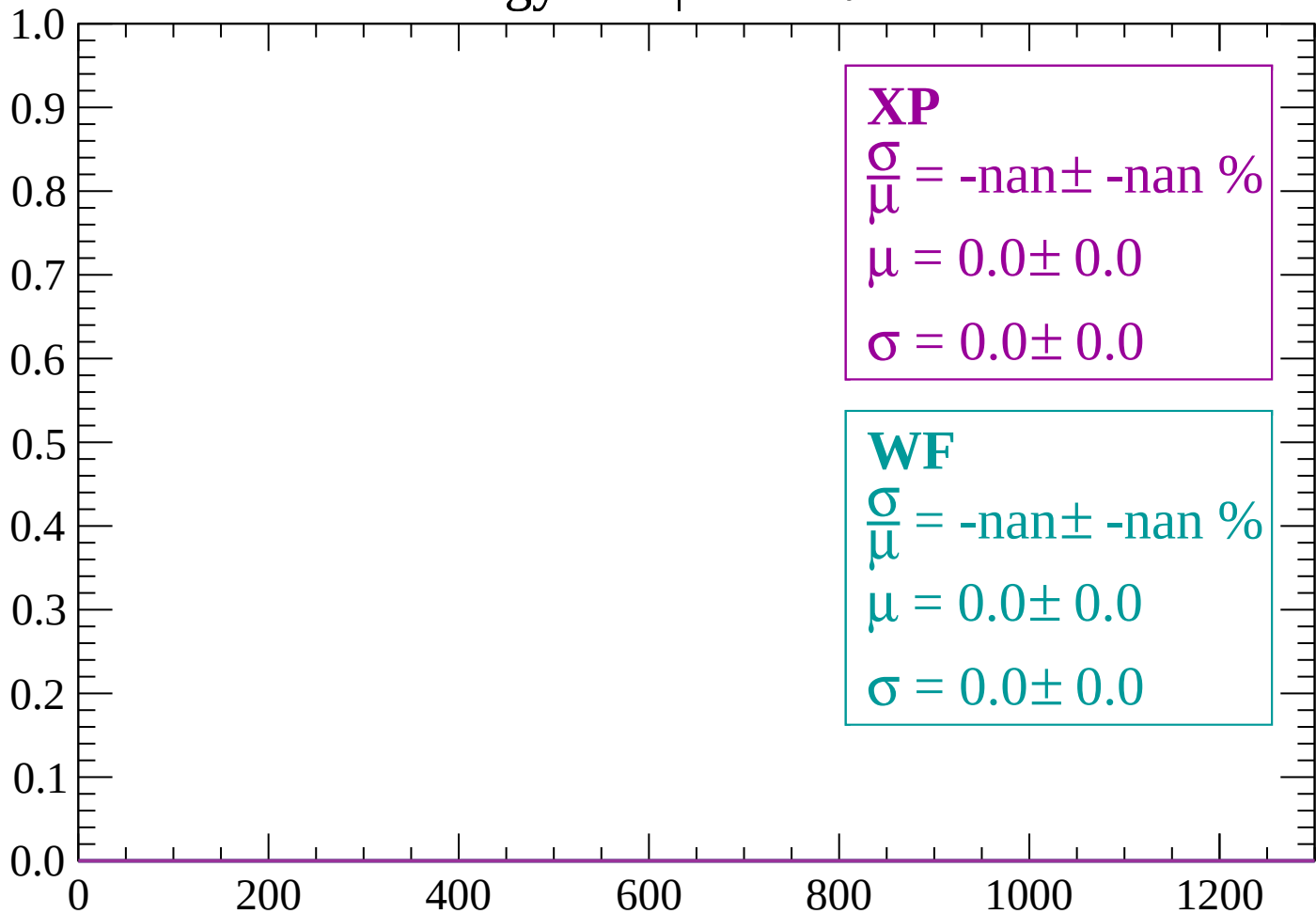
$$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$$

$$\mu = 0.0 \pm 0.0$$

$$\sigma = 0.0 \pm 0.0$$

Energy loss | $-88 < \theta < -86$

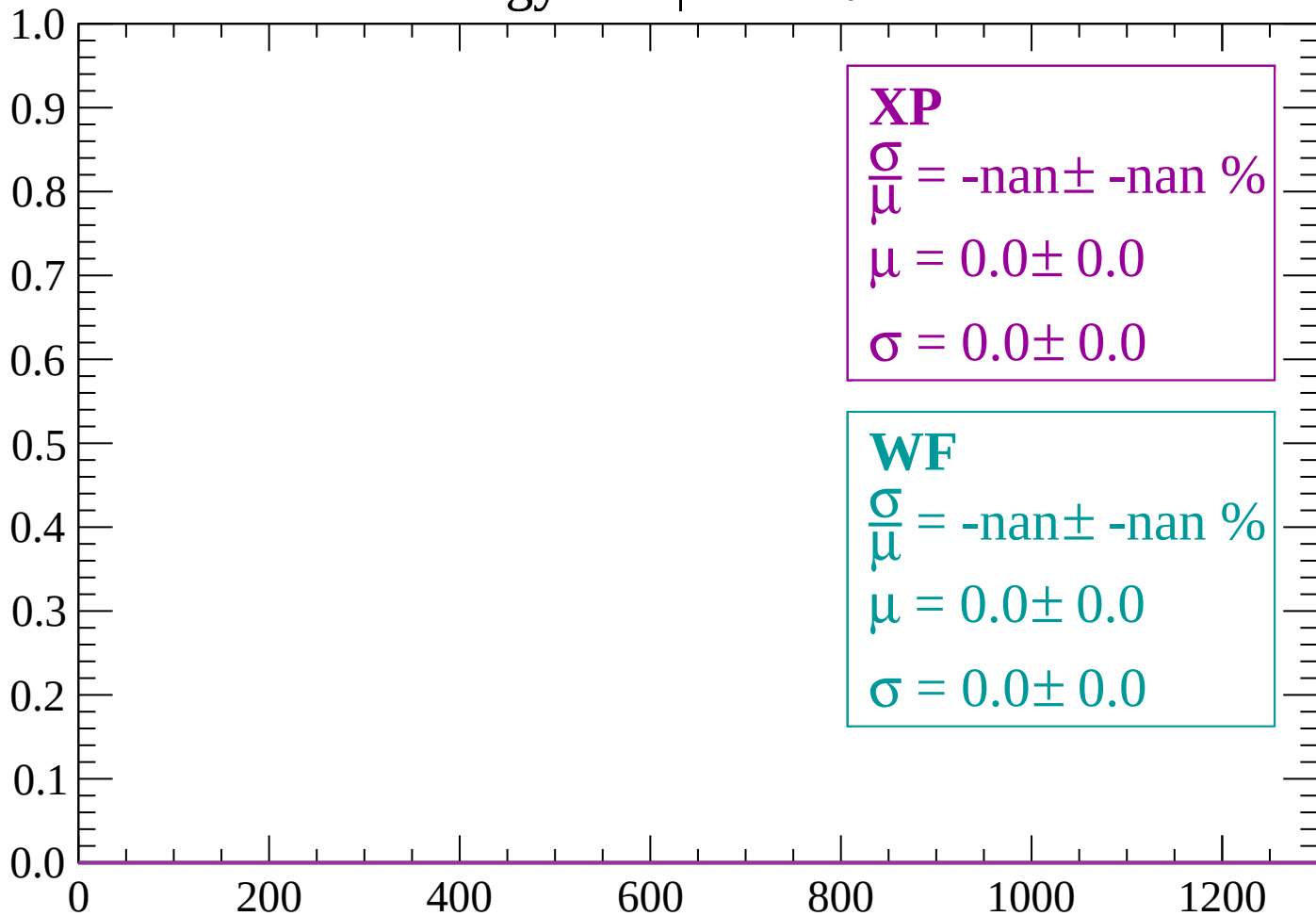
Count



dE/dx (ADC counts/cm)

Energy loss | $-86 < \theta < -84$

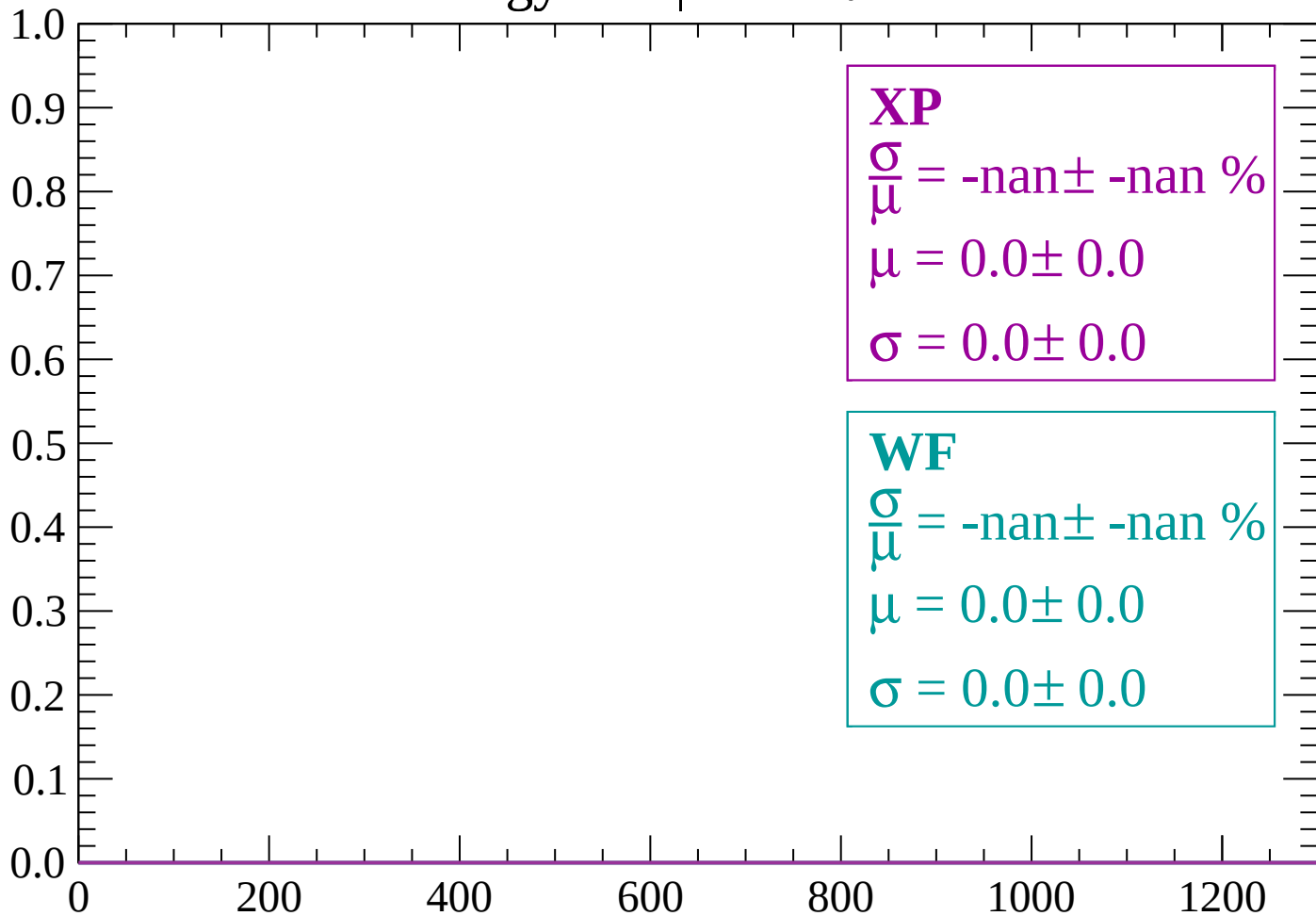
Count



dE/dx (ADC counts/cm)

Energy loss | $-84 < \theta < -82$

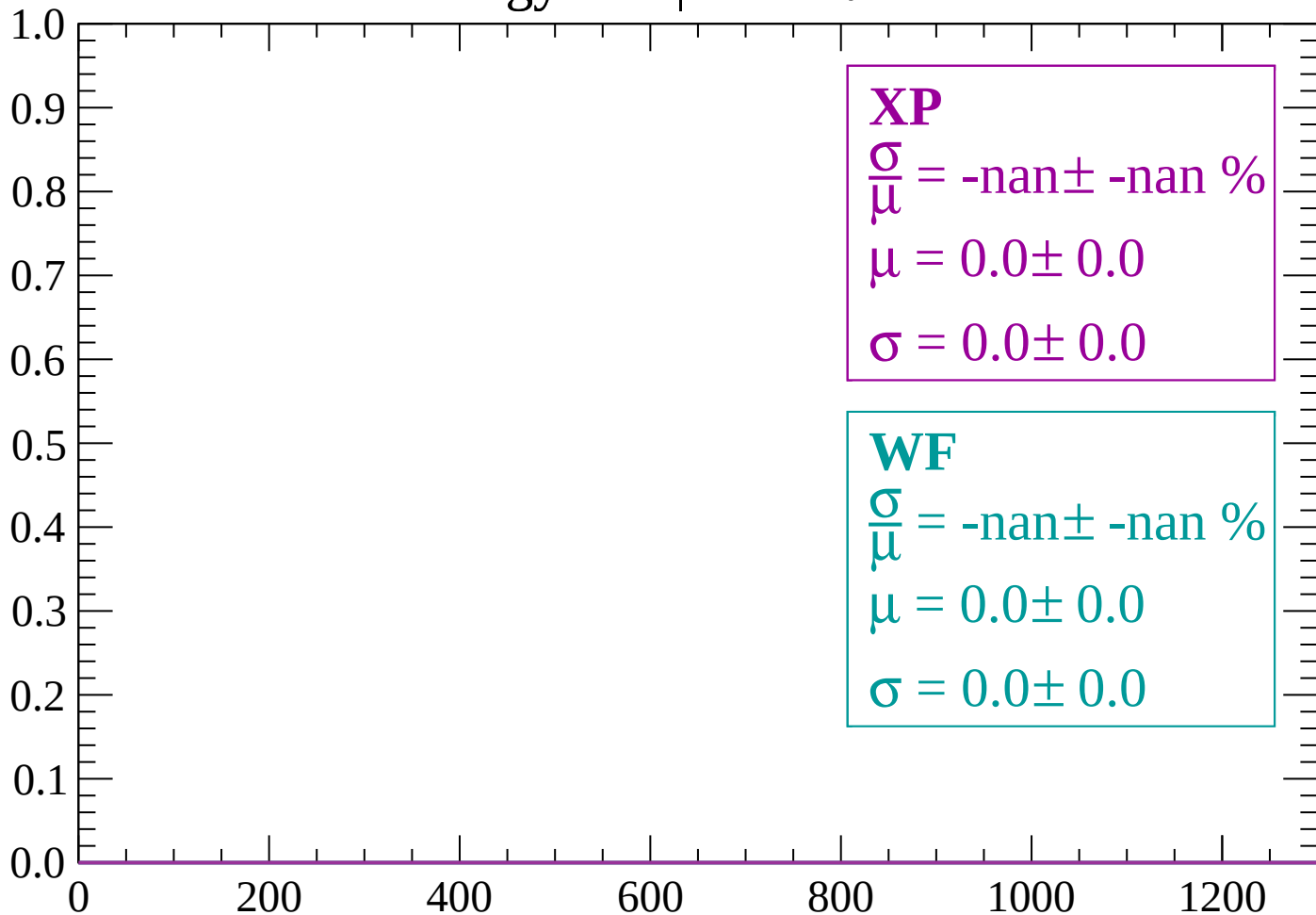
Count



dE/dx (ADC counts/cm)

Energy loss | $-82 < \theta < -80$

Count



dE/dx (ADC counts/cm)

Energy loss | $-80 < \theta < -78$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

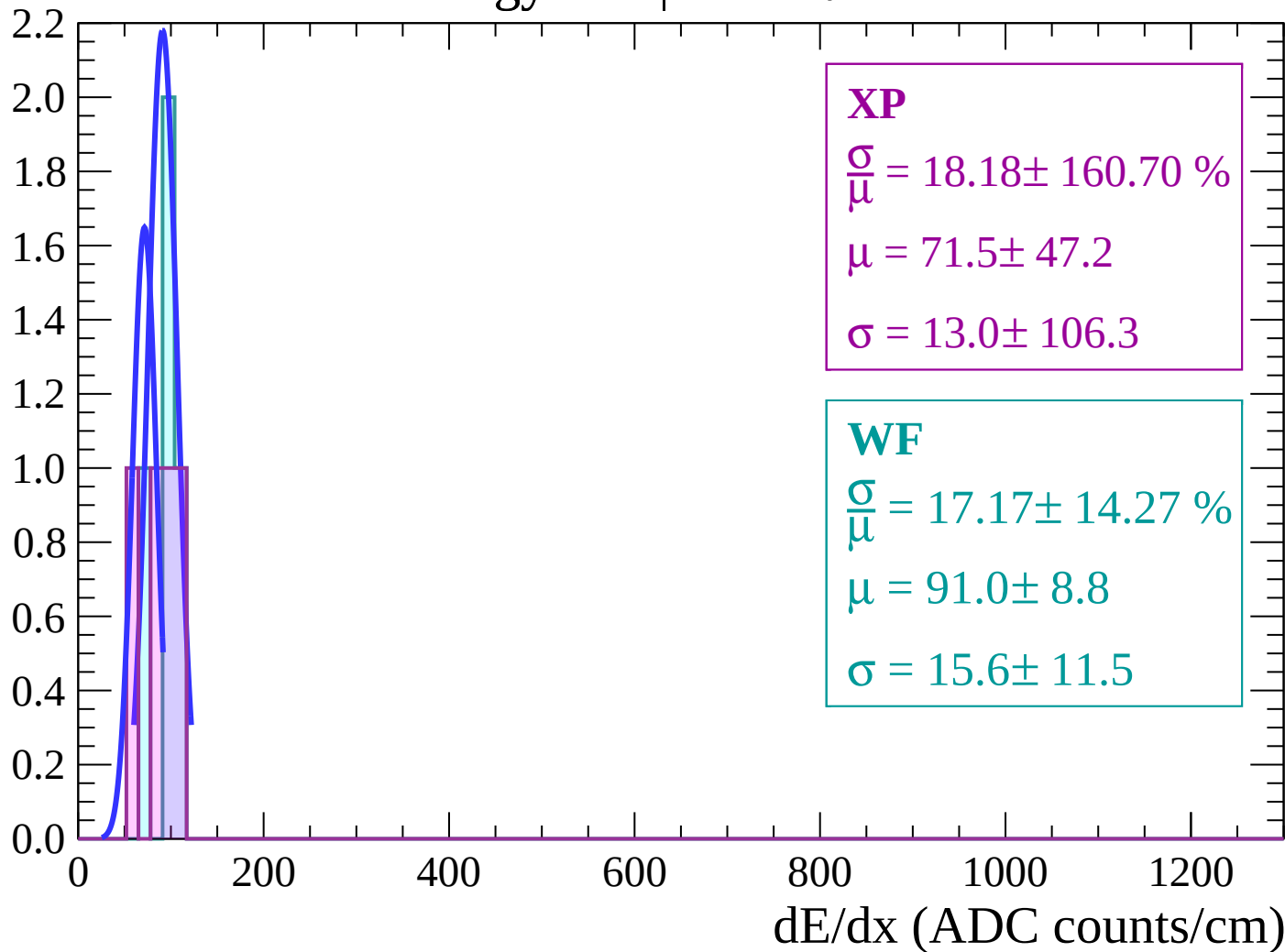
$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

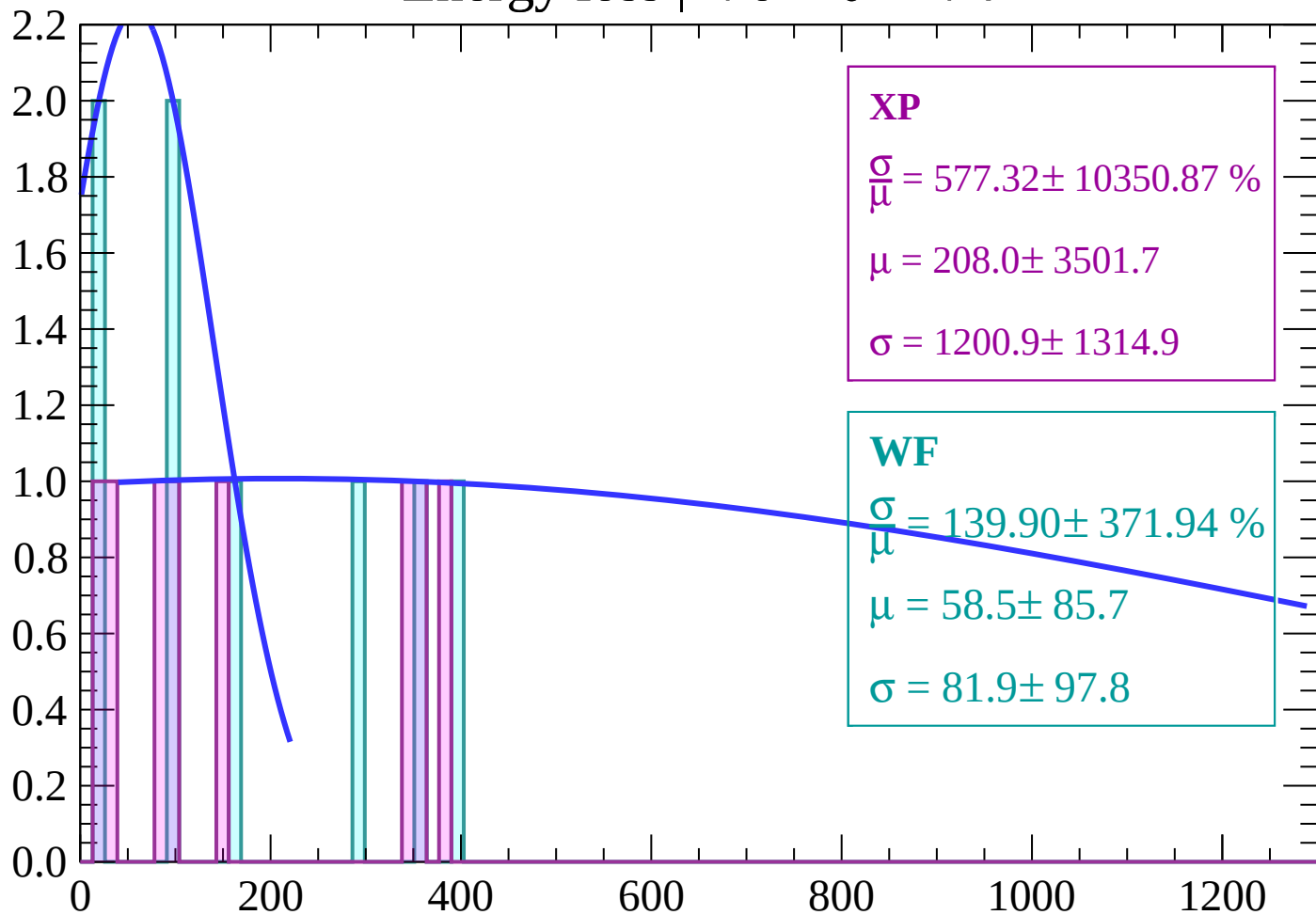
Energy loss | $-78 < \theta < -76$

Count



Energy loss | $-76 < \theta < -74$

Count



XP

$$\frac{\sigma}{\mu} = 577.32 \pm 10350.87 \%$$

$$\mu = 208.0 \pm 3501.7$$

$$\sigma = 1200.9 \pm 1314.9$$

WF

$$\frac{\sigma}{\mu} = 139.90 \pm 371.94 \%$$

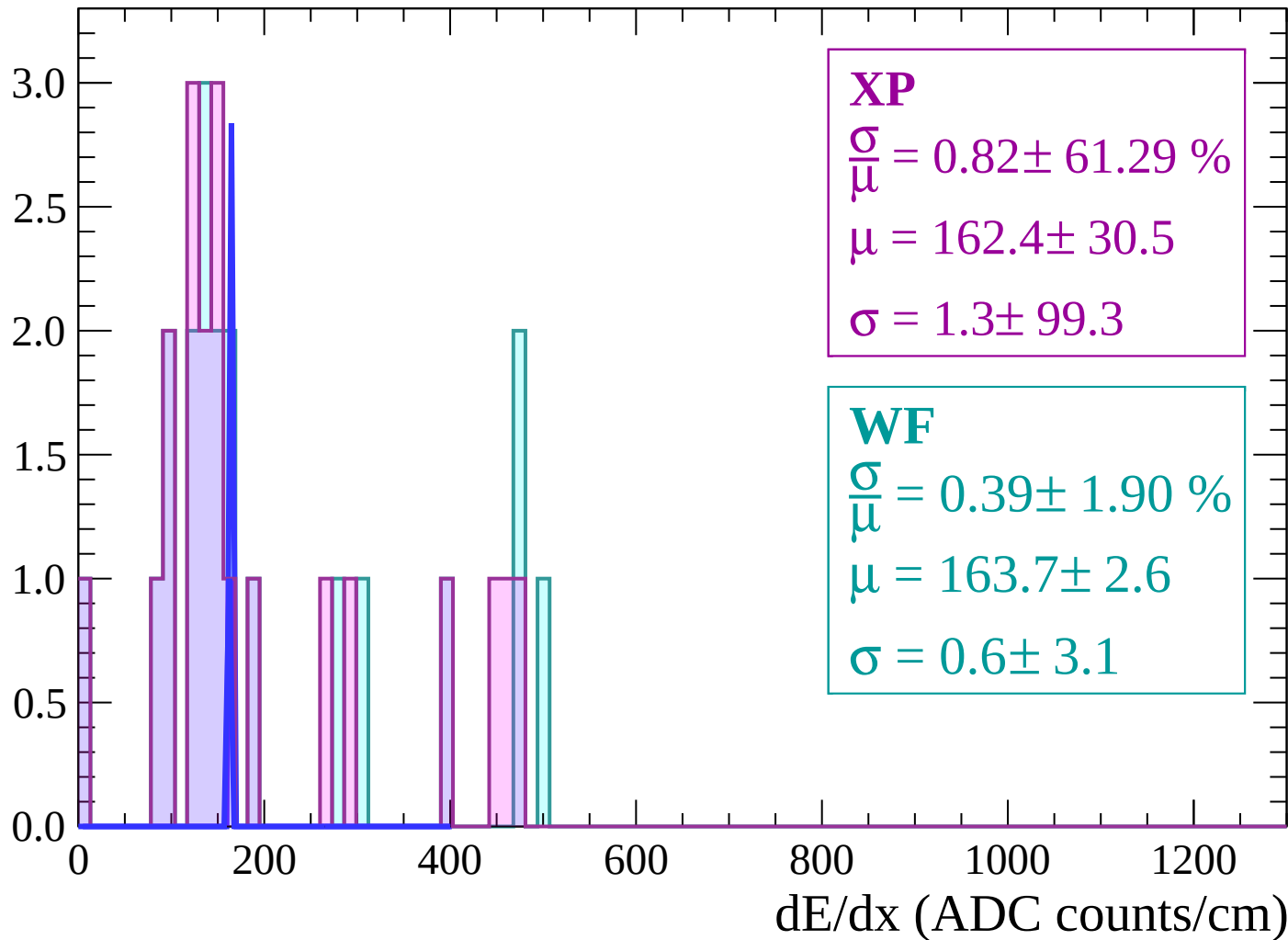
$$\mu = 58.5 \pm 85.7$$

$$\sigma = 81.9 \pm 97.8$$

dE/dx (ADC counts/cm)

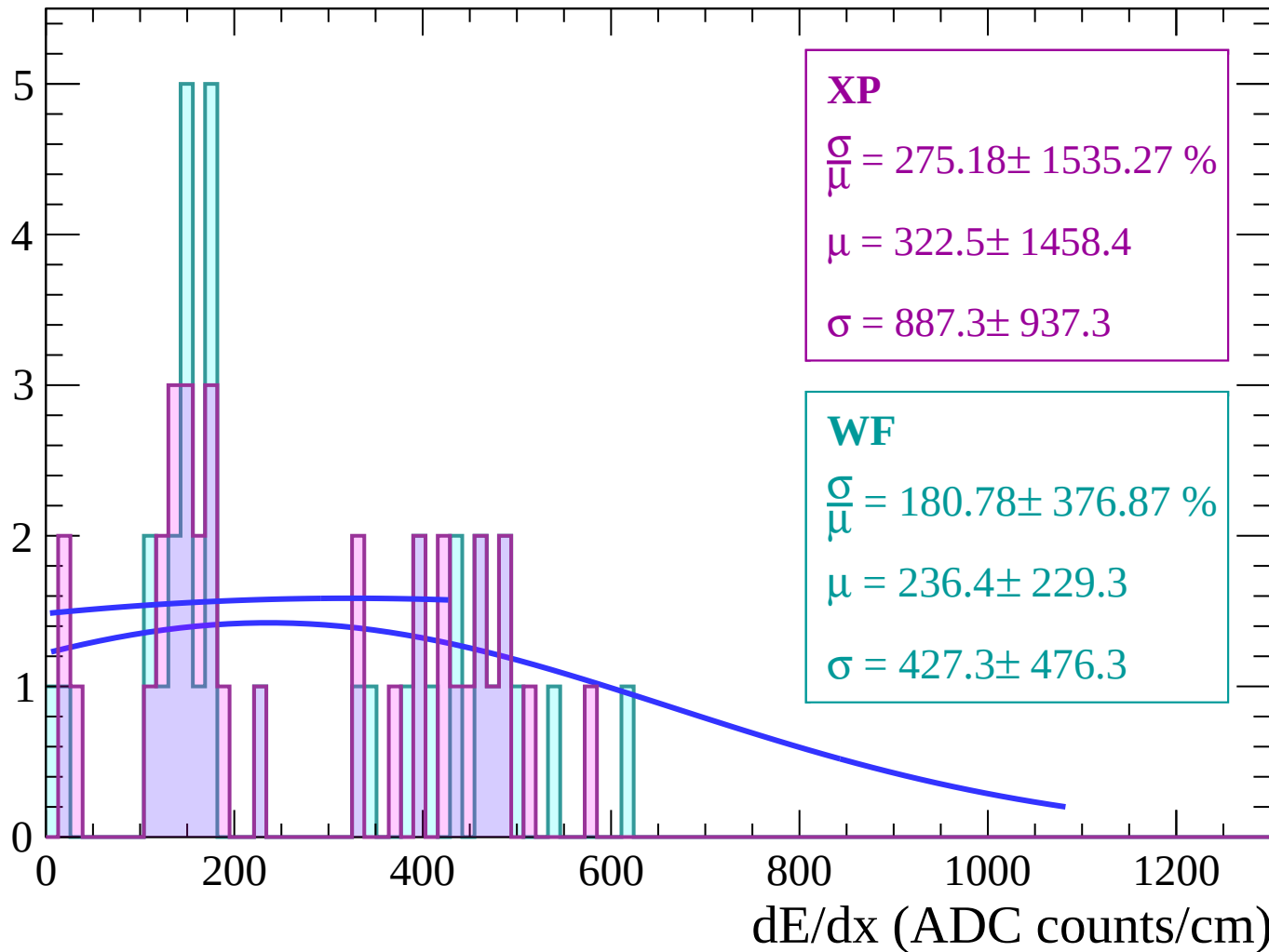
Energy loss | $-74 < \theta < -72$

Count



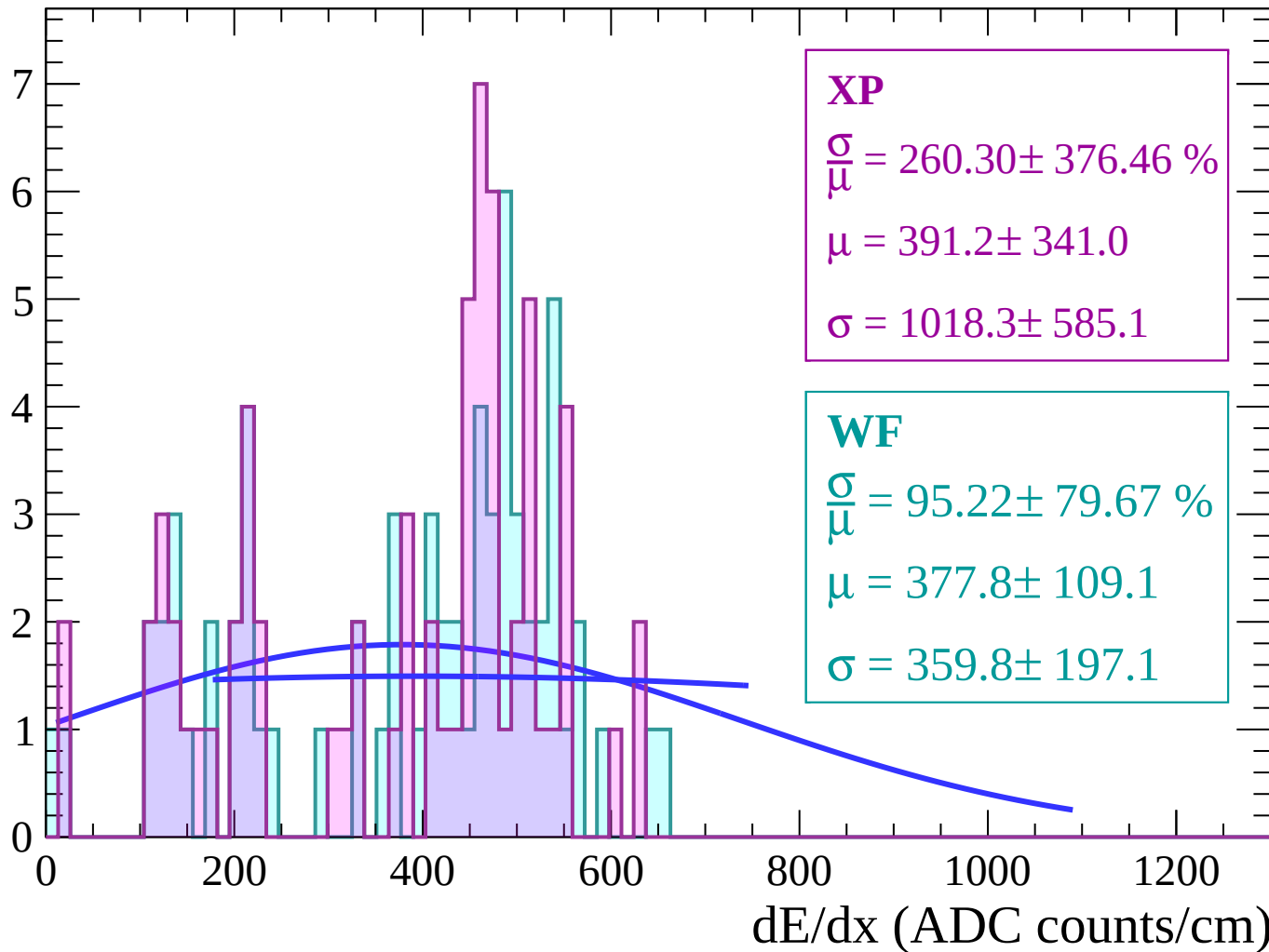
Energy loss | $-72 < \theta < -70$

Count



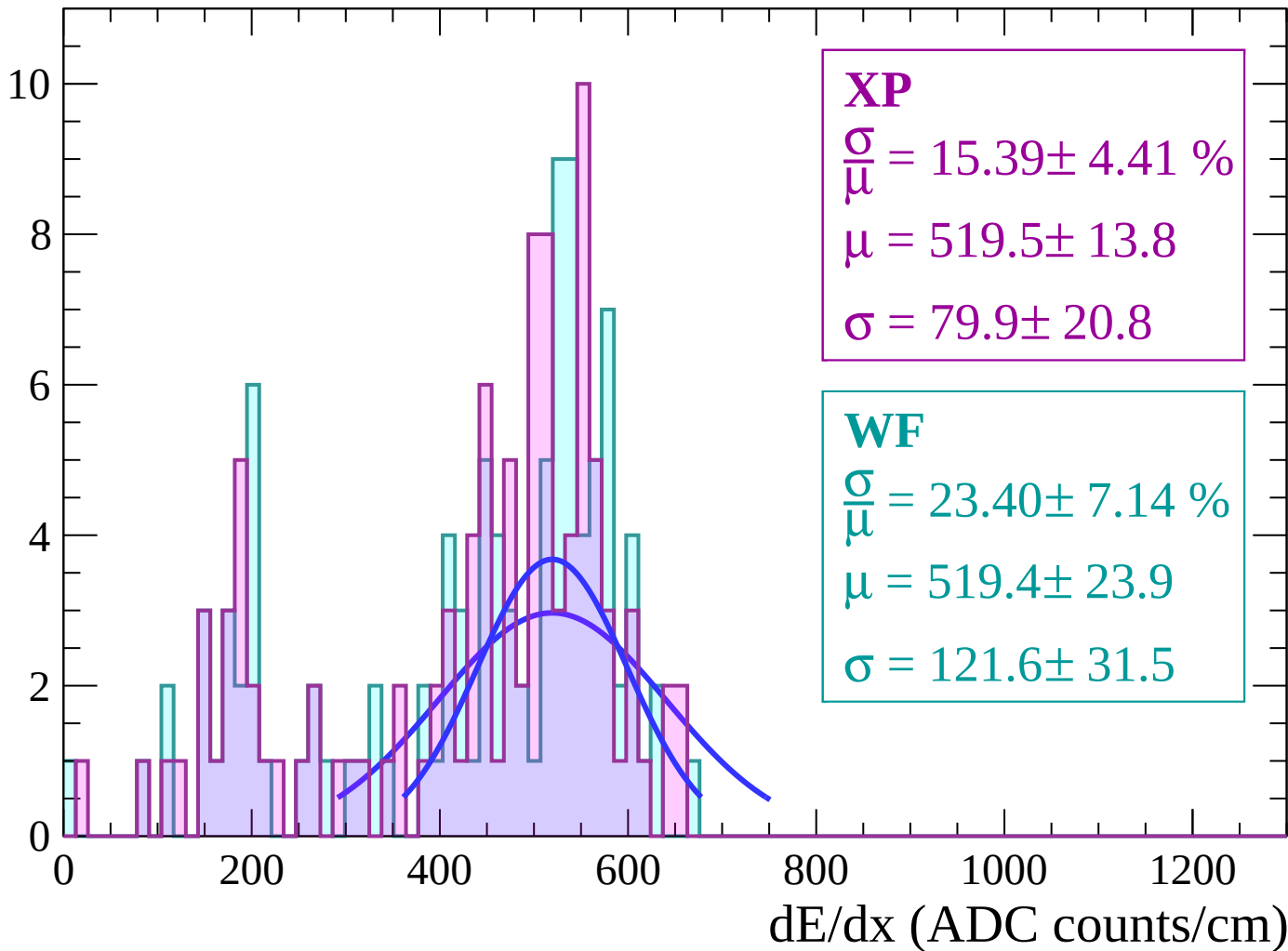
Energy loss | $-70 < \theta < -68$

Count



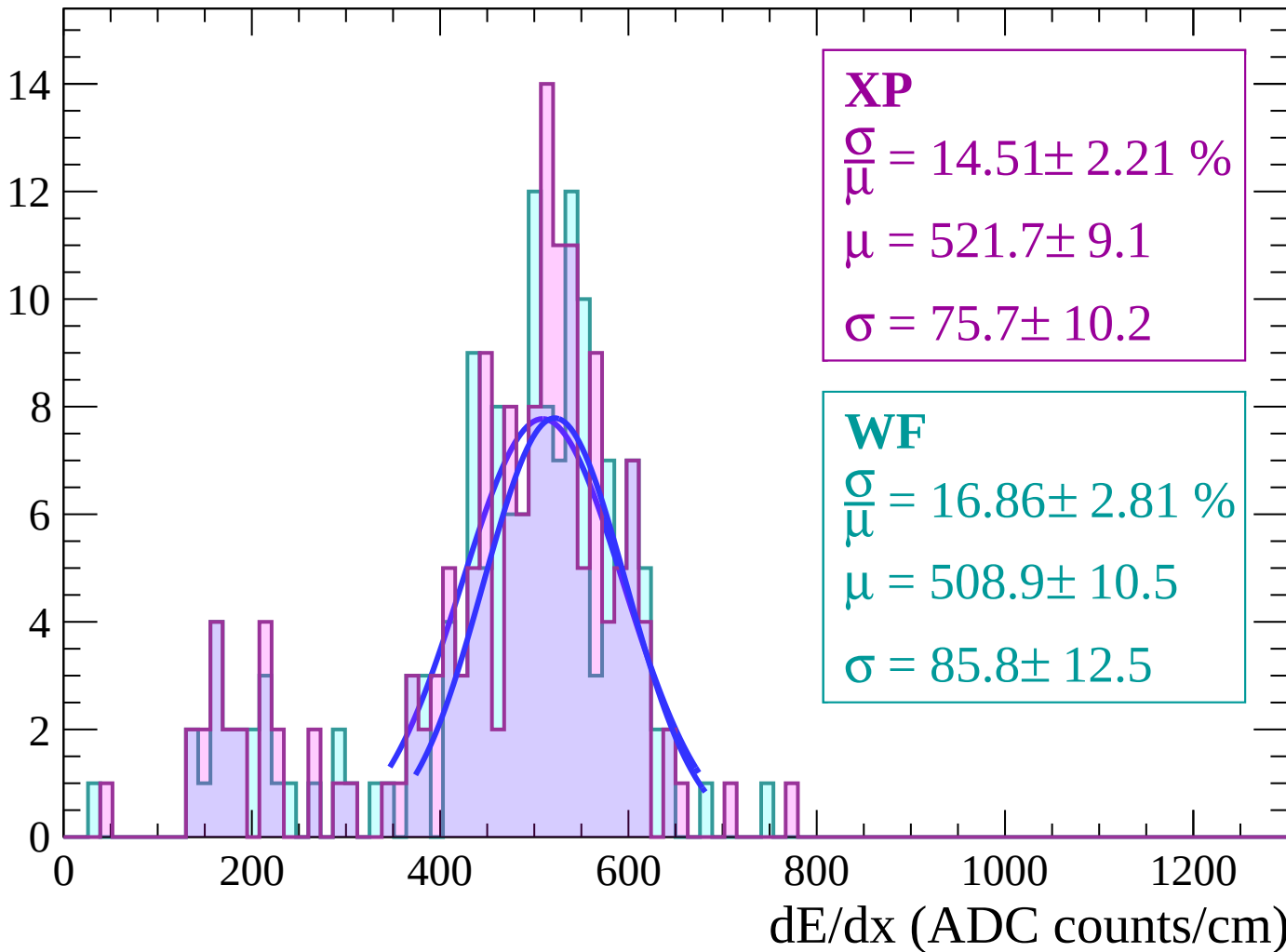
Energy loss | $-68 < \theta < -66$

Count



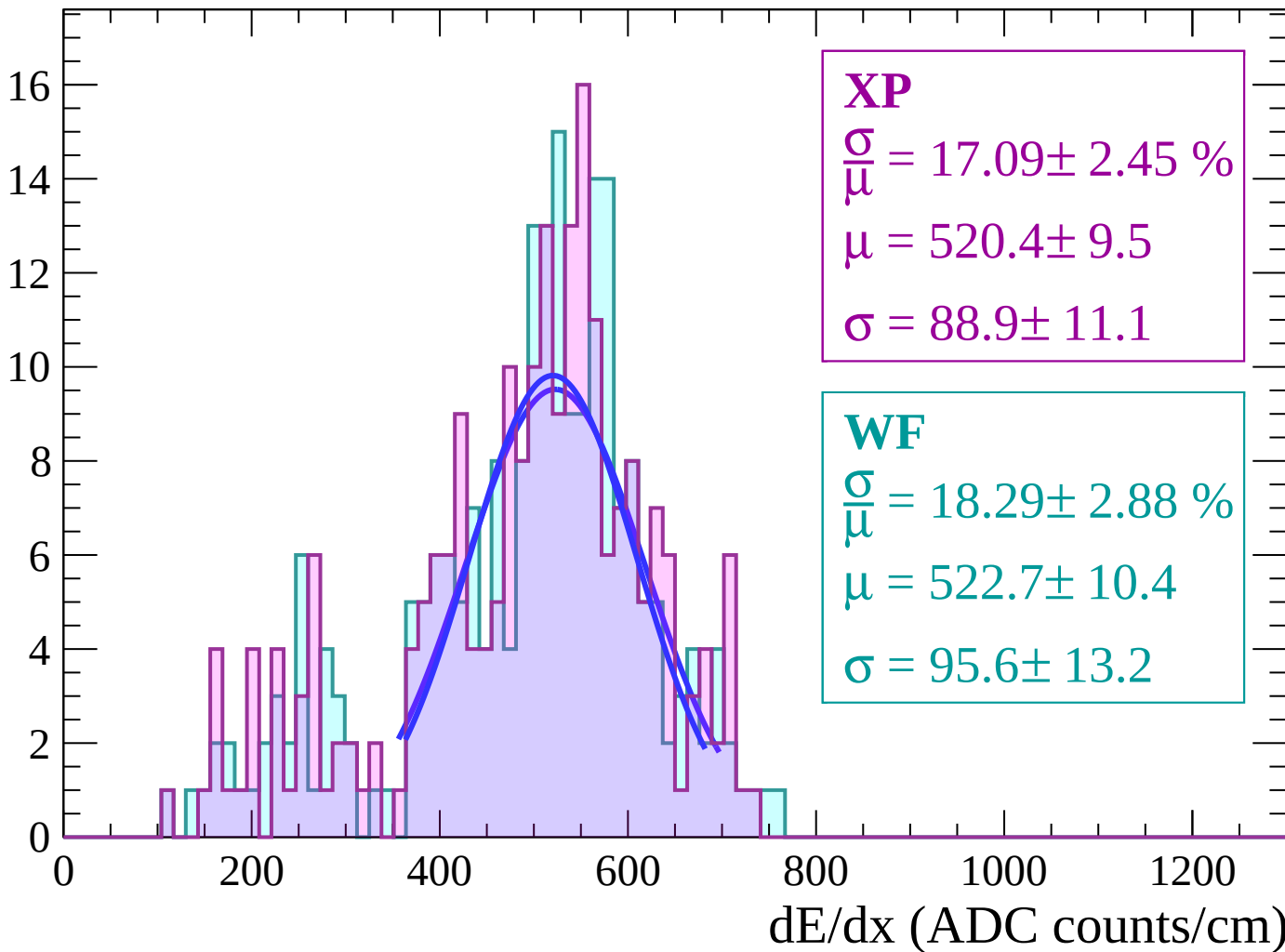
Energy loss | $-66 < \theta < -64$

Count



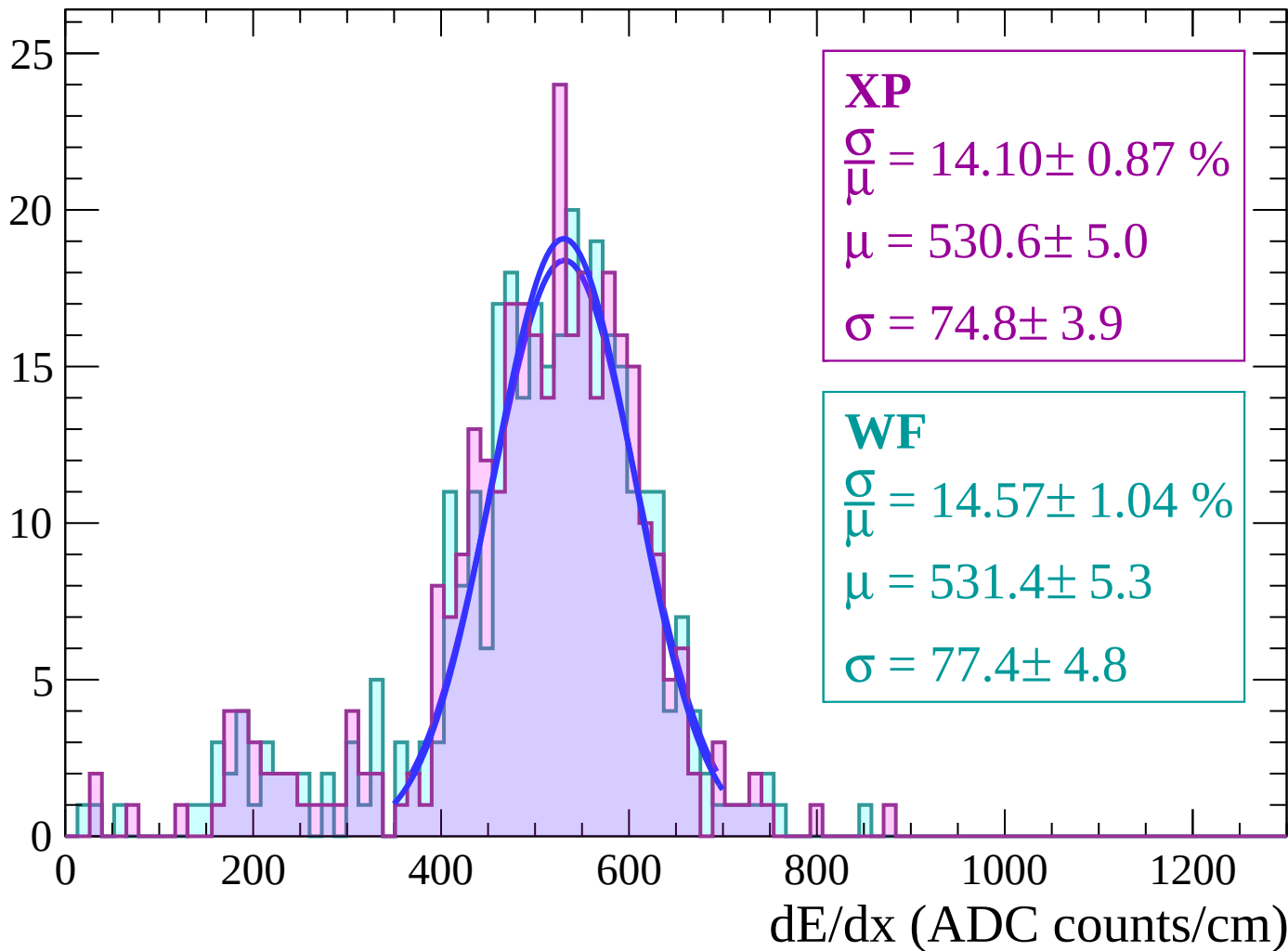
Energy loss | $-64 < \theta < -62$

Count



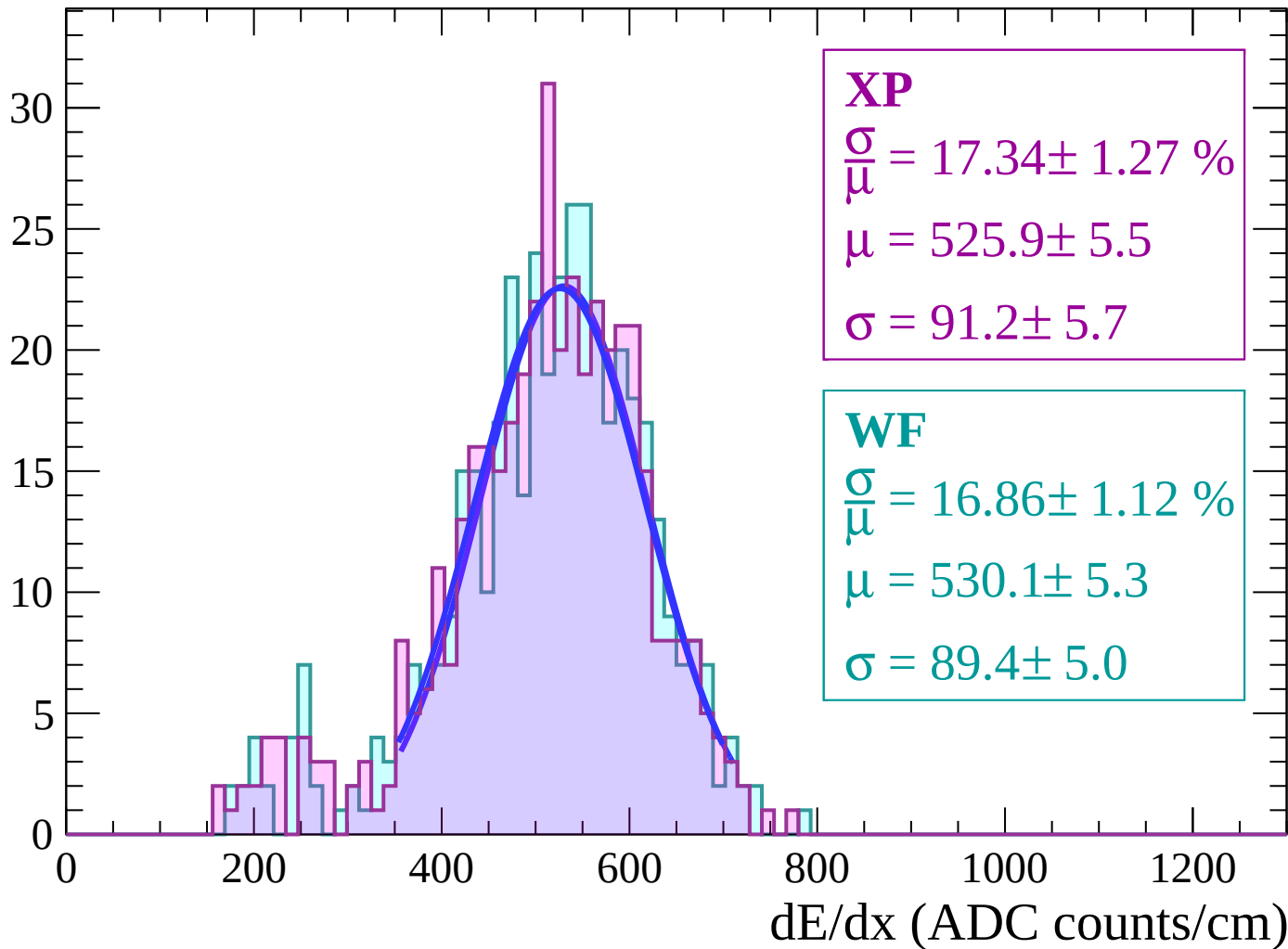
Energy loss | $-62 < \theta < -60$

Count



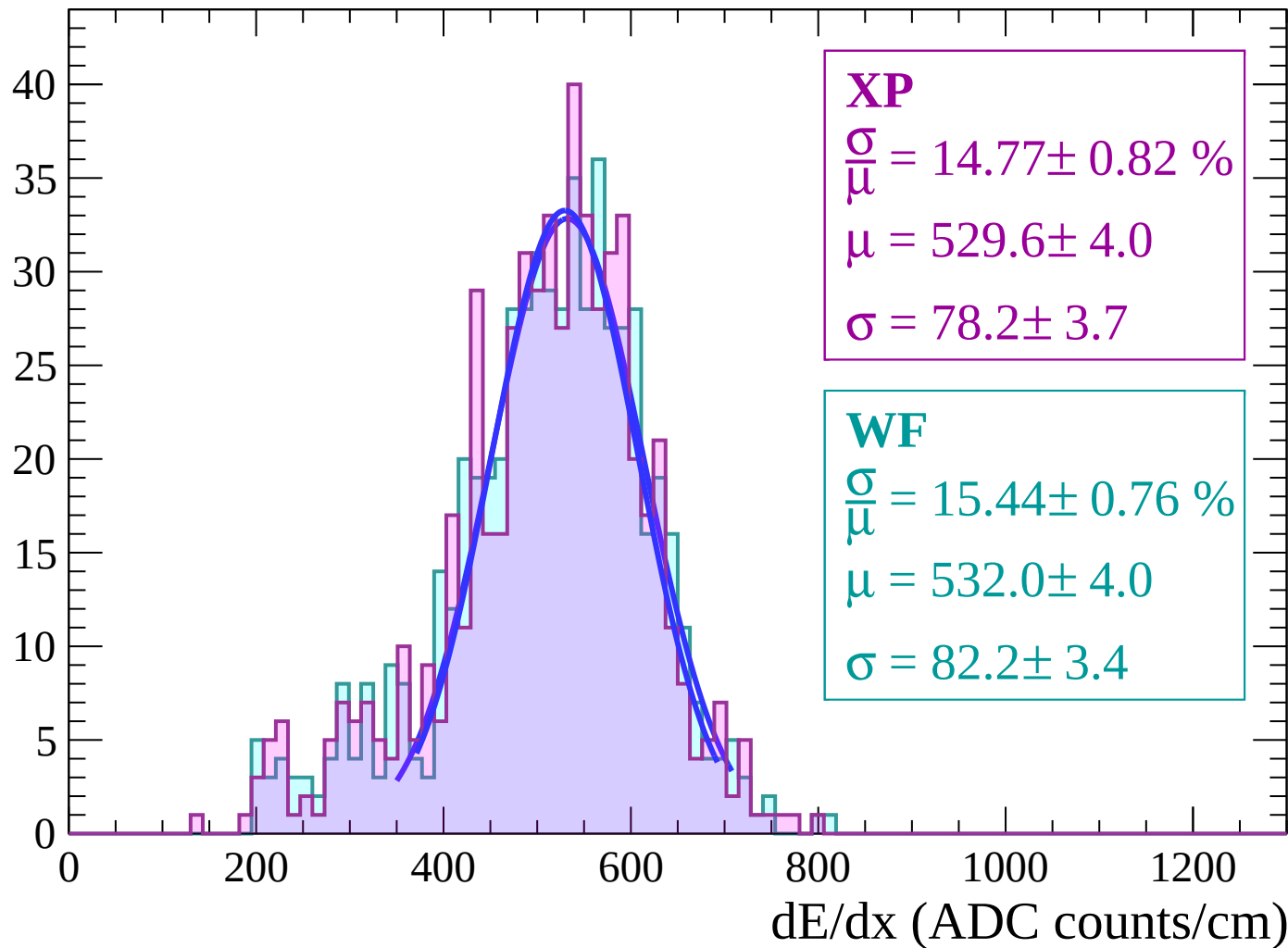
Energy loss | $-60 < \theta < -58$

Count



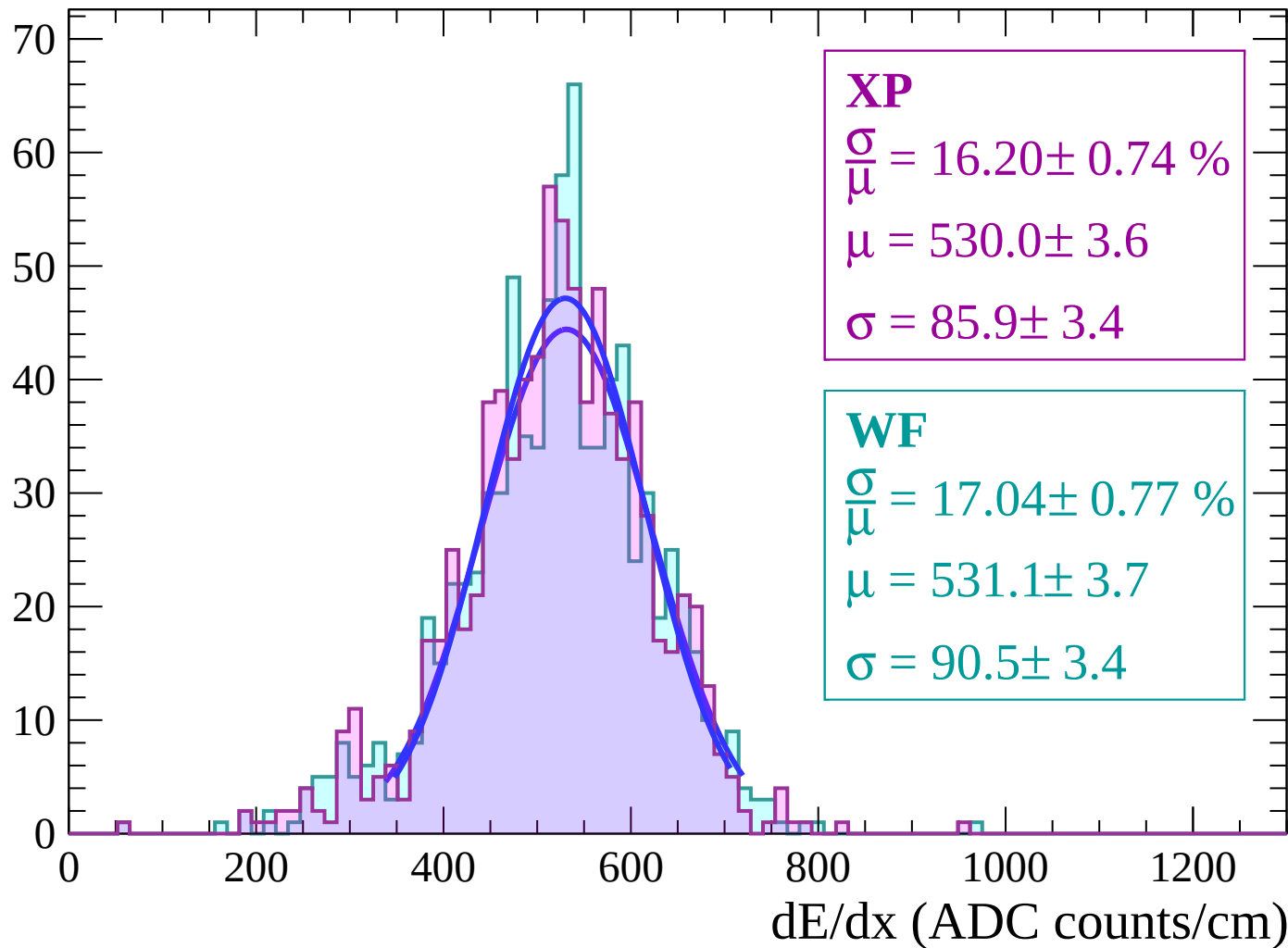
Energy loss | $-58 < \theta < -56$

Count



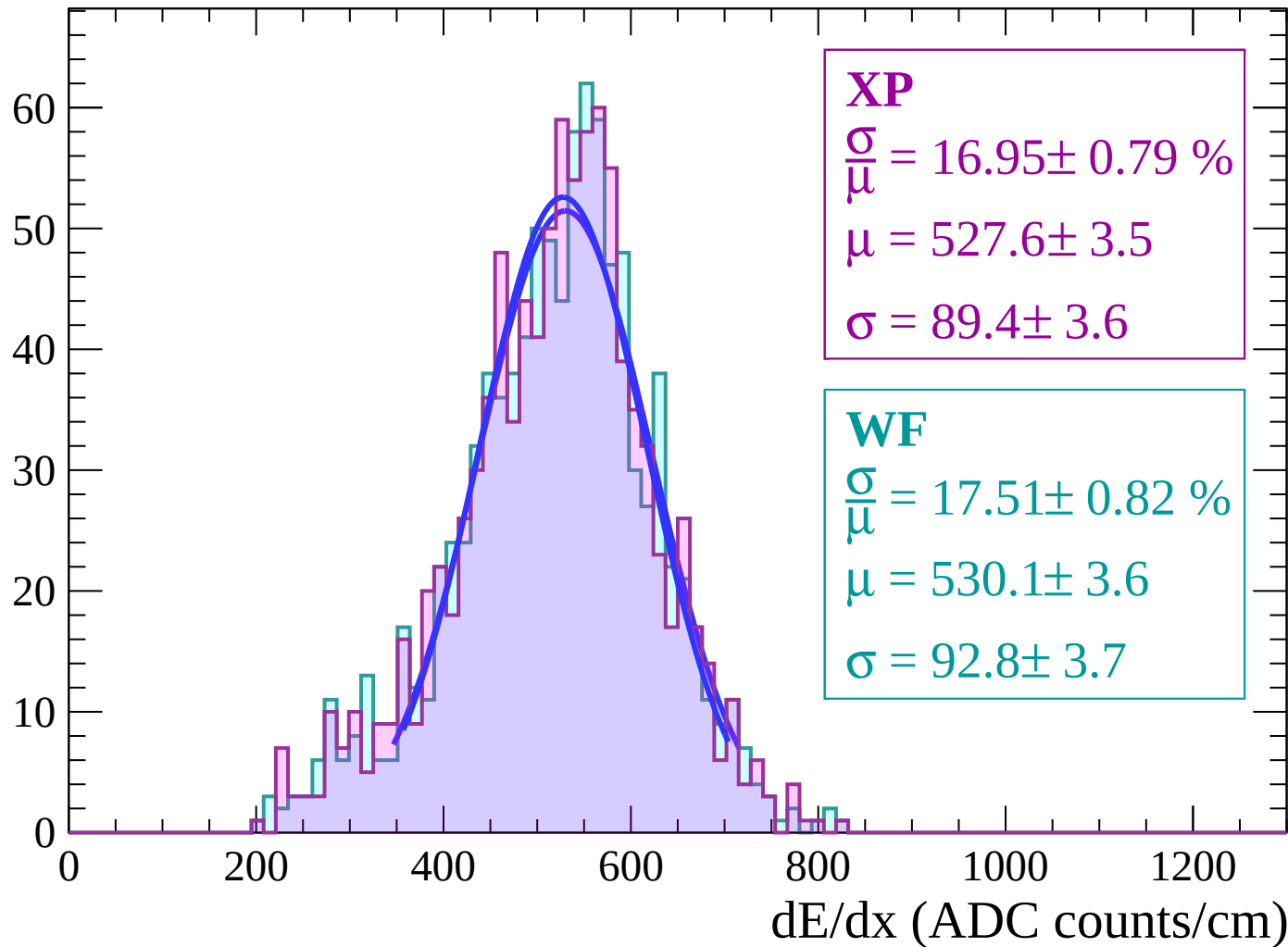
Energy loss | $-56 < \theta < -54$

Count



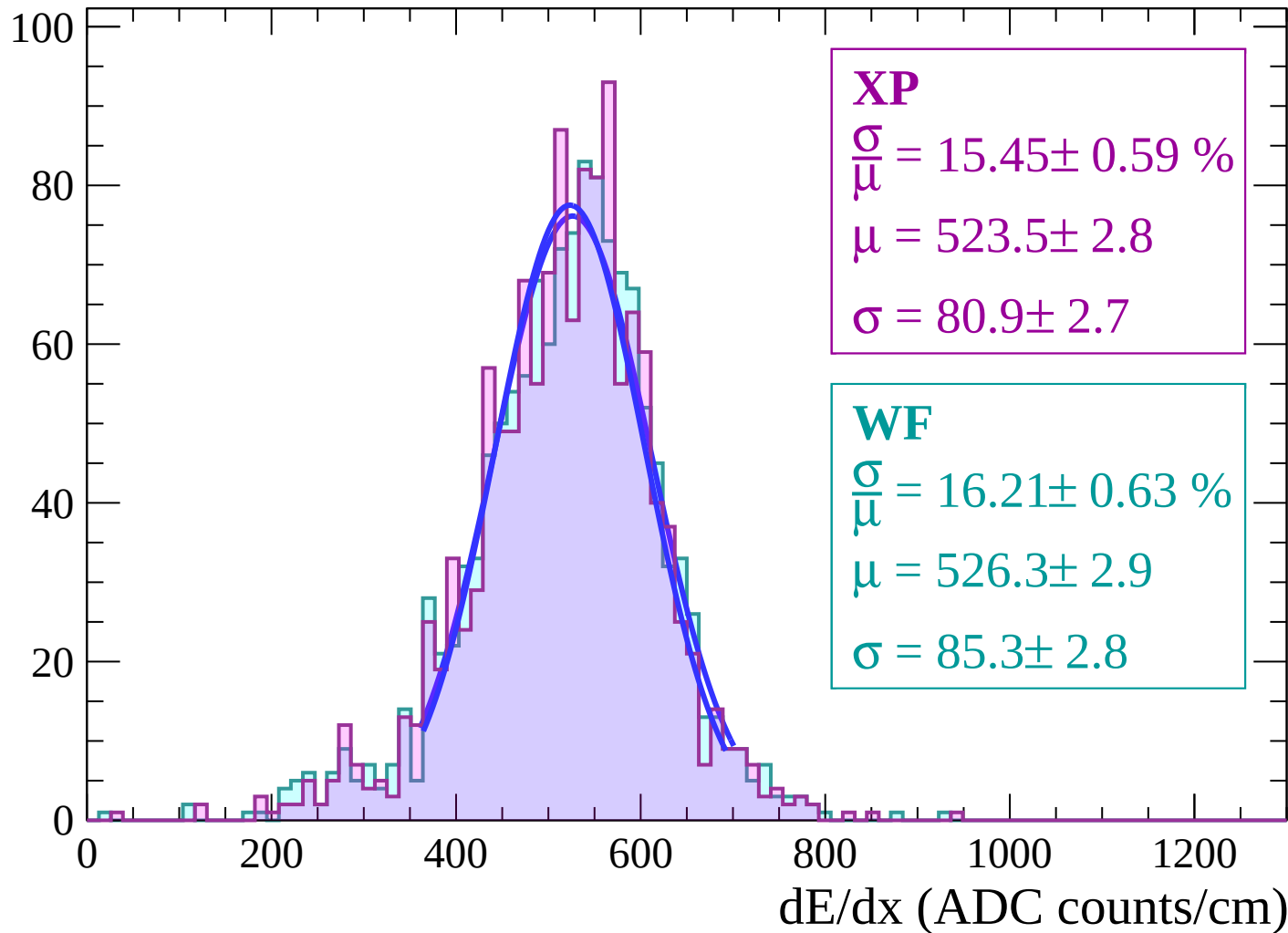
Energy loss | $-54 < \theta < -52$

Count



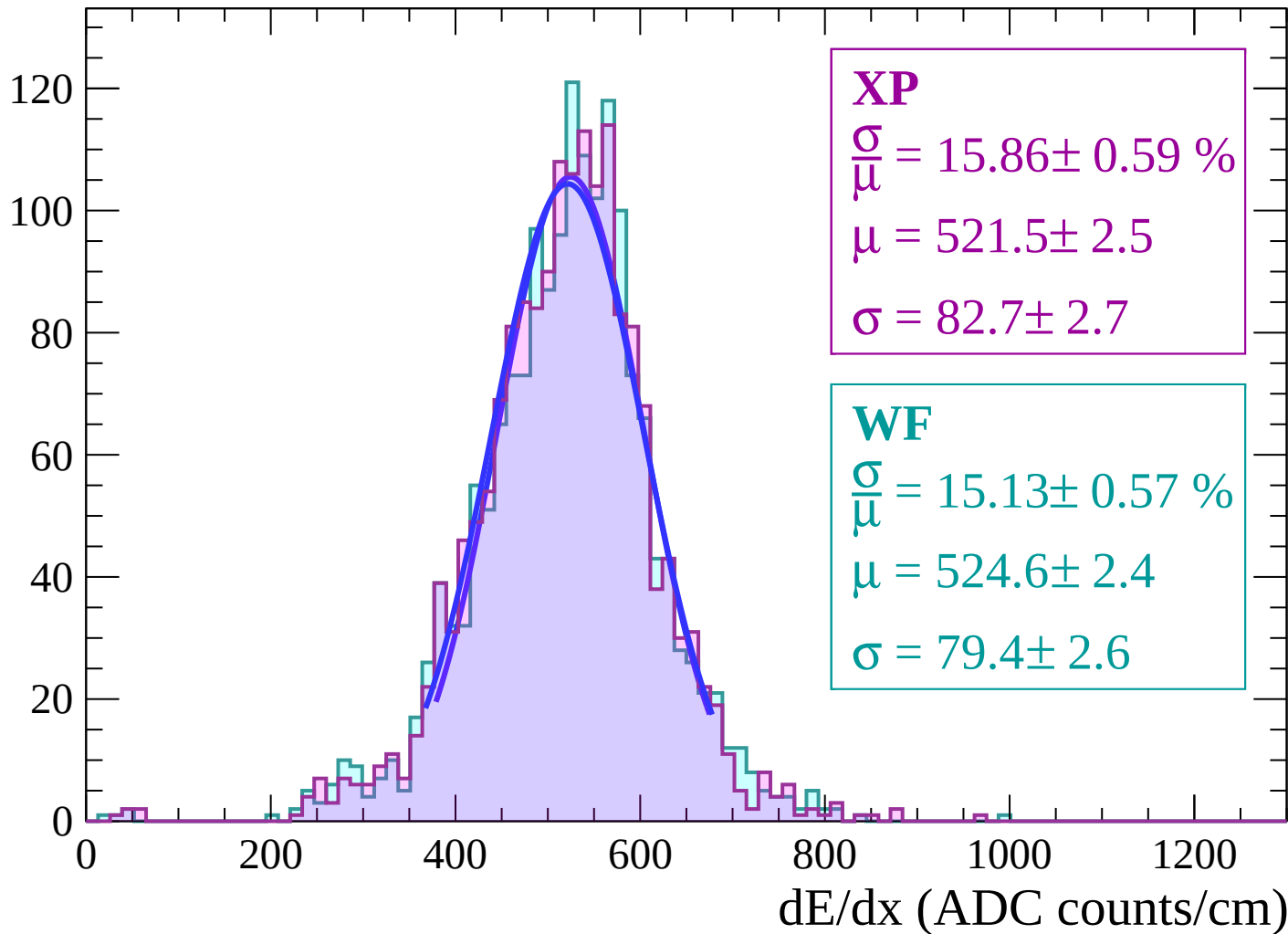
Energy loss | $-52 < \theta < -50$

Count



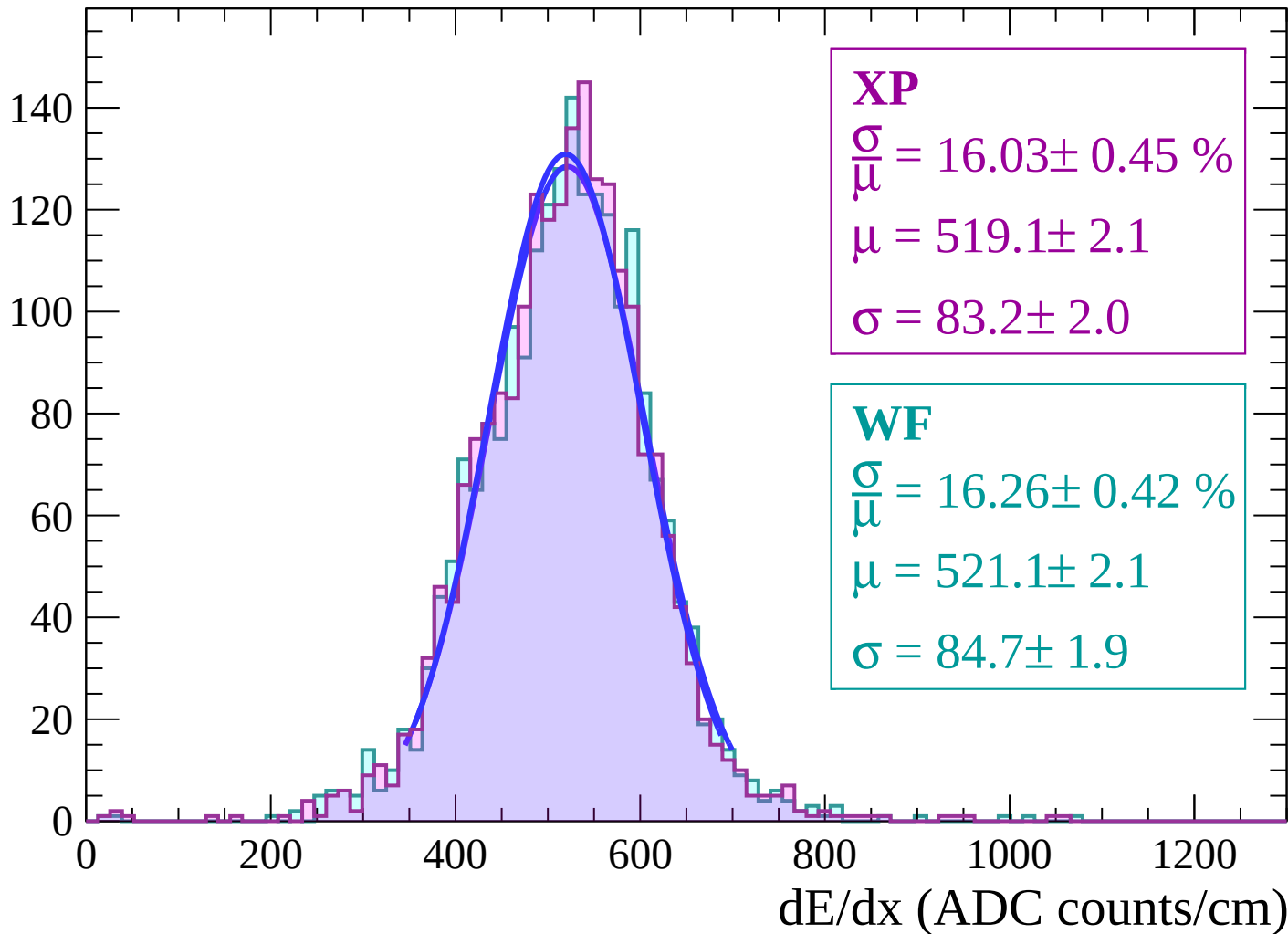
Energy loss | $-50 < \theta < -48$

Count



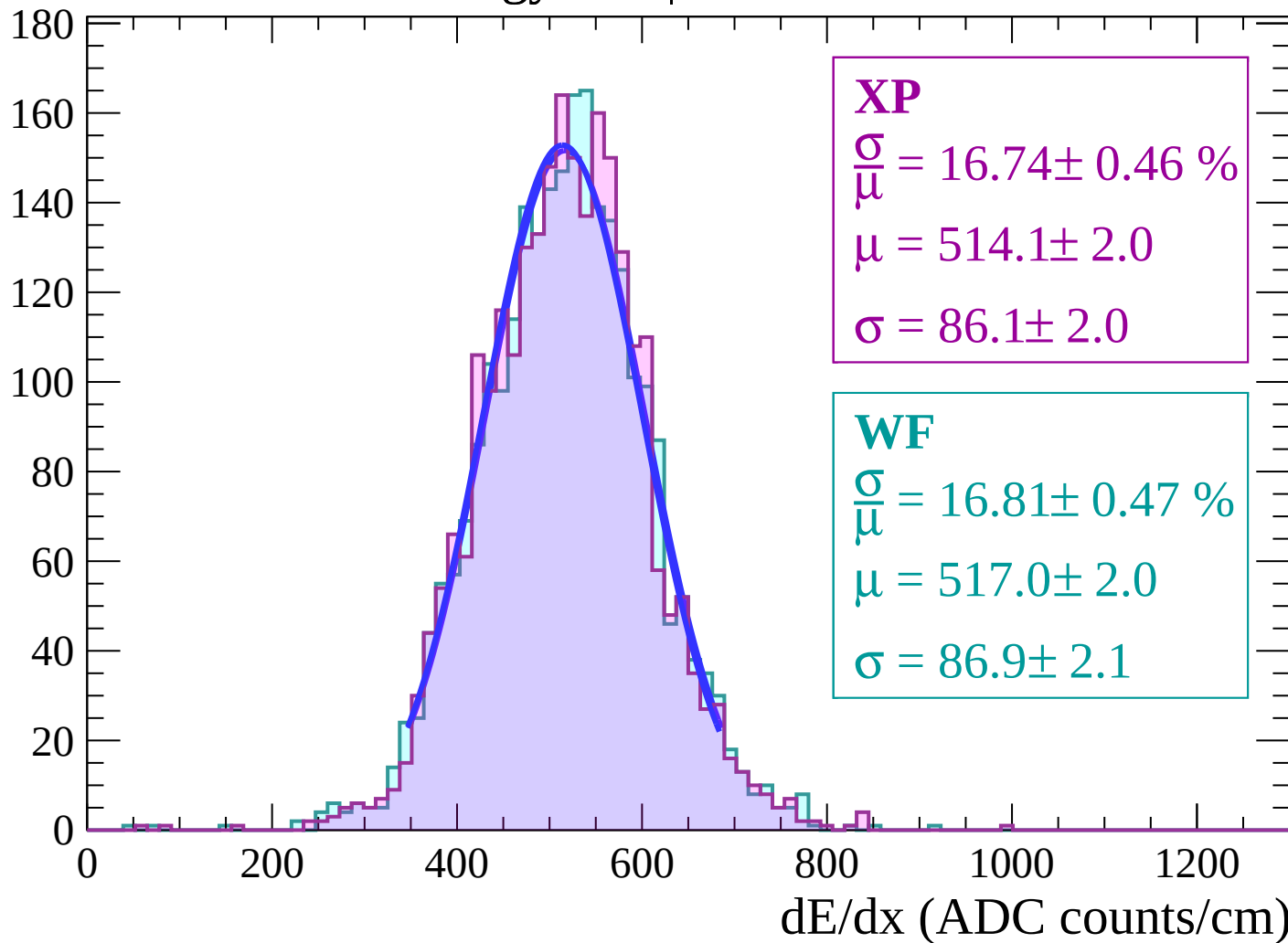
Energy loss | $-48 < \theta < -46$

Count



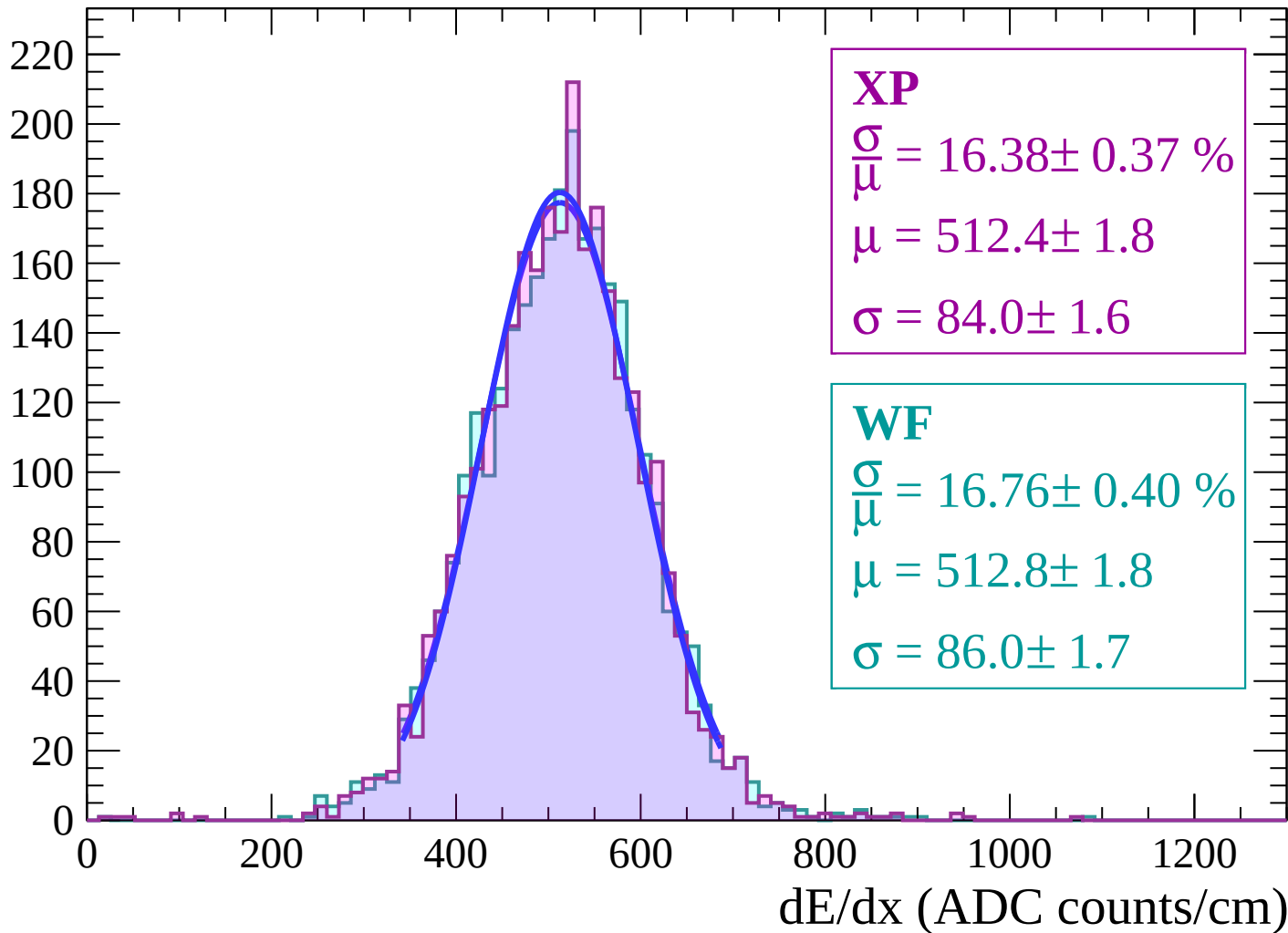
Energy loss | $-46 < \theta < -44$

Count



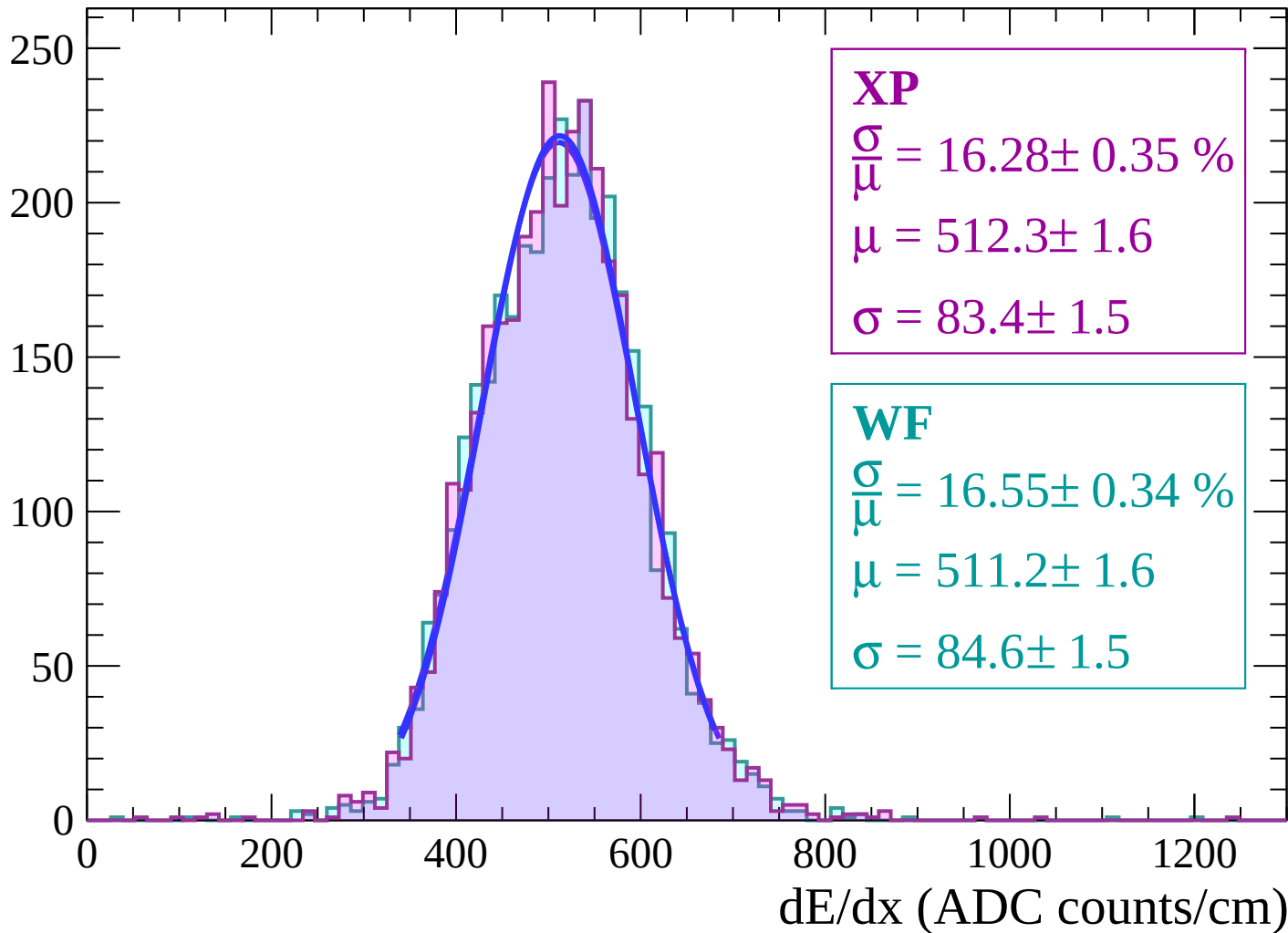
Energy loss | $-44 < \theta < -42$

Count



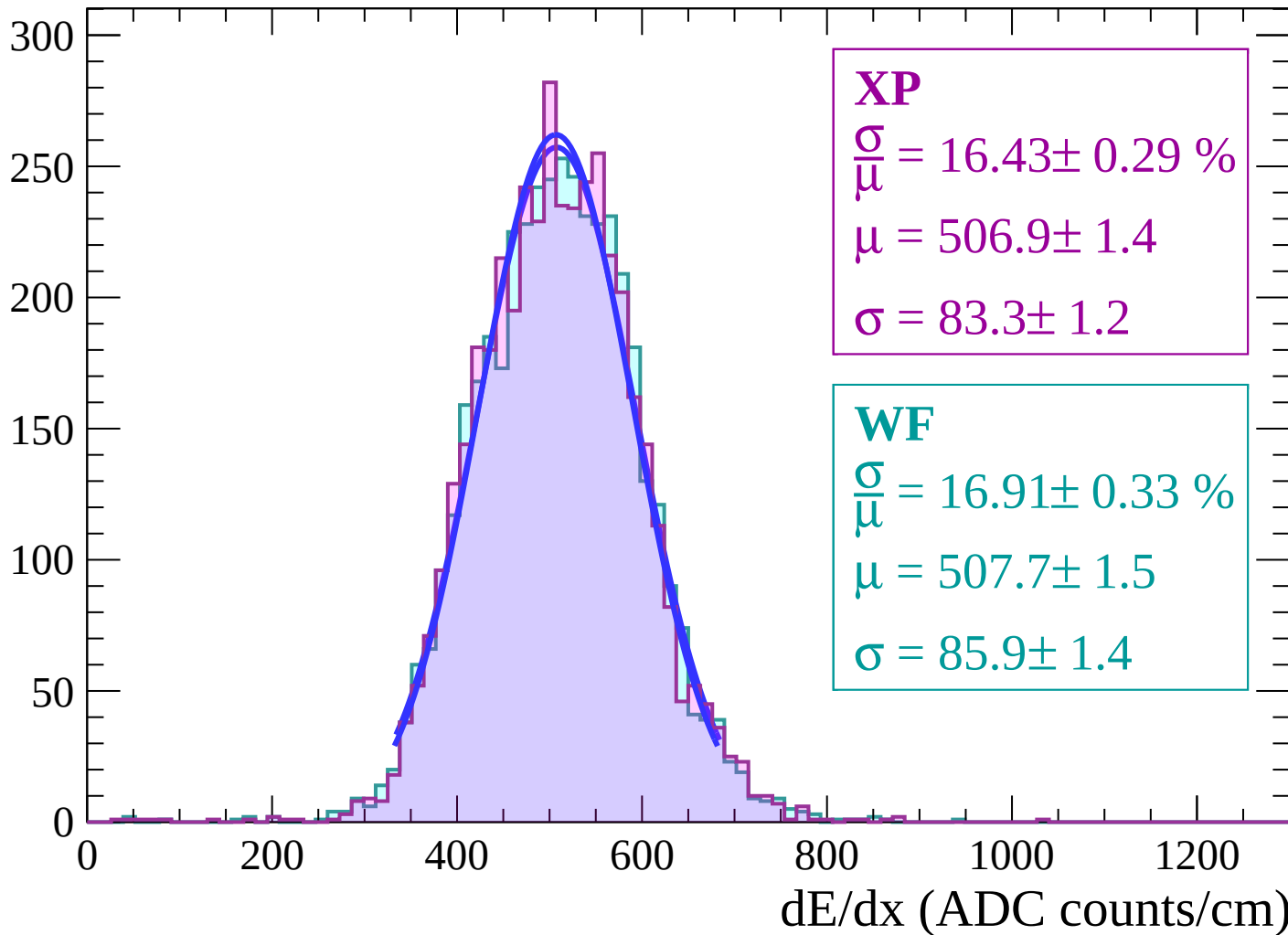
Energy loss | $-42 < \theta < -40$

Count



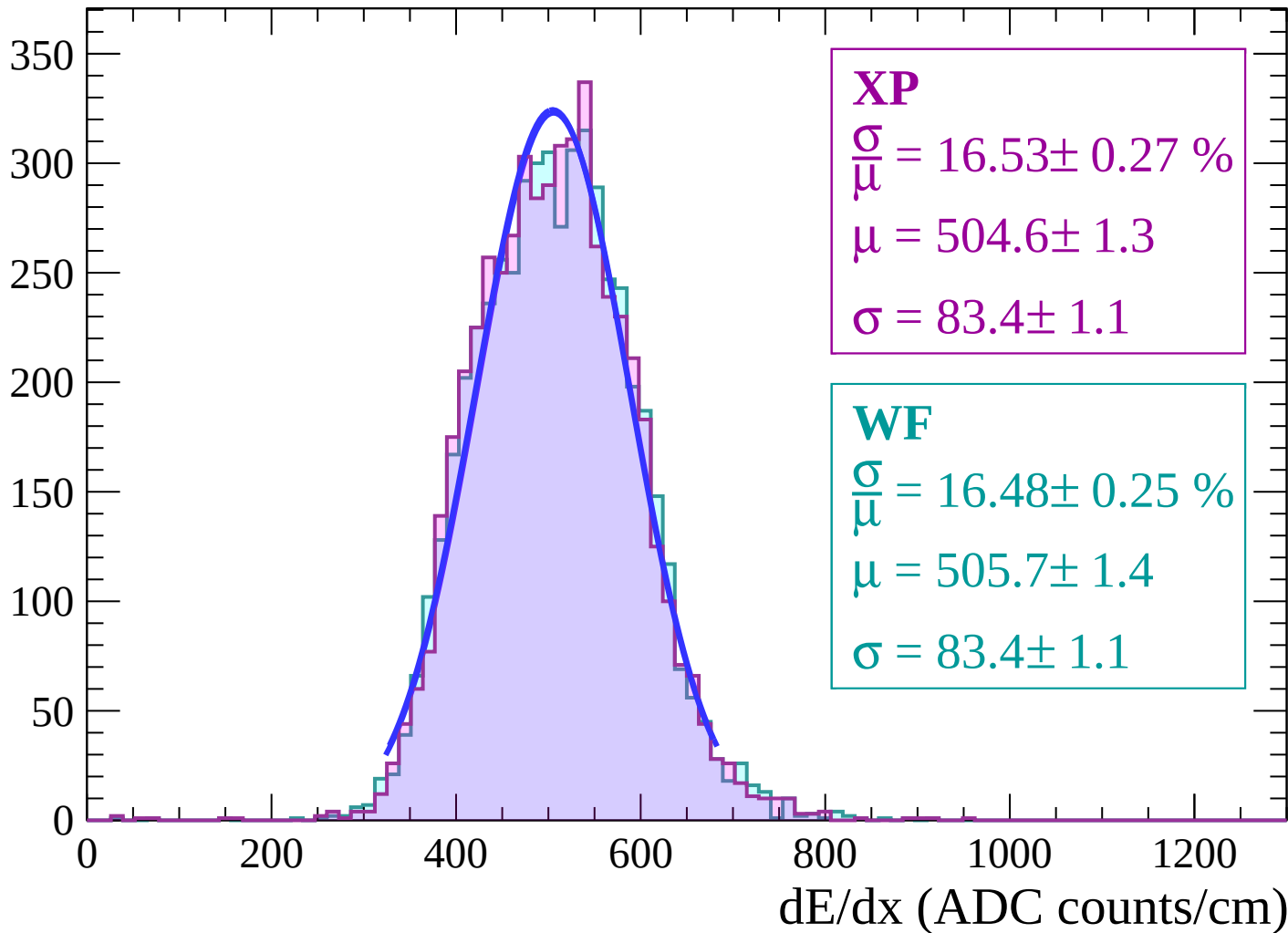
Energy loss | $-40 < \theta < -38$

Count



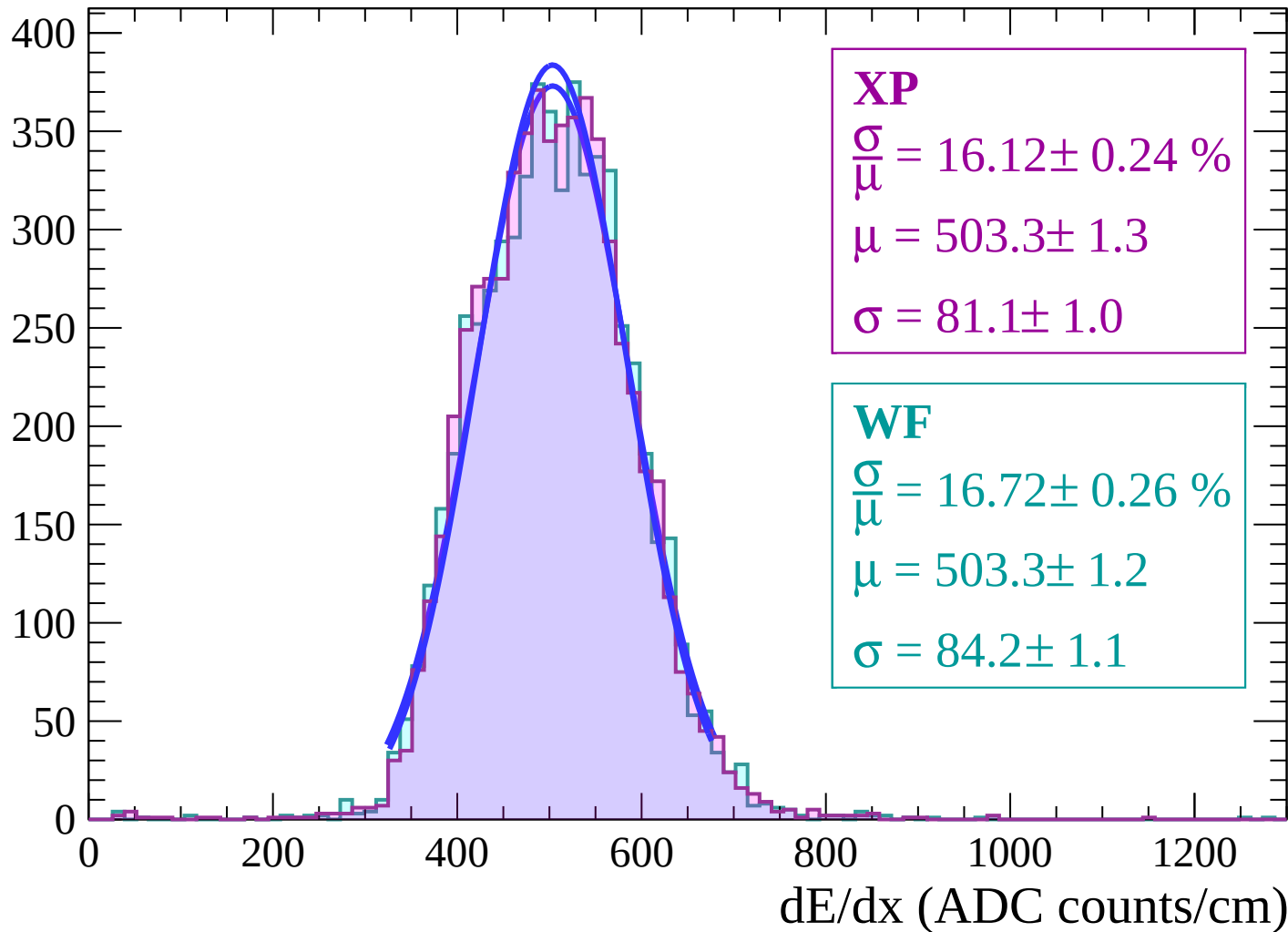
Energy loss | $-38 < \theta < -36$

Count



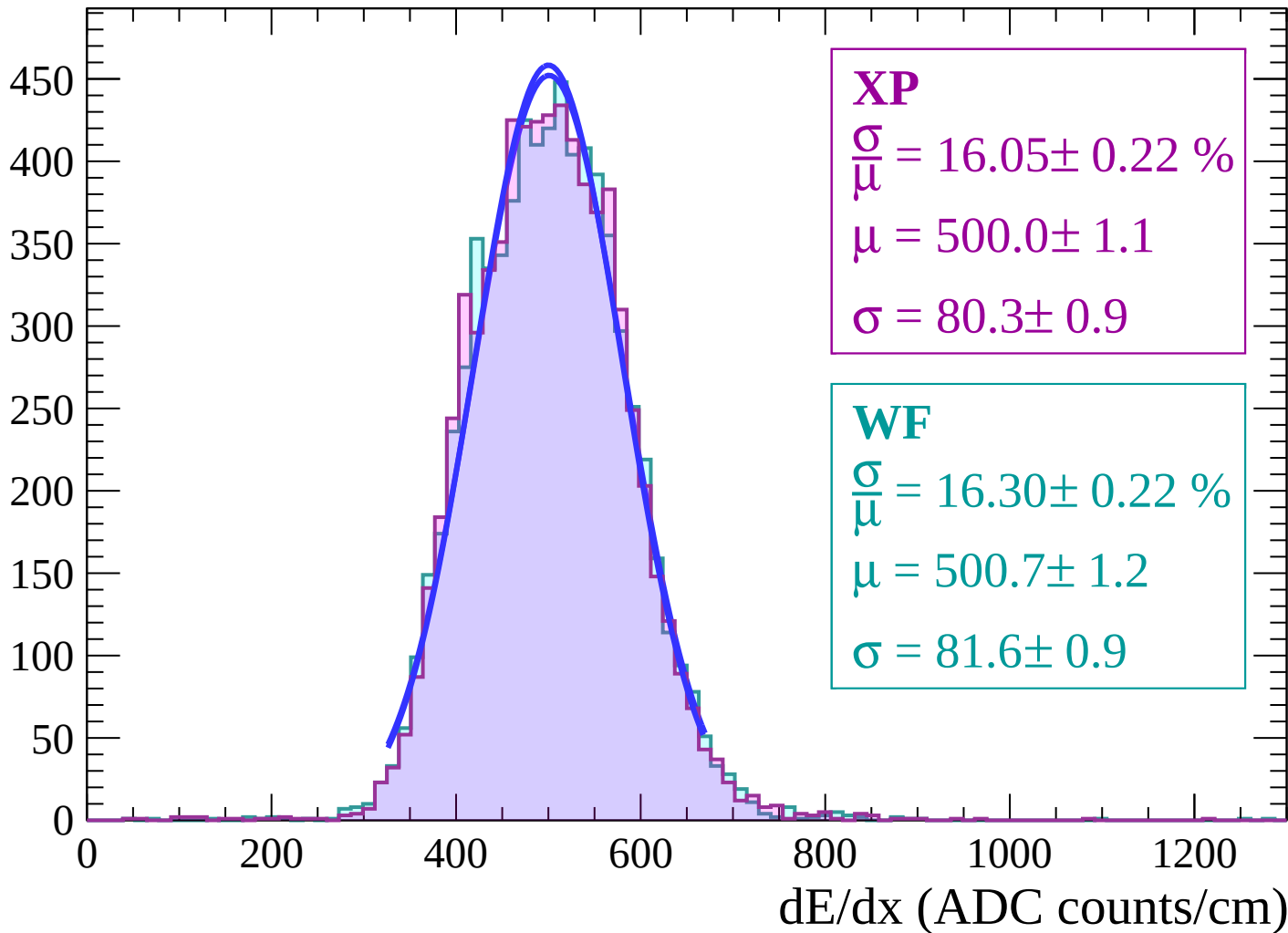
Energy loss | $-36 < \theta < -34$

Count



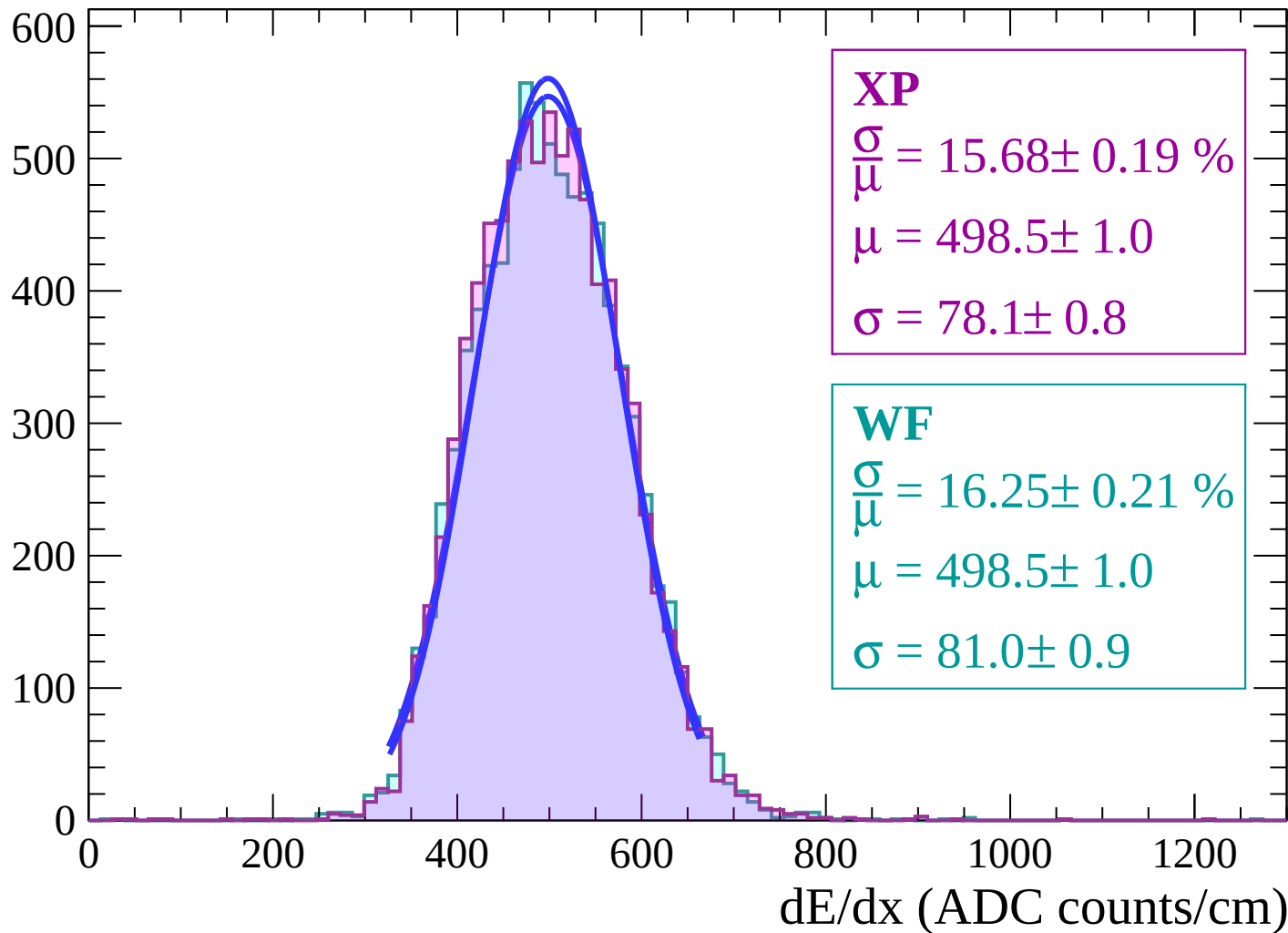
Energy loss | $-34 < \theta < -32$

Count

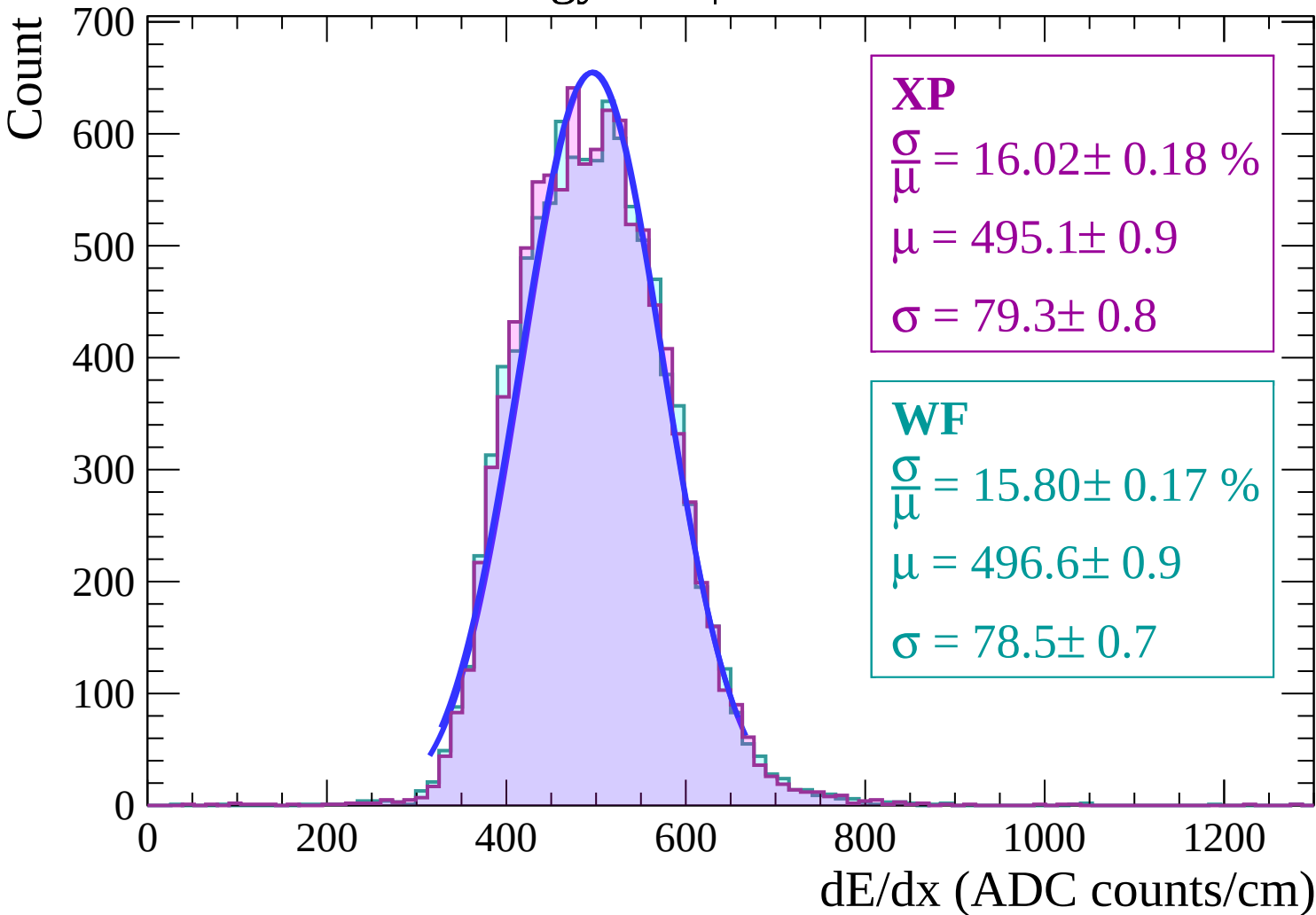


Energy loss | $-32 < \theta < -30$

Count

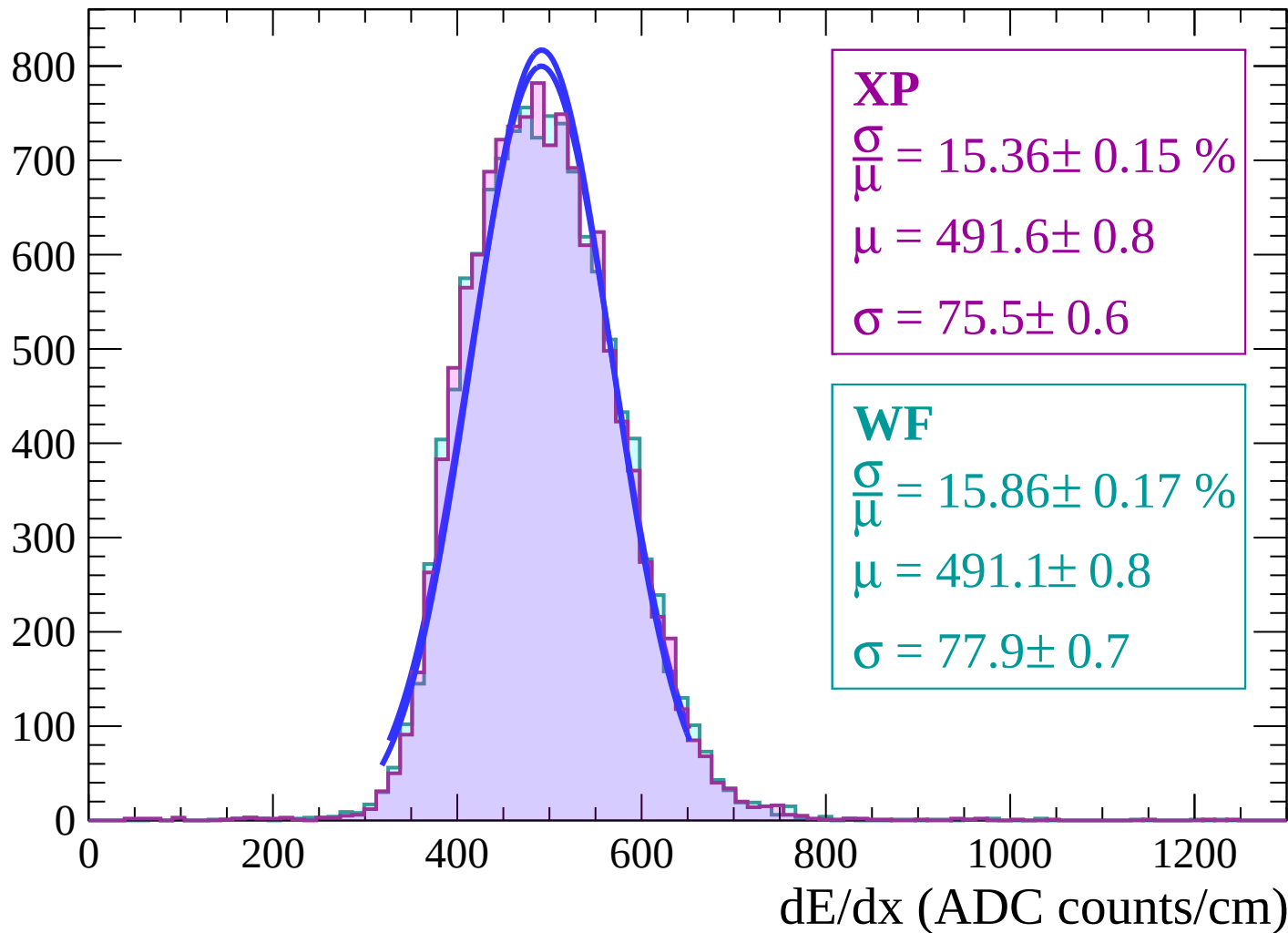


Energy loss | $-30 < \theta < -28$

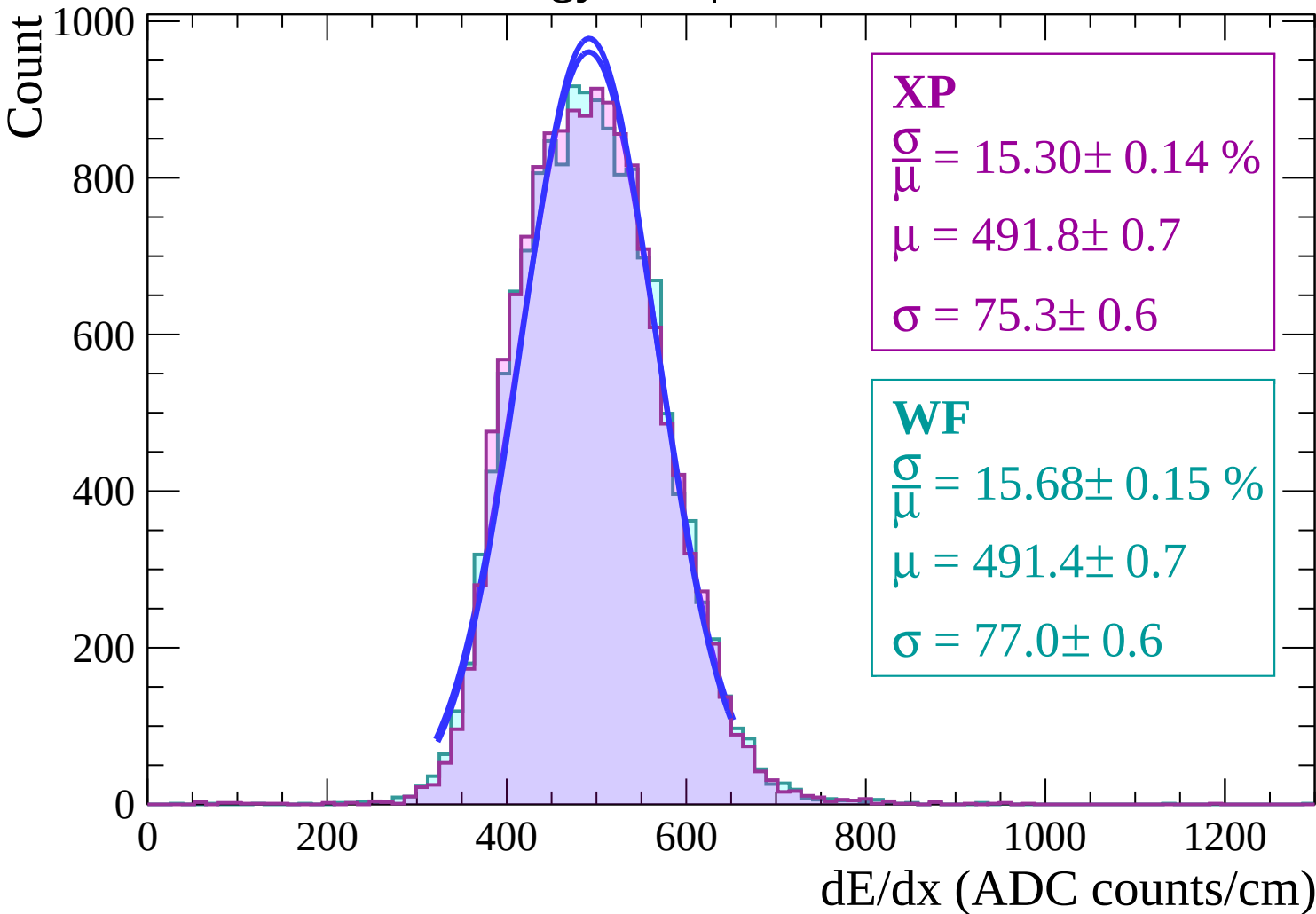


Energy loss | $-28 < \theta < -26$

Count

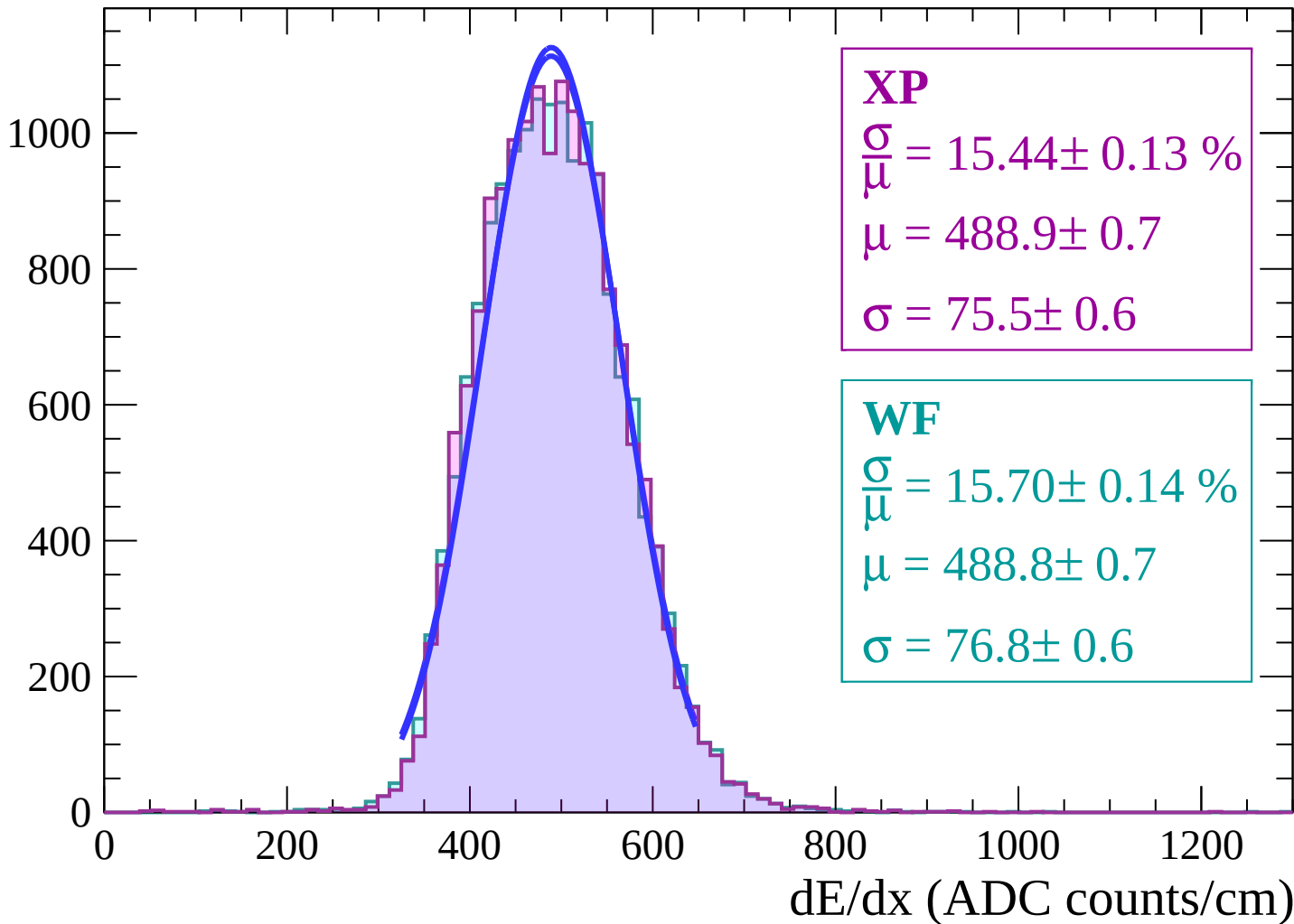


Energy loss | $-26 < \theta < -24$



Energy loss | $-24 < \theta < -22$

Count



Energy loss | $-22 < \theta < -20$

Count

1200

1000

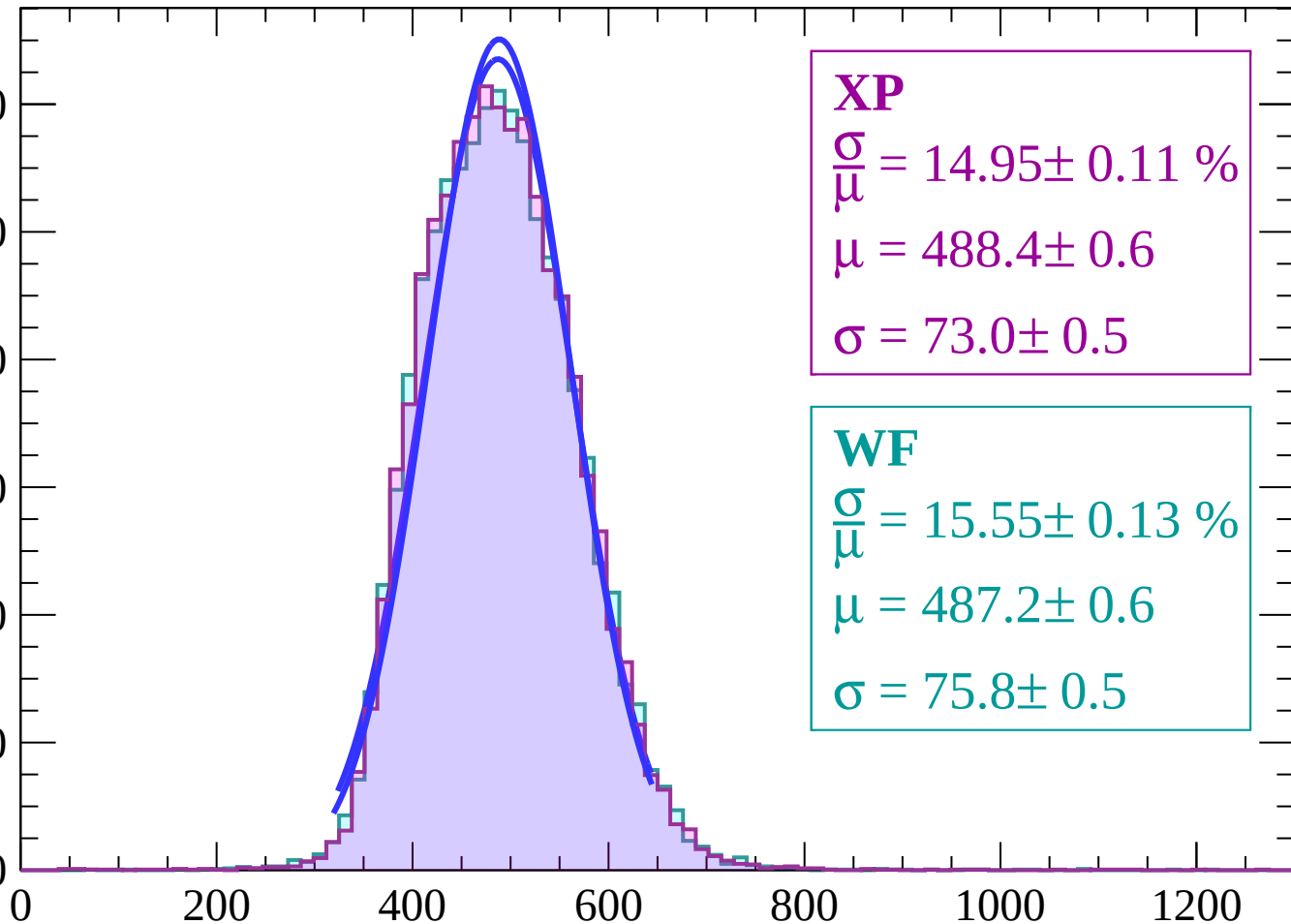
800

600

400

200

0



XP

$$\frac{\sigma}{\mu} = 14.95 \pm 0.11 \%$$

$$\mu = 488.4 \pm 0.6$$

$$\sigma = 73.0 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 15.55 \pm 0.13 \%$$

$$\mu = 487.2 \pm 0.6$$

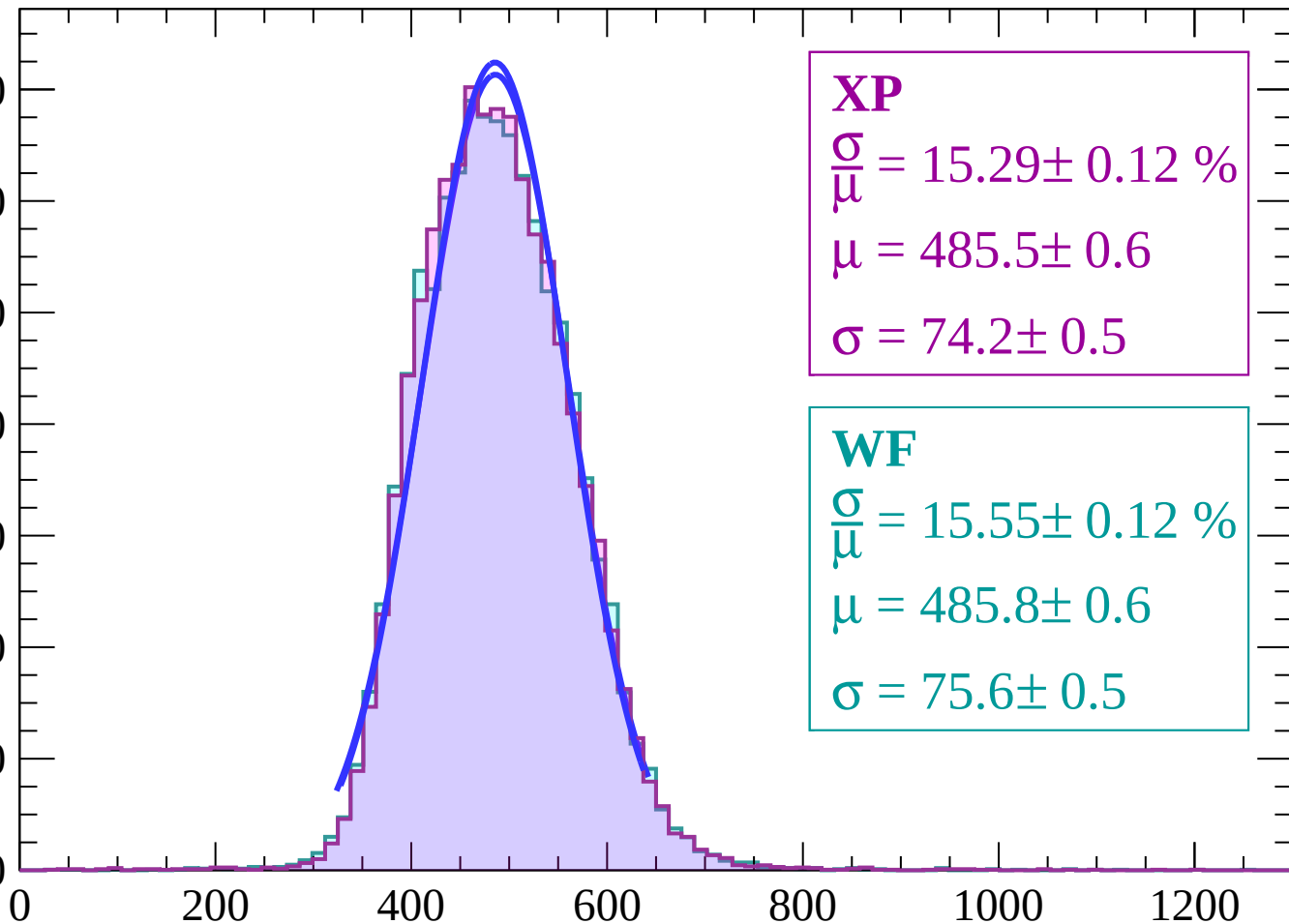
$$\sigma = 75.8 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $-20 < \theta < -18$

Count

1400
1200
1000
800
600
400
200
0



XP

$$\frac{\sigma}{\mu} = 15.29 \pm 0.12 \%$$

$$\mu = 485.5 \pm 0.6$$

$$\sigma = 74.2 \pm 0.5$$

WF

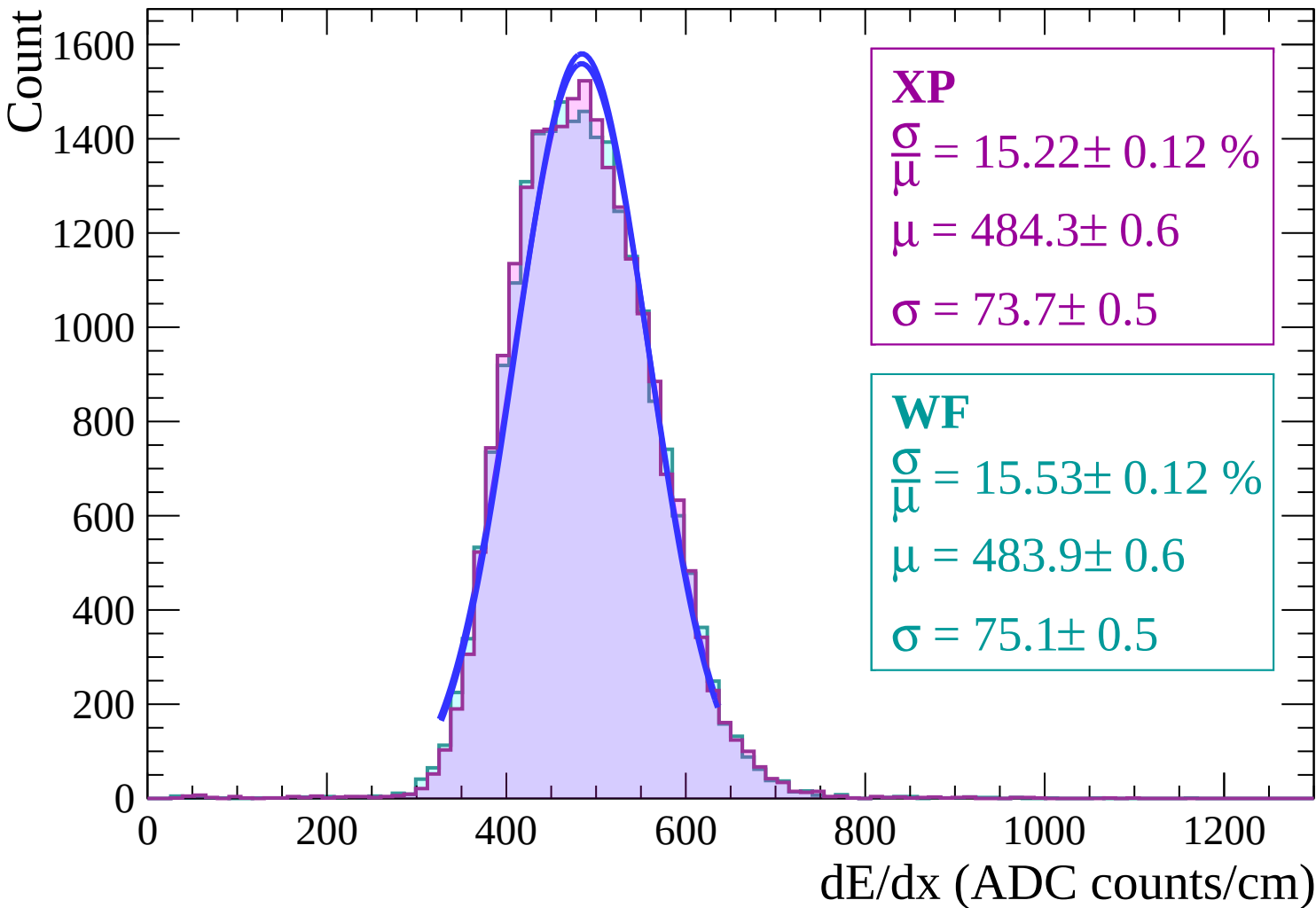
$$\frac{\sigma}{\mu} = 15.55 \pm 0.12 \%$$

$$\mu = 485.8 \pm 0.6$$

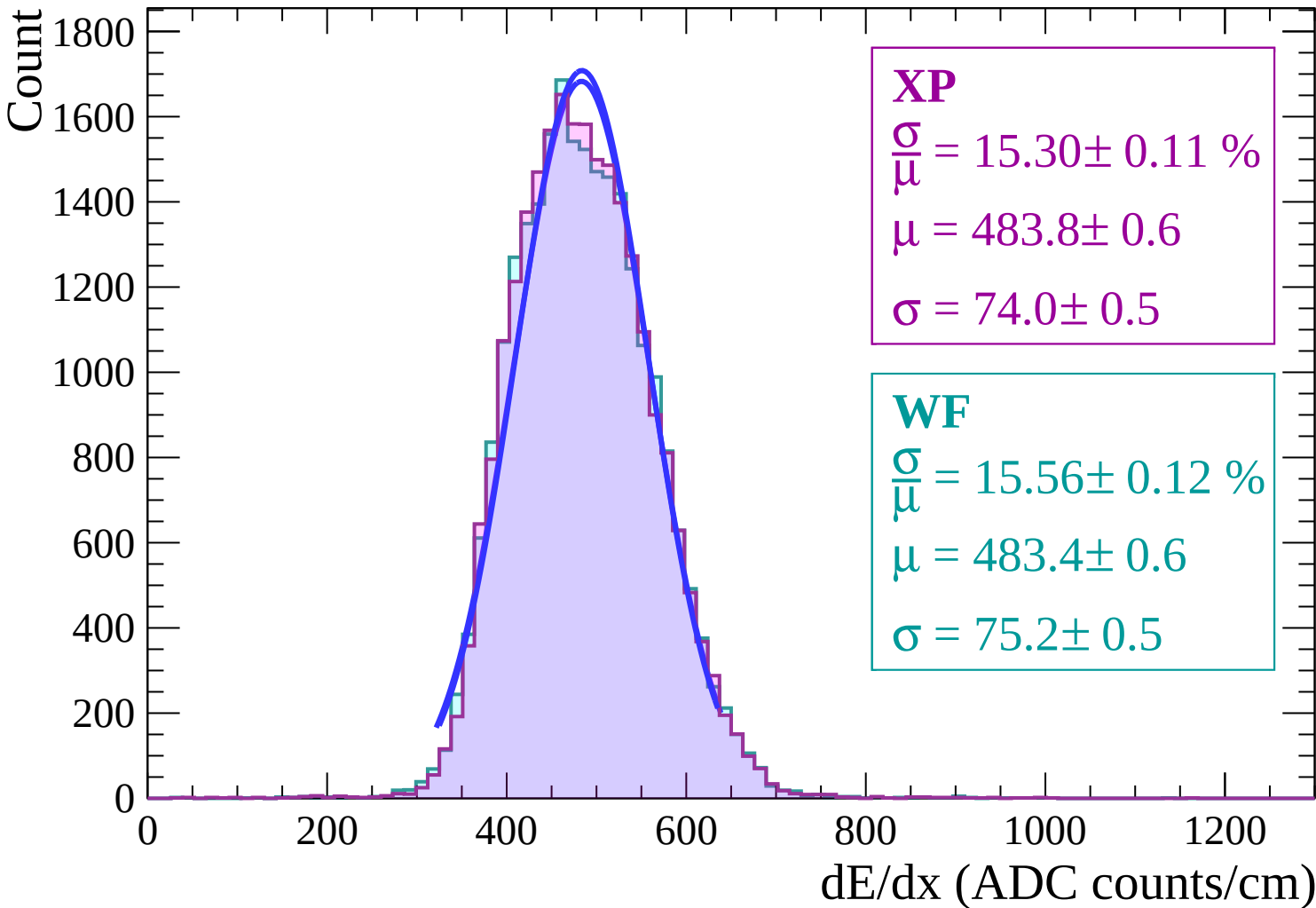
$$\sigma = 75.6 \pm 0.5$$

dE/dx (ADC counts/cm)

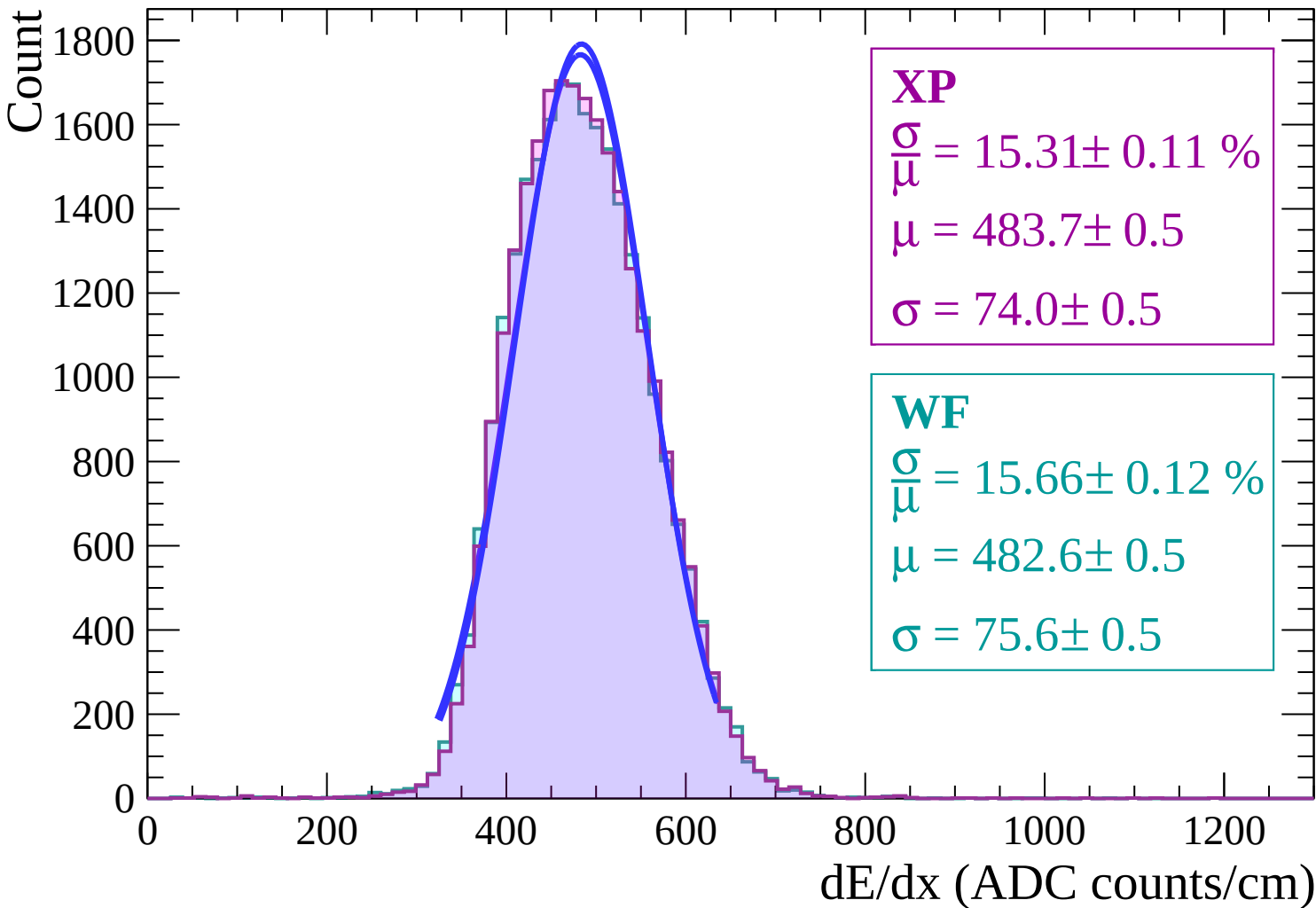
Energy loss | $-18 < \theta < -16$



Energy loss | $-16 < \theta < -14$



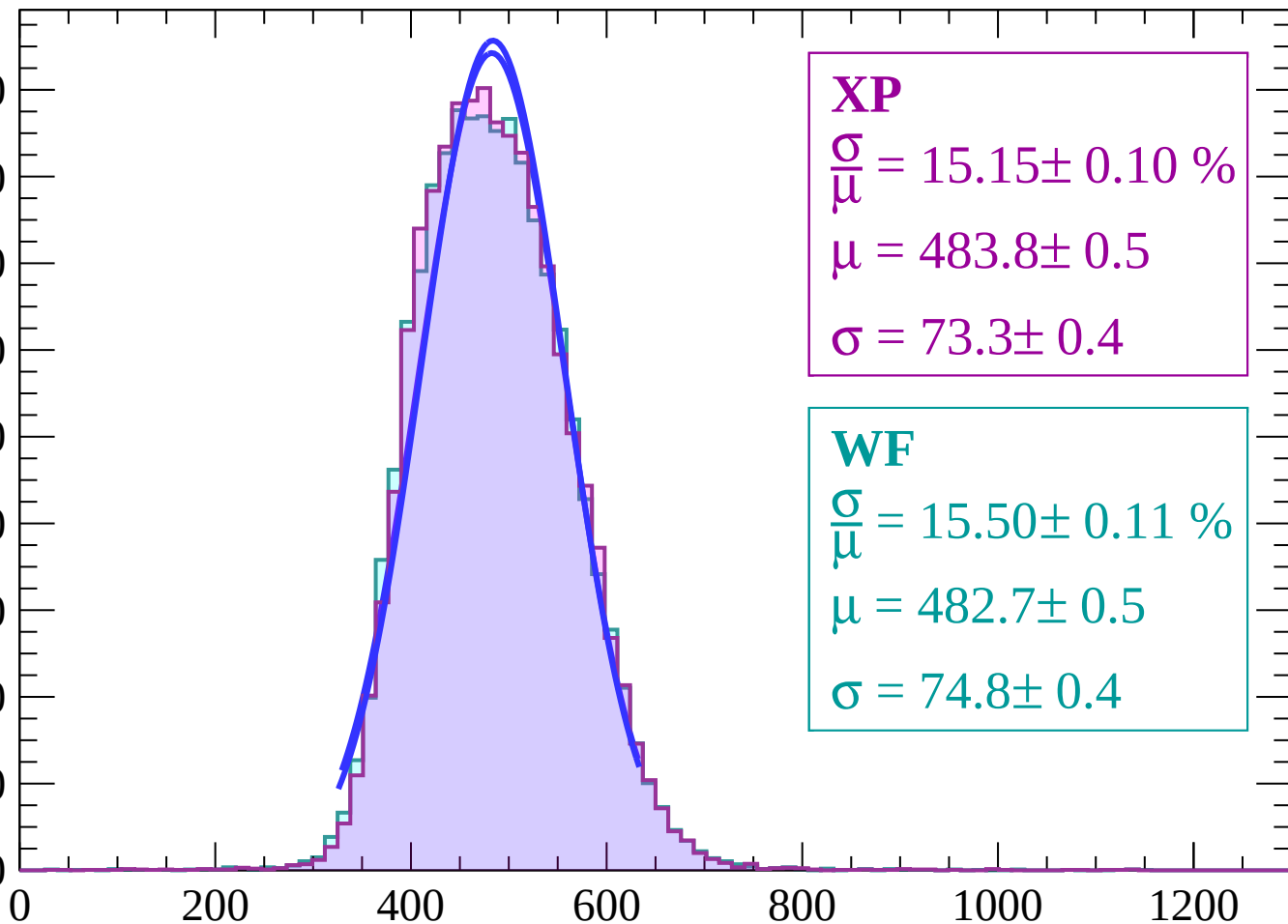
Energy loss | $-14 < \theta < -12$



Energy loss | $-12 < \theta < -10$

Count

1800
1600
1400
1200
1000
800
600
400
200
0



XP

$$\frac{\sigma}{\mu} = 15.15 \pm 0.10 \%$$

$$\mu = 483.8 \pm 0.5$$

$$\sigma = 73.3 \pm 0.4$$

WF

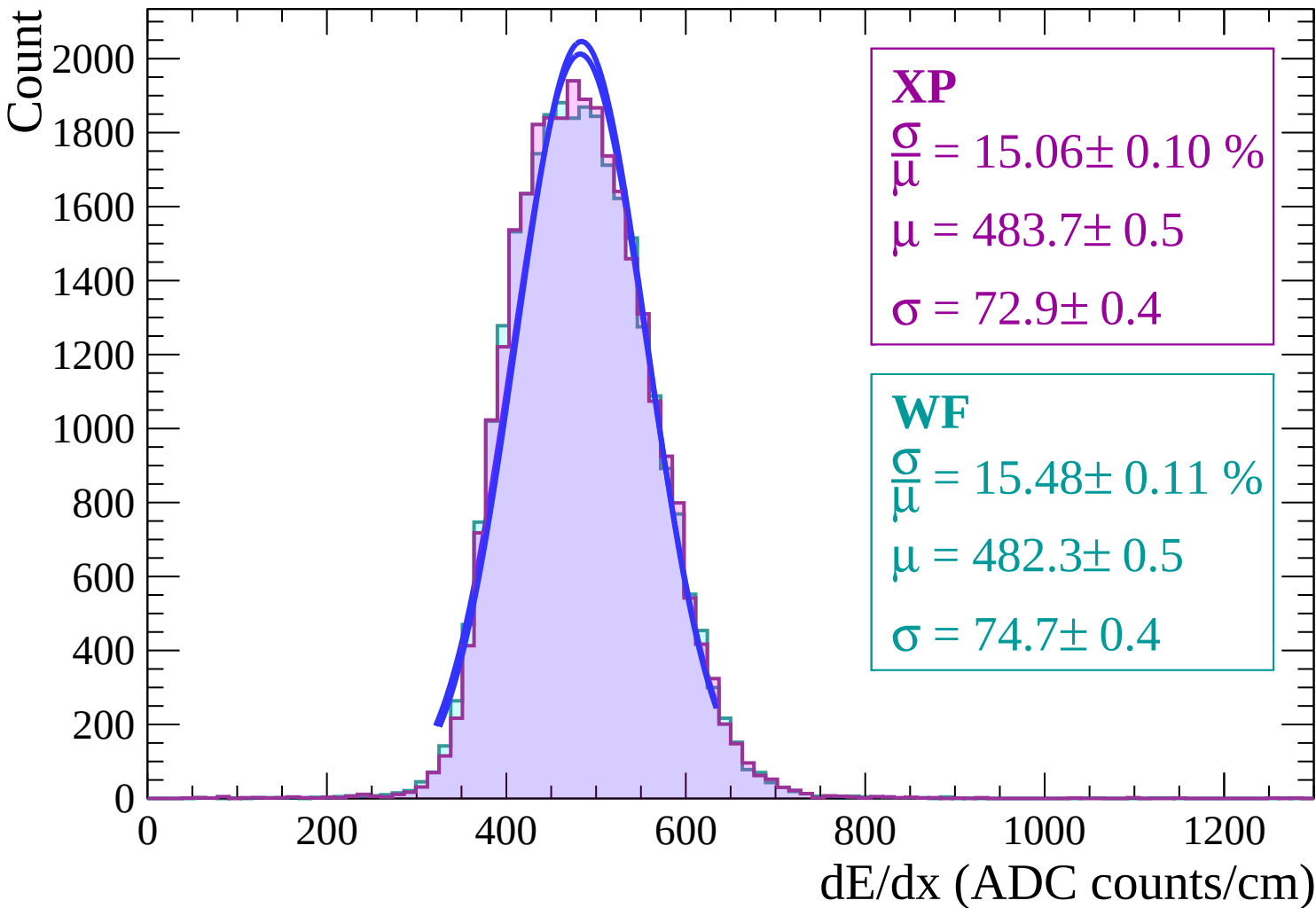
$$\frac{\sigma}{\mu} = 15.50 \pm 0.11 \%$$

$$\mu = 482.7 \pm 0.5$$

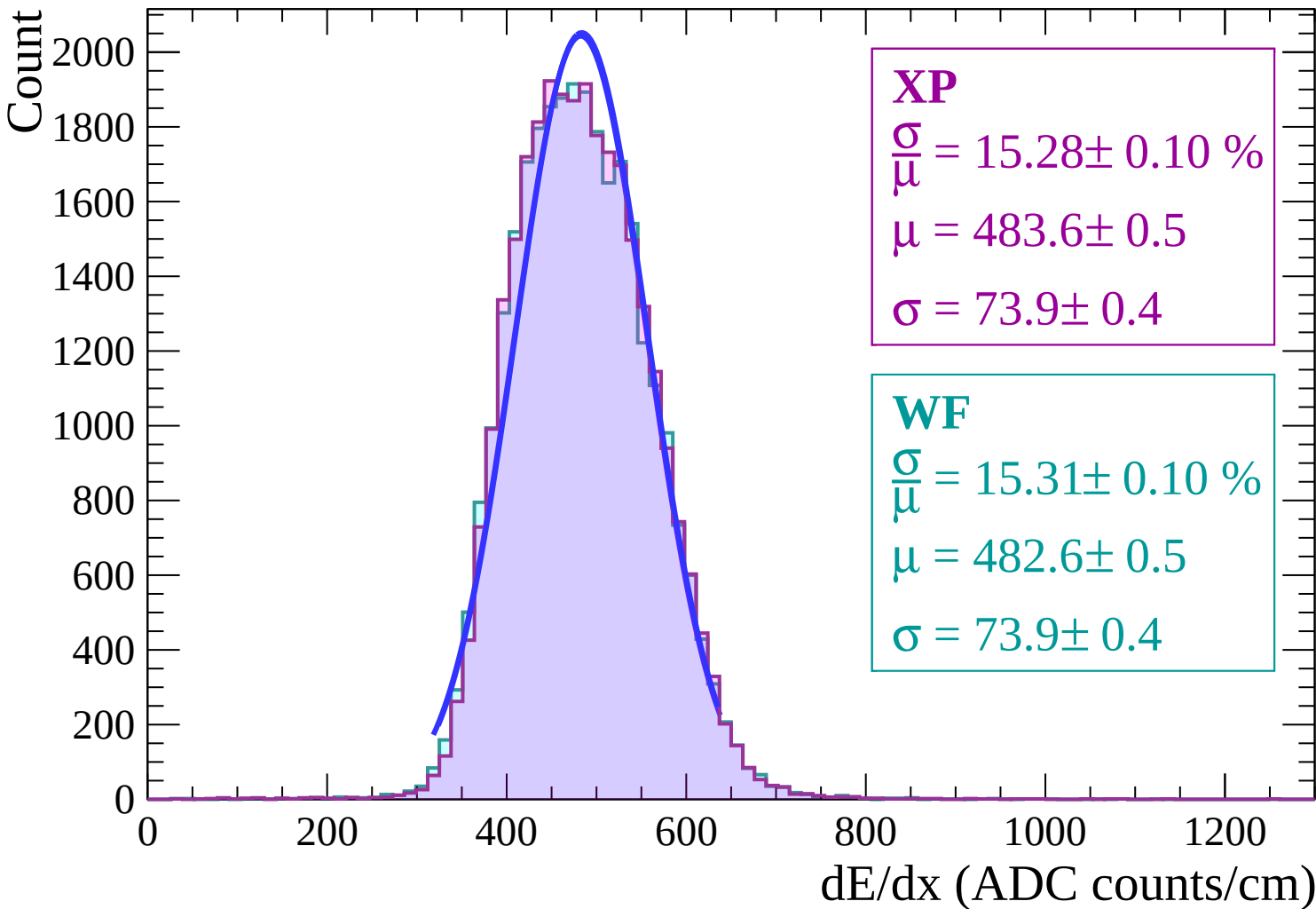
$$\sigma = 74.8 \pm 0.4$$

dE/dx (ADC counts/cm)

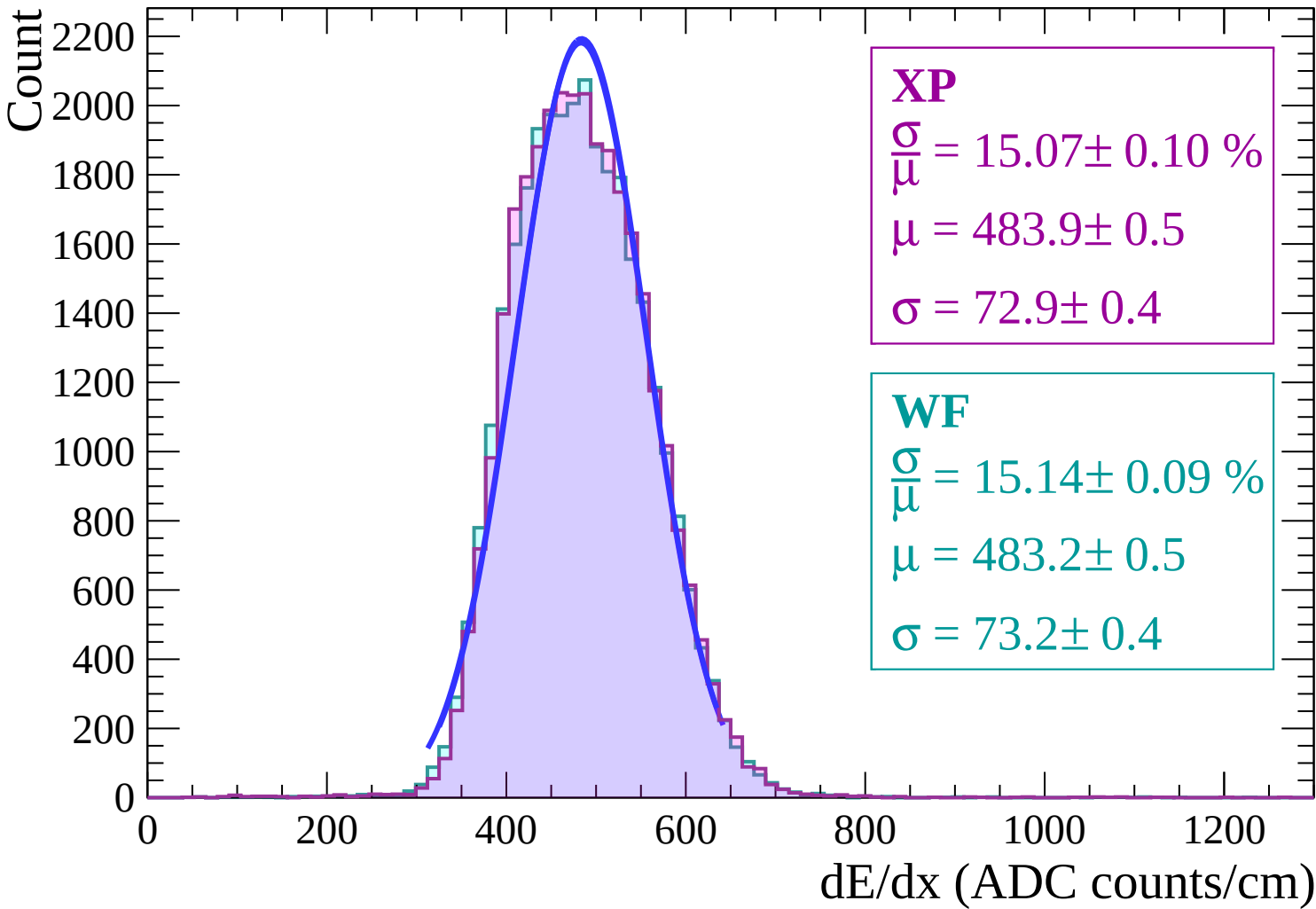
Energy loss | $-10 < \theta < -8$



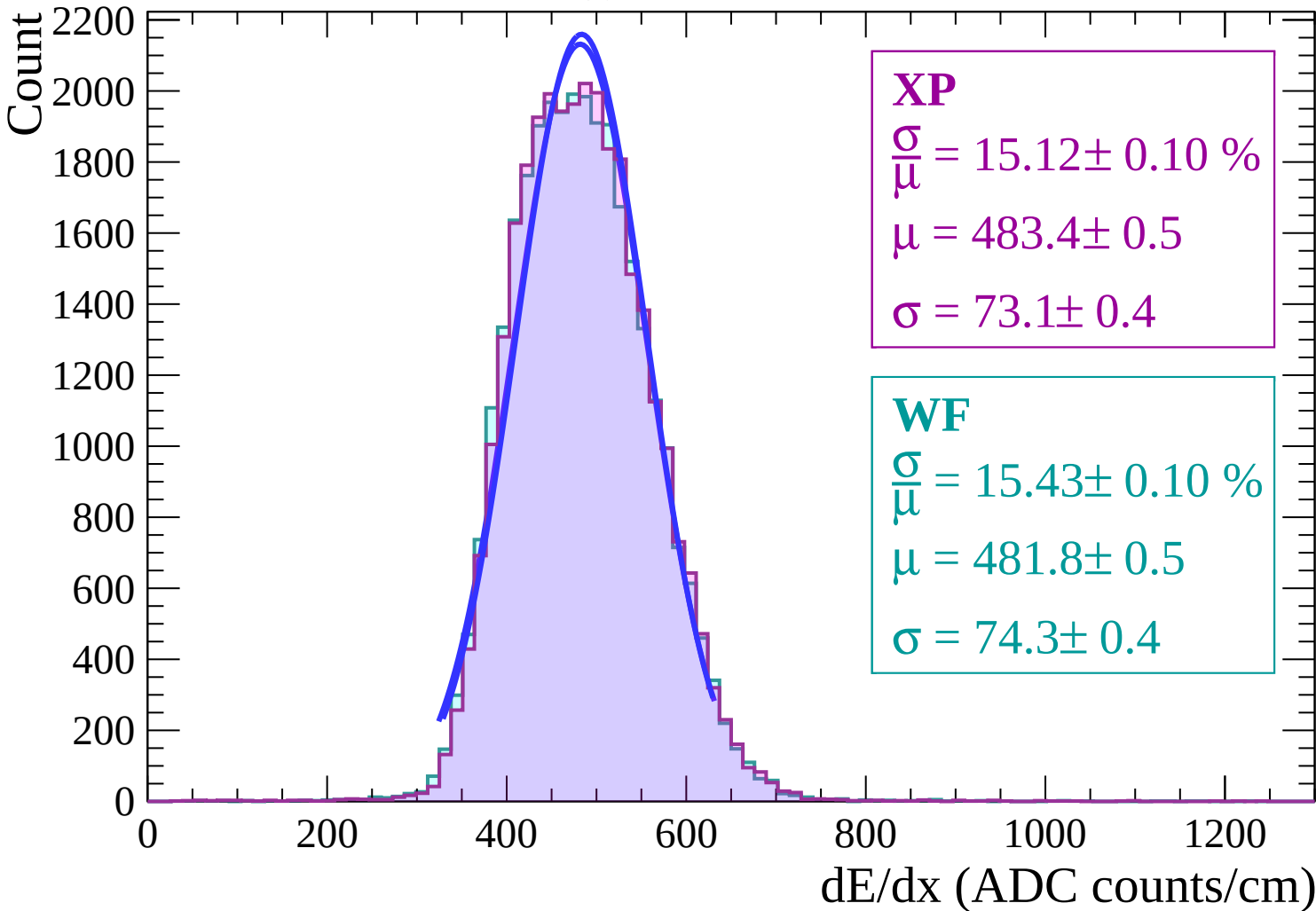
Energy loss | $-8 < \theta < -6$



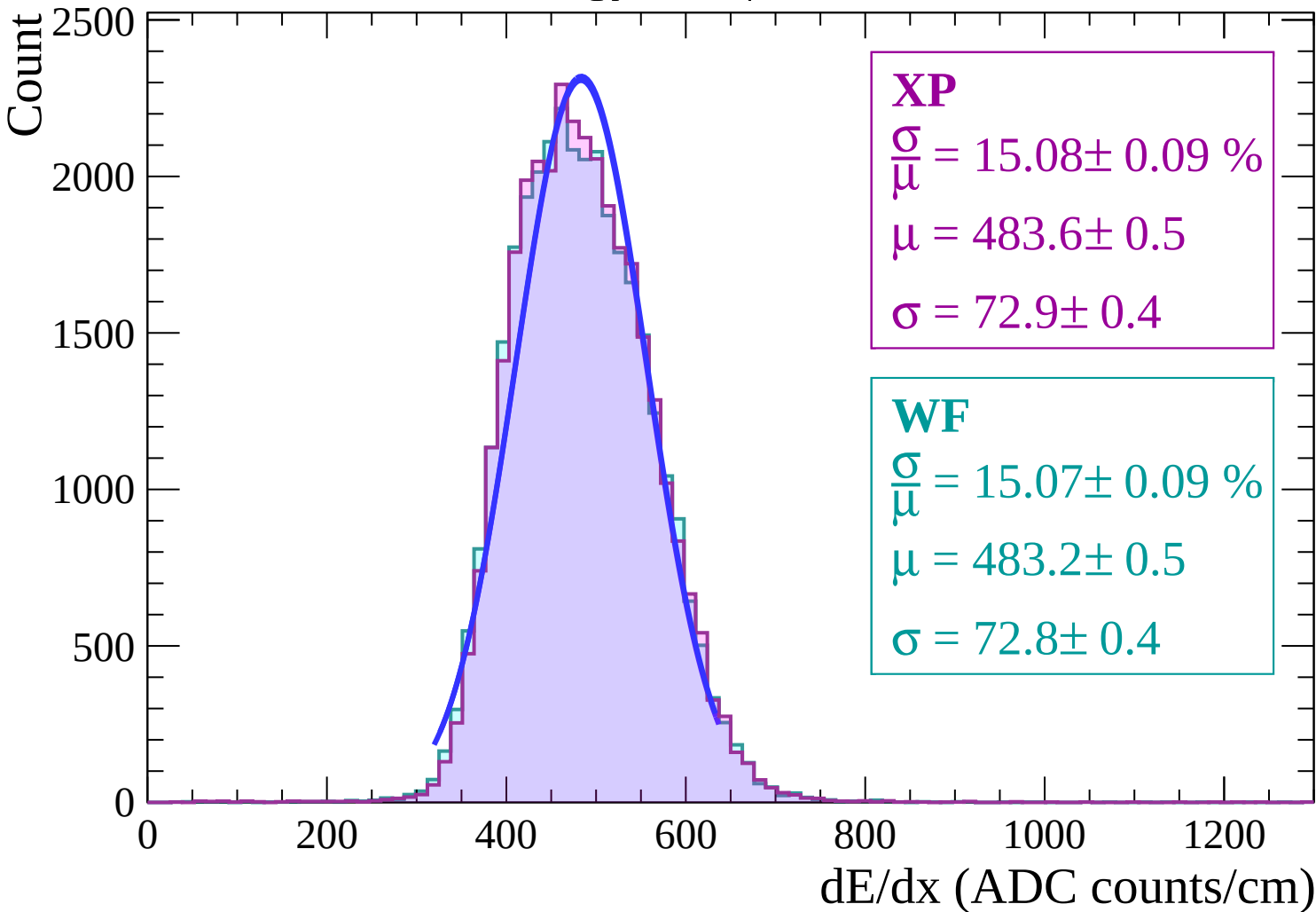
Energy loss | $-6 < \theta < -4$



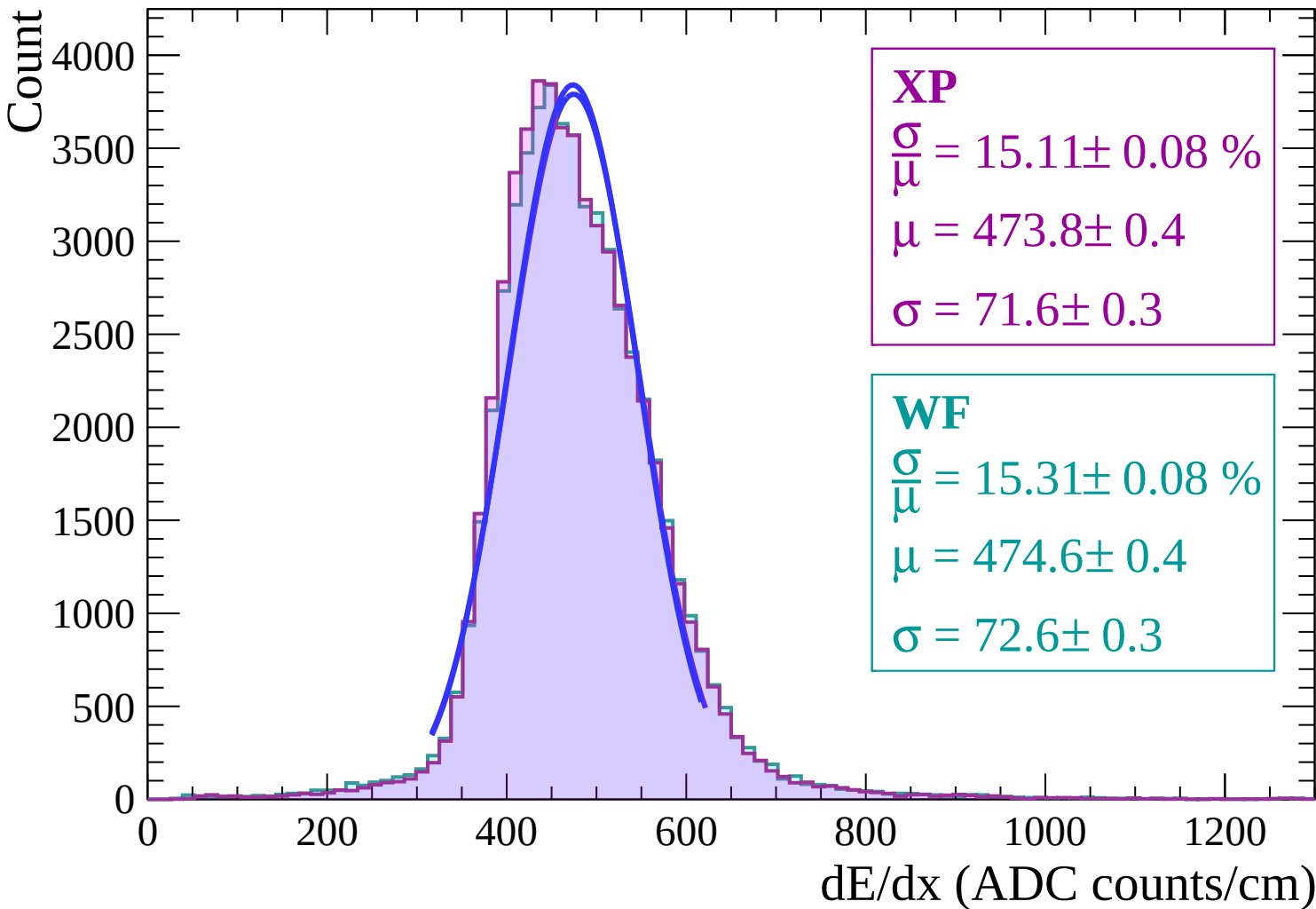
Energy loss | $-4 < \theta < -2$



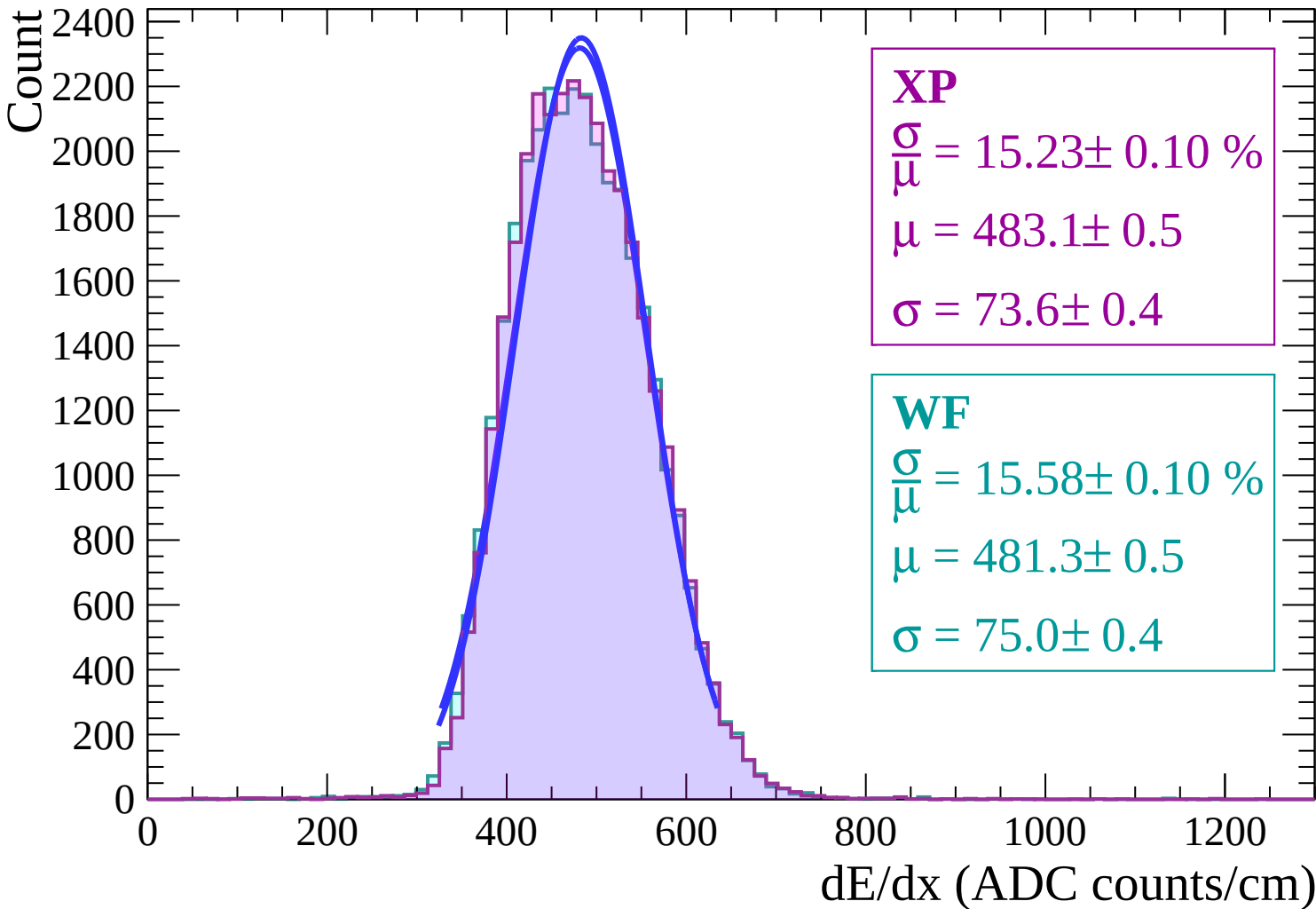
Energy loss | $-2 < \theta < 0$



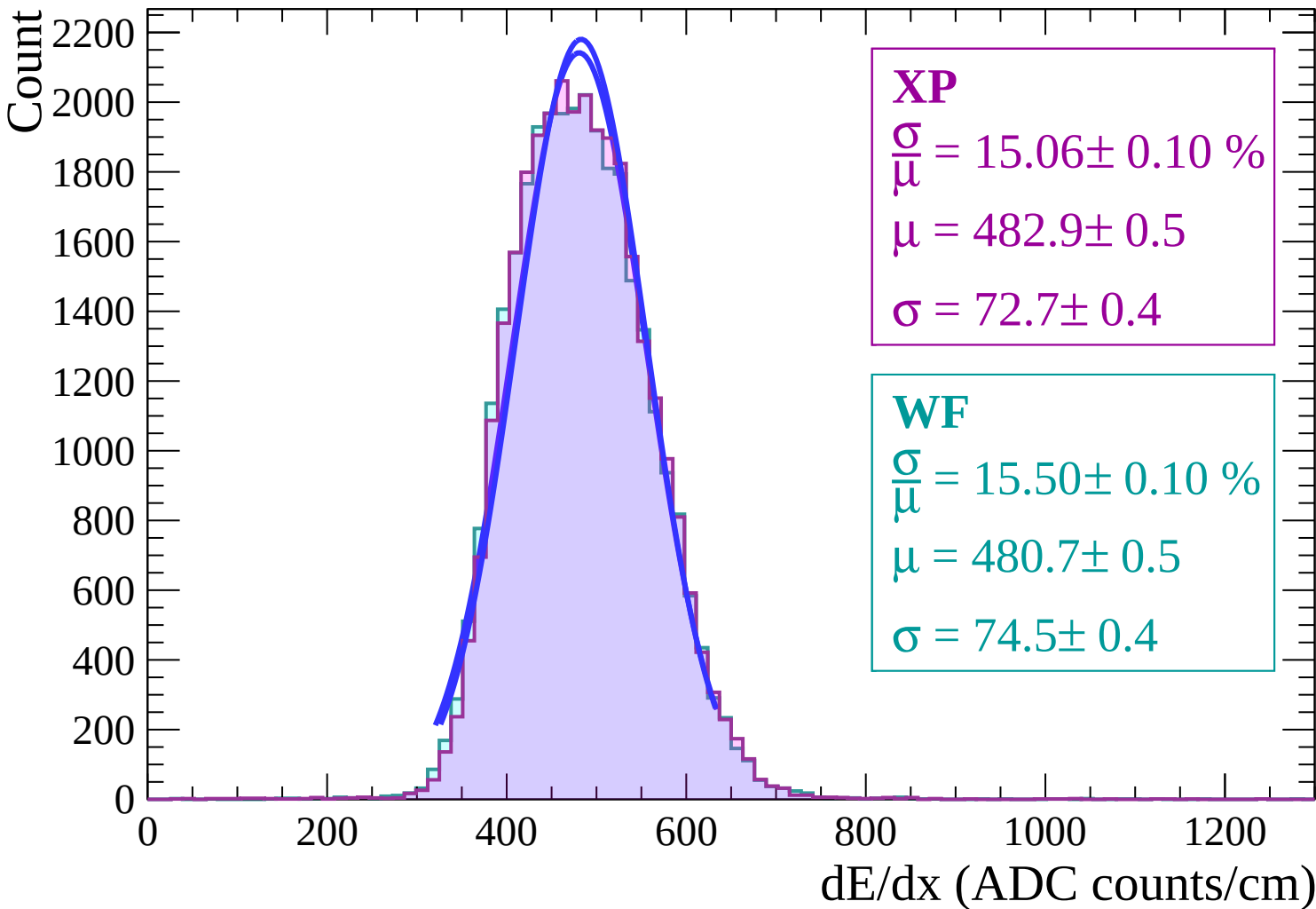
Energy loss | $0 < \theta < 2$



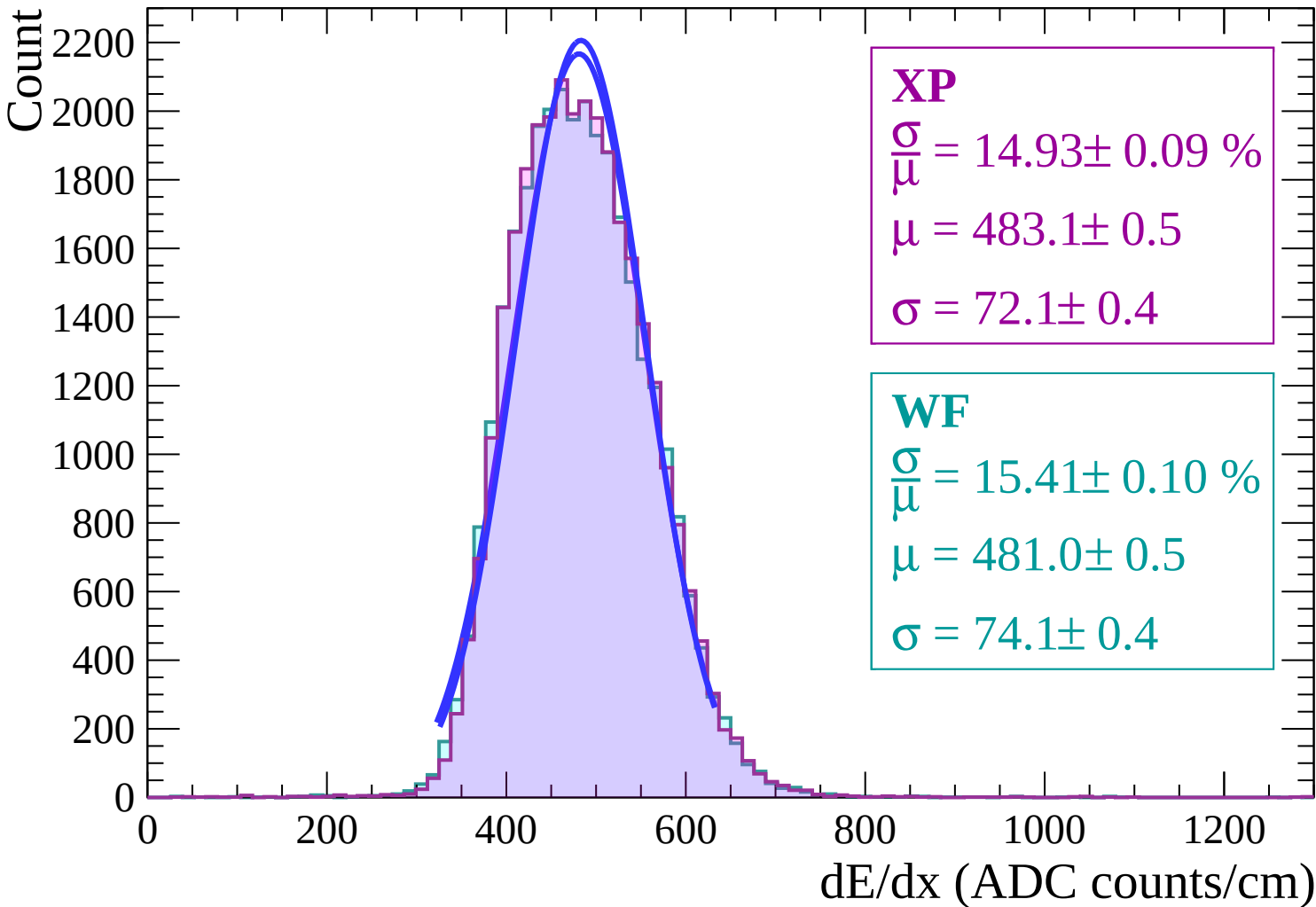
Energy loss | $2 < \theta < 4$



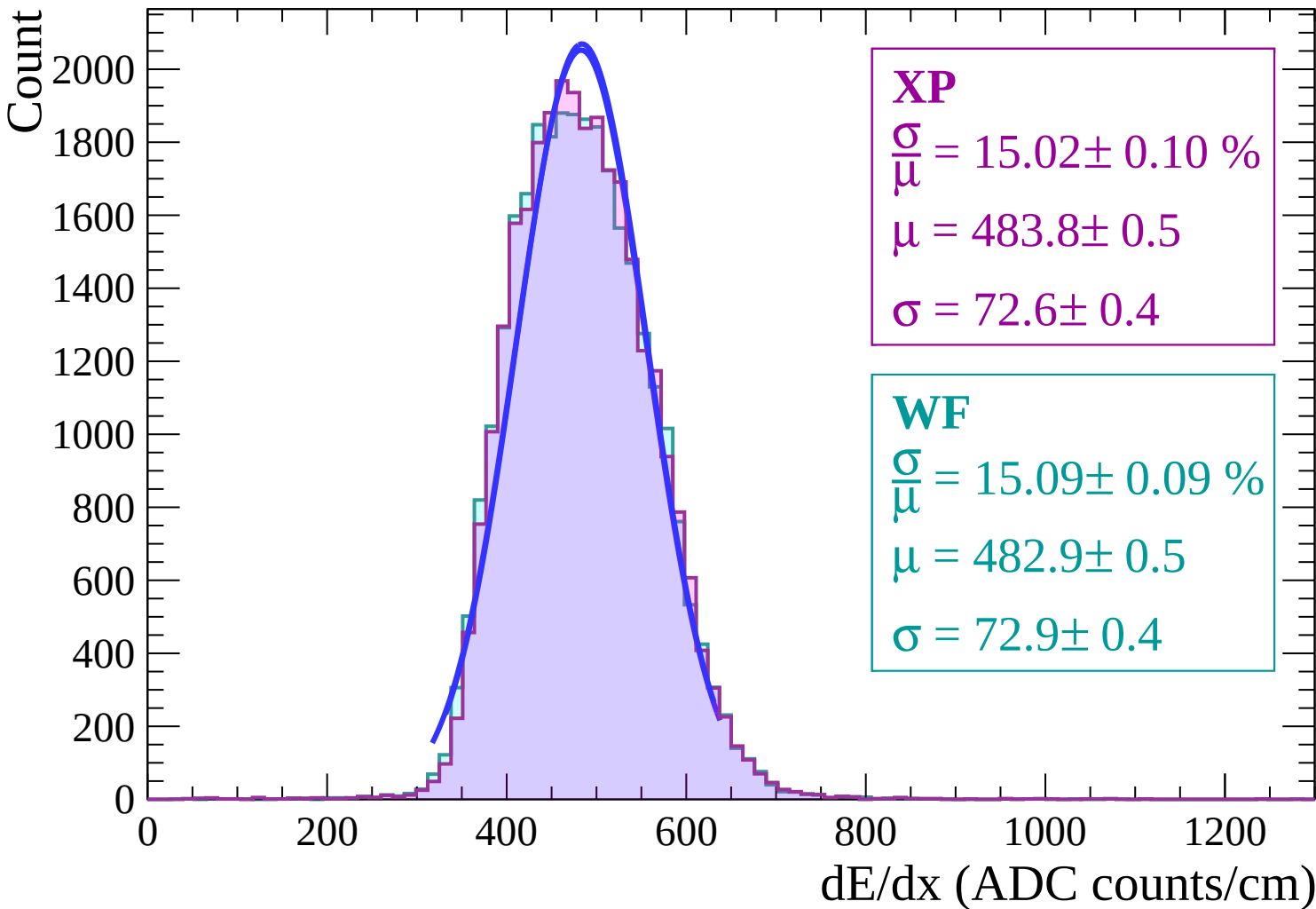
Energy loss | $4 < \theta < 6$



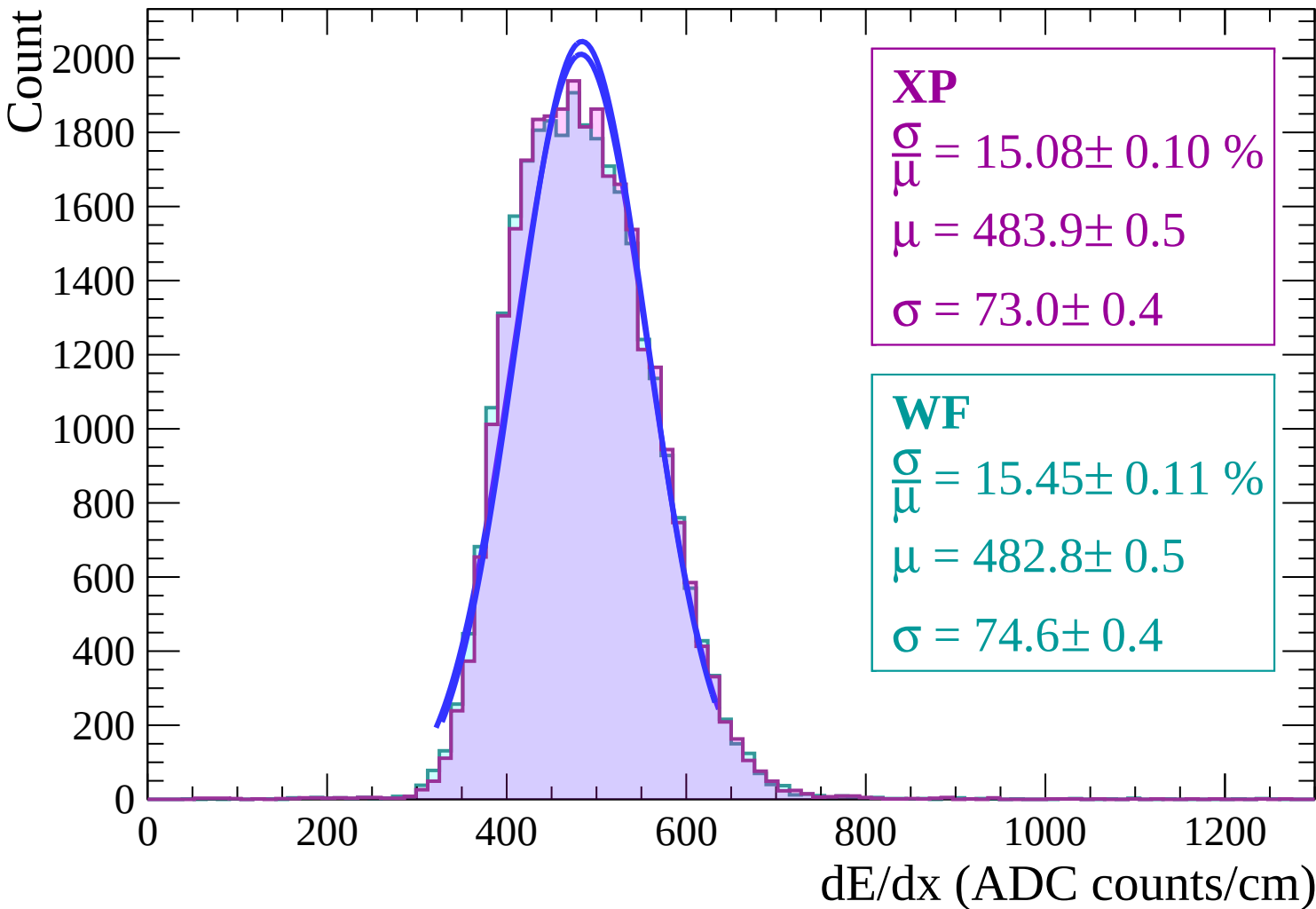
Energy loss | $6 < \theta < 8$



Energy loss | $8 < \theta < 10$

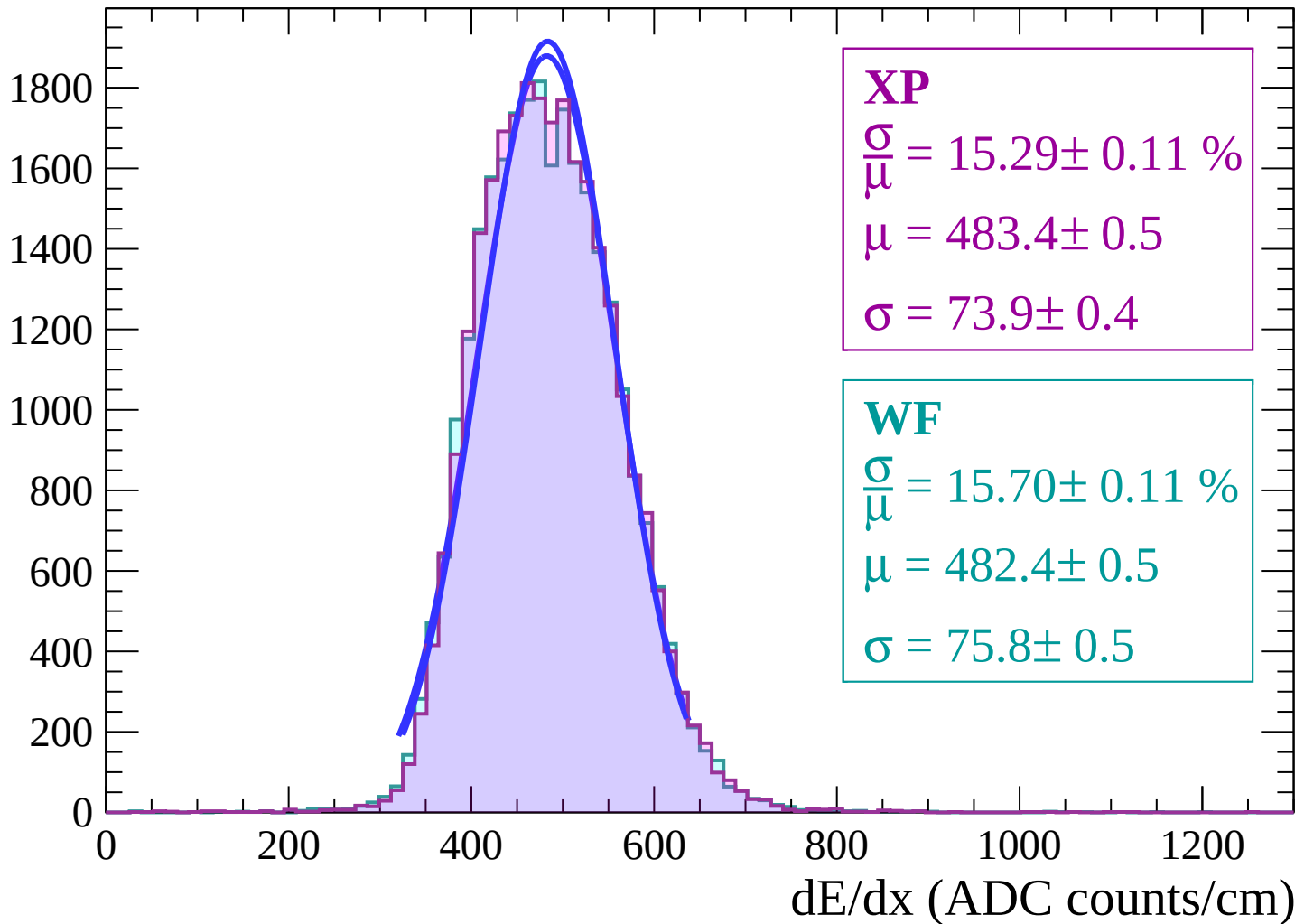


Energy loss | $10 < \theta < 12$

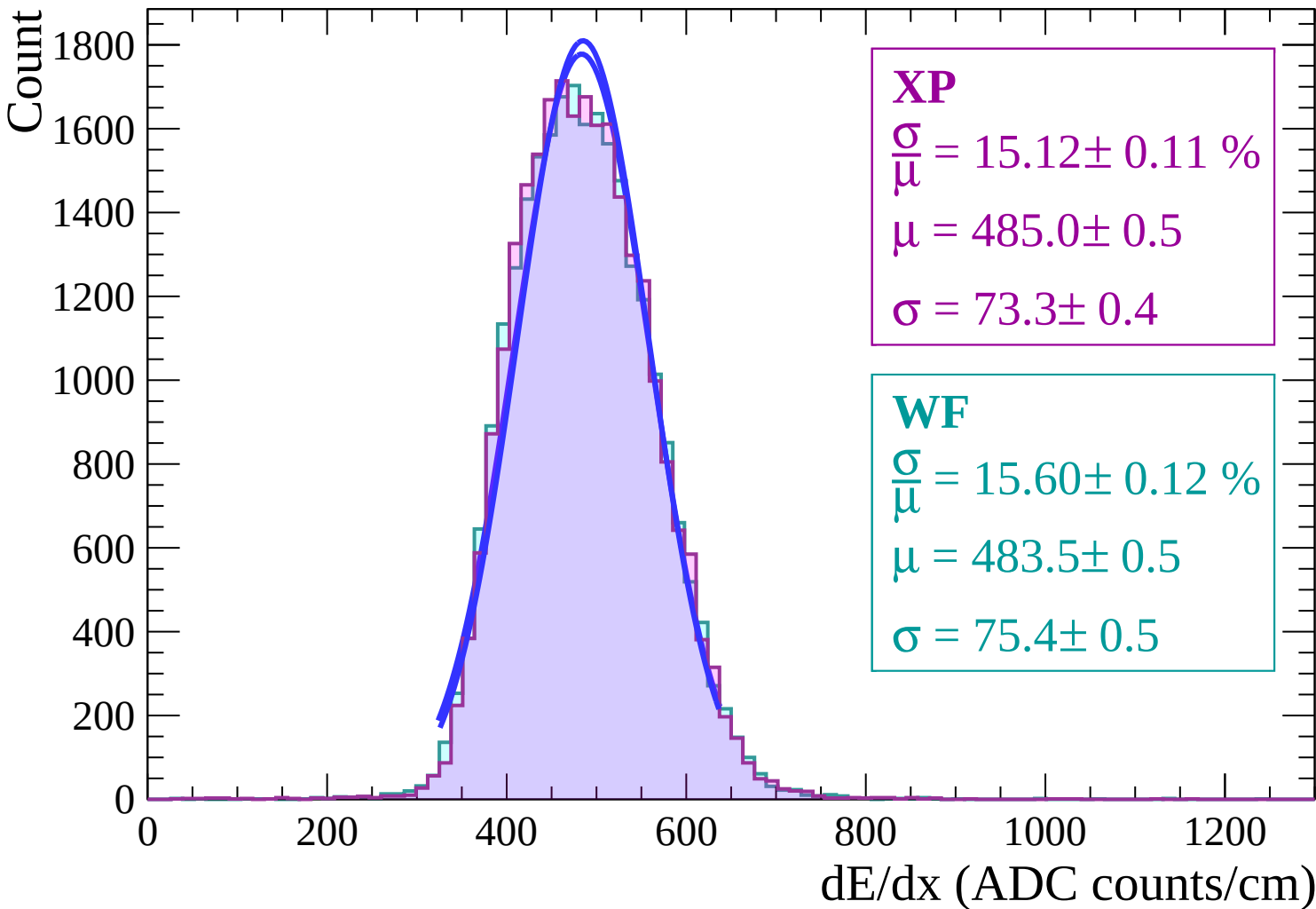


Energy loss | $12 < \theta < 14$

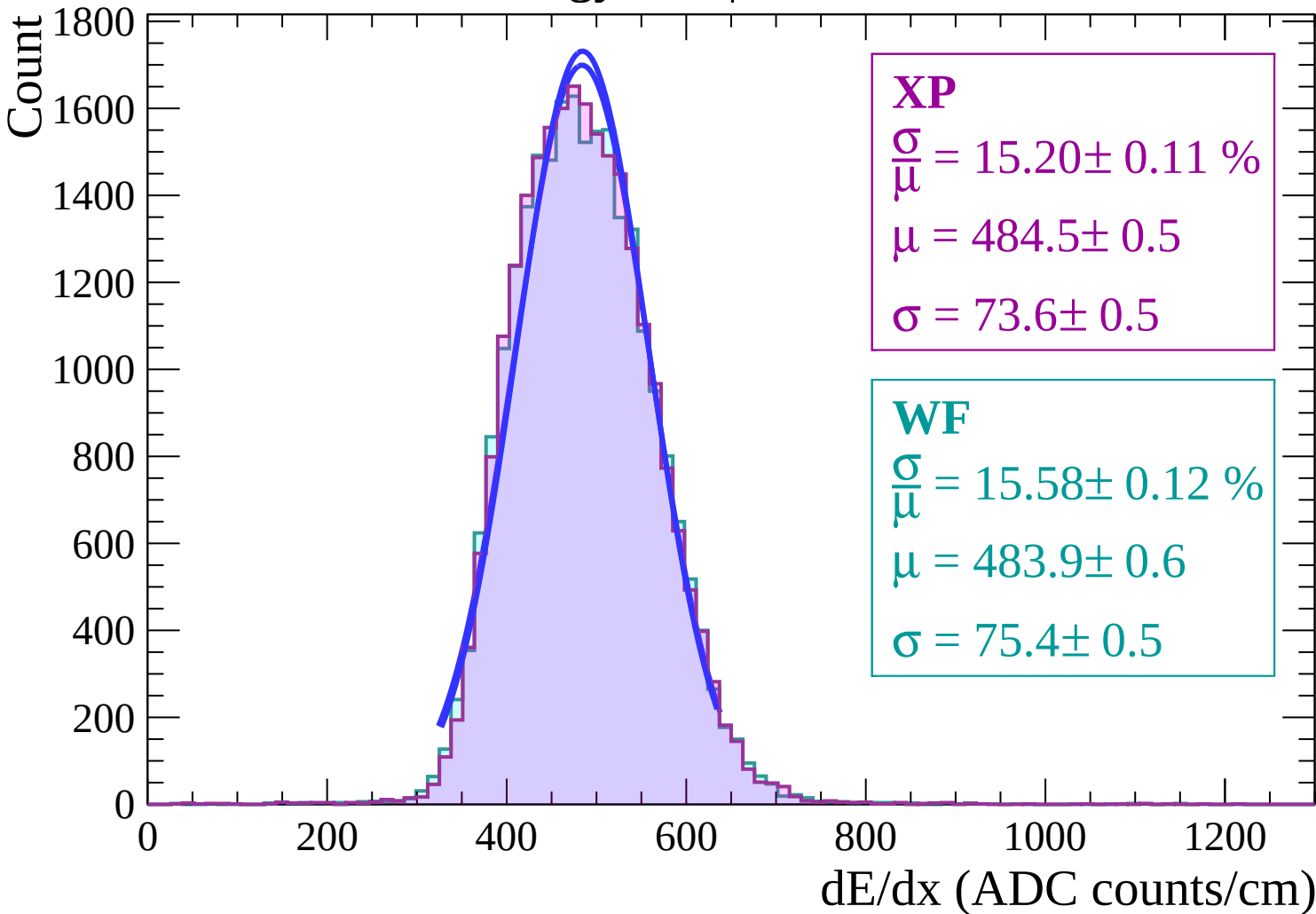
Count



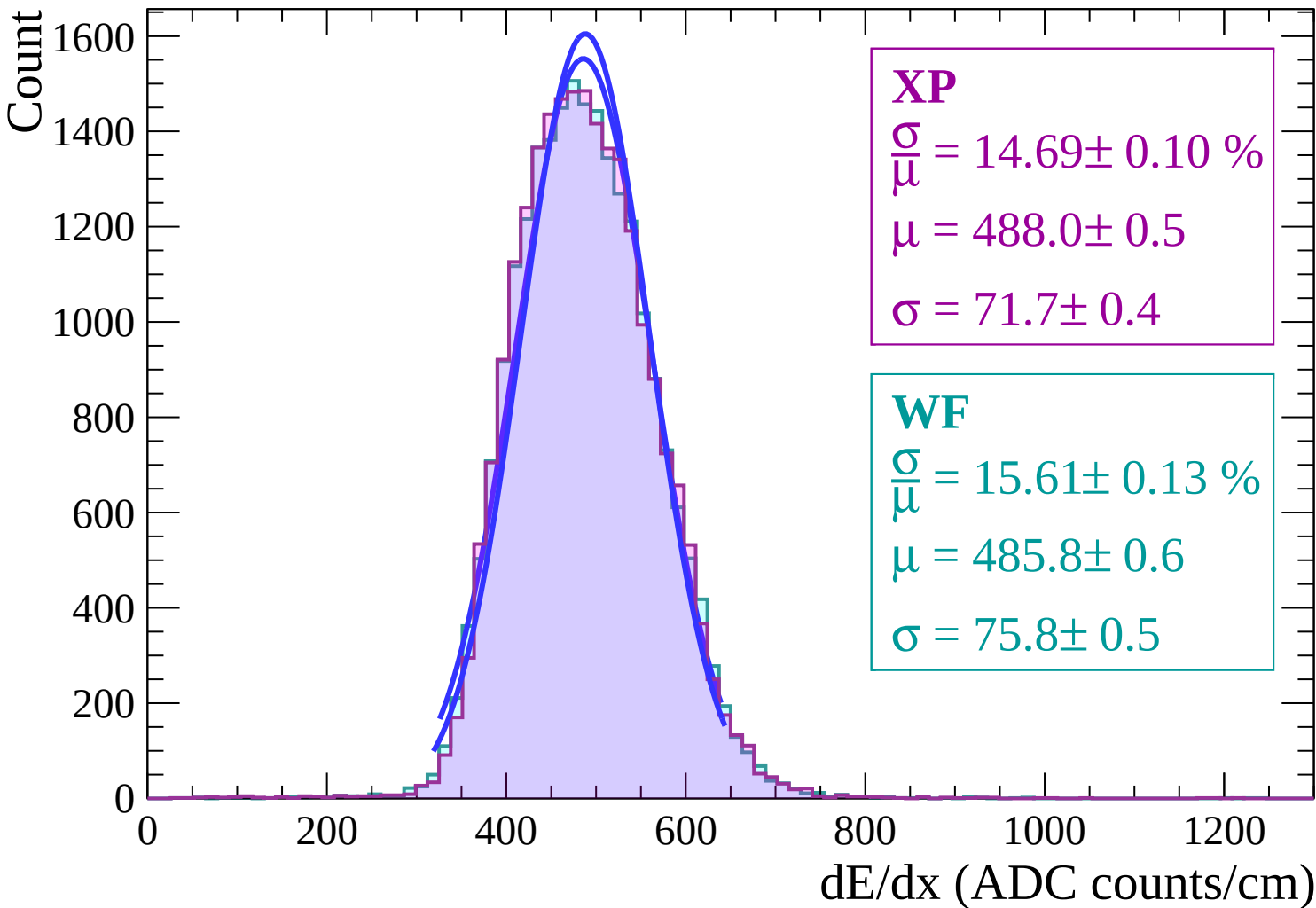
Energy loss | $14 < \theta < 16$



Energy loss | $16 < \theta < 18$



Energy loss | $18 < \theta < 20$



Energy loss | $20 < \theta < 22$

Count

1400
1200
1000
800
600
400
200
0

XP

$$\frac{\sigma}{\mu} = 15.04 \pm 0.12 \%$$

$$\mu = 487.2 \pm 0.6$$

$$\sigma = 73.3 \pm 0.5$$

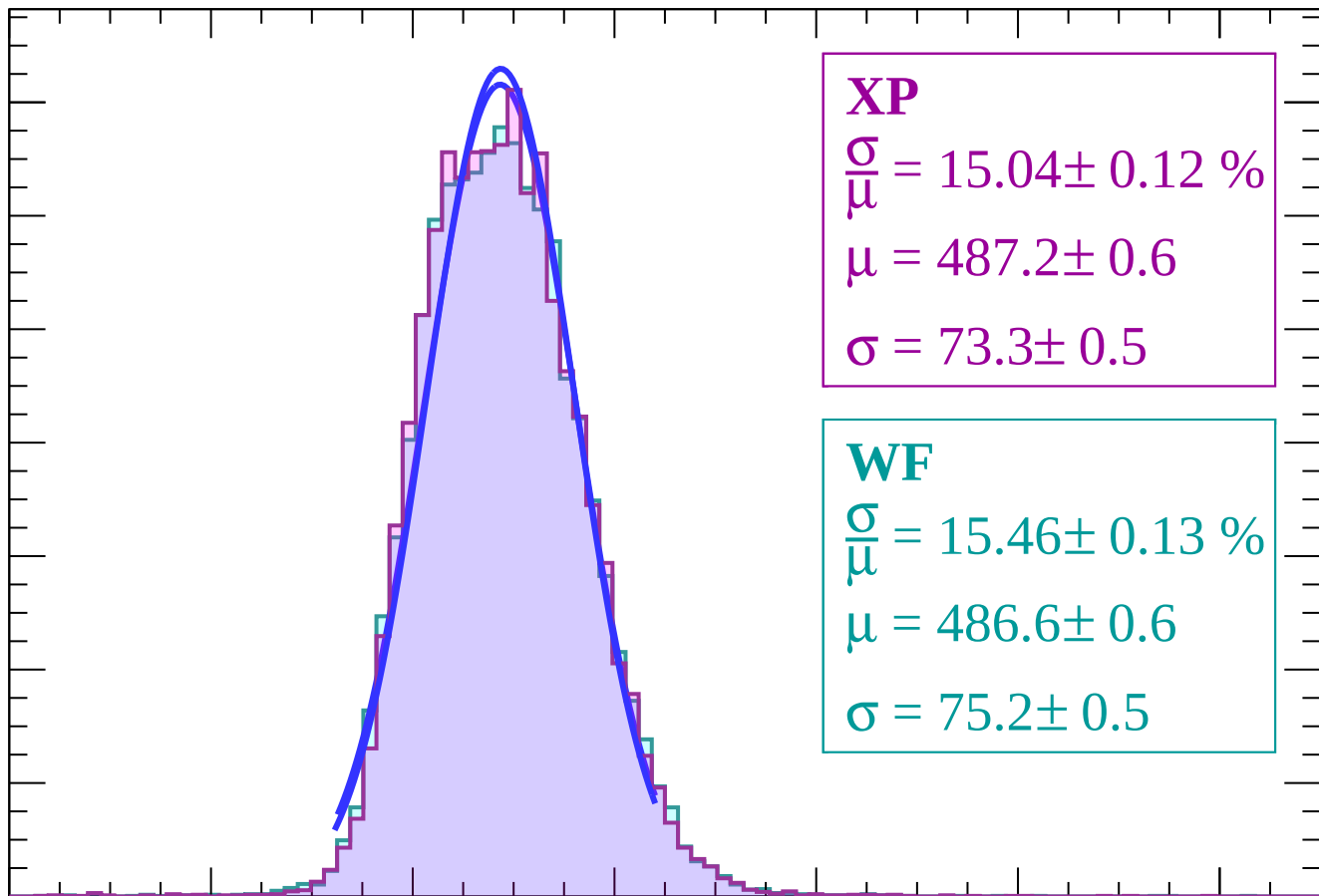
WF

$$\frac{\sigma}{\mu} = 15.46 \pm 0.13 \%$$

$$\mu = 486.6 \pm 0.6$$

$$\sigma = 75.2 \pm 0.5$$

0 200 400 600 800 1000 1200
dE/dx (ADC counts/cm)



Energy loss | $22 < \theta < 24$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP

$$\frac{\sigma}{\mu} = 15.48 \pm 0.13 \%$$

$$\mu = 488.7 \pm 0.6$$

$$\sigma = 75.6 \pm 0.5$$

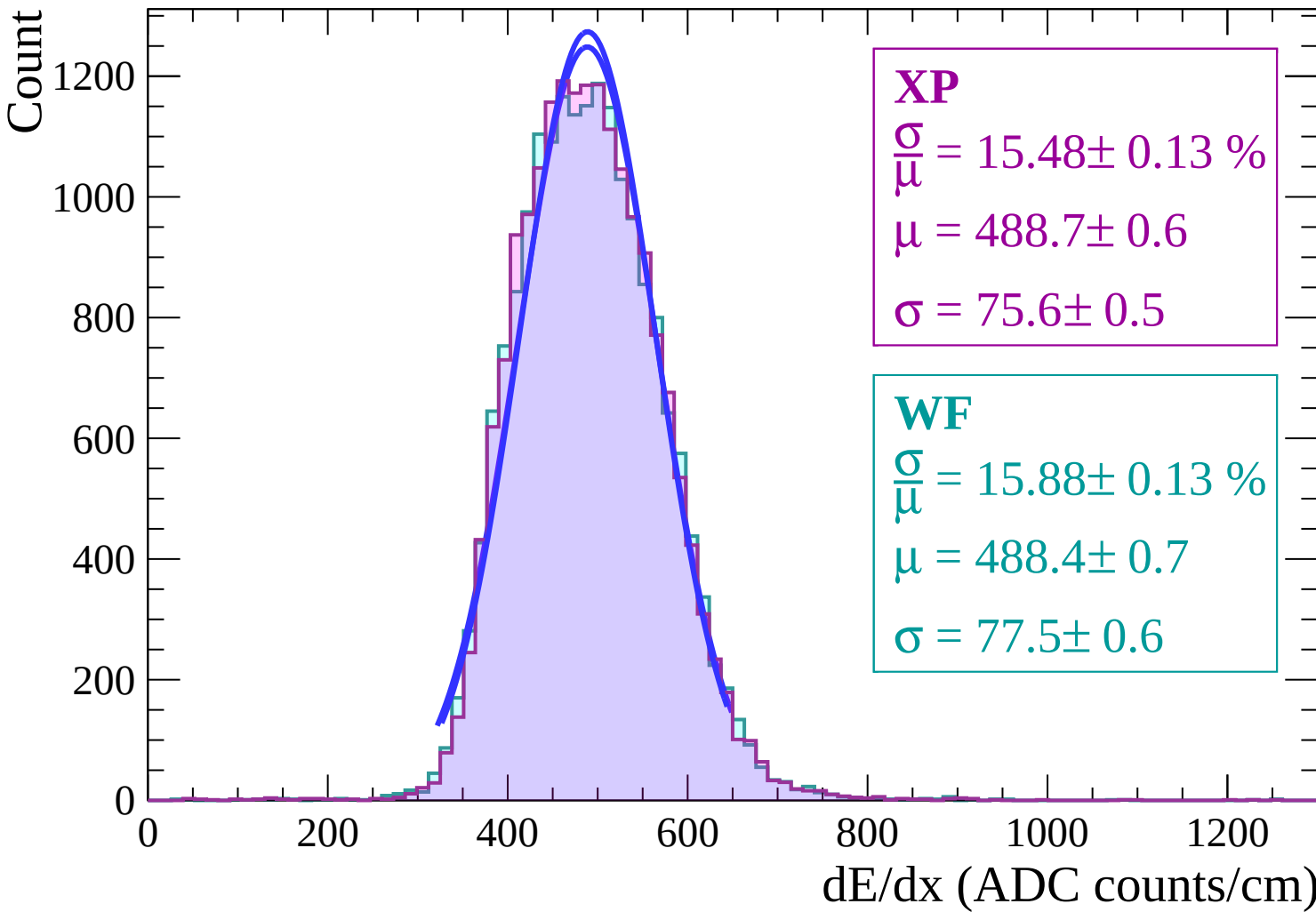
WF

$$\frac{\sigma}{\mu} = 15.88 \pm 0.13 \%$$

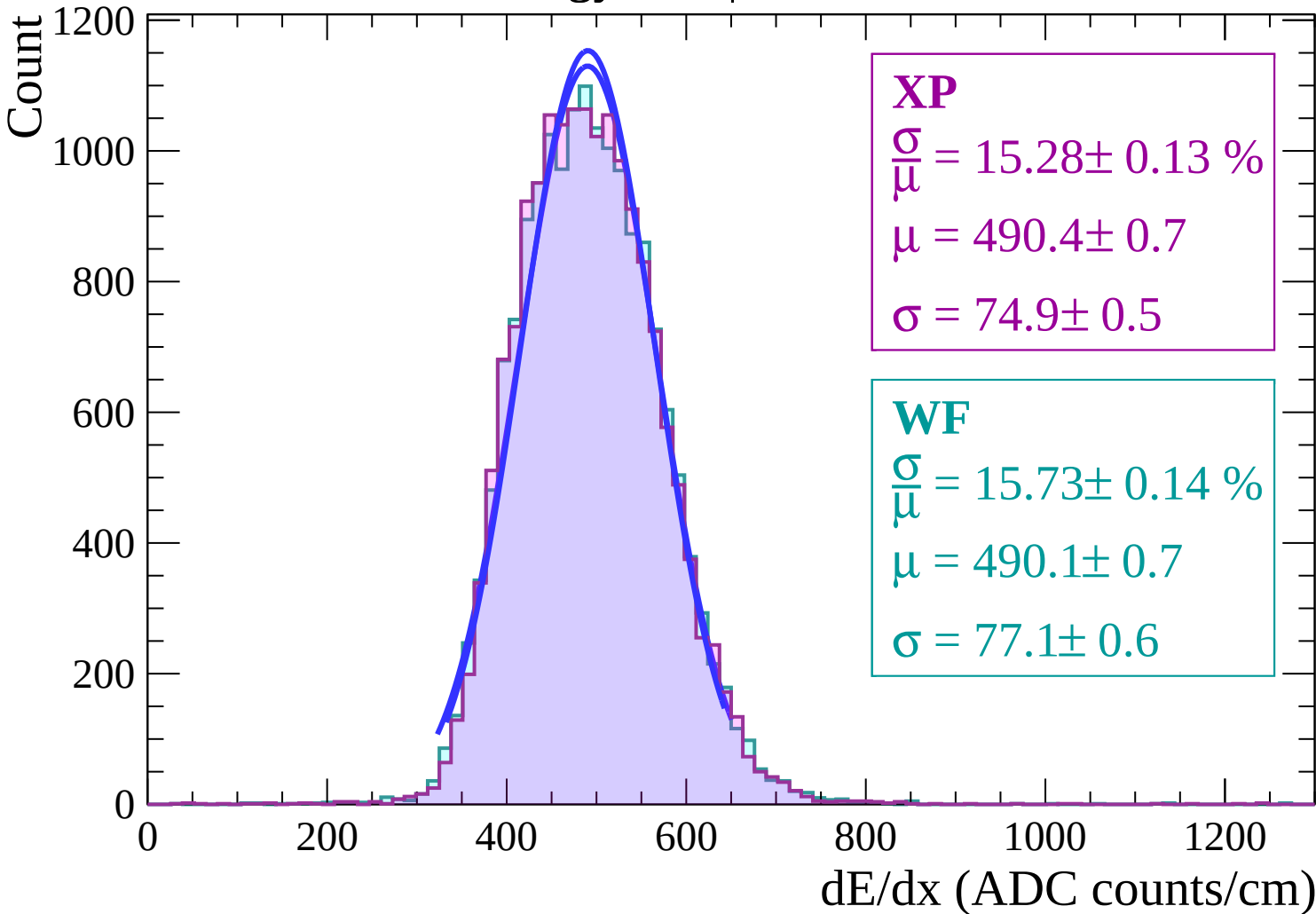
$$\mu = 488.4 \pm 0.7$$

$$\sigma = 77.5 \pm 0.6$$

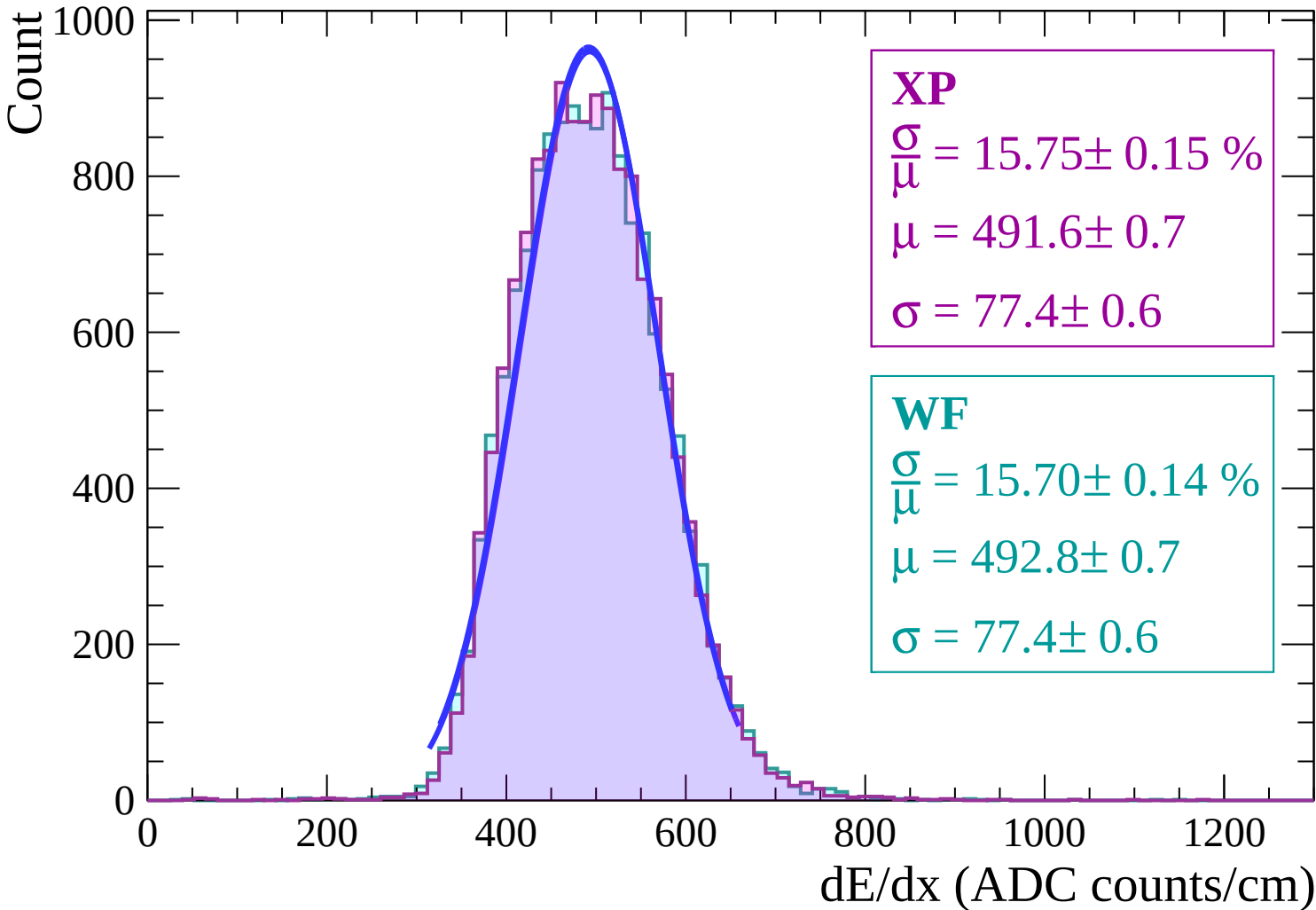
dE/dx (ADC counts/cm)



Energy loss | $24 < \theta < 26$

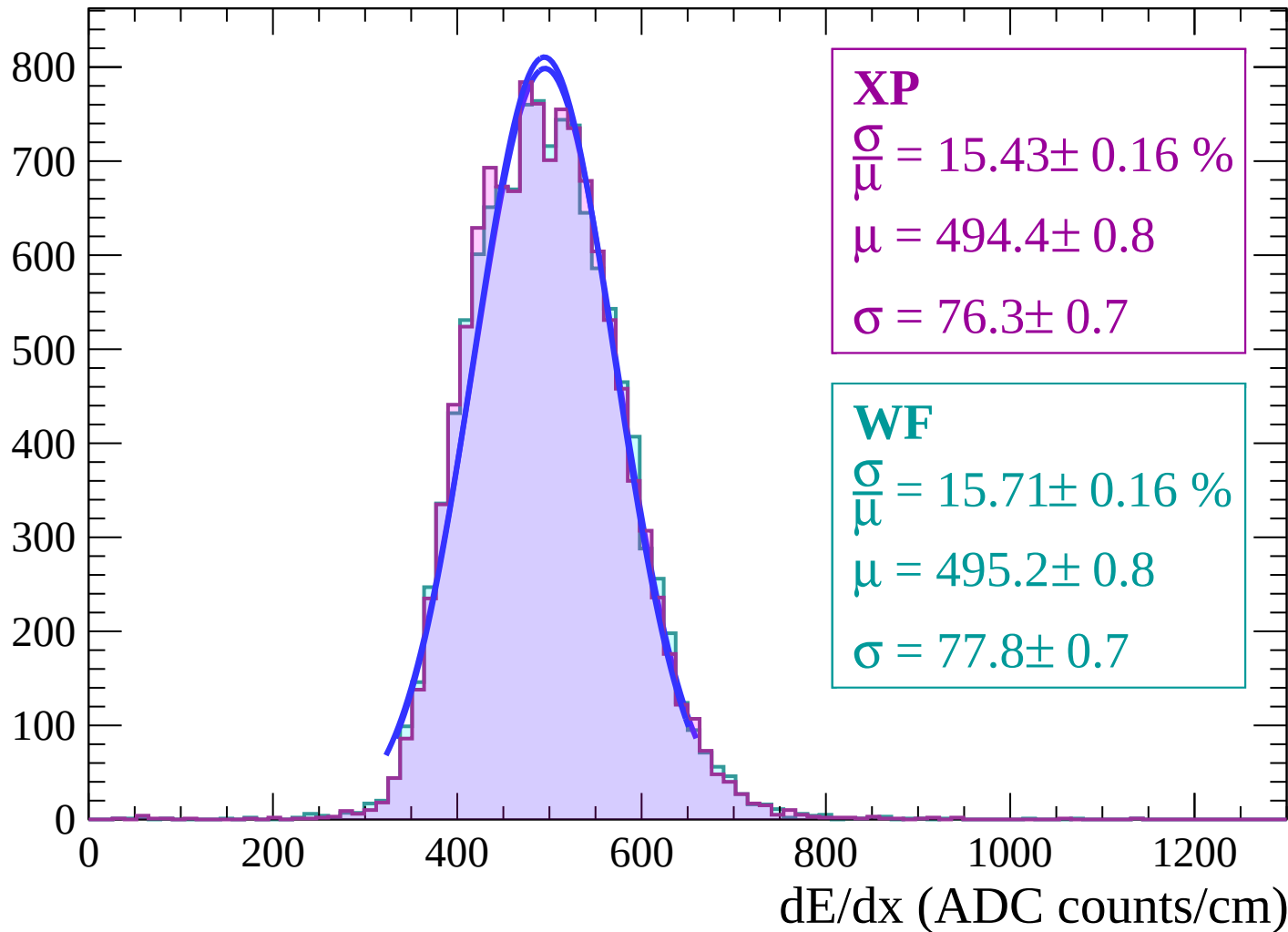


Energy loss | $26 < \theta < 28$



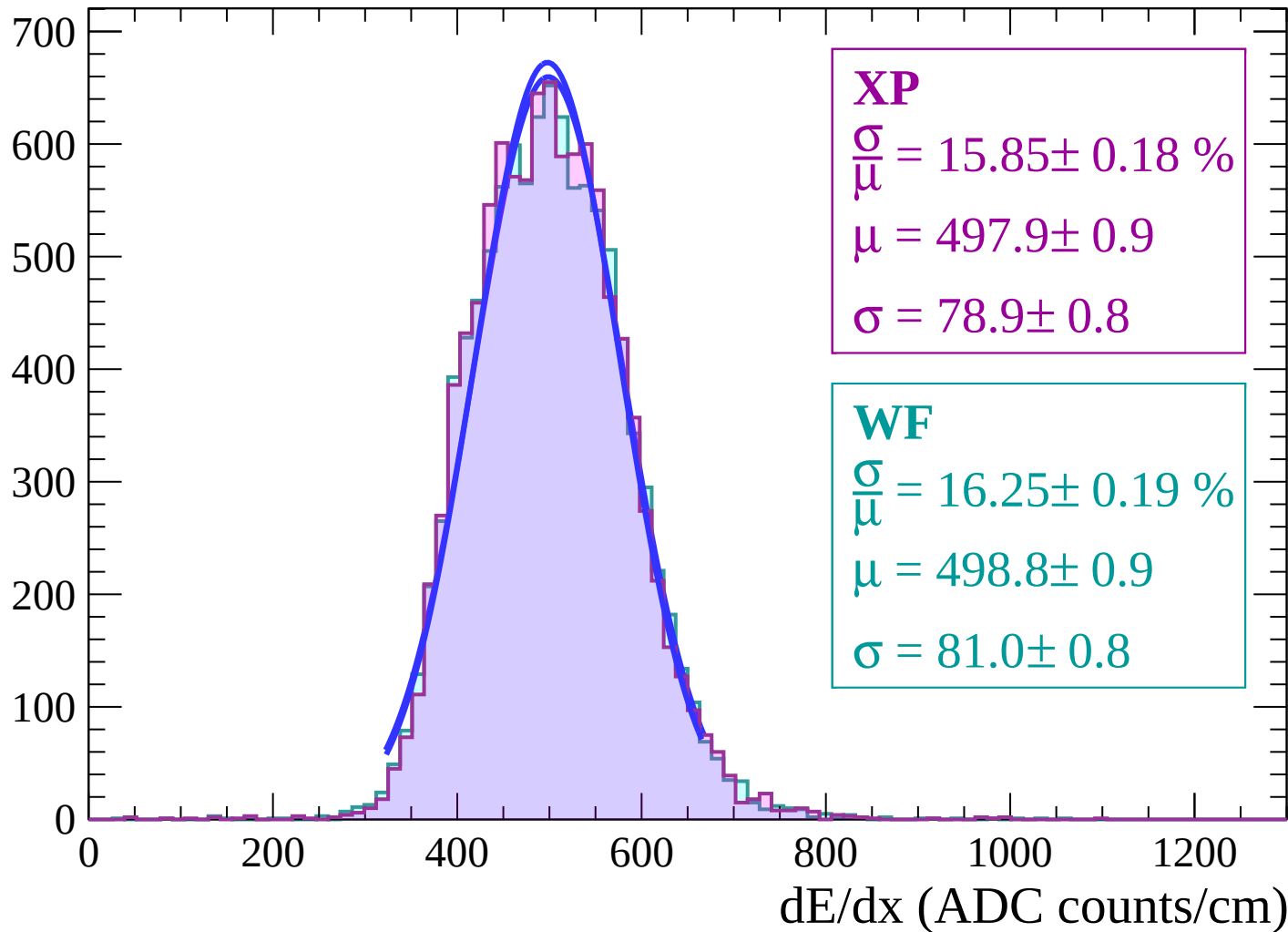
Energy loss | $28 < \theta < 30$

Count



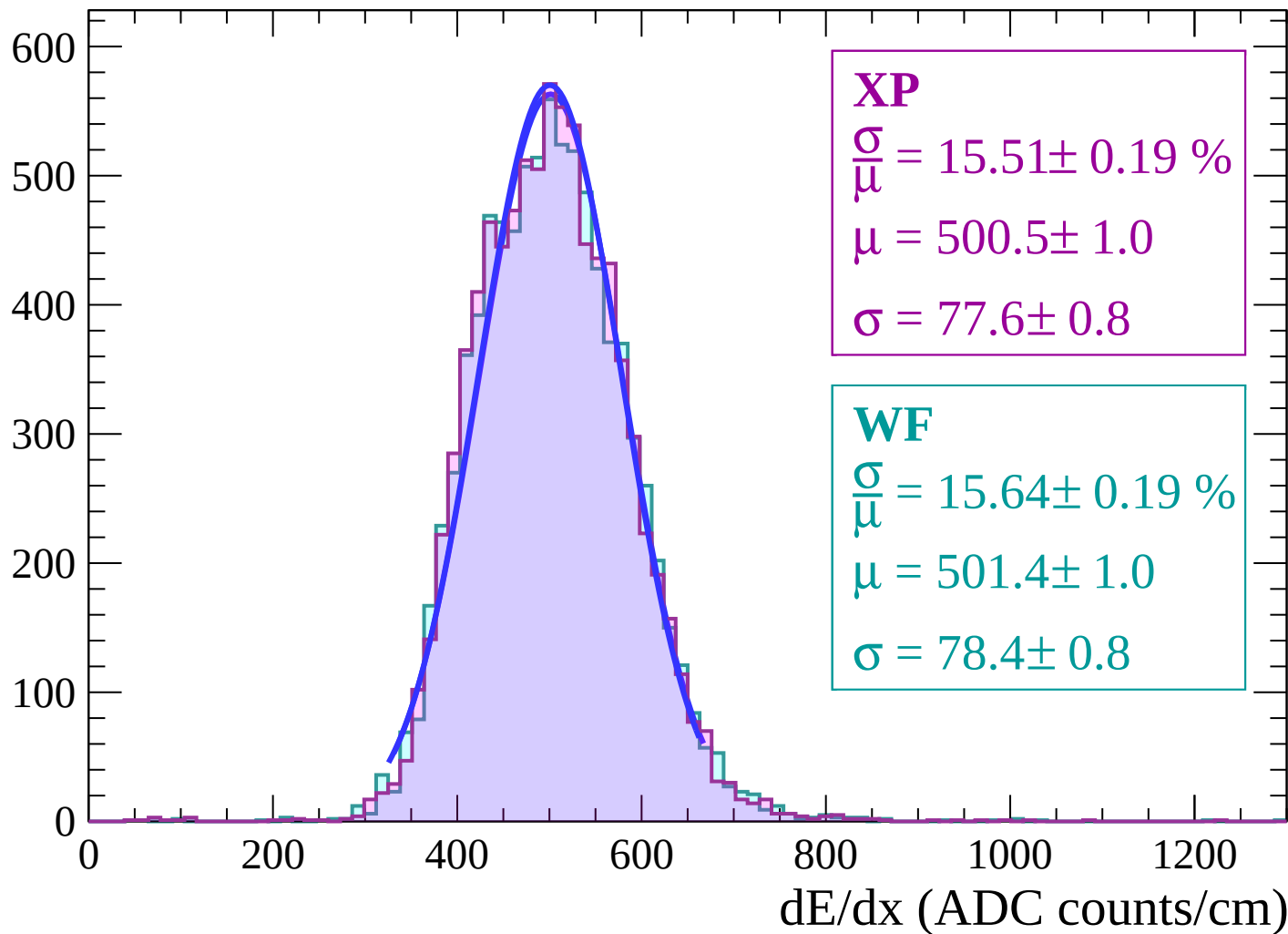
Energy loss | $30 < \theta < 32$

Count



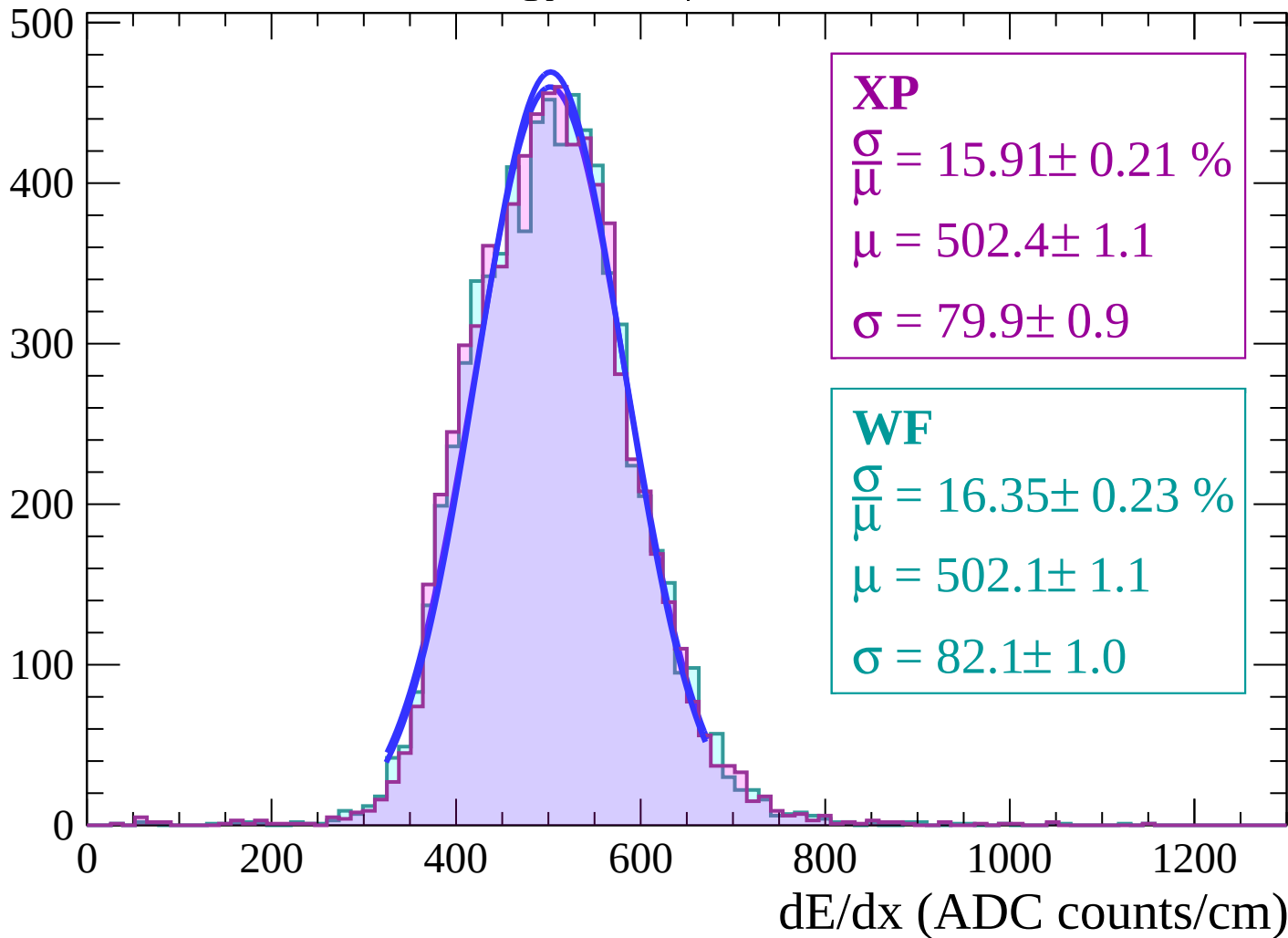
Energy loss | $32 < \theta < 34$

Count



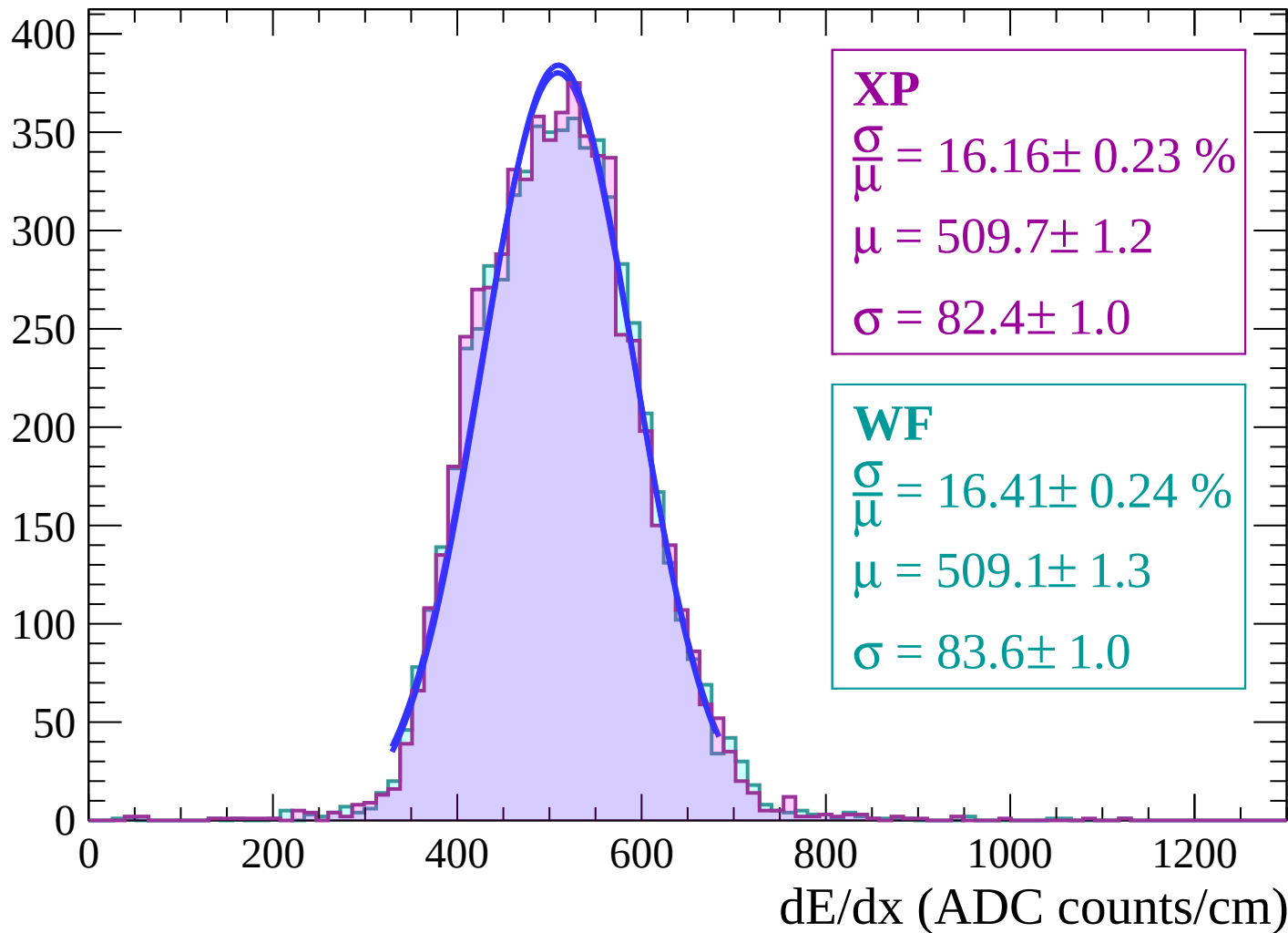
Energy loss | $34 < \theta < 36$

Count



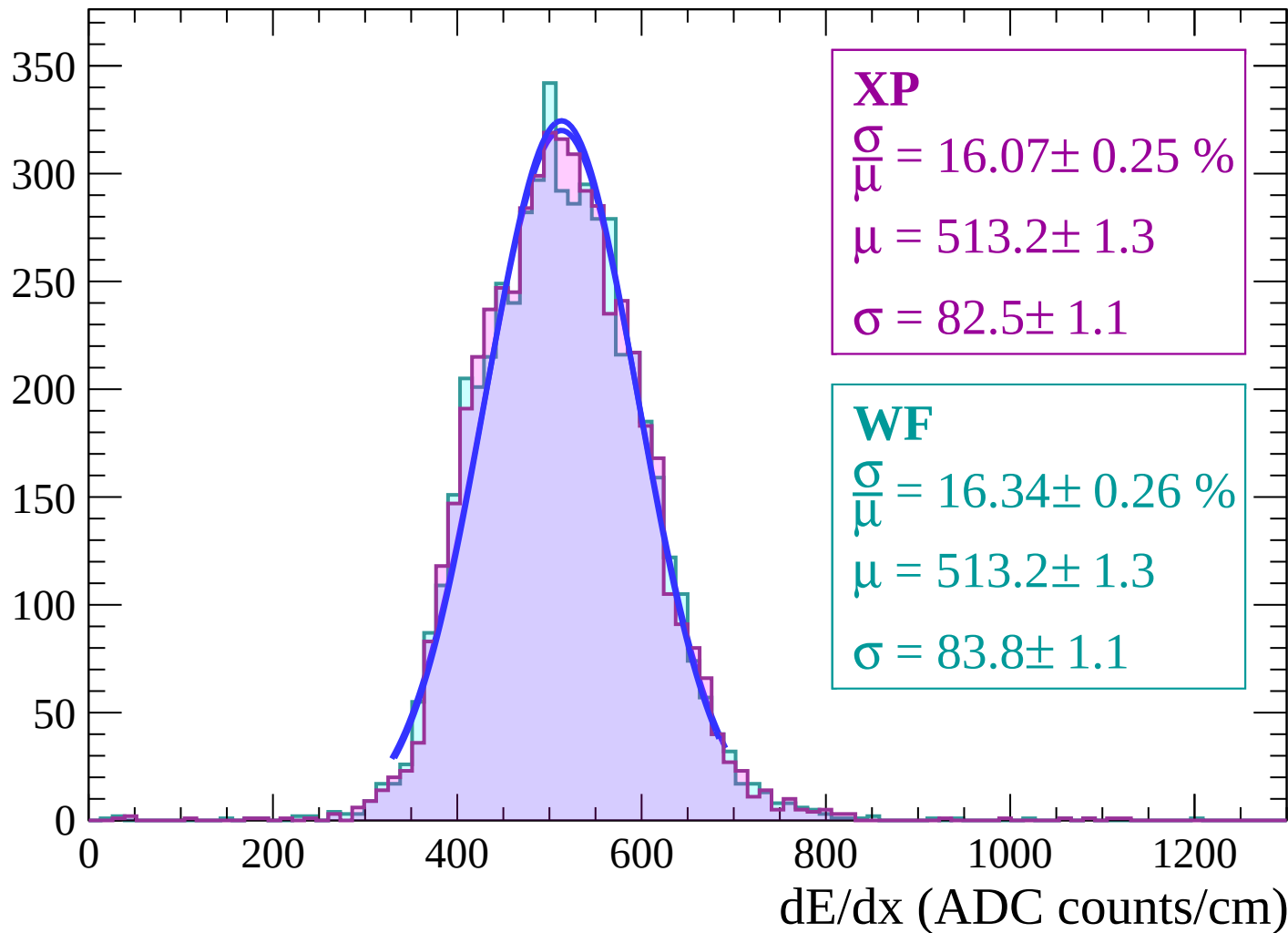
Energy loss | $36 < \theta < 38$

Count



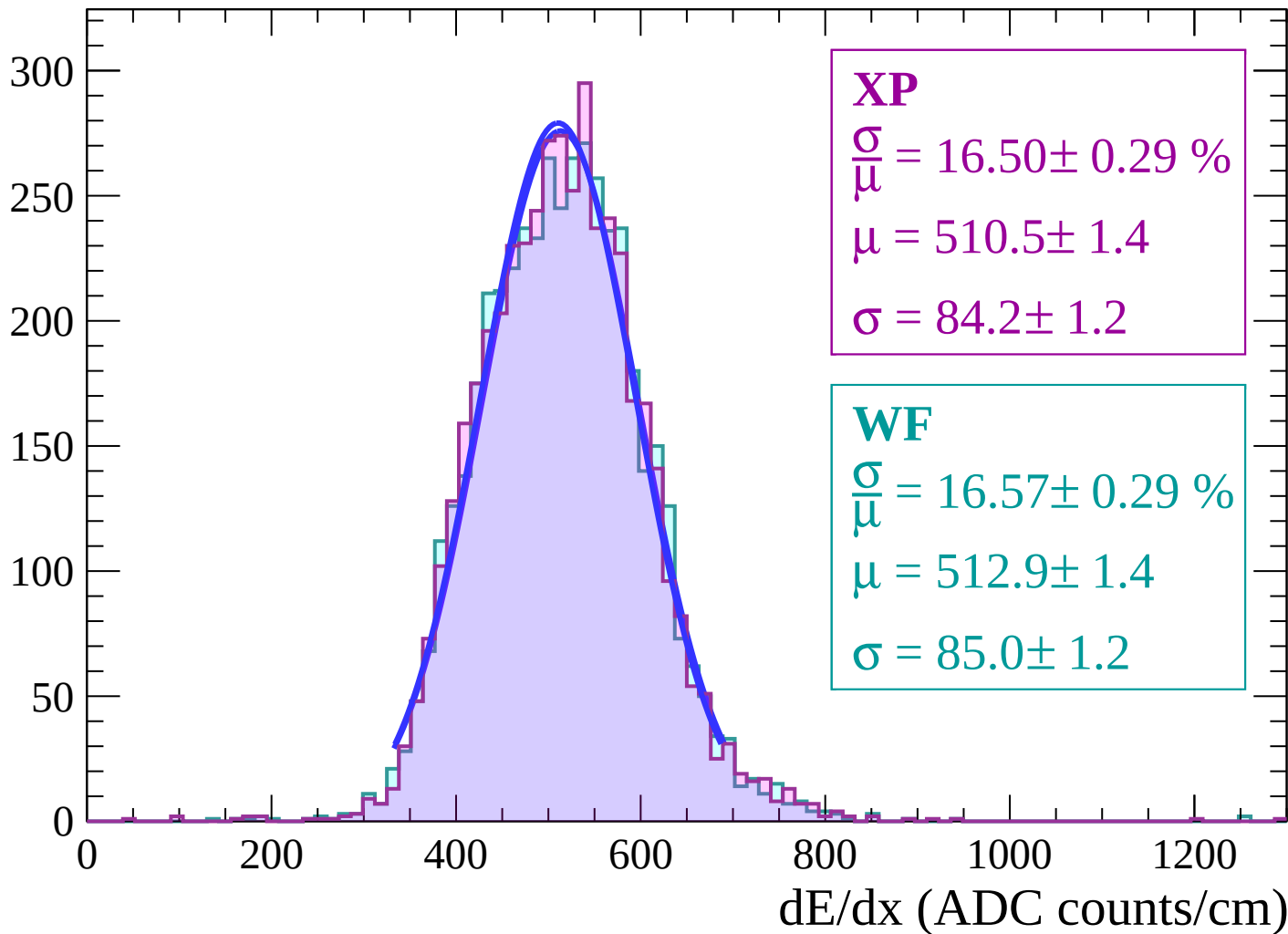
Energy loss | $38 < \theta < 40$

Count



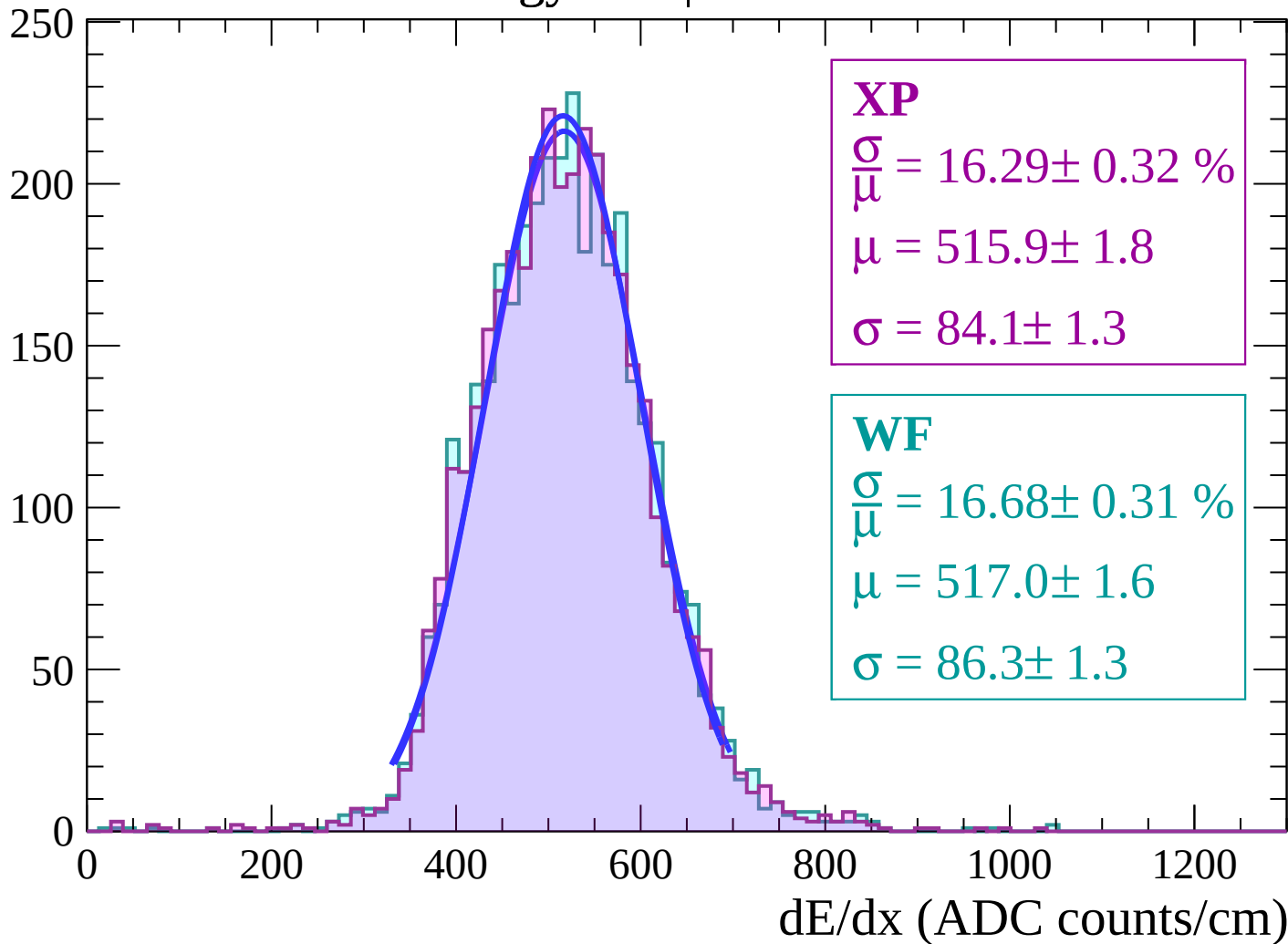
Energy loss | $40 < \theta < 42$

Count

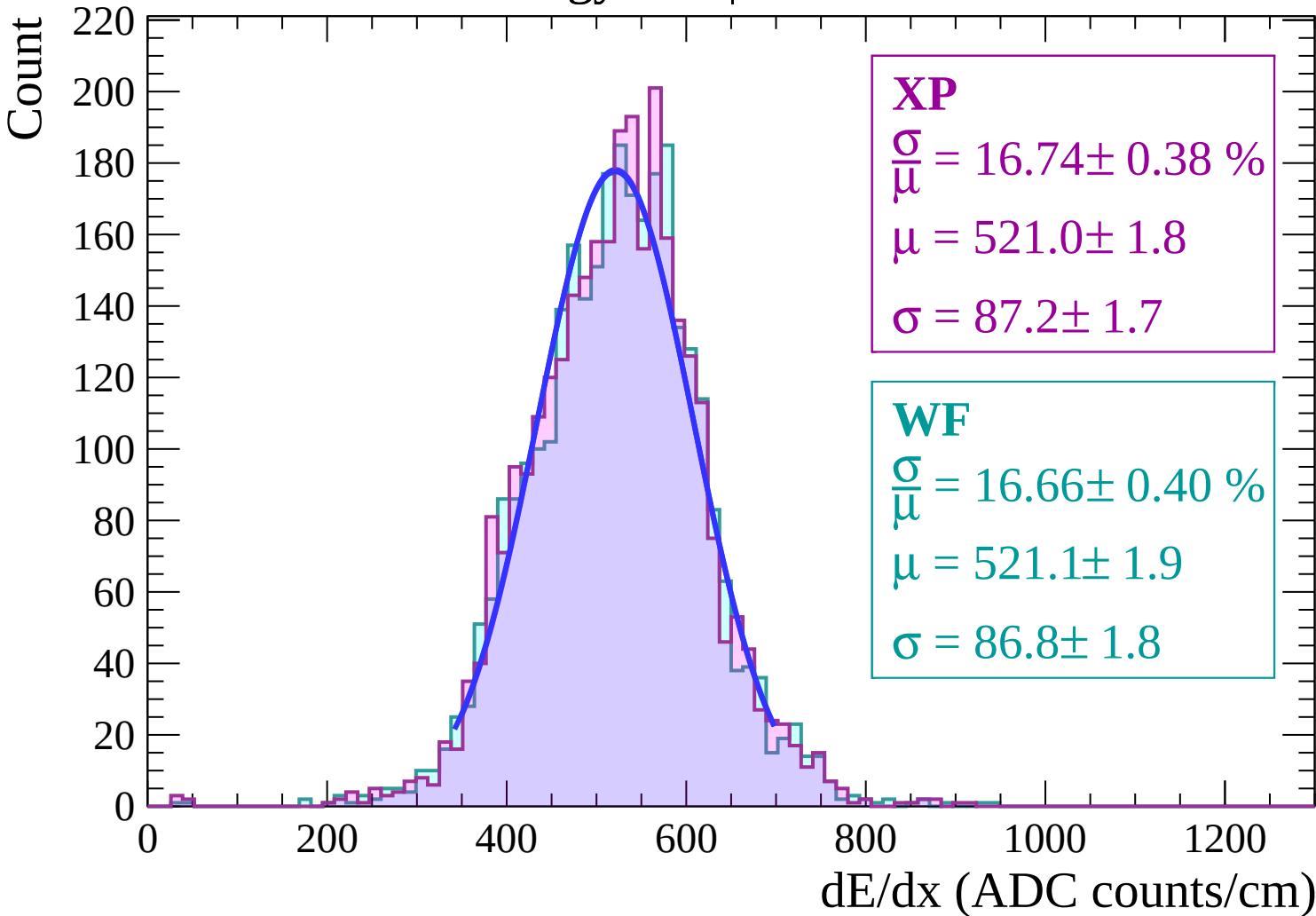


Energy loss | $42 < \theta < 44$

Count

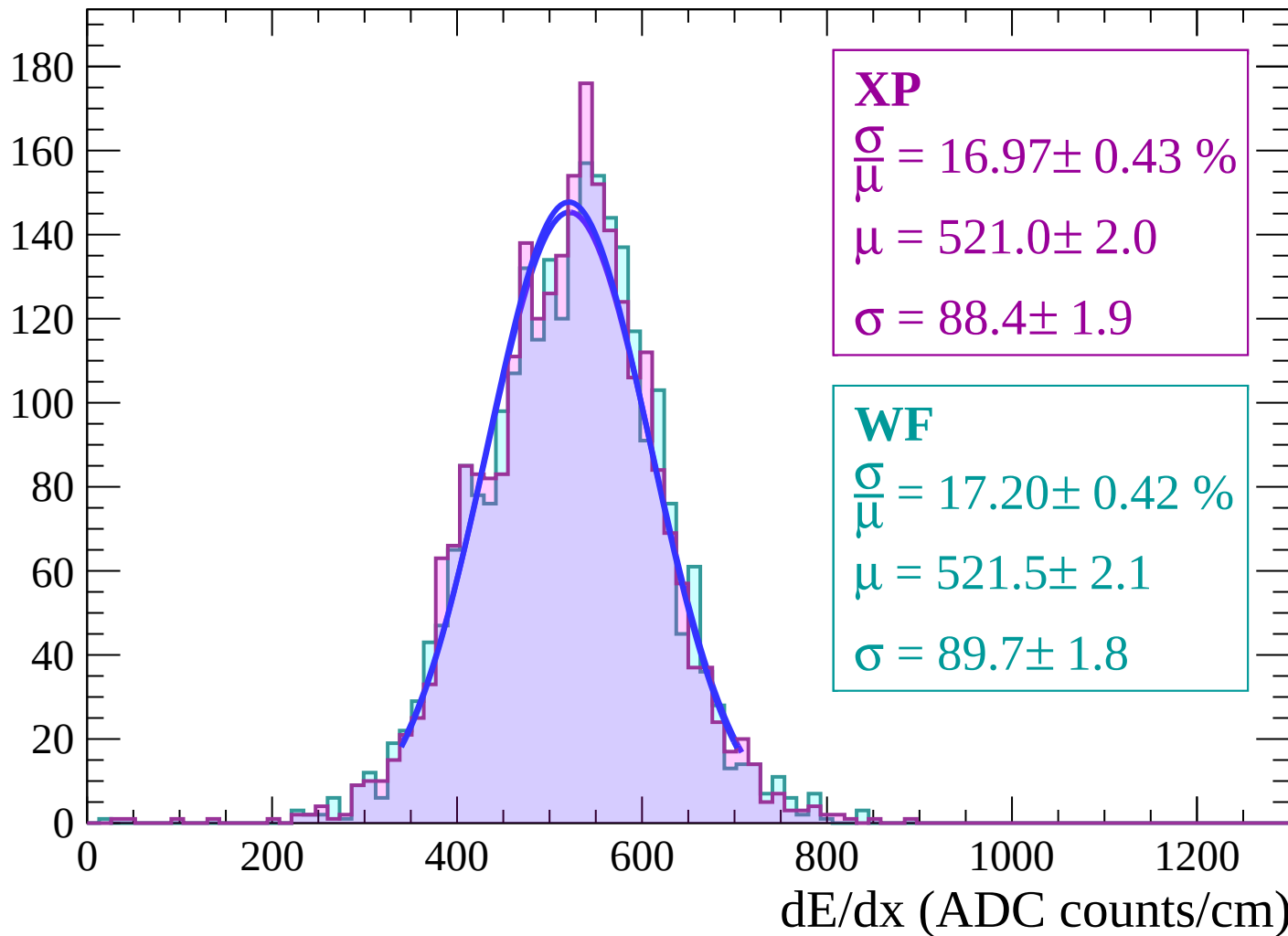


Energy loss | $44 < \theta < 46$



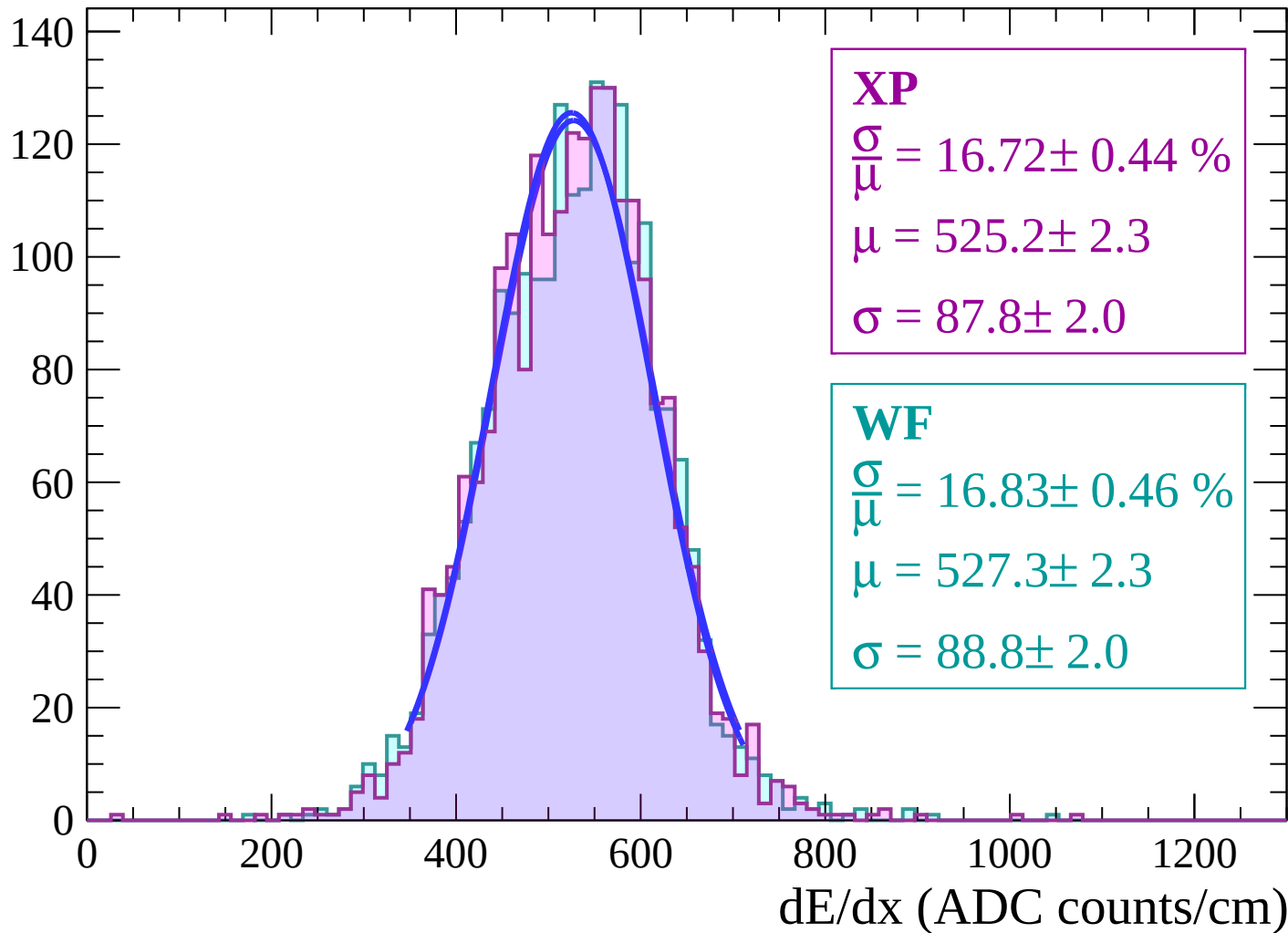
Energy loss | $46 < \theta < 48$

Count



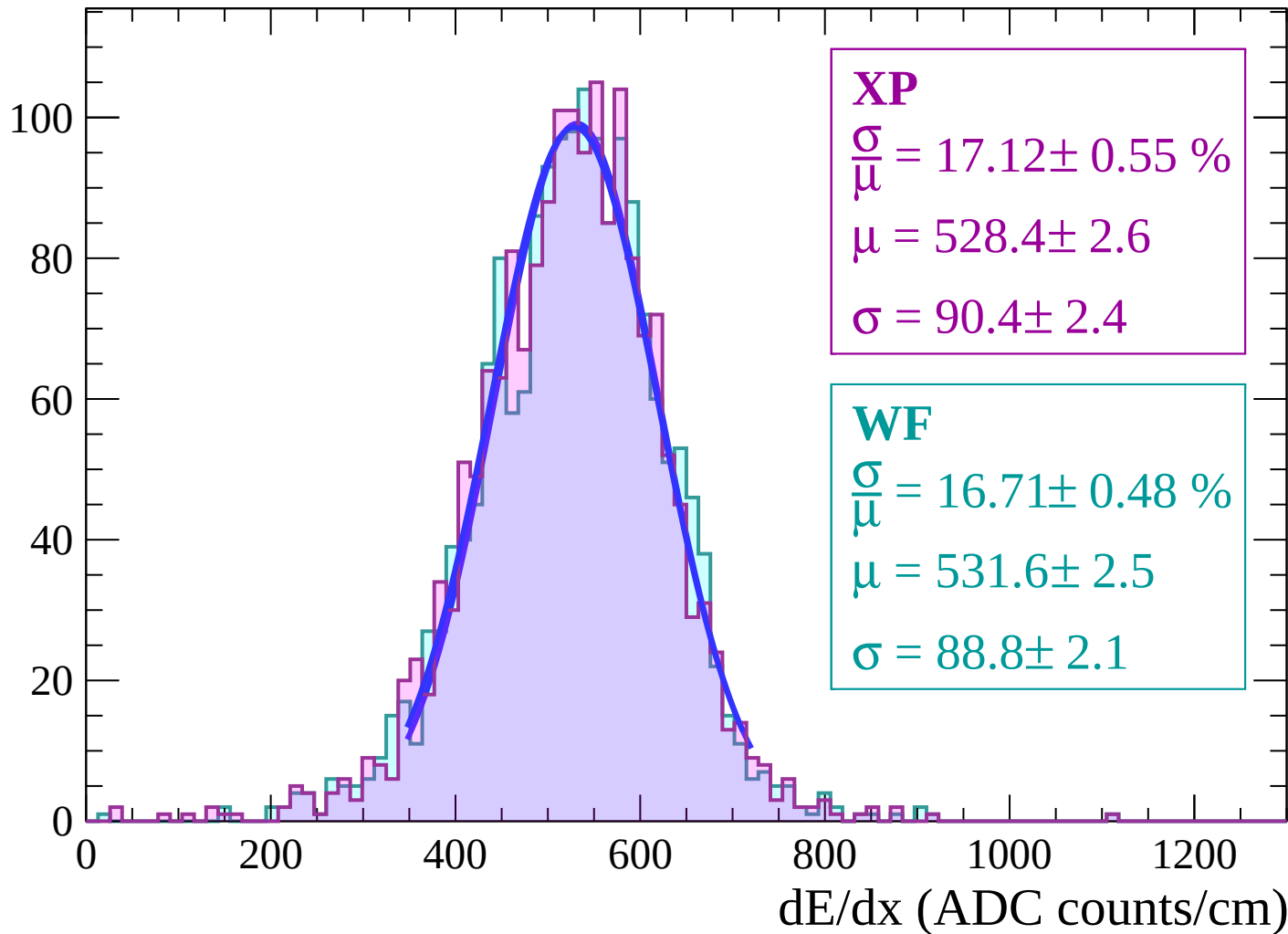
Energy loss | $48 < \theta < 50$

Count



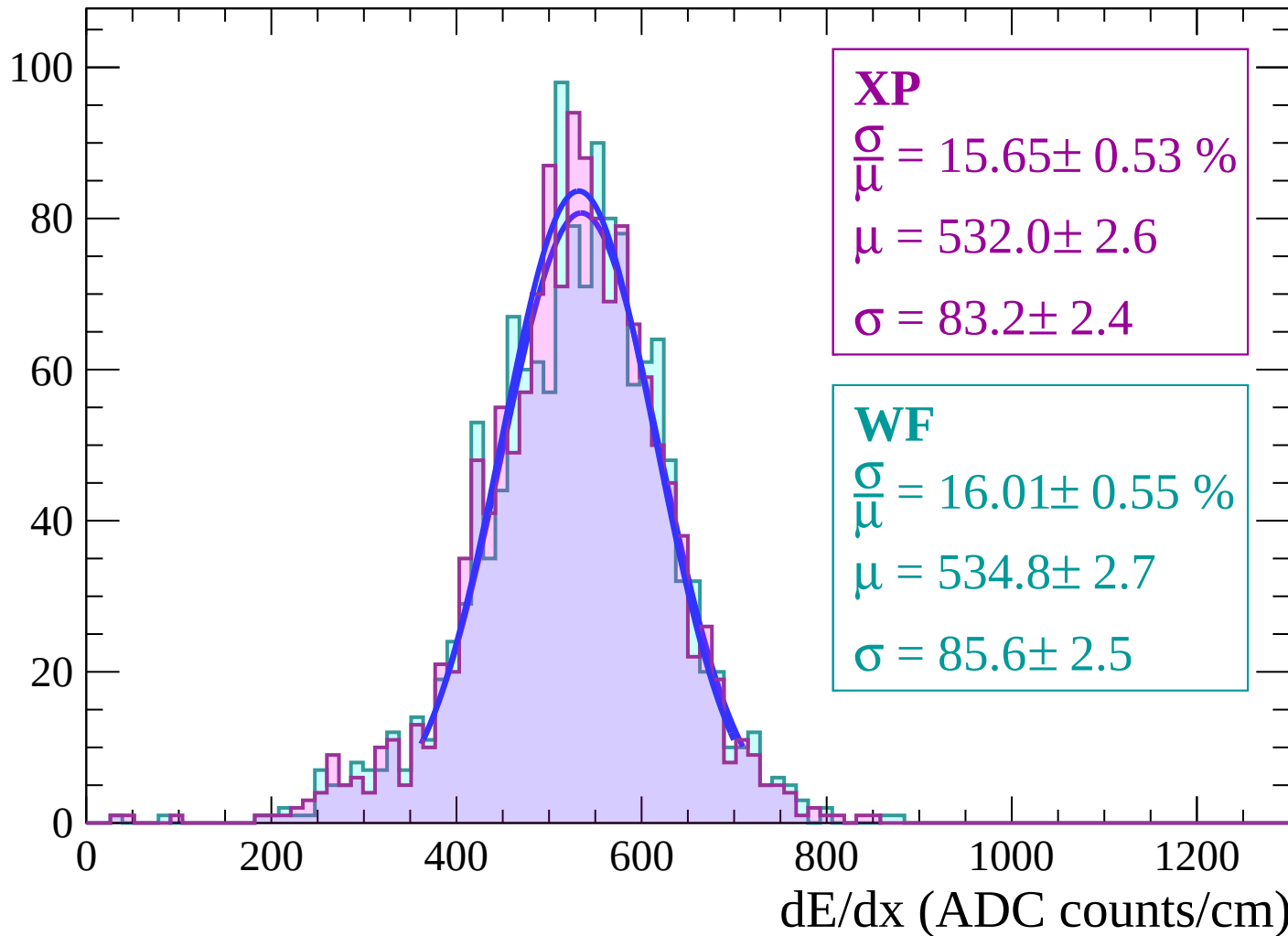
Energy loss | $50 < \theta < 52$

Count



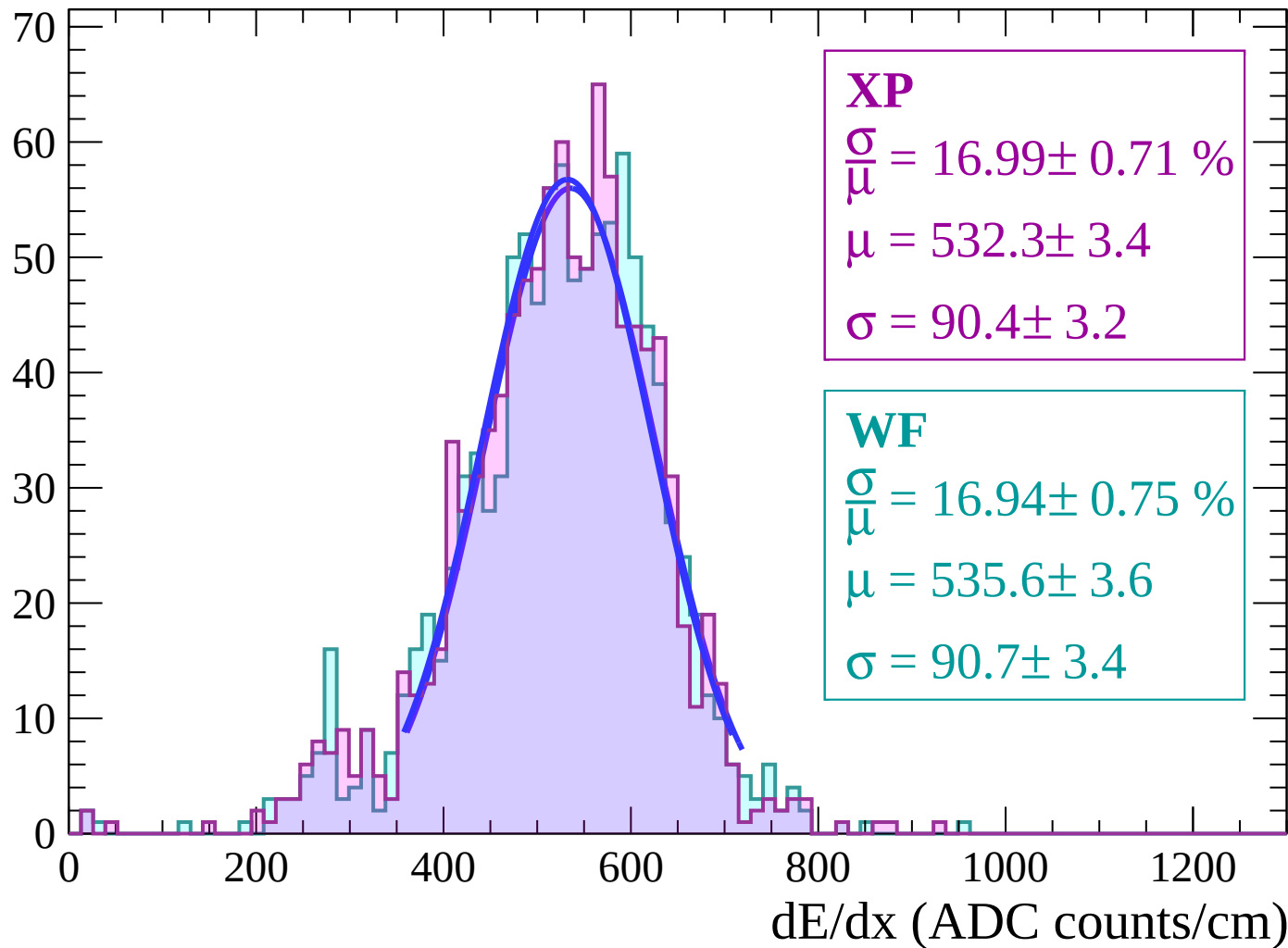
Energy loss | $52 < \theta < 54$

Count



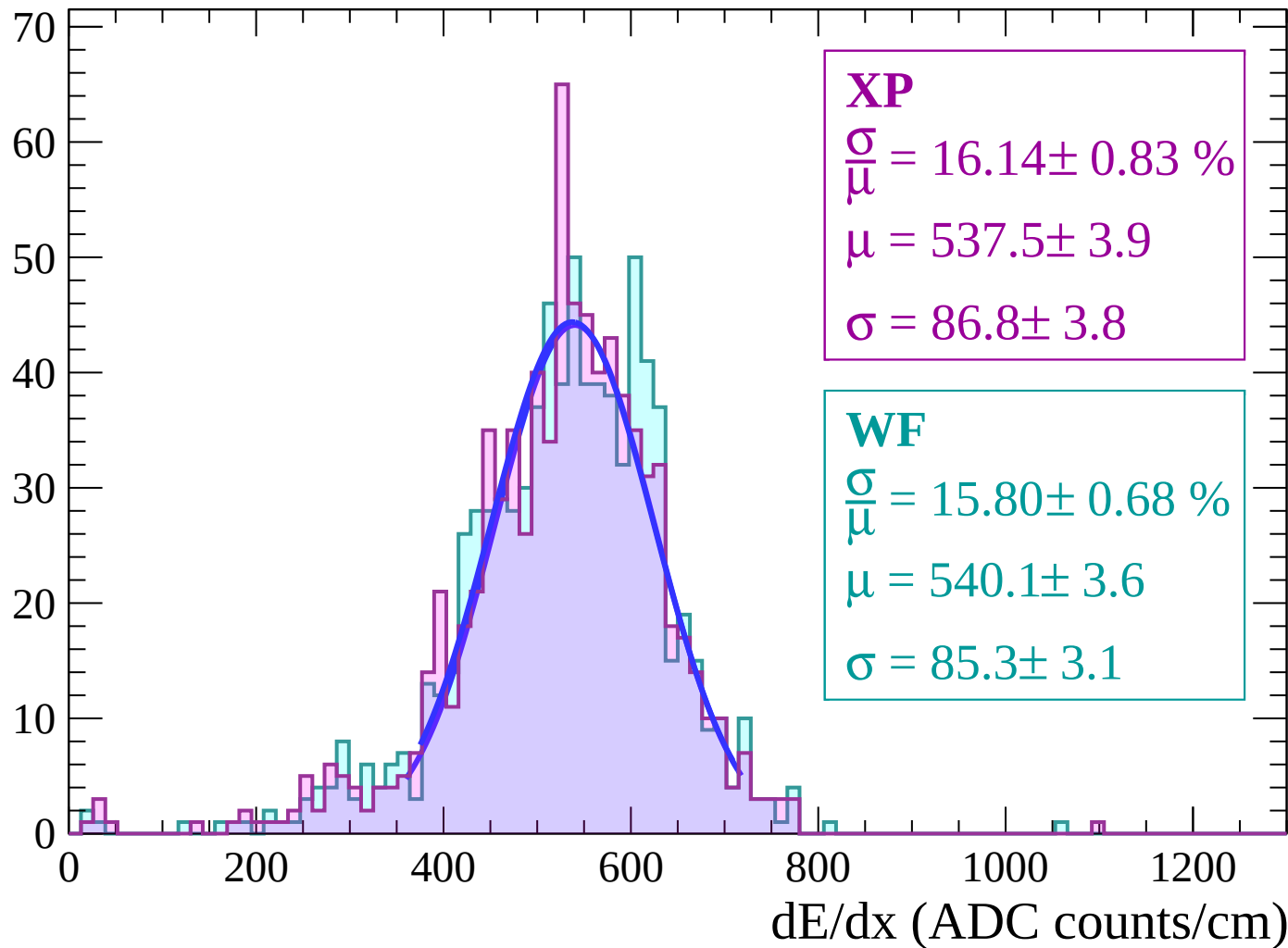
Energy loss | $54 < \theta < 56$

Count



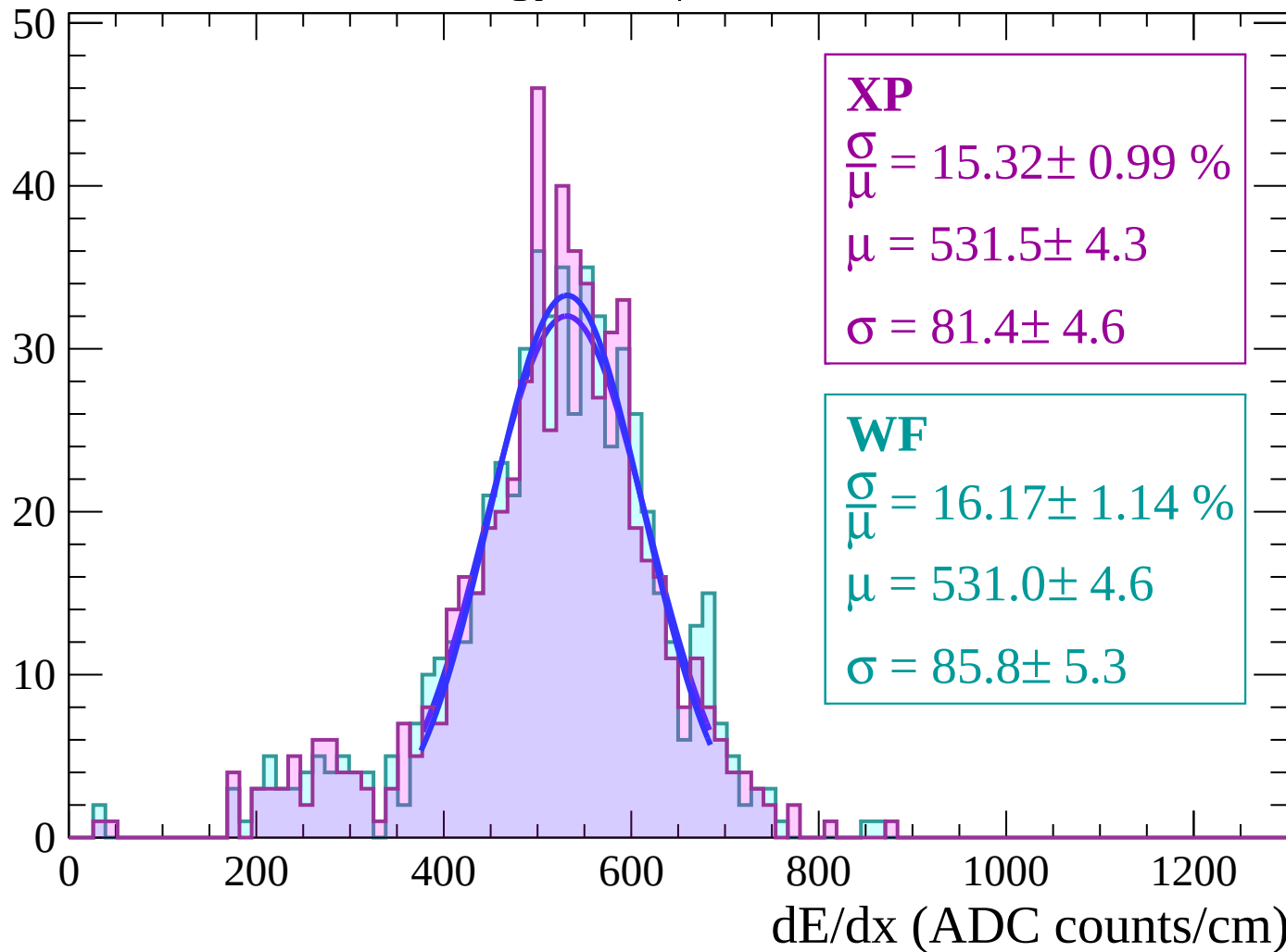
Energy loss | $56 < \theta < 58$

Count



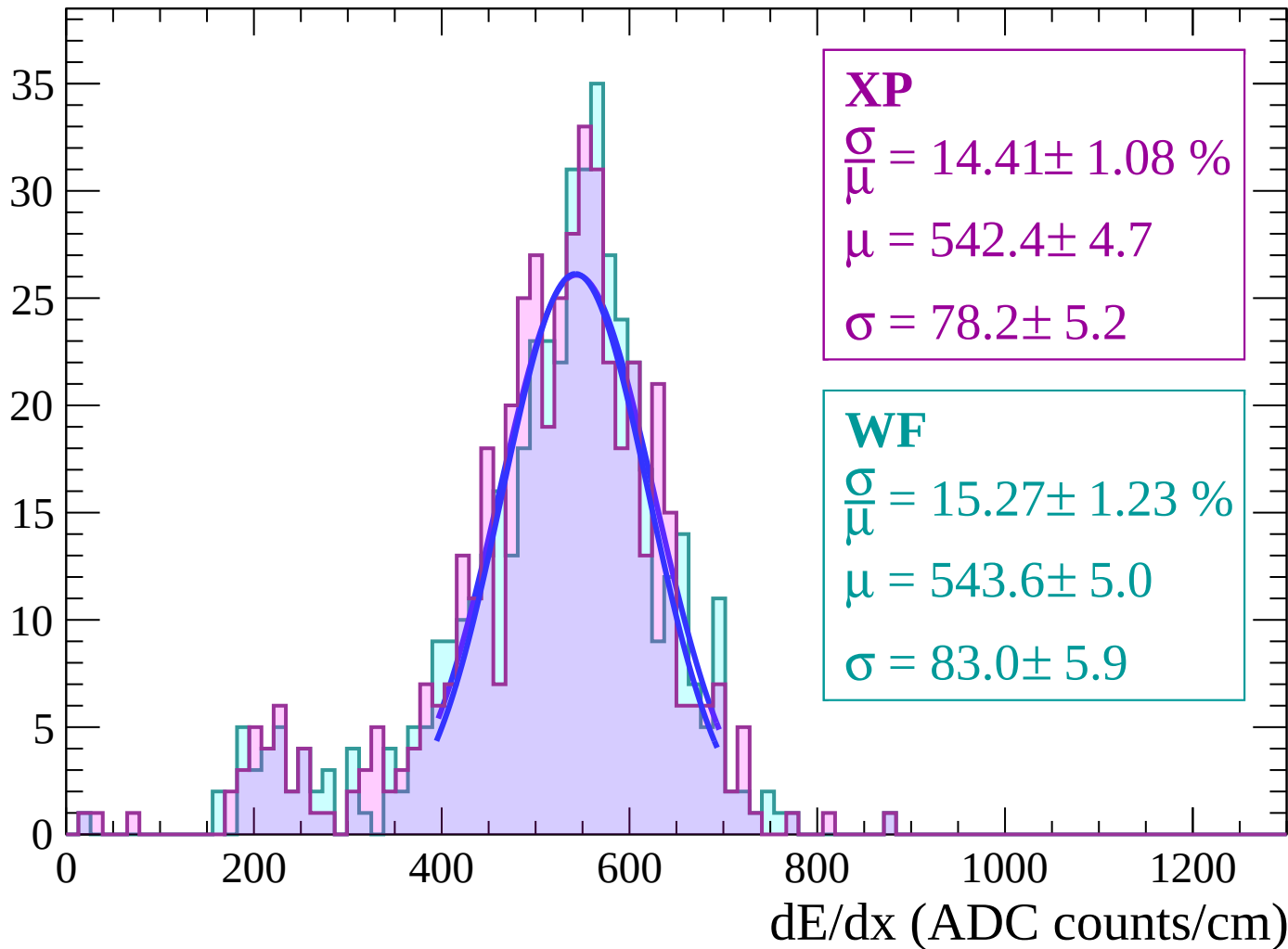
Energy loss | $58 < \theta < 60$

Count



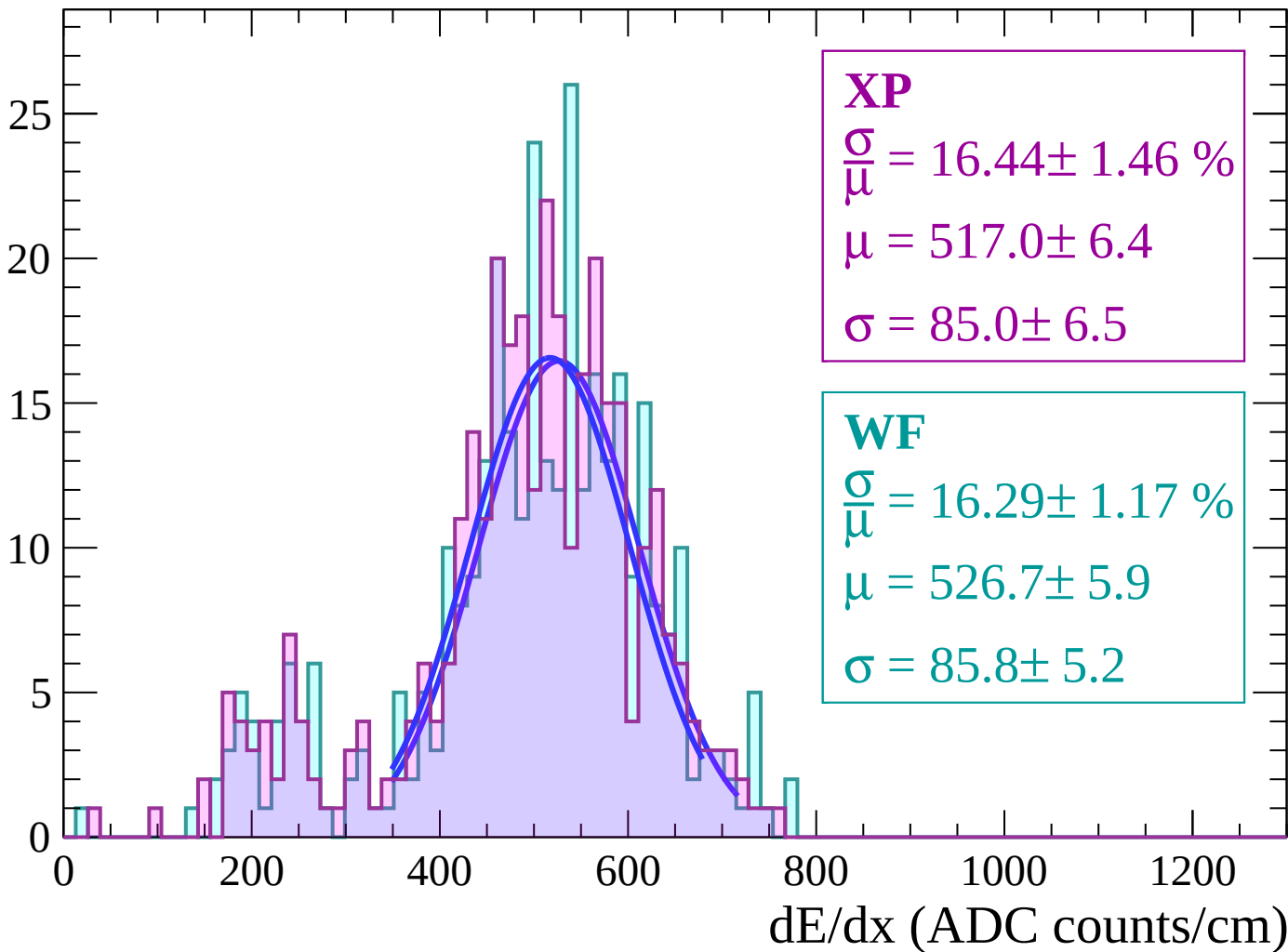
Energy loss | $60 < \theta < 62$

Count



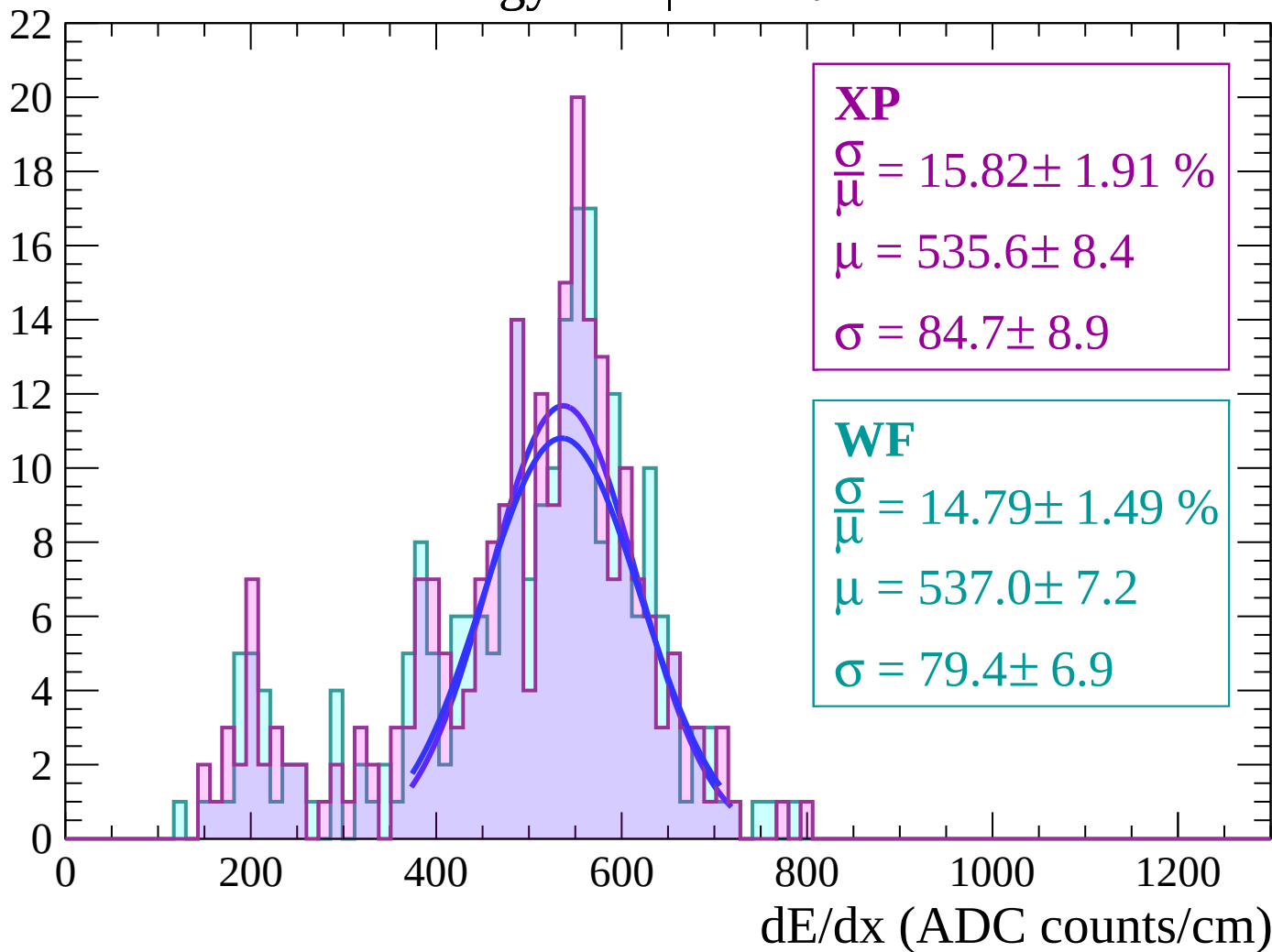
Energy loss | $62 < \theta < 64$

Count



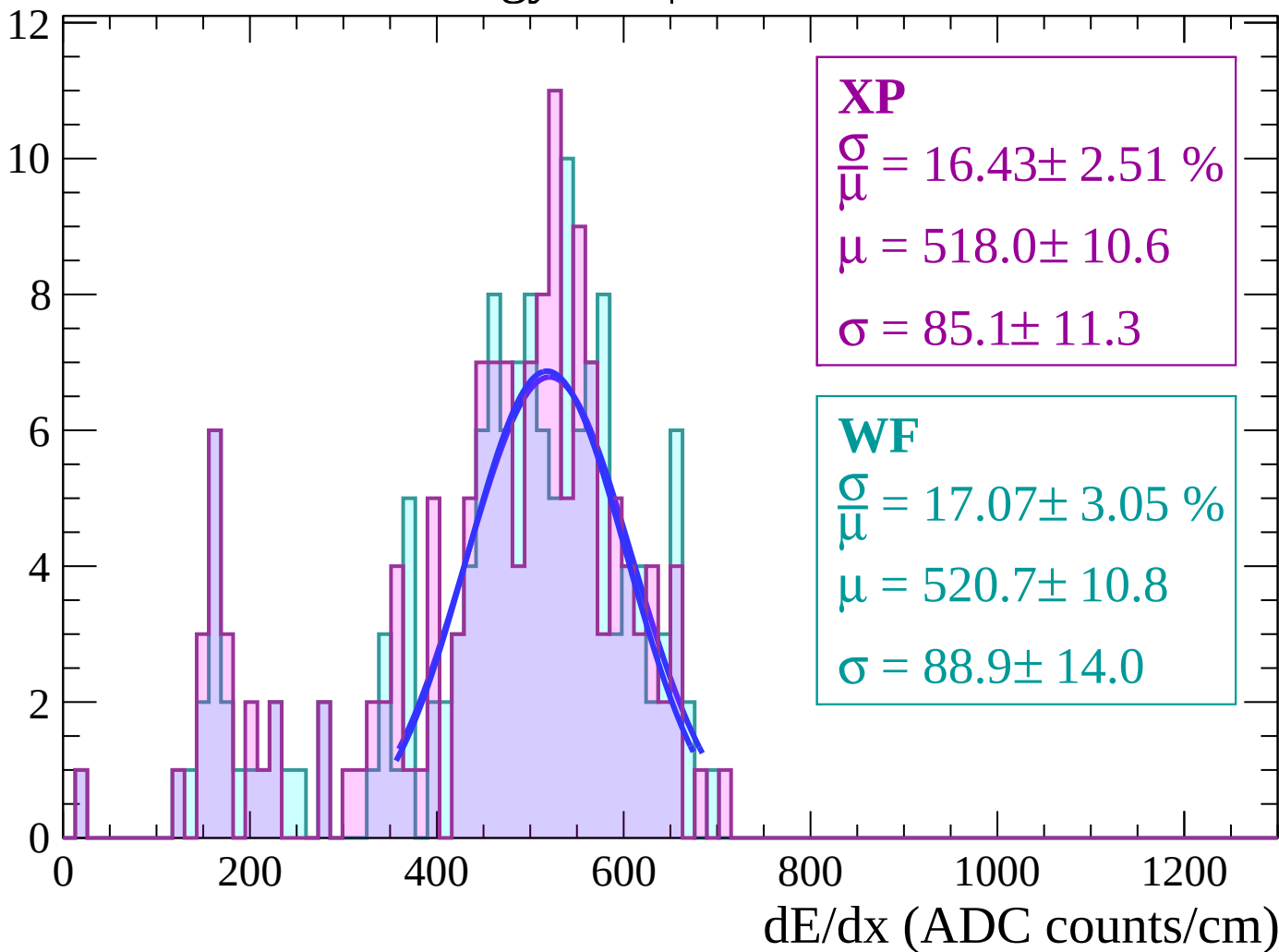
Energy loss | $64 < \theta < 66$

Count



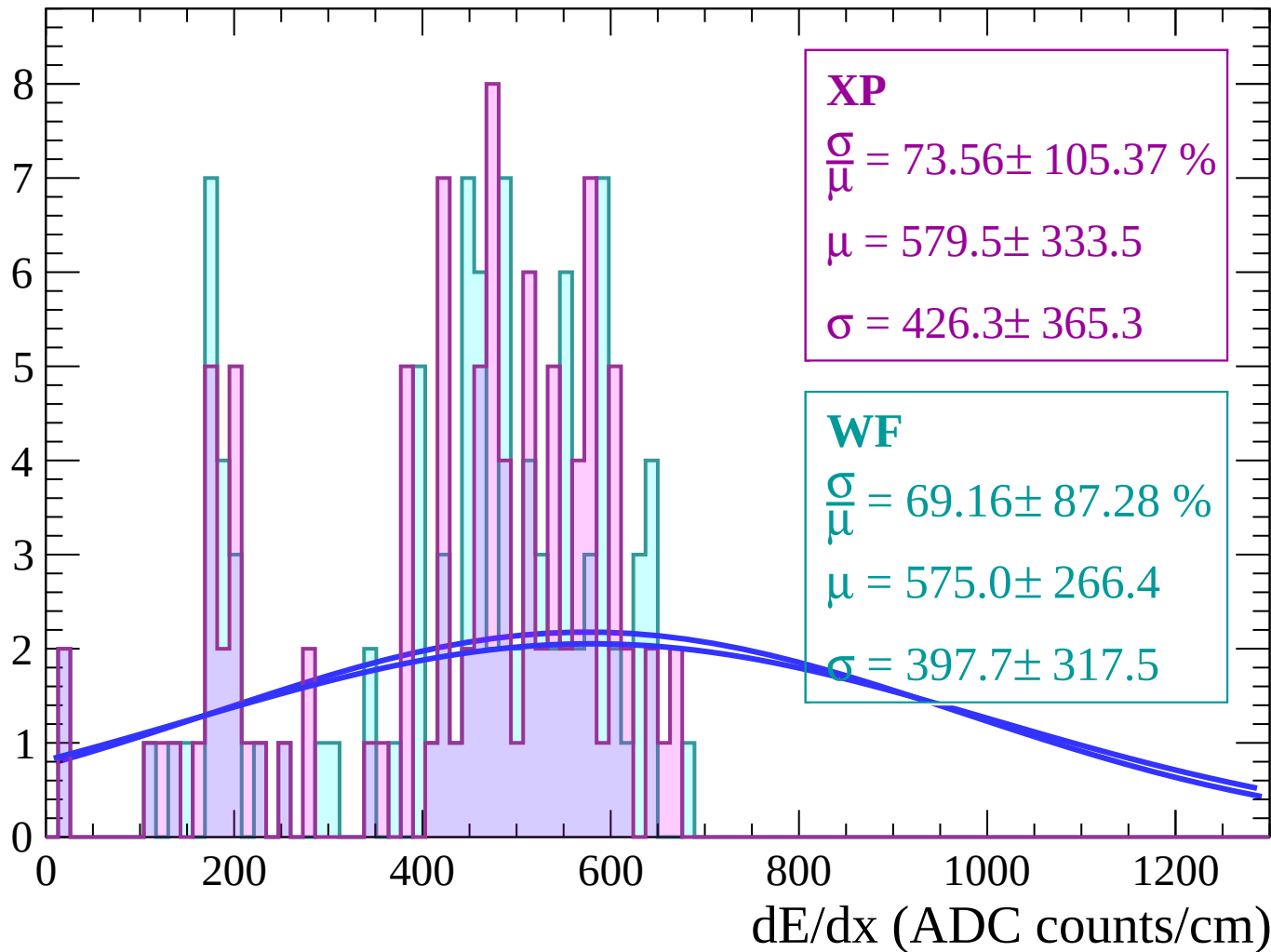
Energy loss | $66 < \theta < 68$

Count



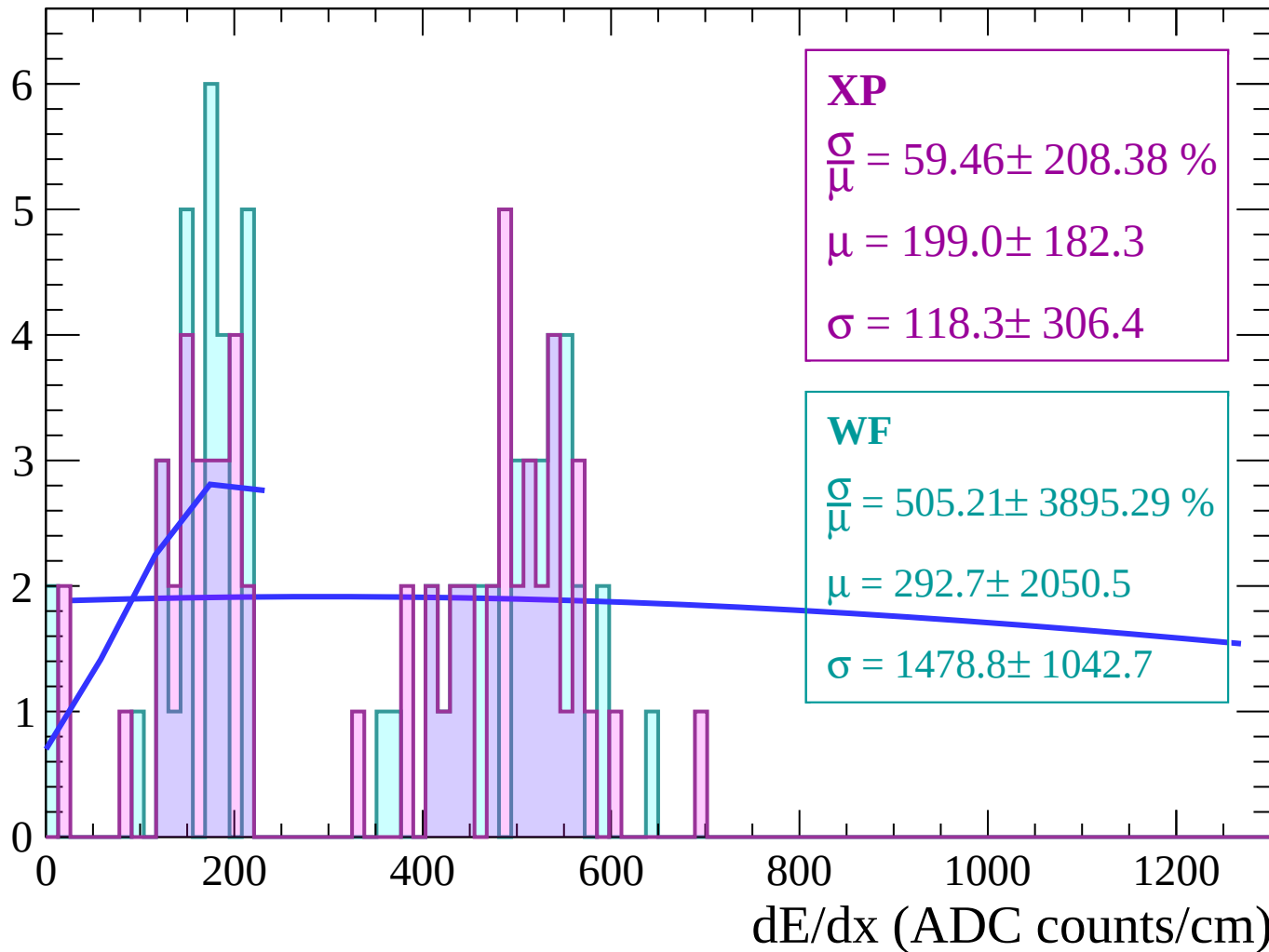
Energy loss | $68 < \theta < 70$

Count



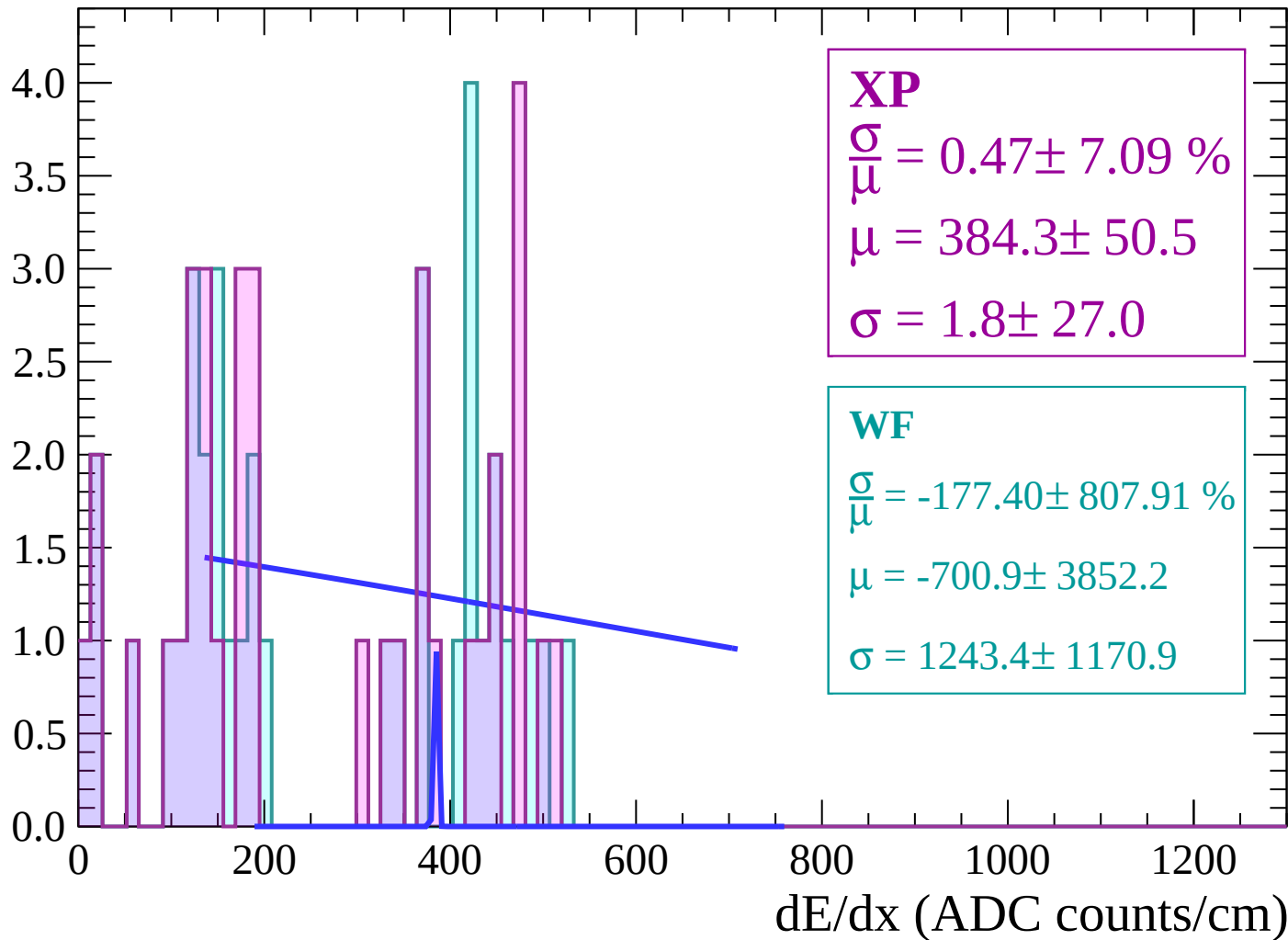
Energy loss | $70 < \theta < 72$

Count



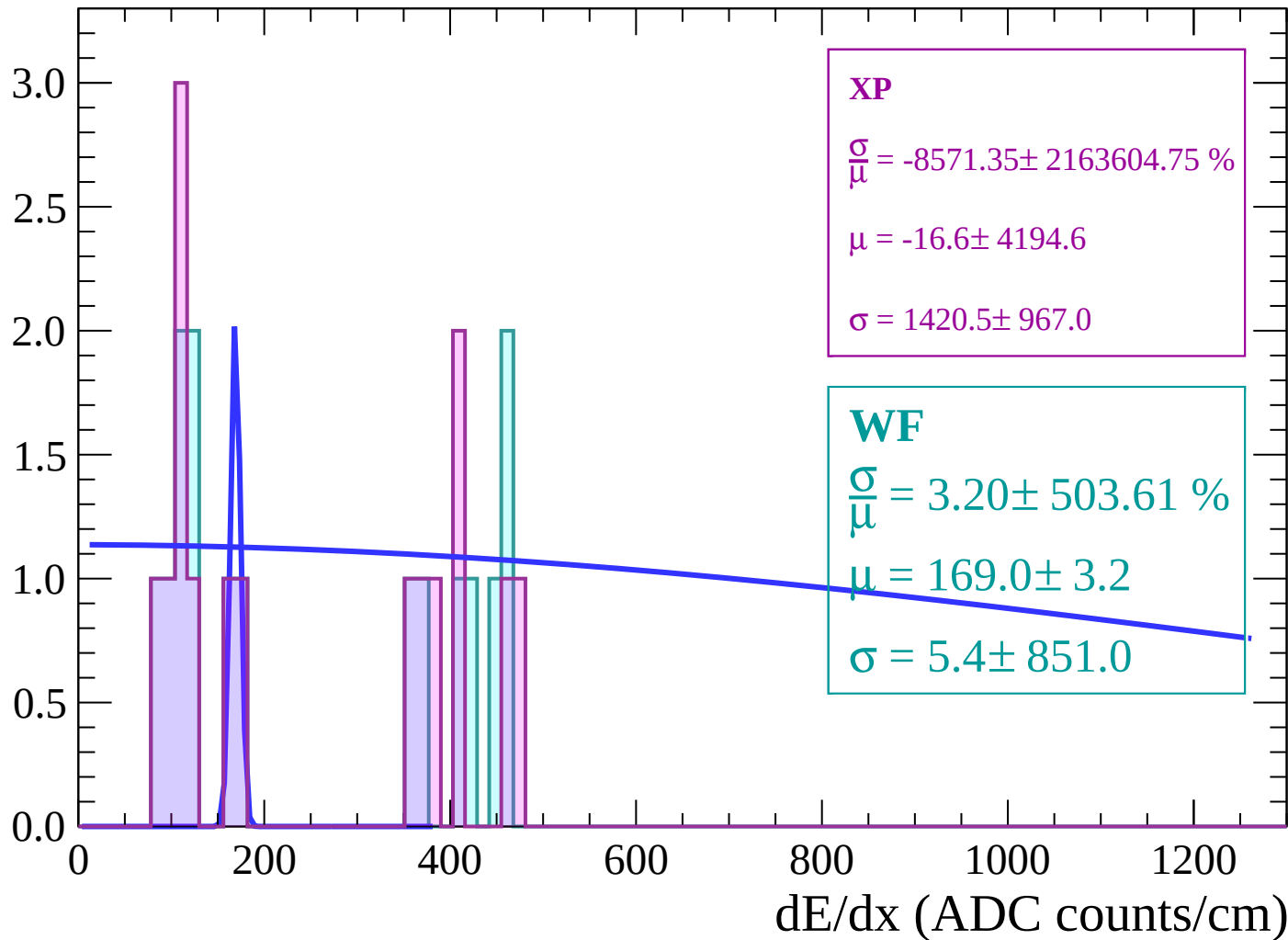
Energy loss | $72 < \theta < 74$

Count



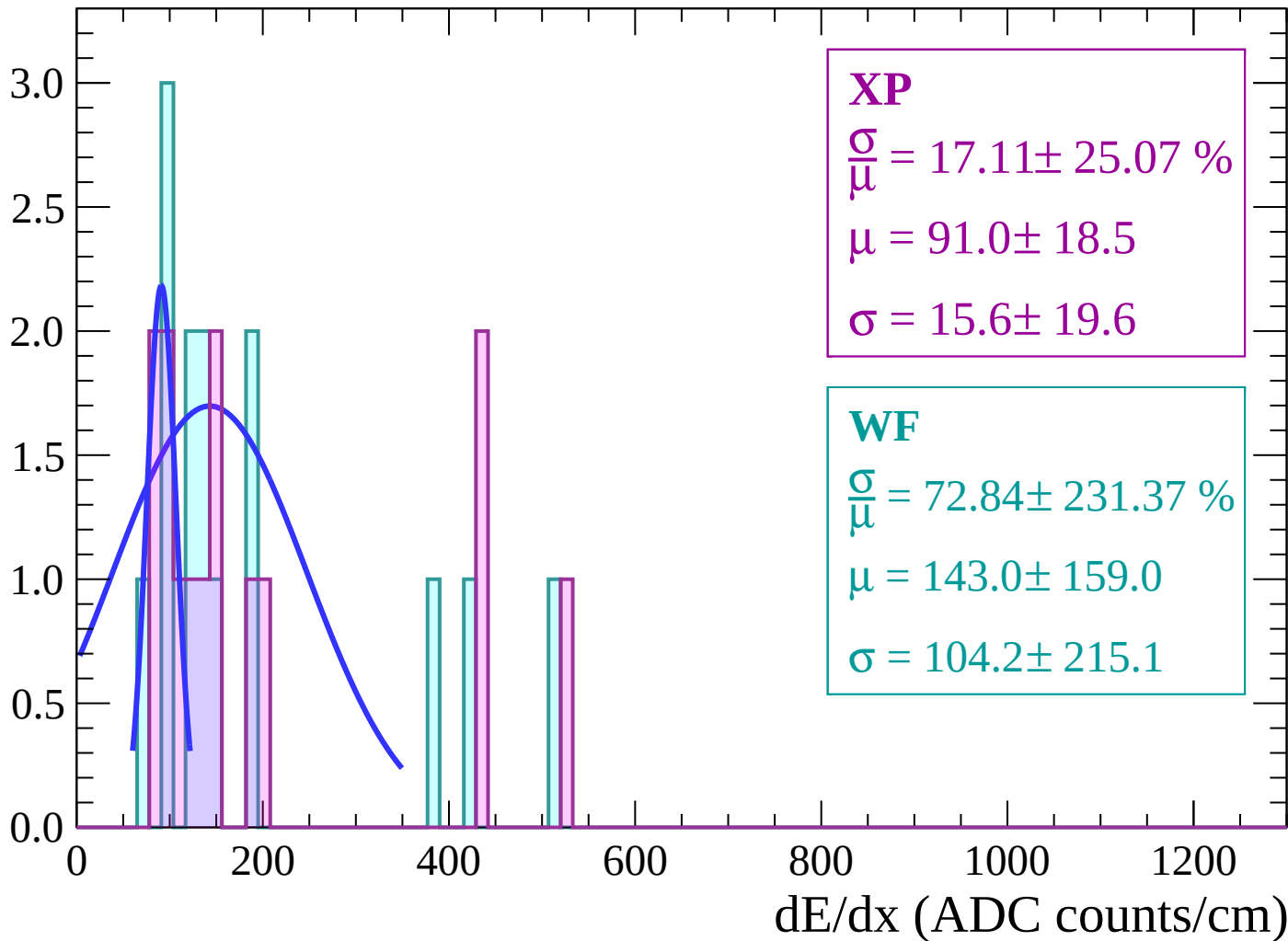
Energy loss | $74 < \theta < 76$

Count



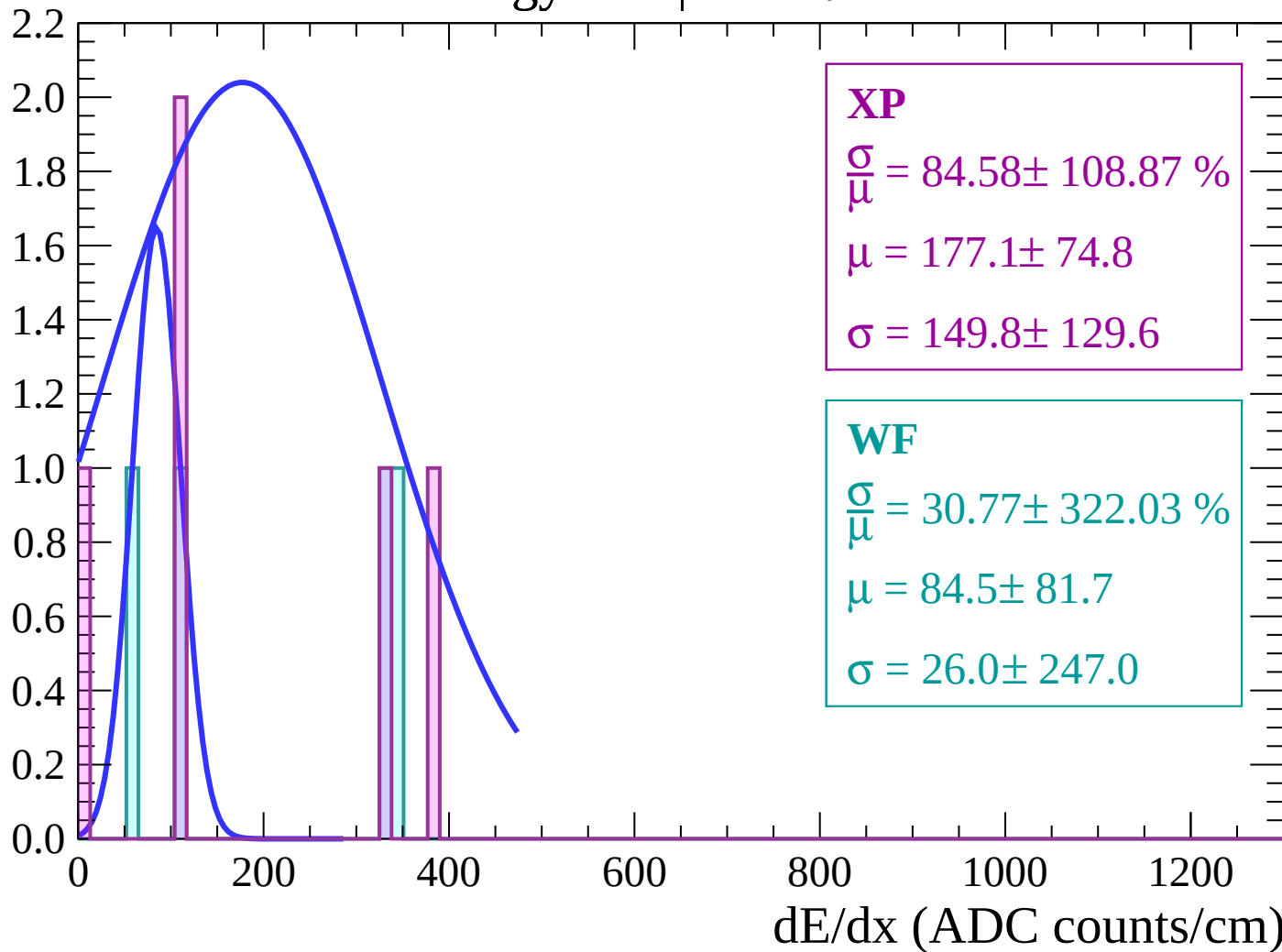
Energy loss | $76 < \theta < 78$

Count



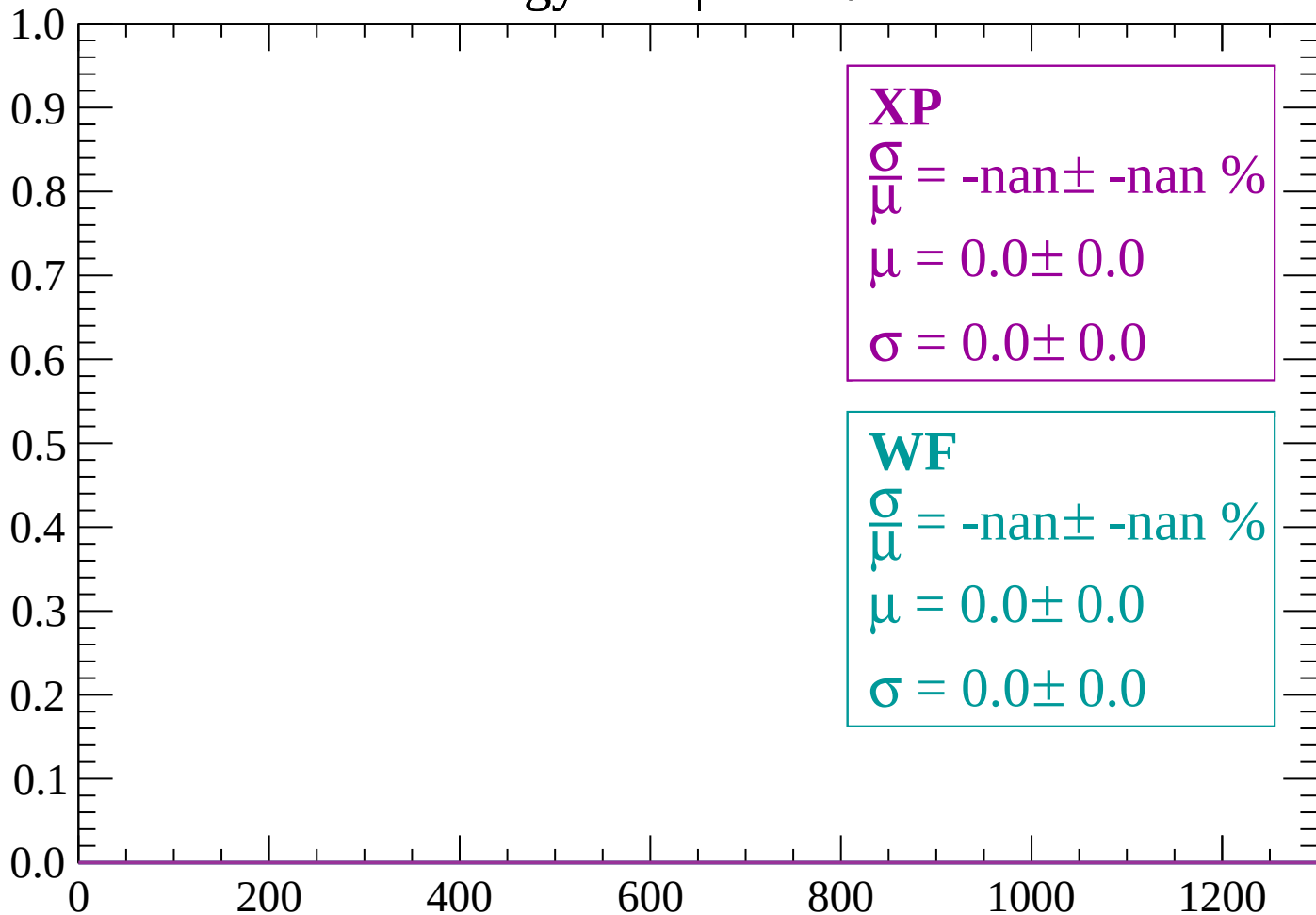
Energy loss | $78 < \theta < 80$

Count



Energy loss | $80 < \theta < 82$

Count



dE/dx (ADC counts/cm)

Energy loss | $82 < \theta < 84$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

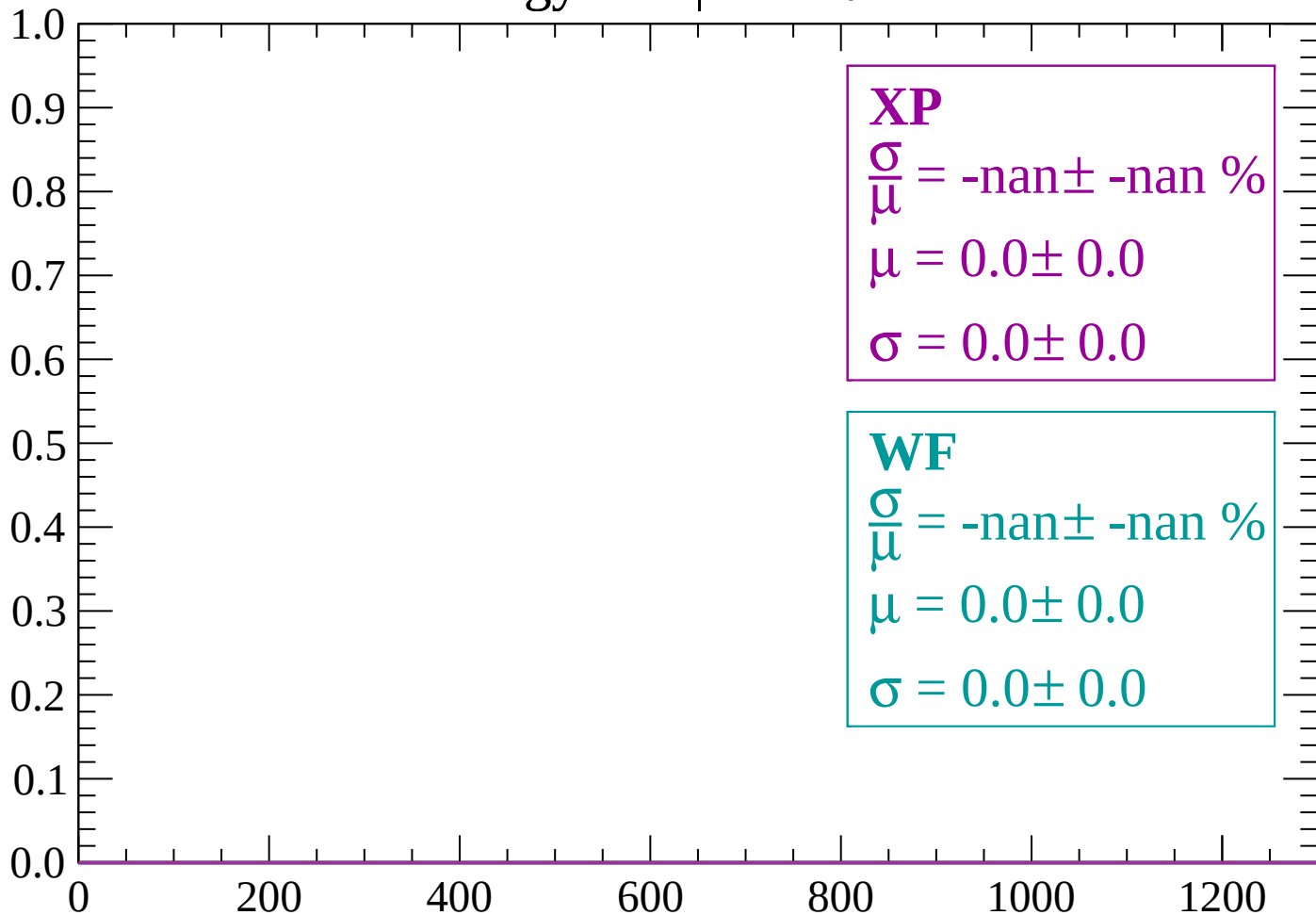
$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $84 < \theta < 86$

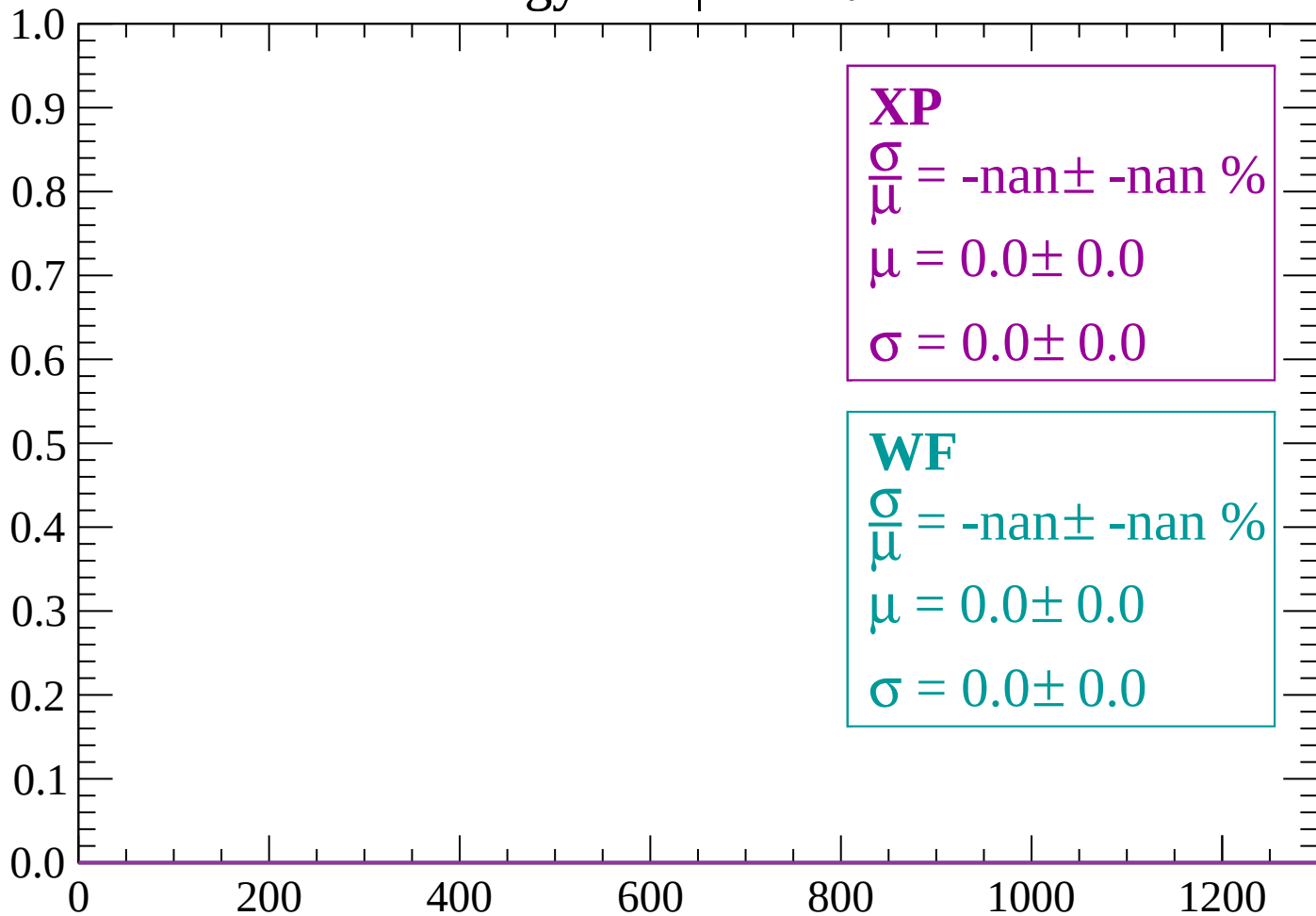
Count



dE/dx (ADC counts/cm)

Energy loss | $86 < \theta < 88$

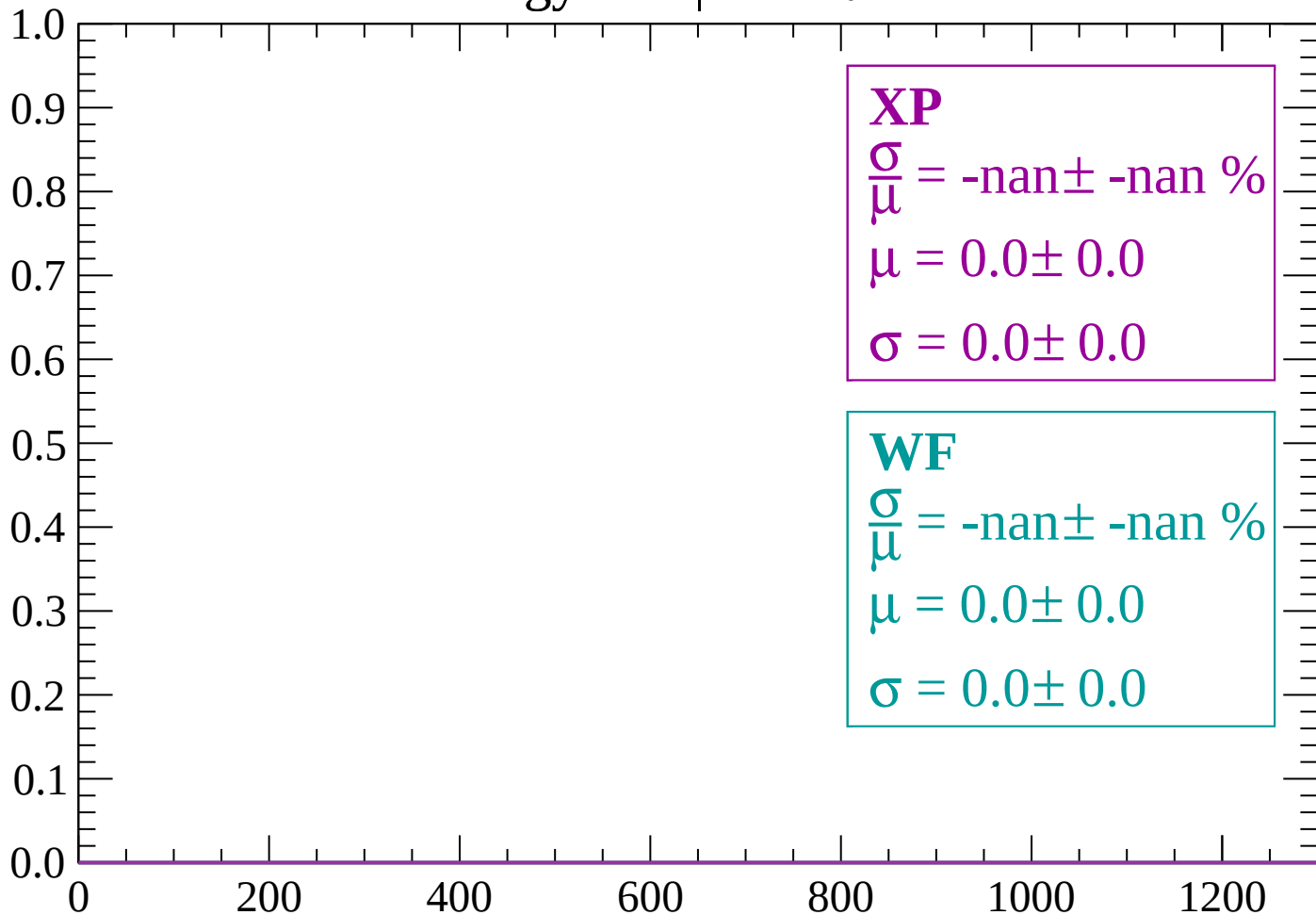
Count



dE/dx (ADC counts/cm)

Energy loss | $88 < \theta < 90$

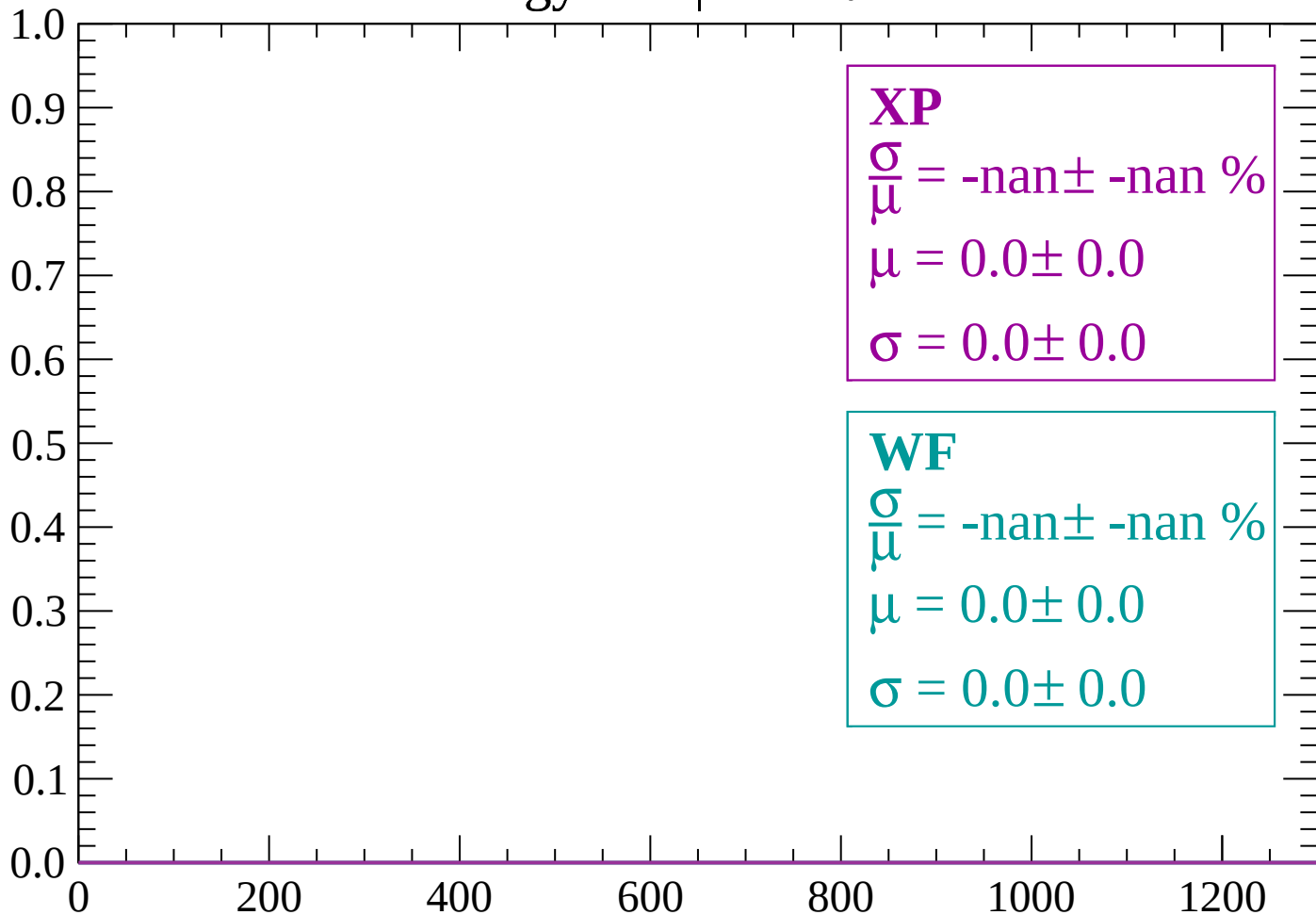
Count



dE/dx (ADC counts/cm)

Energy loss | $90 < \theta < 92$

Count



dE/dx (ADC counts/cm)

